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Corticosteroids for treating sepsis in children and adults (Review)

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[Intervention Review]

Corticosteroids for treating sepsis in children and adults

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ABSTRACT

Background

Sepsis occurs when an infection is complicated by organ failure. Sepsis may be complicated by impaired corticosteroid metabolism. Thus, providing corticosteroids may benefit patients. This is an update of a review originally published in 2004 and previously updated in 2010, 2015 and 2019.

Objectives

To examine the benefits and harms of corticosteroids in children and adults with sepsis.

Search methods

We searched CENTRAL, MEDLINE, Embase, LILACS, ClinicalTrials.gov, ISRCTN and the WHO Clinical Trials Search Portal on 31 December 2023. In addition, we conducted reference checking and citation research, and contacted study authors, to identify additional studies as needed. We updated this search in December 2024, but these results have not yet been incorporated.

Selection criteria

We included randomised controlled trials (RCTs) of corticosteroids versus placebo or usual care (antimicrobials, fluid replacement and vasopressor therapy as needed) in children and adults with sepsis. We also included RCTs of continuous infusion versus intermittent bolus of corticosteroids.

Data collection and analysis

We used the same methods in comparisons of corticosteroids versus placebo or usual care, and of continuous infusion versus intermittent bolus administration of corticosteroids.

The primary outcome was all-cause mortality at 28 days. The most critical secondary outcomes were (i) all-cause mortality in the long term (last follow-up from 90 days to one year) and in the hospital; (ii) length of stay in the intensive care unit and in hospital; (iii) adverse effects, i.e. superinfection and muscle weakness (within 28 days).

All review authors screened and selected studies for inclusion. One review author extracted data, which was checked by the others, and by the lead author of the primary study when possible. For this update, we used Covidence software for screening and selection of studies and abstraction of data by paired review authors, with discrepancies resolved by a third review author. We obtained unpublished data from the authors of some trials.

We assessed the risk of bias in trials using the Cochrane risk of bias tool (RoB 1) and applied GRADE to assess the certainty of evidence.

The review authors did not contribute to the assessment of eligibility or risk of bias, nor to data extraction, for the trials they had participated in.

Main results

We included 87 trials (24,336 participants), of which six included only children, two included children and adults, and the remaining trials included only adults. Seventeen additional trials are ongoing and will be considered in future versions of this review. We judged 25 trials as being at low risk of bias.

Corticosteroids versus placebo or usual care

Compared to placebo or usual care, corticosteroids probably reduce 28-day mortality (risk ratio (RR) 0.89, 95% confidence interval (CI) 0.84 to 0.95; 72 trials, 22,915 participants; moderate-certainty evidence). We downgraded the certainty of evidence for this outcome from high to moderate for inconsistency (significant heterogeneity across trial results). Corticosteroids may result in little to no difference in long-term mortality (RR 0.97, 95% CI 0.91 to 1.03; 12 trials, 8468 participants; low-certainty evidence) and probably reduce in-hospital mortality (RR 0.90, 95% CI 0.84 to 0.97; 40 trials, 17,459 participants; moderate-certainty evidence). Corticosteroids may reduce length of intensive care unit (ICU) stay for all participants (mean difference (MD) -0.86 days, 95% CI -1.67 to -0.05; 25 trials, 8069 participants; low-certainty evidence) and may reduce length of hospital stay for all participants (MD -1.09 days, 95% CI -1.85 to -0.34; 31 trials, 16,954 participants; low-certainty evidence). The evidence is uncertain about the effect of corticosteroids on the risk of muscle weakness (RR 1.09, 95% CI 0.78 to 1.53; 7 trials, 6729 participants; very low-certainty evidence). Corticosteroids may result in little to no difference in the risk of superinfection (RR 0.96, 95% CI 0.86 to 1.07; 36 trials, 7961 participants; low-certainty evidence).

Continuous infusion of corticosteroids versus intermittent bolus

Four trials reported data for this comparison, and the certainty of evidence for all outcomes was very low. We are uncertain about the effects of continuous infusion of corticosteroids compared with intermittent bolus administration on 28-day mortality (RR 1.03, 95% CI 0.81 to 1.32; 3 trials, 310 participants). We downgraded the certainty of evidence to very low due to high risk of bias in all except one trial and due to imprecision. Compared to bolus administration, we are uncertain of the effects of continuous infusion of corticosteroids on long-term mortality (RR 1.36, 95% CI 1.02 to 1.81; 1 trial, 70 participants; very low-certainty evidence), in-hospital mortality (RR 0.92, 95% CI 0.71 to 1.19; 3 trials, 352 participants; very low-certainty evidence), ICU length of stay amongst all participants (MD -0.56 days, 95% CI -3.44 to 2.32; 4 trials, 422 participants; very low-certainty evidence), hospital length of stay amongst all participants (MD -0.21 days, 95% CI -4.72 to 4.30; 4 trials, 422 participants; very low-certainty evidence), risk of muscle weakness (RR 0.89, 95% CI 0.13 to 5.98; 1 trial, 70 participants; very low-certainty evidence) and risk of superinfection (RR 1.12, 95% CI 0.37 to 3.33; 2 trials, 193 participants; very low-certainty evidence).

Authors' conclusions

Moderate-certainty evidence indicates that corticosteroids probably reduce 28-day, 90-day and hospital mortality amongst patients with sepsis. Corticosteroids may shorten ICU and hospital length of stay (low-certainty evidence). There may be little or no difference in the risk of superinfection. The risk of muscle weakness is uncertain. The effects of continuous versus intermittent bolus administration of corticosteroids are uncertain.

PLAIN LANGUAGE SUMMARY

What are the benefits and harms of corticosteroids in the treatment of children and adults with sepsis?

Key messages

• Compared to placebo or usual care, corticosteroids probably reduce the risk of death at 28 days and of in-hospital death, and may have little or no effect on the risk of dying beyond three months.

• We are uncertain about the effects of continuous intravenous infusion of corticosteroids compared with intermittent intravenous bolus (rapid injection) administration.

• Future studies should measure the benefits and harms of corticosteroids for the treatment of specific populations, including children, patients with sepsis without shock or with a mild form of septic shock, patients with acute respiratory distress syndrome (ARDS) and patients with different types of infections. Defining the optimal methods for corticosteroid administration, including the types of molecule - with or without producing sodium and fluid retention and potassium excretion (mineralocorticoid activity), timing of initiation, dose and duration, continuous infusion or intermittent intravenous boluses, and ending of treatment - with or without gradually decreasing the dose to zero, also requires additional studies. Future studies should measure the longer-term effects, i.e. longer than one year, on survival and complications of sepsis.

What is sepsis?

Sepsis is present when an infection is complicated by organ failure. People develop rapid breathing, hypotension (low blood pressure) and mental confusion. Sepsis can interfere with the effectiveness of the body's corticosteroids, which control how your body recognises and defends itself against bacteria, viruses and substances that appear foreign and harmful.

What are corticosteroids (medicine)?

Corticosteroids are medicines used to stop excessive inflammation. However, these medicines may cause unwanted harms, such as increased blood sugar or salt levels, infections, bleeding in the stomach, agitation or delirium, and weakness of the limbs. Corticosteroids have been given for decades to people with infection resulting from various causes.

What did we want to find out?

We wanted to find out:

• the benefits of using corticosteroids in children and adults with sepsis; and

• the best ways to administer this treatment (including the best people to give it to).

We also wanted to know if corticosteroids can cause any unwanted effects.

What did we do?

We searched for studies that compared:

• corticosteroids administered intravenously against a 'dummy' treatment, or sham treatment, which does not contain any medicine but looks or tastes identical to corticosteroids, or usual care; or

• continuous infusion against intermittent rapid intravenous injections (boluses) of corticosteroids administration.

We compared and summarised their results, and rated our confidence in the evidence, based on factors such as study methods and sizes.

What did we find?

This review includes 87 trials (24,336 participants). The biggest study was of 3800 people and the smallest study was of 21 people. Eighty-three trials compared corticosteroids to no corticosteroids (placebo or usual care in 59 and 24 trials, respectively). Four trials also compared continuous administration versus rapid intravenous injection (i.e. bolus) of corticosteroids given at fixed intervals of six or eight hours. Thirty-three trials were conducted in Europe, 15 in North America, 22 in Asia, six in the Middle East, six in Africa and three in Latin America; four were multinational trials.

Three trials were funded by a drug company, 27 by public organisations or through charitable funding and six by both a drug company and public organisations or charitable funding; 25 did not declare the source of funding.

Compared to placebo or usual care, corticosteroids probably reduce the risk of death at 28 days by 10% (72 trials; 22,915 participants), with consistent treatment effects between children and adults. There may be little or no effect of corticosteroids on the risk of dying over the long term (longer than three months), but these results are less certain. Corticosteroids probably reduce the risk of dying in hospital. Corticosteroids may result in a reduction in the length of stay in the intensive care unit (ICU) and in hospital. They may not increase the risk of superinfection (second infection). The evidence is very uncertain about the effect of corticosteroids on the risk of muscle weakness.

We are uncertain about the effects of continuous intravenous infusion of corticosteroids compared with intermittent intravenous bolus (rapid injection) administration. Four studies reported data for this comparison, and the certainty of evidence for all outcomes was very low.

What are the limitations of the evidence?

• Corticosteroids versus placebo or usual care

We have moderate confidence in our findings that corticosteroids reduced 28-day mortality. We found some differences amongst the study populations, the types of corticosteroids and how they were given, and the use of additional medicines.

• Continuous infusion versus bolus administration of corticosteroids

We have very little confidence in our findings about the absence of a difference in 28-day mortality between the administration of corticosteroids continuously and repeated rapid intravenous injections. We found only four studies that enrolled very few patients.

How up-to-date is this evidence?

The evidence provided in this review is current to December 2023.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Corticosteroids versus placebo or usual care

Corticosteroids versus placebo or usual care

Patient or population: sepsis in children and adults
Setting: in-hospital
Intervention: corticosteroids
Comparison: placebo or usual care

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with placebo or usual care	Risk with corticosteroids				
28-day all-cause mortality assessed with: vital status follow-up: range 14 days to 30 days	257 per 1000	**229 per 1000** (216 to 245)	**RR 0.89** (0.84 to 0.95)	22915 (72 RCTs)	⊕⊕⊕⊝ Moderate[a]	Treatment probably reduces 28-day all-cause mortality. The number needed to treat is 36, suggesting that treatment with corticosteroids would save one additional life every 36 treated patients.
Long-term mortality assessed with: vital status follow-up: range 6 months to 12 months	352 per 1000	**342 per 1000** (321 to 363)	**RR 0.97** (0.91 to 1.03)	8468 (12 RCTs)	⊕⊕⊝⊝ Low[b,c]	Treatment may result in little to no difference in long-term mortality.
Hospital mortality assessed with: vital status	332 per 1000	**299 per 1000** (279 to 322)	**RR 0.90** (0.84 to 0.97)	17459 (40 RCTs)	⊕⊕⊕⊝ Moderate[d]	Treatment probably reduces hospital mortality. The number needed to treat is 30, suggesting that treatment with corticosteroids would save one additional life every 30 treated patients.
Length of intensive care unit stay for all participants (ICU length of stay) assessed with: calendar dates Scale from: 0 to 90	The mean length of intensive care unit stay for all participants was **13.7** days	MD **0.86 days lower** (1.67 lower to 0.05 lower)	-	8069 (25 RCTs)	⊕⊕⊝⊝ Low[e,f]	Corticosteroids may reduce the length of intensive care unit stay for all participants. The effect size is small and may or may not represent a patient-important improvement.

Length of hospital stay for all participants (Hospital length of stay) assessed with: calendar dates Scale from: 0 to 90	The mean length of hospital stay for all participants was **20** days	MD **1.09 days lower** (1.85 lower to 0.34 lower)	-	16954 (31 RCTs)	⊕⊕⊝⊝ Low[f,g]	Corticosteroids may reduce the length of hospital stay for all participants. The effect size may represent a small patient-important improvement.
Number of participants with adverse events - superinfection (Superinfection) assessed with: clinical and laboratory variables	182 per 1000	**175 per 1000** (157 to 195)	**RR 0.96** (0.86 to 1.07)	7961 (36 RCTs)	⊕⊕⊝⊝ Low[h,i]	Treatment may result in little to no difference in the number of participants with adverse events (superinfection).
Number of participants with adverse events - muscle weakness (Muscle weakness) assessed with: clinical and laboratory variables	55 per 1000	**60 per 1000** (43 to 84)	**RR 1.09** (0.78 to 1.53)	6729 (7 RCTs)	⊕⊝⊝⊝ Very low[h,i,j]	The evidence is very uncertain about the effect of treatment on the number of participants with adverse events (muscle weakness).

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_447994751227224909.

[a] We downgraded by one level for serious inconsistency (the heterogeneity across trials may be moderate: $Chi^2 = 98.15$, df = 70 (P = 0.01); $I^2 = 29\%$).
[b] We by downgraded one level for serious inconsistency across the trials (trials assessed long-term mortality using fairly variable time points).
[c] We downgraded by one level for serious imprecision (large confidence intervals).
[d] We downgraded by one level for serious inconsistency (the heterogeneity in the results may be moderate; $Chi^2 = 56.57$, df = 39 (P = 0.03); $I^2 = 31\%$).
[e] We downgraded by one level for serious inconsistency (moderate heterogeneity in the results; $Chi^2 = 44.09$, df = 24 (P = 0.007); $I^2 = 46\%$).
[f] We downgraded by one level for serious imprecision (large confidence interval and the upper limit may be perceived barely clinically relevant).
[g] We downgraded by one level for serious inconsistency (the heterogeneity in the results was substantial: $Chi^2 = 89.01$, df = 30 (P < 0.00001); $I^2 = 66\%$).
[h] We downgraded by one level for serious indirectness (the definitions and assessment for this outcome fairly varied across the trials).
[i] We downgraded by one level for serious imprecision (large confidence intervals).
[j] We downgraded by one level for serious inconsistency (the heterogeneity in the results was moderate; $Chi^2 = 8.65$, df = 6 (P = 0.19); $I^2 = 31\%$).

Summary of findings 2. Summary of findings table - Continuous infusion versus bolus administration of corticosteroids

Continuous infusion versus bolus administration of corticosteroids						
Patient or population: sepsis in children and adults **Setting:** in-hospital **Intervention:** continuous intravenous infusion of corticosteroids **Comparison:** intermittent intravenous boluses						
Outcomes	**Anticipated absolute effects* (95% CI)**		**Relative effect (95% CI)**	**№ of participants (studies)**	**Certainty of the evidence (GRADE)**	**Comments**
	Risk with intermittent intravenous boluses	**Risk with continuous intravenous infusion of corticosteroids**				
28-day all-cause mortality assessed with: vital status follow-up: range 14 days to 30 days	444 per 1000	**457 per 1000** (359 to 586)	**RR 1.03** (0.81 to 1.32)	310 (3 RCTs)	⊕⊝⊝⊝ Very low[a,b]	The evidence is very uncertain about the effect of continuous intravenous infusion of corticosteroids on 28-day all-cause mortality.
Long-term mortality assessed with: vital status follow-up: range 6 months to 12 months	636 per 1000	**865 per 1000** (649 to 1000)	**RR 1.36** (1.02 to 1.81)	70 (1 RCT)	⊕⊝⊝⊝ Very low[c,d]	The evidence is very uncertain about the effect of continuous intravenous infusion of corticosteroids on long-term mortality.
Hospital mortality assessed with: vital status	384 per 1000	**353 per 1000** (272 to 457)	**RR 0.92** (0.71 to 1.19)	352 (3 RCTs)	⊕⊝⊝⊝ Very low[a,b]	The evidence is very uncertain about the effect of continuous intravenous infusion of corticosteroids on hospital mortality.
Length of intensive care unit stay for all participants (ICU length of stay) assessed with: calendar dates Scale from: 0 to 90	The mean length of intensive care unit stay for all participants was **11.9** days	MD **0.56 days lower** (3.44 lower to 2.32 higher)	-	422 (4 RCTs)	⊕⊝⊝⊝ Very low[a,b,e]	The evidence is very uncertain about the effect of continuous intravenous infusion of corticosteroids on length of intensive care unit stay for all participants.
Length of hospital stay for all participants (Hospital length of stay) assessed with: calendar dates Scale from: 0 to 90	The mean length of hospital stay for all participants was **17.8** days	MD **0.21 days lower** (4.72 lower to 4.3 higher)	-	422 (4 RCTs)	⊕⊝⊝⊝ Very low[a,b,f]	The evidence is very uncertain about the effect of continuous intravenous infusion of corticosteroids on length of hospital stay for all participants.

Number of participants with adverse events - superinfection (Superinfection) assessed with: clinical and laboratory variables	204 per 1000	**229 per 1000** (76 to 680)	**RR 1.12** (0.37 to 3.33)	193 (2 RCTs)	⊕⊝⊝⊝ Very low[b,g]	The evidence is very uncertain about the effect of continuous intravenous infusion of corticosteroids on number of participants with adverse events (superinfection).
Number of participants with adverse events - muscle weakness (Muscle weakness) assessed with: clinical and laboratory variables	61 per 1000	**54 per 1000** (8 to 362)	**RR 0.89** (0.13 to 5.98)	70 (1 RCT)	⊕⊝⊝⊝ Very low[a,d]	The evidence is very uncertain about the effect of continuous intravenous infusion of corticosteroids on number of participants with adverse events (muscle weakness).

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_448010649701465404.

[a] We downgraded by two levels for very serious study limitations (unclear allocation concealment, open-label).
[b] We downgraded by two levels for very serious imprecision (large confidence interval, very small sample size).
[c] We downgraded by two levels for very serious study limitations (open-label).
[d] We downgraded by two levels for extremely serious imprecision (this outcome was available for only one small size study).
[e] We downgraded by one level for serious inconsistency (the heterogeneity in the results was substantial: $Chi^2 = 6.93$, df = 3 (P = 0.07); $I^2 = 57\%$).
[f] We downgraded by one level for serious inconsistency (the heterogeneity in the results was substantial: $Chi^2 = 6.75$ df = 3 (P = 0.08); $I^2 = 56\%$).
[g] We downgraded by one level for serious inconsistency (the heterogeneity in the results was substantial: $Chi^2 = 3.90$, df = 1 (P = 0.05); $I^2 = 74\%$).

BACKGROUND

Description of the condition

Sepsis carries a heavy social, economic and health burden worldwide. This life-threatening dysfunction of organs is triggered by an abnormal immune response to infection (Singer 2016), related to bacteria, viruses (e.g. COVID-19), parasites and fungi. Amongst them, *S pneumoniae, S aureus, E coli, K pneumoniae* and *P aeruginosa* account for more than two-thirds of cases, and these pathogens have a high prevalence of multidrug resistance (WHO 2018). The dysregulated host response is usually defined by the presence of a sequential organ failure assessment (SOFA) score of 2 or higher (Singer 2016; Vincent 1996). Septic shock is a subset of sepsis in which particularly profound circulatory, cellular and metabolic abnormalities are associated with a greater risk of mortality than with sepsis alone. Patients with septic shock can be clinically identified by a vasopressor requirement to maintain a mean arterial pressure of 65 mmHg or greater and serum lactate levels greater than 2 mmol/L (> 18 mg/dL) in the absence of hypovolaemia. The dysregulated response may result in systemic inflammation and organ damage, or in immune paresis and secondary infection (van der Poll 2017). Sepsis, whilst a major threat for European and US populations, disproportionally affects the most vulnerable, and the populations of low- and middle-income countries, with ~50 million cases (3.4 in the EU) and ~11 million deaths annually (0.68 in the EU) (WHO 2018). This epidemiology highlights the impact of socioeconomic inequities on sepsis incidence and outcomes. With the growing and ageing of the world's population, the number of people with sepsis will double by 2050 (WHO 2018). Healthcare-associated sepsis occurs in ~1.5% of hospitalisations, with 20% to 30% mortality, and one-third of hospital survivors died within a year (WHO 2018). According to a recent retrospective cohort study of adult patients admitted to 409 academic, community and federal hospitals in the USA from 2009 to 2014, sepsis was present in 6% of adult hospitalisations (Rhee 2017). Another study of electronic health records from 27 academic hospitals in the USA reported an annual incidence of septic shock of about 19 per 1000 hospitalisations in 2014 (Kadri 2017). People with sepsis usually die from hypotension or progressive multiple organ failure (Angus 2013; Annane 2003; Annane 2005; Parrillo 1993).

There is no current diagnostic test for sepsis. Its standard management includes control of the source of infection with antibiotics and surgery whenever needed, as well as control of tissue oxygenation with fluid replacement, oxygen with or without respiratory support, and vasopressors whenever needed (Evans 2021; Rhodes 2017). No specific interventions are yet available to control immune responses to invading pathogens (Evans 2021; Rhodes 2017).

The financial burden of sepsis on the healthcare system has been calculated to be > 24 billion USD, representing 6.2% of total hospital costs in 2013 (WHO 2018). Studies in Europe and Canada estimated the daily costs of hospital care for a septic patient in 2000 to be between EUR 710 and EUR 1033 (equivalent to about USD 645 and USD 939, respectively) (WHO 2018). In France, 2019 incidence was 403/100,000 (357 in 2015), mortality 23%, disability 15% and cost ~EUR 16,000 per patient (Dupuis 2020; Pandolfi 2022). New health problems occur in three-quarters of three-year survivors, with annual costs of EUR 6.8 billion (Fleischmann-Struzek 2021). Effective access to the best clinical care, and preventive and curative strategies, is a top priority.

Description of the intervention

Corticosteroids include the natural steroid hormones produced by adrenocortical cells and a broad variety of synthetic analogues. These substances have various effects that may be grossly classified into glucocorticoid and mineralocorticoid effects. Glucocorticoid effects include mainly the regulation of carbohydrate, lipid and protein metabolism, as well as the regulation of inflammation. Mineralocorticoid effects include mainly the regulation of electrolytes and water metabolism. At the molecular level, glucocorticoids have non-genomic and genomic effects (Annane 2017a; Cain 2017). Rapid (within minutes) non-genomic effects of glucocorticoids include a decrease in platelet aggregation, cell adhesion and intracellular phosphotyrosine kinases, and they include an increase in annexin 1 externalisation (Lowenberg 2005). These effects may result from the interaction of glucocorticoids with specific membrane sites (Norman 2004). Glucocorticoids have indirect genomic effects, called trans-repression (Rhen 2005). These occur within a few hours following exposure of cells to glucocorticoids. They result from the physical interaction between the monomeric glucocorticoid-glucocorticoid receptor (G-GR)α complex and various nuclear transcription factors, such as nuclear factor (NF)-κB and activator protein (AP)-1. Subsequently, these nuclear transcription factors are sequestrated in the cytosol and cannot enter the nucleus, preventing the expression of gene encoding for most if not all pro-inflammatory mediators. Glucocorticoids also have direct genomic effects, called transactivation. These effects require only a few days of cell exposure to glucocorticoids. Indeed, conformational changes (i.e. dimerisation of the G-GRα complex) are needed before this complex can migrate to the nucleus to interact with glucocorticoid-responsive elements; that is, parts of gene encoding for regulators of the termination of inflammation. Then, key anti-inflammatory factors are up-regulated, leading to phagocytosis, chemokinesis and anti-oxidative processes. The net effect of glucocorticoids involves reprogramming rather than inhibiting immune cell function (Erschen 2007). Glucocorticoids induce specific activated anti-inflammatory monocyte subtypes that migrate quickly to inflamed tissues (Varga 2008). They prolong survival of this subtype of monocyte via A3 adenosine receptor-triggered anti-apoptotic effects (Barczyk 2010). Overall, these molecular mechanisms of action of glucocorticoids are appropriate for counteracting the uncontrolled inflammation that can characterise sepsis.

How the intervention might work

Researchers have explored the biological mechanisms of sepsis to investigate potential interventions. Corticosteroids have been a topic of particular focus because of their influence on the immune response (Cain 2017). In sepsis, the hypothalamic-pituitary gland hormonal pathway to the adrenal glands stimulates corticosteroid production (Annane 2017a; Chrousos 1995; Cooper 2003; Heming 2018). These hormones affect inflammation through the production of white blood cells, cytokines (proteins that influence the immune response) and nitric oxide. In sepsis, cytokines may suppress adrenocorticotropin hormone synthesis (Annane 2017a; Polito 2011; Sharshar 2003), along with the cortisol response to exogenous adrenocorticotropin hormone (Annane 2017a; Hotta 1986; Jaattela 1991). Likewise, sepsis may be associated with alterations in scavenger receptor B1-mediated cholesterol delivery (Cai 2008). This causes poor adrenal activity in almost half of patients (Annane 2000; Lipiner 2007; Marik 2008; Rothwell 1991), as well as possible resistance of body tissues to corticosteroids due to fewer corticosteroid receptors or receptors with lower affinity (Barnes 1995; Huang 1987; Meduri 1998a; Molijn 1995). Alterations of corticosteroid receptor numbers and in binding capacity may be related at least in part to nitric oxide (Duma 2004; Galigniana 1999). Recent work suggests that immune cells - not steroid-secreting cells - are key regulators of the interaction between the immune system and the adrenals (Kanczkowski 2013). In addition, acute illness such as sepsis may be associated with decreased cortisol clearance from plasma (Boonen 2013; Melby 1958), likely resulting from altered hepatic and renal inactivation of cortisol (Boonen 2013). Early studies showed that a pharmacological dose of corticosteroids prolonged survival amongst animals with sepsis (Fabian 1982). More recent studies in rodents have demonstrated that lower doses of corticosteroids, for example, 0.1 mg/kg of dexamethasone, improved haemodynamic and organ function, favourably modulated the inflammatory response, and prolonged survival (di Villa Bianca 2003; Heller 2003; Tsao 2004; Vachharajani 2006).

The protective effects of these glucocorticoids against sepsis may be mediated in part by the endothelial glucocorticoid receptor (Goodwin 2013). In healthy volunteers challenged with endotoxin, a low dose of corticosteroids, for example, 10 mg of prednisolone, blocked the release of pro-inflammatory cytokines, prevented endothelial cell and neutrophil activation, and inhibited the acute phase response without altering coagulation and fibrinolysis balance (de Kruif 2007). Studies in patients with septic shock have shown that a short course of corticosteroids may result in a rebound in the systemic inflammatory response (Briegel 1994; Keh 2003). In addition, it is now recognised that increased pro-inflammatory cytokine release can be sustained for longer than a week in patients with sepsis (Kellum 2007). Likewise, the timing of initiation of corticosteroids may be an important factor in the response to treatment. Indeed, in observational studies, short-term mortality increased with delayed initiation of hydrocortisone (Katsenos 2014; Park 2012). For these reasons, we would anticipate that corticosteroid treatment is beneficial for patients with sepsis, and that differences in dose, timing or duration of corticosteroid treatment may differentially affect patient response to treatment. Several authors have argued that in patients with sepsis, hydrocortisone should be given as a continuous infusion rather than as intermittent boluses to reduce the risk of metabolic complications (Rhodes 2017). In sepsis trials, continuous infusion of hydrocortisone was variably associated with better outcomes or worse outcomes than intermittent intravenous boluses (Loisa 2007; Tilouche 2019).

The scientific community is competing to identify markers delineating between patients who draw survival benefit from corticosteroids and those who may be harmed. Studies suggest that a transcriptomic signature based on 100 genes may identify a subset of paediatric sepsis (endotype A) that has an increased risk of death when exposed to corticosteroids (Wong 2015). However, patients with an endotype A at intensive care unit (ICU) day 1 that shifted to an endotype B later on gained benefit from corticosteroids (Wong 2018). Another study found transcriptomic-based sepsis response signatures (SRS) associated with immune paresis (SRS1) or with systemic inflammation (SRS 2) (Antcliffe 2019). In this study, patients with an SRS 2 transcriptomic signature had significantly higher mortality when treated with hydrocortisone. Apart from transcriptomic profiles, the effects of corticosteroids on mortality from sepsis may be modified by hormonal factors. In adults with sepsis, a corticotropin-stimulated cortisol-to-corticosterone ratio of 32.2 or more predicted in-hospital mortality (area under the curve 0.8, CI 0.70 to 0.88; sensitivity 83% and specificity 78%) (Briegel 2022). In another study, researchers found that adults with sepsis and a low serum IFNγ/IL10 ratio predicted increased survival when treated with hydrocortisone, whereas those with a high ratio had a greater risk of death (König 2021).

Why it is important to do this review

This is an update of a Cochrane review first published in 2004, and previously updated in 2010, 2015 and 2019. The update in 2010 included nine additional trials with no change in the conclusion of the review. The update in 2015 included the modification of the title to 'Corticosteroids for treating sepsis' owing to changes in the definition of sepsis (Singer 2016), the exclusion of quasi-randomised trials and an update of the definition of 'low dose' and 'long course of corticosteroids'. The addition of nine new trials suggested moderate evidence for reduced 28-day mortality with corticosteroids in the primary analysis and confirmed significant interactions between the relative risk of dying at 28 days and treatment modalities used (lower doses and longer duration yielded better chance of survival) and the patient case mix included (patients with septic shock, sepsis-related acute respiratory distress syndrome (ARDS) or community-acquired pneumonia may be more likely to benefit from corticosteroids). The update in 2019 included 25 additional trials and confirmed the direction and magnitude of the point estimate for 28-day mortality with narrow confidence interval limits. The review found evidence of a significant reduction in ICU and hospital mortality, and in ICU and hospital length of stay, and included evidence of a reduction in midterm (90-day) to long-term (up to one year) mortality with corticosteroids.

Initially, researchers used high doses of corticosteroids, usually given as a single bolus, in an attempt to block potential bursts in pro-inflammatory cytokines. Two systematic reviews and meta-analyses of trials of corticosteroids in sepsis or in septic shock included 10 (Lefering 1995) and nine (Cronin 1995) randomised controlled trials (RCTs), respectively. These systematic reviews showed no significant effect on the relative risk of death, gastrointestinal bleeding or superinfection associated with the use of corticosteroids.

Subsequently, most clinicians will not recommend the use of high doses of corticosteroids in sepsis (Annane 2017b; Evans 2021; Rhodes 2017). The potential benefits of a lower dose (≤ 400 mg hydrocortisone or equivalent per day) and a longer duration at full dose (≥ 3 days) of treatment have been investigated in numerous RCTs over the past three decades (Annane 2017b; Lamontagne 2018; Pitre 2023; Rochwerg 2018). Previous clinical practice guidelines recommended against the use of corticosteroids in sepsis, except in patients with septic shock and a poor response to fluid replacement and vasopressor therapy (Annane 2017b; Nishida 2018; Rhodes 2017; Tavaré 2017). Another guideline made a conditional recommendation for the use of corticosteroids in patients with sepsis regardless of the presence of shock or not (Lamontagne 2018). Some guidelines suggest that corticosteroids should be given as a continuous infusion rather than in intermittent boluses (Annane 2017b; Rhodes 2017). In the year 2018, five different systematic reviews and meta-analyses addressed the effects of corticosteroids in sepsis (Allen 2018; Fang 2018; Ni 2018;

Rochwerg 2018; Rygard 2018). The number of included trials was different in all reviews and ranged from 14 to 42. The risk ratio of death in the short term varied from 0.91 to 0.96, and the upper limit of the 95% confidence interval (CI) varied from 0.98 to 1.03. Another systematic review and meta-analysis of one randomised trial and 17 observational studies examined the risk of acquired muscle weakness associated with exposure to corticosteroids in patients in the intensive care unit (ICU) (Yang 2018). This review found an odds ratio for acquired muscle weakness of 1.84 (95% CI 1.26 to 2.67) with corticosteroids compared to control. An individual patient data meta-analysis has evaluated the effects of hydrocortisone in adults with septic shock (Pirracchio 2023). Of 24 eligible trials (n = 8528), 17 (n = 7882) provided individual patient data and seven (n = 5929) provided 90-day mortality data. The marginal relative risk (RR) for 90-day mortality with hydrocortisone versus placebo was 0.93 (95% CI 0.82 to 1.04; P = 0.22; moderate certainty). This was 0.86 (95% CI 0.79 to 0.92) for hydrocortisone with fludrocortisone and 0.96 (95% CI 0.82 to 1.12) without fludrocortisone. Another recent meta-analysis of 45 trials involving 9563 patients suggests that corticosteroids probably reduce short-term mortality (RR 0.93, 95% CI 0.88 to 0.99; moderate certainty) and increase shock reversal at seven days (RR 1.24, 95% CI 1.11 to 1.38; high certainty) (Pitre 2023). The latest update of the Surviving Sepsis Campaign suggests the use of intravenous hydrocortisone in adults with vasopressor therapy-dependent sepsis (Evans 2021). The 2024 focused update of guidelines on the use of corticosteroids in critically ill patients, developed by the Society of Critical Care Medicine, suggested administering corticosteroids to adult patients with septic shock (conditional recommendation, low certainty), recommended against administration of high-dose/short-duration corticosteroids (defined as > 400 mg/day of hydrocortisone equivalent for < 3 days) for adult patients with septic shock (strong recommendation, moderate certainty) and made no recommendation for corticosteroid use in paediatric patients with sepsis (Chaudhuri 2024).

Therefore, we aim to systematically review the effects of corticosteroids in children and adults with sepsis.

OBJECTIVES

To examine the benefits and harms of corticosteroids in children and adults with sepsis.

METHODS

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials (RCTs) with no methodological restrictions, i.e. parallel design with two or more arms, cross-over design, factorial design, cluster-randomisation. We used data from cross-over or cluster-randomised trials only for qualitative assessment and did not pool them in quantitative meta-analysis. We excluded quasi-randomised trials (i.e. trials assigning patients to treatment arms based on systematic methods, such as alternation, assignment based on date of birth, case record number and date of presentation).

Types of participants

We included children (except neonates) and adults (from 18 years old with no upper limit for age) with sepsis as defined by the Sepsis 3 criteria (Singer 2016) or by the following criteria (ACCP/SCCM 1992; Vincent 2013).

- Suspected or documented infection defined as culture or Gram stain of blood, sputum, urine or normally sterile body fluid that is positive for a pathogenic microorganism; or a focus of infection identified by visual inspection (e.g. ruptured bowel with the presence of free air or bowel contents in the abdomen found at the time of surgery; wound with purulent drainage).
- At least two symptoms of a systemic inflammatory response syndrome, such as fever (body temperature > 38 °C) or hypothermia (< 36 °C), tachycardia (> 90 beats per minute), tachypnoea (> 20 breaths per minute) or hyperventilation (arterial carbon dioxide tension ($PaCO_2$) < 32 mmHg), and abnormal white blood cell count (> 12,000 cells/mL or < 4000 cells/mL) or more than 10% immature band of neutrophils.
- At least one sign of organ dysfunction; that is, metabolic acidosis, arterial hypoxaemia (arterial oxygen tension (PaO_2):fractional inspired oxygen (FiO_2) < 250 mmHg), oliguria (< 30 mL/h for ≥ 3 hours), coagulopathy or encephalopathy.

Septic shock is defined by the presence of sepsis and of hypotension (persisting systolic arterial pressure < 90 mmHg) that is refractory to fluid resuscitation and requires vasopressor support (i.e. > 5 μg/kg of body weight per minute of dopamine or any dose of epinephrine or norepinephrine).

We included data from trials of community-acquired pneumonia or acute respiratory distress syndrome (ARDS) when separate data were available for participants with sepsis, or when contact with study authors resulted in provision of the data.

Types of interventions

Corticosteroids versus placebo or usual care

Intervention

Systemic treatment provided as any type of corticosteroid preparation (e.g. cortisone, fludrocortisone, hydrocortisone, methylprednisolone, betamethasone, dexamethasone). We also consider a combination of hydrocortisone plus vitamin C and thiamine.

Low-dose corticosteroid treatment was defined as a total dose per day of 400 mg or less of hydrocortisone (or equivalent); otherwise, the dose of corticosteroid was considered high. A long course for the intervention was defined as a full-dose treatment duration of three or more days; otherwise, treatment was considered as a short course.

Control

Standard therapy (usual care) was provided, which may have included antimicrobials, fluid replacement, inotropic or vasopressor therapy, mechanical ventilation or renal replacement therapy, or placebo.

Continuous infusion versus bolus administration of corticosteroids

Intervention

Continuous infusion was defined as intravenous infusion of corticosteroids with or without an initial loading dose.

Control

Bolus administration was defined as intermittent intravenous injections with a duration of less than 30 minutes.

Types of outcome measures

Primary outcomes

- 28-day all-cause mortality

Indeed, this has been the primary outcome measure in most of the RCTs on sepsis conducted since 1992 (Annane 2009b). Most studies performed before 1992 looked at 14-day or hospital mortality rates. We used these data to compute the pooled analysis for 28-day mortality, unless we could obtain actual 28-day mortality rates from primary study authors.

Secondary outcomes

- Long-term all-cause mortality (last follow-up beyond three months to one year).
- Hospital all-cause mortality.
- 90-day all-cause mortality. This was the primary outcome in the two most recent and largest trials on corticosteroids for sepsis. We did not extrapolate mortality by 90 days from another time point.
- ICU all-cause mortality.

In-ICU and in-hospital mortality outcomes provide the location of death, which adds context to the primary outcome.

- Length of stay in the ICU (for all participants and for survivors only). This outcome is expressed in the mean (standard deviation (SD)) number of days, and is calculated by the difference between dates of ICU discharge and ICU admission, with first and last days of ICU stays counted as full ICU days regardless of the time of admission and time of discharge. We considered only the first ICU stay if a patient had repeated ICU admissions during follow-up.
- Length of hospital stay (for all participants and for survivors only). This outcome is expressed in the mean (SD) number of days, and is calculated by the difference between dates of hospital discharge and hospital admission, with first and last days of hospital stays counted as full hospital days regardless of time of admission and time of discharge. We considered only the first hospital stay if a patient had repeated hospital admissions during follow-up.
- Number of participants with shock reversal (as defined by stable haemodynamic status ≥ 24 hours after withdrawal of vasopressor therapy) on day seven and on day 28. We considered only the first episode of shock if a patient had repeated shock episodes.
- Number of organs affected and severity of organ dysfunction on day seven, in individual patients, as measured by the Sequential Organ Failure Assessment (SOFA) score (Vincent 1996). This scale scores from 0 (normal function) to 4 (most severe) for the dysfunction of six organ systems (Respiration, Coagulation, Liver, Cardiovascular, Central nervous system, Renal). It ranges from 0 (no organ failure) to 24 (most severe organ dysfunction). We did not extrapolate the SOFA score by seven days from another time point.
- Adverse events (i.e. superinfection, muscle weakness, gastrointestinal bleeding, hyperglycaemia, hypernatraemia, neuropsychiatric events, stroke, cardiac events or any other adverse effects or complications of corticosteroid treatment). Each adverse event is expressed as the number (%) of patients with at least one episode of this event (within 28 days), as defined in individual studies, except for hyperglycaemia and hypernatraemia. Whenever possible, hyperglycaemia was defined by values > 180 mg/dL and hypernatraemia by values > 149 mmol/L.

Search methods for identification of studies

We attempted to identify all relevant studies regardless of language or publication status (e.g. published, unpublished, in press, in progress).

Electronic searches

We searched the Cochrane Central Register of Controlled Trials (CENTRAL; 2023, Issue 12), in the Cochrane Library, using the search terms 'sepsis', 'septic shock', 'steroids' and 'corticosteroids' (for the detailed search strategy, see Appendix 1).

We also searched (to 31 December 2023) MEDLINE ALL (Ovid SP), Embase (Ovid SP) and Latin American Caribbean Health Sciences Literature (LILACS) using the topic search terms in combination with the search strategy for identifying trials developed by Cochrane (Higgins 2024). For detailed search strategies, see Appendix 2 (MEDLINE), Appendix 3 (Embase) and Appendix 4 (LILACS).

Finally, we searched for ongoing RCTs (to 31 December 2023) in ClinicalTrials.gov, International Standard Randomised Controlled Trials Number (ISRCTN) and the World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP) using the search terms 'septic shock', 'sepsis', 'steroids', 'corticosteroids', 'adrenal cortex hormones' and 'glucocorticoids'.

We performed a further search in December 2024. These results have been added to 'Studies awaiting classification' and will be incorporated into the review at the next update.

Searching other resources

We checked the reference lists and citations of all trials and relevant systematic reviews identified by the electronic searches, and we contacted study authors to request additional published or unpublished data. We also searched the proceedings of annual meetings of major critical care medicine symposia; that is, Society of Critical Care Medicine, American Thoracic Society, International Symposium on Intensive Care and Emergency Medicine, American College of Chest Physicians and European Society of Intensive Care Medicine (1998 to 2023).

Data collection and analysis

Selection of studies

Using Covidence (systematic review management software), pairs of review authors (amongst DA, JB, PEB, DG, RP) screened titles and abstracts and subsequently full-text manuscripts independently and in duplicate. We decided which trials met the inclusion criteria. We resolved disagreements between review authors by discussion (with a third review author amongst EB and BR) until we reached consensus. The review authors did not contribute to the decision regarding the inclusion of trials in which they had participated.

One review author (DA) contacted the study authors for clarification, when necessary.

Data extraction and management

One review author (DA) drew up a standard data extraction form, and four other review authors (PEB, JB, DK, YK) amended and validated the design of the form before data abstraction (Appendix 5). Review authors (DA, PEB, JB, DG, DK, RP, BR) independently extracted data, except those from trials in which they had participated. For this update, we used Covidence for data extraction (Covidence). Two review authors (between DA, JB and DG) independently abstracted data with discrepancies resolved by a third review author (BR). Data were subsequently exported to RevMan (RevMan 2024).

One review author (DA) systematically contacted the authors of trials to request missing data when possible.

One review author (DA) and one member of this author's research staff independently extracted and entered data into RevMan (RevMan 2024). All review authors checked the accuracy of data entered against the original articles.

Assessment of risk of bias in included studies

Pairs of review authors (amongst DA, JB, PEB, DG, RP) assessed the risk of bias within individual trials using the Cochrane risk of bias tool (RoB 1) (Guyatt 2013; Higgins 2024). We considered the following domains: selection bias, performance bias, detection bias, attrition bias, reporting bias and any other bias. For studies published prior to 2019 and included in the last iteration of this review, we assessed performance and detection bias together. For studies published since 2019 and included in this most recent review, we assessed these domains separately. We judged selection bias on the basis of how the random sequence was generated and how allocation was concealed. We judged performance bias and detection bias on the basis of who was blinded and how, amongst participants, caregivers, pharmacists, data collectors, outcome assessors and data analysts (Devereaux 2001). In judging attrition bias, we considered how many participants were lost to follow-up or were not included in analyses (and the reasons why). When available, we compared outcomes reported in trial protocols versus actual results reported, to identify potential selective reporting bias. We resolved disagreements between review authors by discussion (with a third review author between EB and BR) until we reached consensus.

One review author (DA) contacted the study authors for clarification, when necessary.

The review authors did not contribute to the assessment of the risk of bias in any trial in which they had participated.

Measures of treatment effect

- We performed intention-to-treat (ITT) analyses. We performed all statistical calculations using RevMan 2024 or Stata 2015, as appropriate.
- We calculated a weighted treatment effect across trials. We expressed results as risk ratios (RRs) with 95% confidence intervals (CIs) for dichotomous outcomes and as mean differences (MDs, 95% CIs) for continuous outcomes.

Unit of analysis issues

In this review, we used data from trials in which the unit of randomisation was the individual, and in which parallel groups were assessed. Data from cluster-randomisation or cross-over trials were used only for qualitative and not quantitative analysis. We did not pool results from these trials types with data from trials with a parallel-group design.

When trials included more than two arms (e.g. comparing versus control two different corticosteroids or two different modes of administration of the same corticosteroid), we (i) committed groups that were not relevant to the comparison of interest, and (ii) combined multiple groups that are eligible as the experimental or comparator intervention to create a single pair-wise comparison.

For events that may occur repeatedly, such as receiving vasopressor therapy or staying in the ICU, we used only the first occurrence of the event.

Dealing with missing data

We systematically tried to contact primary authors of original trials to obtain missing information and unpublished data. We obtained additional data from the primary authors of 28 trials, including access to individual patient data for 16 trials (Appendix 6). This information is provided for each trial in the notes section of Characteristics of included studies.

For the primary outcome of this review (28-day all-cause mortality), we systematically contacted trial authors when needed to obtain data for participants who dropped out. When trials did not report 28-day all-cause mortality, and contact with trial authors failed to yield actual 28-day mortality rates, we used the available mortality data closest to 28 days.

When trials reported length of stay in the ICU or in hospital only as median and interquartile range (IQR), and when contact with trial authors failed to elicit means and SDs, we did not include these trials in the analysis.

If missing data were still a concern after contact with the trial's primary author, we factored this into the risk of bias assessments and certainty ratings.

Assessment of heterogeneity

We assessed statistical heterogeneity across trials using the I^2 statistic, visual inspection of forest plots and the Chi^2 test for homogeneity (Higgins 2024). We considered poor overlap between confidence intervals by visual inspection of forest plots, suggesting statistical heterogeneity. We used the I^2 statistic to measure heterogeneity amongst the trials in each analysis but acknowledge that there is substantial uncertainty in the value of I^2 when there are only a few studies. Therefore, we also considered the P value from the Chi^2 test, with a significance level of the P value set to 0.10. If we identified substantial heterogeneity, we investigated and reported the potential reasons for this by prespecified subgroup analysis. We considered that heterogeneity might not be important if the I^2 value was from 0% to 40%, moderate if the I^2 value was from 30% to 60%, substantial if the I^2 value was from 50% to 90%, and considerable if the I^2 value was from 75% to 100%.

Assessment of reporting biases

We sought evidence of publication bias by using the funnel plot method. We used Stata 2015 to prepare a contour-enhanced funnel plot (Peters 2008). This graphical analysis used the standard error of the log of the RR. We plotted contours, illustrating the statistical significance of study effect estimates by using a two-tailed test.

Data synthesis

We used methods based on the random-effects model for all analyses. Indeed, we suspected that we would observe heterogeneity across studies, as they were conducted over a wide period of time (almost half a century between the first and last trials) and the rationale for which studies were designed varied greatly over time, with marked differences in treatment strategies and in populations between studies conducted before and after the early 1990s.

Subgroup analysis and investigation of heterogeneity

To identify potential sources of heterogeneity, we sought, a priori, to conduct subgroup analyses based on four factors related to the intervention.

- Dose and duration; that is, a long course (≥ 3 days at full dose) of low-dose (≤ 400 mg/day) hydrocortisone or equivalent. This subgroup analysis allowed evaluation of a strategy based on developments in our understanding of the role of corticosteroids in host response to sepsis, as tested in trials performed after 1992. Older trials most often used a short course (one to four bolus doses within 24 hours) of high-dose corticosteroids (> 400 mg of hydrocortisone or equivalent) and trials conducted after 1992 most often used low-dose corticosteroids at full dose over a longer period (≥ 3 days).
- Types of corticosteroids.
- The method of corticosteroid administration: intravenous bolus versus continuous infusion.
- The method of corticosteroid termination: without versus with tapering off.

We also conducted a subgroup analysis based on factors related to participants.

- Age: children (except neonates) less than 18 years old and adults (of 18 years old or more with no upper limit).
- Targeted population: sepsis, only septic shock, sepsis with ARDS, community-acquired pneumonia, COVID-19-related sepsis and sepsis with critical illness-related corticosteroid insufficiency (Annane 2017a).

To further explore the putative interaction between corticosteroid dose and duration and the magnitude of effect, we considered performing a meta-regression analysis using 28-day all-cause mortality as the dependent variable, and dosage and duration of corticosteroids as predictors. We also tested, a priori, the interaction between baseline severity of illness and magnitude of effect in a meta-regression analysis using mortality rates in controls as predictors.

We performed subgroup analyses only for the primary outcome, which is all-cause mortality at 28 days. We compared subgroups using the formal 'test for subgroup differences' in RevMan 2024. We performed meta-regression analyses using Stata 2015.

We assessed the validity of subgroup analyses on the basis of the following criteria.

- Subgroup comparisons within rather than between studies.
- Hypothesis preceding the analysis.
- One of very few hypotheses.
- Large and consistent differences across studies.
- External evidence supporting the results (Guyatt 2008b).

When subgroup analyses met these criteria and were found to be statistically significant, we applied the GRADE criteria to evaluate the certainty of evidence (GRADEpro GDT 2015; Guyatt 2008a), and we used ICEMAN to evaluate the credibility of effect modification (Schandelmaier 2020)

Sensitivity analysis

We conducted sensitivity analyses based on generation of allocation sequence, concealment of allocation and blinding, and for trials judged at low risk of bias.

Post hoc analysis: in the current update, we performed a sensitivity analysis based on trials published after 2016, i.e. after the 2016 consensus definition of Sepsis 3 (Singer 2016).

Summary of findings and assessment of the certainty of the evidence

In the comparison of corticosteroids versus placebo or usual care, and of continuous infusion versus intermittent bolus administration of corticosteroids, for an assessment of the overall certainty of evidence for each outcome that included pooled data from RCTs only, we downgraded the evidence from 'high certainty' by one level for serious (or by two for very serious) study limitations (according to risk of bias evaluation), indirectness of evidence, serious inconsistency (i.e. when $I^2 > 30\%$), imprecision of effect estimates (large 95% confidence intervals or small treatment effects) or potential publication bias.

We exported data from RevMan (RevMan 2024) to GRADEpro GDT 2015 to create a summary of findings table for the comparison of the effects of corticosteroids versus placebo or usual care, and for the comparison of the effects of a continuous infusion versus intermittent bolus administration of corticosteroids. We included the following patient-centred outcomes in the summary of findings tables.

- 28-day all-cause mortality.
- Long-term all-cause mortality (longest follow-up beyond three months and up to one year).
- In-hospital all-cause mortality.
- Length of stay in the ICU.
- Length of hospital stay.
- Number of participants with superinfection within 28 days.
- Number of participants with muscle weakness within 28 days.

The GRADE assessment was made by two review authors (amongst DA, DG, BR) with disagreements resolved by discussion between three review authors (DA, DG, BR).

RESULTS

Description of studies

Results of the search

Our search process is detailed in Figure 1.

Figure 1.

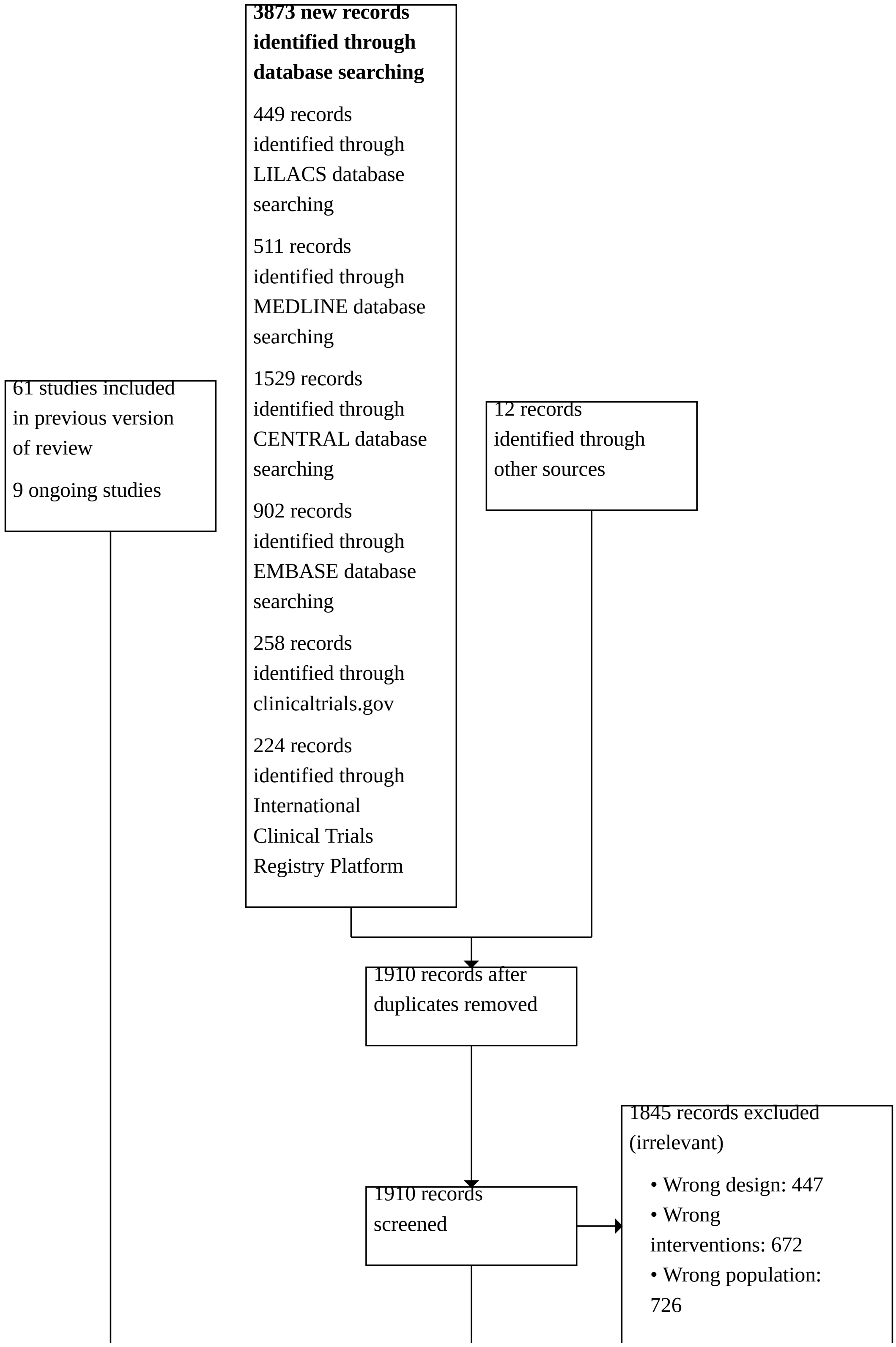
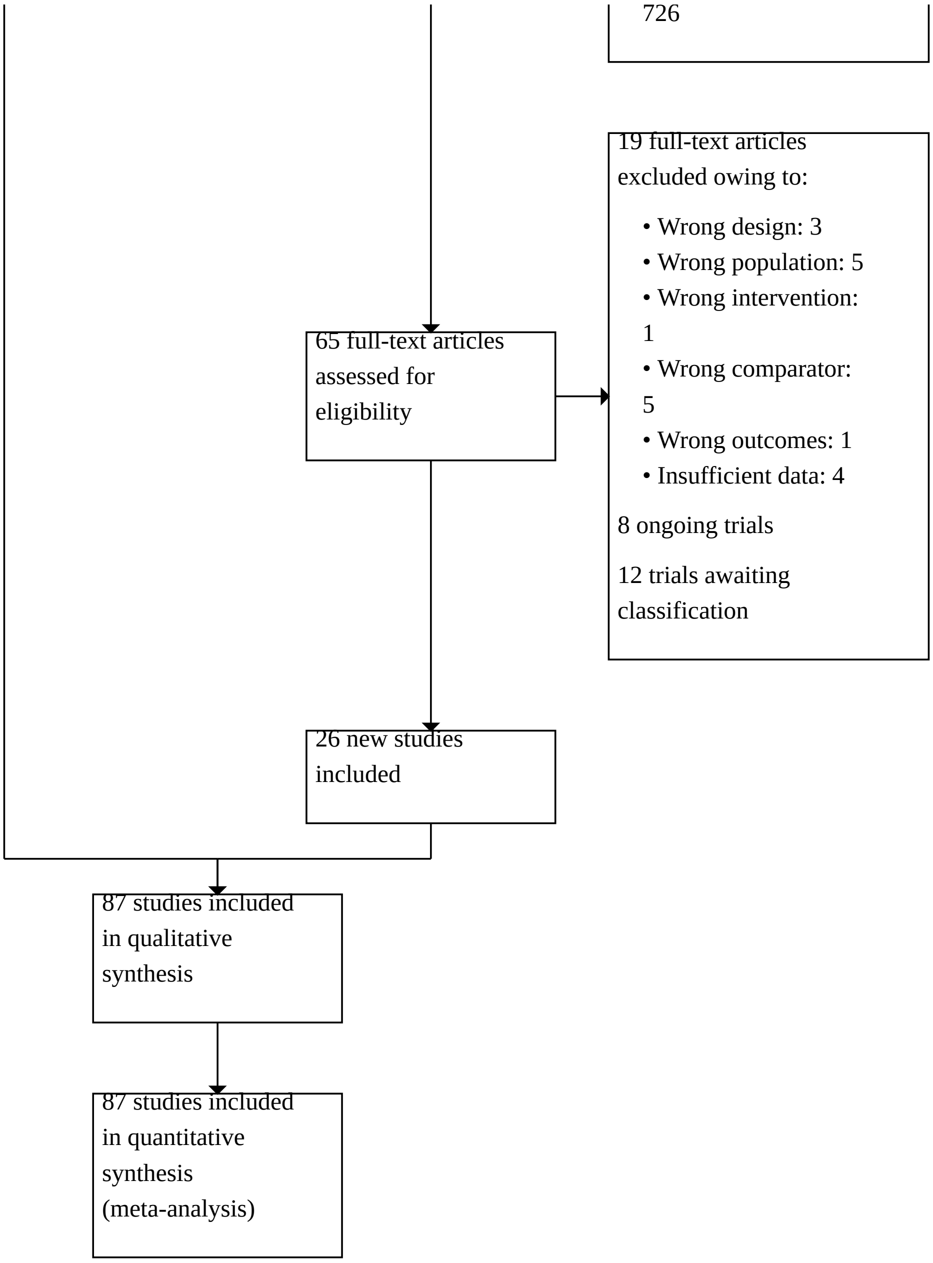

3873 new records identified through database searching
449 records identified through LILACS database searching
511 records identified through MEDLINE database searching
1529 records identified through CENTRAL database searching
902 records identified through EMBASE database searching
258 records identified through clinicaltrials.gov
224 records identified through International Clinical Trials Registry Platform
61 studies included in previous version of review
9 ongoing studies
12 records identified through other sources
1910 records after duplicates removed
1910 records screened
1845 records excluded (irrelevant)
• Wrong design: 447
• Wrong interventions: 672
• Wrong population: 726

Figure 1. (Continued)



In the 2019 update, we included 61 trials, excluded 24 trials and nine trials were ongoing. The current search of the literature yielded 65 additional trials that evaluated corticosteroids in sepsis, of which 26 were included, 19 were excluded based on the full manuscript and eight trials are still ongoing. Thus, a total of 87 trials have been included in this update; 43 have been excluded. One of the nine ongoing trials in the 2019 update was completed and published (Walsham 2024). This trial is now awaiting classification. There are therefore 16 ongoing trials in the current update.

Twelve study reports from an updated search in December 2024 have been added to awaiting classification (Characteristics of studies awaiting classification).

Included studies

Since the last update in 2019, we have included 26 additional trials, making a total of 87 trials (24,336 participants). We describe these trials below.

Source of information

In addition to data extracted from these publications, we obtained unpublished information from 33 trials by contacting the primary authors (Angus 2020; Annane 2002; Annane 2010; Annane 2018; Arabi 2011; Bollaert 1998; Briegel 1999; Chawla 1999; Cicarelli 2007; Confalonieri 2005; Gordon 2014; Gordon 2016; Dequin 2020; Dequin 2023; Hyvernat 2016; Keh 2003; Keh 2016; Liu 2012; Meduri 2007;

Meduri 2009; Meduri 2022; Meijvis 2011; Mirea 2014; Oppert 2005; Rinaldi 2006; Sprung 1984; Sprung 2008; Tandan 2005; Tilouche 2019; Torres 2015; Villar 2020; Yildiz 2002) (Appendix 6). We did not contact the authors of 25 trials, mainly because of the absence of contact details (Characteristics of included studies). For the remaining trials, contact with the study authors did not lead to the provision of additional information (Characteristics of included studies).

Trial centres

Thirty-three trials were multi-centre (i.e. more than two centres) (Angus 2020; Annane 2002; Annane 2010; Annane 2018; Blum 2015; Bone 1987; Confalonieri 2005; Corral-Gudino 2021; CSG 1963; Dequin 2020; Dequin 2023; Ghanei 2021; Gordon 2014; Gordon 2016; Horby 2021; Hyvernat 2016; Keh 2016; Meduri 2007; Meduri 2022; Menon 2017; Mohamed 2023; Moskowitz 2020; Sabry 2011; Sevransky 2021; Sprung 1984; Sprung 2008; Tagaro 2017; Tomazini 2020; Torres 2015; VASSCSG 1987; Venkatesh 2018; Villar 2020;Wittermans 2021) (Characteristics of included studies).

Thirty-two trials were conducted in Europe (Aboab 2008; Annane 2002; Annane 2010; Annane 2018; Blum 2015; Bollaert 1998; Briegel 1999; Confalonieri 2005; Corral-Gudino 2021; Dequin 2020; Dequin 2023; Fernández-Serrano 2011; Gordon 2014; Gordon 2016; Horby 2021; Hyvernat 2016; Keh 2003; Keh 2016; Loisa 2007; McHardy 1972; Meijvis 2011; Mirea 2014; Nagy 2013; Oppert 2005; Rinaldi 2006; Snijders 2010; Tagaro 2017; Torres 2015; Villar 2020; Wittermans 2021; Yildiz 2002; Yildiz 2011), 16 in North America (Birudaraju 2022; Bone 1987; Chawla 1999; CSG 1963; Iglesias 2020; Kurungundla 2008; Luce 1988; Meduri 2007; Meduri 2009; Meduri 2022; Menon 2017; Moskowitz 2020; Schumer 1976; Sevransky 2021; Sprung 1984; VASSCSG 1987), 23 in Asia (Agarwal 2022; Chang 2020; Doluee 2018; Ghanei 2021; Hu 2009; Huang 2014; Huh 2007; Jamshidi 2021; Jiang 2019; Li 2016; Liu 2012; Lv 2017; Lyu 2022; Mohamed 2020; Ngaosuwan 2018; Ram 2022; Sai 2021; Sui 2013; Tandan 2005; Tongyoo 2016; Valoor 2009; Wani 2020; Zhou 2015), three in the Middle East (Arabi 2011; El Ghamrawy 2006; Mohamed 2023), six in Africa (El-Nawawy 2017; Labib 2022; Nafae 2013; Rezk 2013; Sabry 2011; Tilouche 2019), and three in Latin America (Cicarelli 2007; Pierre de Oliveira 2023; Tomazini 2020). Four multinational trials were conducted in North America and Africa (Slusher 1996), in Europe and the Middle East (Sprung 2008), in Australia, New Zealand, Europe and the Middle East (Venkatesh 2018), and in North America, Latin America, Europe, Africa, Asia and Australia and New Zealand (Angus 2020) (Characteristics of included studies).

Trial design

Eighty-five trials were performed on parallel groups. Five trials had more than two intervention arms, including two platform trials (Angus 2020; Horby 2021) and three trials with three interventions, i.e. placebo or usual care and two active interventions (Mirea 2014; Schumer 1976; Sprung 1984). One trial had a factorial design (Annane 2010) and another trial had a cross-over design (Keh 2003).

Age of participants

Two trials enrolled both children and adults (CSG 1963; McHardy 1972). Six trials included only children (El-Nawawy 2017; Menon 2017; Nagy 2013; Slusher 1996; Tagaro 2017; Valoor 2009). All the remaining trials included only adults.

Description of participants

Thirteen trials included both participants with sepsis and individuals with septic shock (Agarwal 2022; Bone 1987; Chang 2020; Huang 2014; Iglesias 2020; Meduri 2009; Sai 2021; Sevransky 2021; Slusher 1996; VASSCSG 1987; Wani 2020; Yildiz 2002; Yildiz 2011). Three trials included participants with sepsis without shock (Fernández-Serrano 2011; Keh 2016; Rinaldi 2006). Seventeen trials targeted participants with community-acquired pneumonia-related sepsis (Blum 2015; Confalonieri 2005; Dequin 2023; El Ghamrawy 2006; Fernández-Serrano 2011; Li 2016; McHardy 1972; Meduri 2022; Meijvis 2011; Nafae 2013; Nagy 2013; Sabry 2011; Snijders 2010; Sui 2013; Tagaro 2017; Torres 2015; Wittermans 2021). Six trials focused on participants with ARDS and sepsis (Liu 2012; Meduri 2007; Rezk 2013; Tongyoo 2016; Villar 2020; Zhou 2015). Seven trials included participants with COVID-19-related sepsis (Angus 2020; Corral-Gudino 2021; Dequin 2020; Ghanei 2021; Horby 2021; Pierre de Oliveira 2023; Tomazini 2020). Thirty-eight trials focused on participants with septic shock treated by a vasopressor (Annane 2002; Annane 2010; Annane 2018; Arabi 2011; Bollaert 1998; Briegel 1999; Chawla 1999; Cicarelli 2007; CSG 1963; Doluee 2018; El-Nawawy 2017; Gordon 2014; Gordon 2016; Hu 2009; Hyvernat 2016; Jamshidi 2021; Jiang 2019; Keh 2003; Kurungundla 2008; Labib 2022; Loisa 2007; Luce 1988; Lv 2017; Lyu 2022; Menon 2017; Mirea 2014; Mohamed 2020; Mohamed 2023; Moskowitz 2020; Ngaosuwan 2018; Oppert 2005; Ram 2022; Schumer 1976; Sprung 1984; Sprung 2008; Tilouche 2019; Valoor 2009; Venkatesh 2018). Four trials included only participants with sepsis or septic shock and adrenal insufficiency as defined by a cortisol increment of less than 9 µg/dL after a corticotropin bolus (Aboab 2008; Birudaraju 2022; Huh 2007; Tandan 2005). In 20 trials, investigators systematically performed a short corticotropin test at baseline (Aboab 2008; Annane 2002; Annane 2010; Annane 2018; Birudaraju 2022; Bollaert 1998; Doluee 2018; El-Nawawy 2017; Huang 2014; Huh 2007; Hyvernat 2016; Keh 2016; Kurungundla 2008; Liu 2012; Meduri 2007; Oppert 2005; Sprung 2008; Tandan 2005; Yildiz 2002; Yildiz 2011).

Control

Comparison of corticosteroids versus placebo or usual care

Twenty-one studies did not use a placebo and compared corticosteroid therapy versus usual care; that is, antimicrobials, fluid resuscitation and vasopressor when needed (Agarwal 2022; Angus 2020; El Ghamrawy 2006; Ghanei 2021; Horby 2021; Hu 2009; Huang 2014; Jamshidi 2021; Jiang 2019; Li 2016; McHardy 1972; Mohamed 2020; Mohamed 2023; Ngaosuwan 2018; Rinaldi 2006; Sui 2013; Tomazini 2020; Villar 2020; Wani 2020; Zhou 2015). In one study, only one centre used a placebo (Sprung 1984). Two trials that compared hydrocortisone versus hydrocortisone plus fludrocortisone did not use a placebo of fludrocortisone for technical reasons (Annane 2010; Labib 2022). Another trial compared the duration of hydrocortisone treatment (i.e. three days versus seven days) and did not use a placebo (Huh 2007). The remaining trials compared corticosteroid therapy to placebo (Aboab 2008; Annane 2002; Annane 2018; Arabi 2011; Birudaraju 2022; Blum 2015; Bollaert 1998; Bone 1987; Briegel 1999; Chang 2020; Chawla 1999; Cicarelli 2007; Confalonieri 2005; Corral-Gudino 2021; CSG 1963; Dequin 2020; Dequin 2023; Doluee 2018; El-Nawawy 2017; Fernández-Serrano 2011; Gordon 2014; Gordon 2016; Iglesias 2020; Keh 2003; Keh 2016; Kurungundla 2008; Liu 2012; Luce 1988; Lv 2017; Luce 1988; Lyu 2022; Meduri 2007; Meduri

2009; Meduri 2022; Meijvis 2011; Menon 2017; Moskowitz 2020; Nafae 2013; Nagy 2013; Oppert 2005; Pierre de Oliveira 2023; Rezk 2013; Sabry 2011; Sai 2021; Schumer 1976; Sevransky 2021; Slusher 1996; Snijders 2010; Sprung 2008; Tagaro 2017; Tandan 2005; Tongyoo 2016; Torres 2015; Valoor 2009; VASSCSG 1987; Venkatesh 2018; Wittermans 2021; Yildiz 2002; Yildiz 2011).

Comparison of continuous infusion versus bolus administration of corticosteroids

Four trials compared continuous infusion versus bolus administration of hydrocortisone (Hyvernat 2016; Loisa 2007; Ram 2022; Tilouche 2019). One trial had three parallel arms, including continuous infusion of 200 mg hydrocortisone daily for seven days, an intravenous bolus of 50 mg hydrocortisone every six hours for seven days and usual care (Mirea 2014).

Corticosteroid dose and treatment course

Thirty-five trials tested the effects of a long course (three or more days at full dose) of low-dose hydrocortisone (Agarwal 2022; Angus 2020; Arabi 2011; Bollaert 1998; Briegel 1999; Chawla 1999; Confalonieri 2005; Dequin 2020; Dequin 2023; Doluee 2018; El Ghamrawy 2006; El-Nawawy 2017; Gordon 2014; Gordon 2016; Hu 2009; Jiang 2019; Keh 2003; Keh 2016; Liu 2012; Loisa 2007; Lv 2017; Meduri 2009; Menon 2017; Mirea 2014; Nafae 2013; Oppert 2005; Ram 2022; Rinaldi 2006; Sabry 2011; Sprung 2008; Tandan 2005; Tilouche 2019; Tongyoo 2016; Valoor 2009; Venkatesh 2018). In one trial, investigators compared hydrocortisone 50 mg intravenously every six hours when given for three days versus seven days (Huh 2007). One trial had three parallel groups, including continuous infusion of 200 mg hydrocortisone per day for seven days, intravenous bolus of 50 mg hydrocortisone every six hours for seven days and usual care (Mirea 2014). Two trials compared two doses of hydrocortisone (Hyvernat 2016; Ngaosuwan 2018). In the first trial, the investigators compared two doses of hydrocortisone given intravenously for five days (50 mg bolus every six hours, versus a continuous infusion of 300 mg per day) (Hyvernat 2016). In the second trial, participants were randomised to receive 100 mg per day or 200 mg per day of hydrocortisone, then stepwise down-titrated and discontinued on day eight (Ngaosuwan 2018). Two trials compared seven-day treatment with hydrocortisone versus seven-day treatment with the combination of hydrocortisone plus fludrocortisone (Annane 2010; Labib 2022). One trial compared a short course (two days at full dose) of low-dose intravenous hydrocortisone (300 mg on day one and 250 mg on day two) versus placebo (CSG 1963). Another study used a cross-over design to compare a three-day course of low-dose hydrocortisone versus placebo (Keh 2003).

Four trials tested the effects of a long course of low-dose hydrocortisone plus fludrocortisone (Aboab 2008; Annane 2002; Annane 2018; Sai 2021).

Six trials tested the effects of a long course of low-dose prednisone or prednisolone (Blum 2015; Ghanei 2021; McHardy 1972; Snijders 2010; Yildiz 2002; Yildiz 2011).

Six trials tested the effects of a long course of low-dose dexamethasone (Cicarelli 2007; Horby 2021; Meijvis 2011; Tomazini 2020; Villar 2020; Wittermans 2021). Two trials tested the effects of a short course (two days at full dose) of low-dose dexamethasone (Slusher 1996; Tagaro 2017).

Twelve studies tested the effects of a long course of low-dose intravenous methylprednisolone (Birudaraju 2022; Corral-Gudino 2021; Fernández-Serrano 2011; Li 2016; Meduri 2007; Meduri 2022; Nagy 2013; Pierre de Oliveira 2023; Rezk 2013; Sui 2013; Torres 2015; Zhou 2015).

Three trials tested the effects of a short course of a large dose of methylprednisolone (Bone 1987; Luce 1988; VASSCSG 1987). Two trials have three parallel arms, i.e. tested the effects of a short course of a large dose of methylprednisolone, a short course of a large dose of dexamethasone and placebo (Schumer 1976; Sprung 1984).

One trial did not report the type of corticosteroids given (Kurungundla 2008).

Eight trials tested the effects of a long course of low-dose hydrocortisone combined with vitamin C and thiamine (Chang 2020; Iglesias 2020; Lyu 2022; Mohamed 2020; Mohamed 2023; Moskowitz 2020; Sevransky 2021; Wani 2020).

Outcomes

Overall, data from 74 trials were used to inform 28-day mortality rates. One trial provided data for both the comparison of corticosteroids versus placebo or usual care and the comparison of continuous infusion versus bolus administration of corticosteroids (Mirea 2014). Two trials provided data for the comparison of continuous infusion versus bolus administration of corticosteroids (Hyvernat 2016; Tilouche 2019). We used 72 trials in the comparison of corticosteroids versus placebo or usual care. Of these trials, 51 explicitly reported on 28-day mortality (Agarwal 2022; Angus 2020; Annane 2002; Annane 2018; Arabi 2011; Birudaraju 2022; Bollaert 1998; Briegel 1999; Chang 2020; Chawla 1999; Cicarelli 2007; Corral-Gudino 2021; Dequin 2020; Dequin 2023; Doluee 2018; Ghanei 2021; Gordon 2014; Gordon 2016; Huang 2014; Horby 2021; Iglesias 2020; Jamshidi 2021; Jiang 2019; Keh 2016; Li 2016; Liu 2012; Lv 2017; Lyu 2022; Meduri 2009; Meduri 2022; Meijvis 2011; Mohamed 2020; Mohamed 2023; Moskowitz 2020; Oppert 2005; Ram 2022; Pierre de Oliveira 2023; Sai 2021; Sevransky 2021; Snijders 2010; Sprung 2008; Tagaro 2017; Tandan 2005; Tomazini 2020; Tongyoo 2016; Venkatesh 2018; Villar 2020; Wani 2020; Wittermans 2021; Yildiz 2002; Yildiz 2011). Contact with the primary author of four additional trials led to the recording of 28-day mortality rates (Confalonieri 2005; Meduri 2007; Rinaldi 2006; Sprung 1984). Thus, actual 28-day mortality rates were computed for 55 trials. For 19 additional trials, 28-day mortality rates were extrapolated from hospital mortality rates (Aboab 2008; El Ghamrawy 2006; Fernández-Serrano 2011; Luce 1988; McHardy 1972; Menon 2017; Nafae 2013; Schumer 1976; Slusher 1996; Torres 2015), ICU mortality rates (Hu 2009; Sabry 2011), 14-day mortality rates (Bone 1987; VASSCSG 1987) and short-term mortality rates (Kurungundla 2008; Mirea 2014; Nagy 2013; Rezk 2013; Valoor 2009).

Twenty-one trials explicitly reported ICU mortality rates (Annane 2002; Annane 2018; Arabi 2011; Bollaert 1998; Briegel 1999; Confalonieri 2005; Gordon 2014; Gordon 2016; Hu 2009; Iglesias 2020; Keh 2016; Lyu 2022; Meduri 2007; Menon 2017; Moskowitz 2020; Sabry 2011; Sai 2021; Sevransky 2021; Sprung 2008; Venkatesh 2018; Villar 2020), and the primary authors of three additional trials reported this outcome (Chawla 1999; Rinaldi 2006; Torres 2015).

Hospital mortality rates were available for 40 trials (Angus 2020; Annane 2002; Annane 2018; Arabi 2011; Bollaert 1998; Briegel 1999; Chawla 1999; Confalonieri 2005; El Ghamrawy 2006; Fernández-Serrano 2011; Ghanei 2021; Gordon 2014; Gordon 2016; Horby 2021; Iglesias 2020; Jamshidi 2021; Keh 2016; Luce 1988; Lv 2017; Lyu 2022; McHardy 1972; Meduri 2007; Meduri 2022; Meijvis 2011; Mohamed 2020; Mohamed 2023; Moskowitz 2020; Nafae 2013; Pierre de Oliveira 2023; Rinaldi 2006; Sai 2021; Schumer 1976; Sprung 1984; Sprung 2008; Tandan 2005; Torres 2015; Venkatesh 2018; Villar 2020; Wani 2020; Yildiz 2002).

Mortality rates at 90 days were reported for 13 trials (Angus 2020; Annane 2002; Annane 2018; Briegel 1999; Dequin 2023; Keh 2016; Lyu 2022; Meduri 2007; Meduri 2022; Mohamed 2020; Tongyoo 2016; Venkatesh 2018; Villar 2020).

Mortality rates in the long term were reported for 12 trials (Angus 2020; Annane 2002; Annane 2018; Blum 2015; Briegel 1999; Keh 2016; Meduri 2007; Meduri 2022; Mohamed 2023; Sevransky 2021; Sprung 2008; Venkatesh 2018).

Rates of shock reversal on day seven were reported in 18 trials (Annane 2002; Annane 2018; Arabi 2011; Bollaert 1998; Bone 1987; Briegel 1999; Chawla 1999; Gordon 2014; Hu 2009; Huang 2014; Iglesias 2020; Menon 2017; Mohamed 2020; Oppert 2005; Sabry 2011; Sprung 1984; Sprung 2008; Venkatesh 2018).

Rates of shock reversal on day 28 were reported in 16 trials (Agarwal 2022; Annane 2002; Annane 2018; Bollaert 1998; Briegel 1999; Chawla 1999; Doluee 2018; Gordon 2014; Gordon 2016; Liu 2012; Lv 2017; Lyu 2022; Mohamed 2020; Sprung 2008; Tandan 2005; Venkatesh 2018).

Seventeen trials reported the numbers of dysfunctional organs, that is, SOFA scores, on day seven (Annane 2002; Annane 2018; Cicarelli 2007; Confalonieri 2005; Gordon 2014; Iglesias 2020; Mirea 2014; Mohamed 2020; Mohamed 2023; Moskowitz 2020; Oppert 2005; Rinaldi 2006; Sabry 2011; Sprung 2008; Tomazini 2020; Villar 2020; Wani 2020).

Twenty-five trials reported on length of ICU stay (Agarwal 2022; Annane 2002; Annane 2018; Arabi 2011; Bollaert 1998; Briegel 1999; Chawla 1999; Confalonieri 2005; Fernández-Serrano 2011; Gordon 2014;;Gordon 2016; Hu 2009; Iglesias 2020; Keh 2016; Lv 2017; Meduri 2007; Meduri 2009; Menon 2017; Mirea 2014; Mohamed 2020; Pierre de Oliveira 2023; Rinaldi 2006; Sprung 2008; Torres 2015; Venkatesh 2018).

Thirty-one trials reported on length of hospital stay (Annane 2002; Annane 2018; Arabi 2011; Blum 2015; Bollaert 1998; Chawla 1999; Confalonieri 2005; Corral-Gudino 2021; Fernández-Serrano 2011; Ghanei 2021; Gordon 2014; Gordon 2016; Horby 2021; Iglesias 2020; Keh 2016; Lv 2017; Meduri 2007; Meijvis 2011; Menon 2017; Mirea 2014; Mohamed 2020; Nagy 2013; Pierre de Oliveira 2023; Sevransky 2021; Snijders 2010; Sprung 2008; Torres 2015; Venkatesh 2018; Wani 2020; Wittermans 2021; Yildiz 2002).

Thirty-six trials reported on superinfection (Agarwal 2022; Annane 2002; Annane 2018; Arabi 2011; Blum 2015; Bollaert 1998; Bone 1987; Briegel 1999; Chang 2020; Chawla 1999; Cicarelli 2007; Confalonieri 2005; Corral-Gudino 2021; Dequin 2023; Gordon 2014; Jiang 2019; Keh 2016; Luce 1988; Meduri 2022; Meijvis 2011; Menon 2017; Mohamed 2023; Moskowitz 2020; Pierre de Oliveira 2023; Rezk 2013; Schumer 1976; Snijders 2010; Sprung 1984; Sprung 2008; Tomazini 2020; Tongyoo 2016; Torres 2015; VASSCSG 1987; Villar 2020; Yildiz 2002; Yildiz 2011).

Seven trials reported on muscle weakness (Annane 2002; Annane 2018; Confalonieri 2005; Keh 2016; Meduri 2022; Sprung 2008; Venkatesh 2018).

Excluded studies

We excluded 43 trials (see Characteristics of excluded studies for details). Ten trials have tested the effects of corticosteroids in non-septic patients (Asehnoune 2014; Huang 2015; Mikami 2007; Newberry 2017; Peeters 2018; Roquilly 2011; Tam 2012; van Woensel 2003; Venet 2015; Xi 2021). Eight trials reported on the effects of corticosteroids in a mixed population and separate data for participants with sepsis were not available (Bernard 1987; Cicarelli 2006; Khan 2021; McKee 1983; Meduri 1998b; Schwingshackl 2016; Steinberg 2006; Weigelt 1985). In six trials, they used comparators irrelevant for this review (ACTRN12621000247875 2021; CTRI/2021/11/038268 2021; Fujii 2020; Huang 2014a; Hussein 2021; Malaska 2022). Five quasi-randomised trials were excluded (Fan 2019; Klastersky 1971; Lucas 1984; Raghu 2021; Wagner 1955). Six trials were prematurely terminated without providing any relevant data for this review (Munch 2021; NCT02602210 2015; NCT03335124 2017; NCT03828929 2019; Rogers 1970; Thompson 1976). Six additional trials reported only acute effects of corticosteroids on clinical or laboratory parameters (Hahn 1951; Hughes 1984; Kaufman 2008; Lan 2015; Luo 2014; Marik 1993). One trial evaluated a bundle of measures including corticosteroids (Lloyd 2019). One trial with a cross-over design had no relevant information on the outcomes for this review (NCT04134403 2019).

Ongoing studies

From trial registries, we identified 16 additional RCTs of corticosteroids for sepsis. These trials are still recruiting patients, and we will follow the status of these trials to include them in a future update of the review whenever data become publicly available.

Studies awaiting classification

There have been 12 trials published in the past year that are awaiting classification. Six trials have compared different treatment modalities for corticotherapy in patients with COVID-19 (Corral Gudino 2023; Franco-Moreno 2024; Laikitmongkhon 2024; Sahraei 2023; Salton 2023; Taher 2023). Three trials have investigated the benefits and risks of the combination of hydrocortisone with vitamin C and thiamine (Schlapbach 2024; Sharma 2024; Wang 2023). One trial has compared the benefits and risks of a continuous infusion of hydrocortisone to intermittent intravenous bolus administration (Salhotra 2024). One trial has compared the effects of increasing doses of fludrocortisone in adults with septic shock treated with hydrocortisone (Walsham 2024). One trial has compared the effects of intravenous bolus of hydrocortisone in adults with sepsis and adrenal insufficiency, defined by low serum levels of random total cortisol (Mohanty 2024).

Risk of bias in included studies

We report the detailed risk of bias assessments for individual trials in the risk of bias tables in Figure 2.

Figure 2. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.

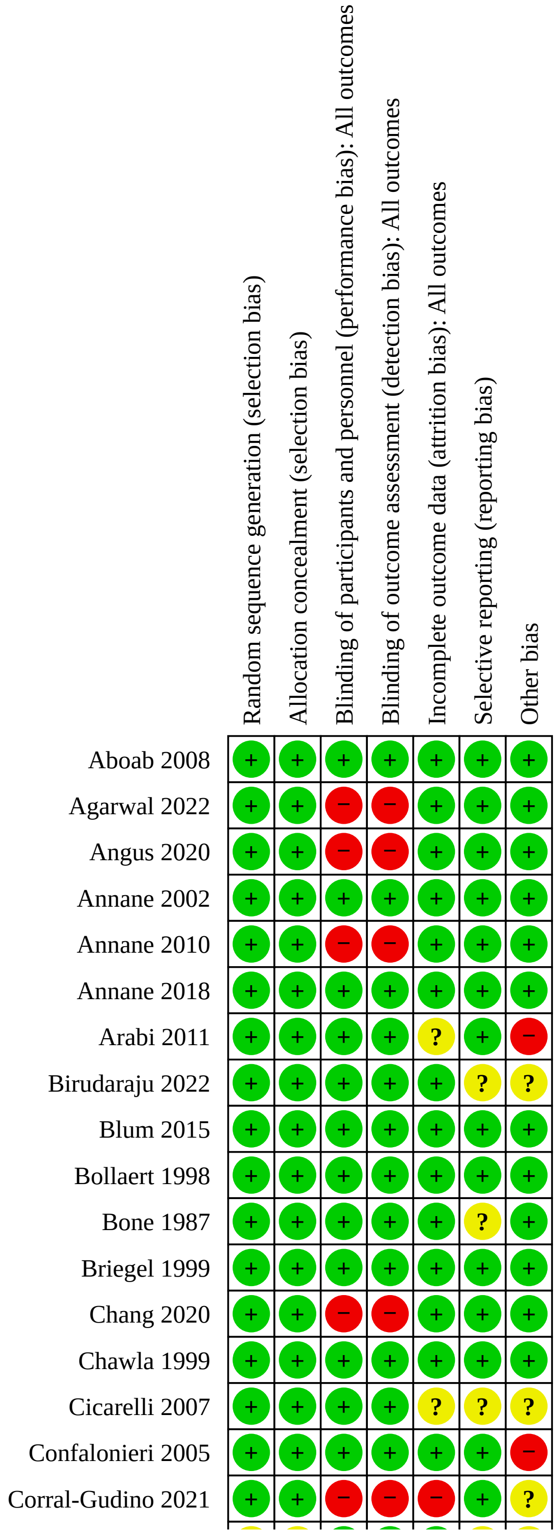
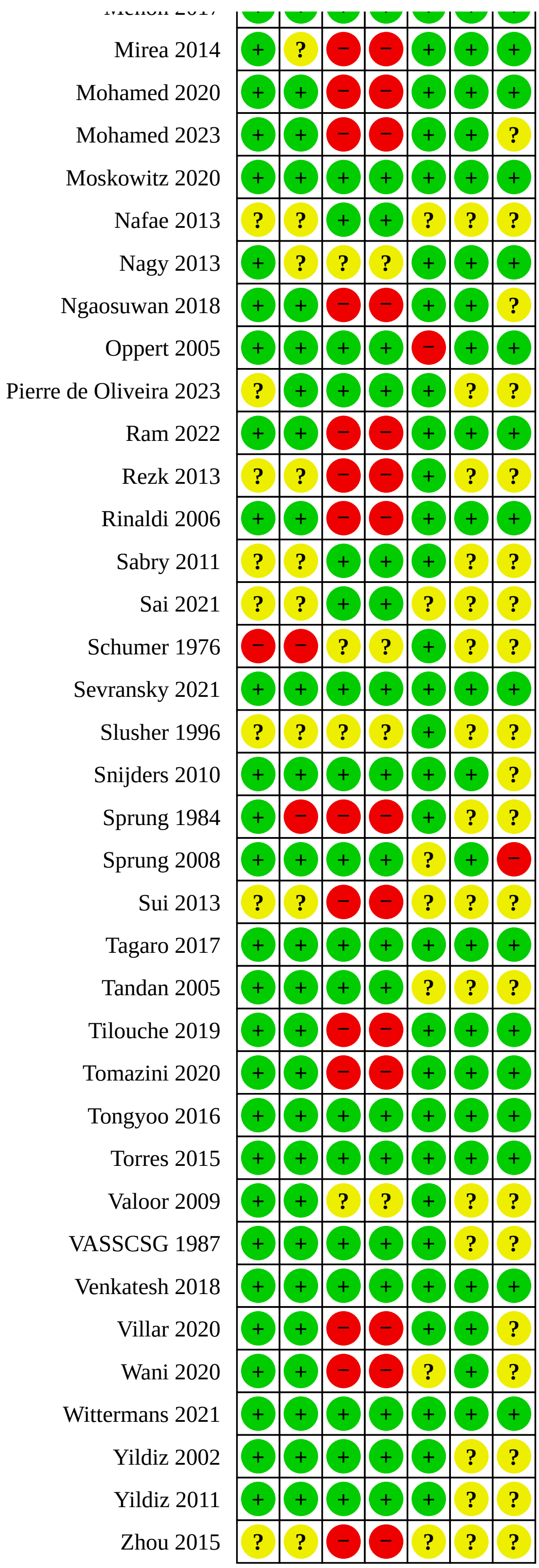


Figure 2. (Continued)

Corral-Gudino 2021	+	+	−	−	−	+	?
CSG 1963	?	?	+	+	+	?	?
Dequin 2020	+	+	+	+	+	+	+
Dequin 2023	+	+	+	+	+	+	+
Doluee 2018	?	+	+	+	+	+	?
El Ghamrawy 2006	?	?	?	?	+	?	?
El-Nawawy 2017	?	?	+	+	+	?	?
Fernández-Serrano 2011	+	+	+	+	−	+	?
Ghanei 2021	?	+	−	−	?	+	+
Gordon 2014	+	+	+	+	+	+	+
Gordon 2016	+	+	+	+	+	+	+
Horby 2021	+	+	−	−	+	+	?
Hu 2009	?	?	?	?	+	?	?
Huang 2014	+	?	?	?	+	?	?
Huh 2007	+	?	−	−	+	?	?
Hyvernat 2016	+	+	+	?	+	+	+
Iglesias 2020	+	+	+	+	+	+	+
Jamshidi 2021	+	?	−	−	+	+	?
Jiang 2019	?	?	−	−	+	?	?
Keh 2003	+	+	+	+	+	+	+
Keh 2016	+	+	+	?	+	+	+
Kurungundla 2008	?	?	?	?	?	?	?
Labib 2022	+	+	−	−	+	?	+
Li 2016	?	?	?	?	?	?	?
Liu 2012	+	?	?	?	?	?	?
Loisa 2007	?	?	−	−	?	?	?
Luce 1988	+	+	+	+	−	?	?
Lv 2017	+	+	+	+	+	+	+
Lyu 2022	+	+	+	+	+	+	+
McHardy 1972	?	+	−	−	+	?	?
Meduri 2007	+	+	+	+	+	+	−
Meduri 2009	+	+	+	+	+	+	?
Meduri 2022	+	+	+	+	+	+	?
Meijvis 2011	+	+	+	+	+	+	+
Menon 2017	+	+	+	+	+	+	+
Mirea 2014	+	?	−	−	+	+	+

Figure 2. (Continued)



We assessed 25 trials as having a low risk of bias for the five domains (allocation, blinding, incomplete data, selective reporting and other potential sources of bias). We judged these 25 trials as being at overall low risk of bias (Aboab 2008; Annane 2002; Annane 2018; Blum 2015; Bollaert 1998; Briegel 1999; Chawla 1999; Dequin 2023; Gordon 2014; Gordon 2016; Hyvernat 2016; Iglesias 2020; Keh 2003; Keh 2016; Lv 2017; Lyu 2022; Meijvis 2011; Menon 2017; Moskowitz 2020; Sevransky 2021; Tagaro 2017; Tongyoo 2016; Torres 2015; Venkatesh 2018; Wittermans 2021).

Random sequence generation

In one trial, the method of generation of allocation sequence was based on a card system (Schumer 1976); we judged this trial to be at high risk of selection bias. For 19 trials, the method was unclear (CSG 1963; Doluee 2018; El Ghamrawy 2006; El-Nawawy 2017; Ghanei 2021; Hu 2009; Jiang 2019; Kurungundla 2008; Li 2016; Loisa 2007; McHardy 1972; Nafae 2013; Pierre de Oliveira 2023; Rezk 2013; Sabry 2011; Sai 2021; Slusher 1996; Sui 2013; Zhou 2015). We judged the remaining trials to have a low risk of bias in this domain (Aboab 2008; Agarwal 2022; Angus 2020; Annane 2002; Annane 2010; Annane 2018; Arabi 2011; Birudaraju 2022; Blum 2015; Bollaert 1998; Bone 1987; Briegel 1999; Chang 2020; Chawla 1999; Cicarelli 2007; Confalonieri 2005; Corral-Gudino 2021; Dequin 2020; Dequin 2023; Fernández-Serrano 2011; Gordon 2014; Gordon 2016; Horby 2021; Huang 2014; Huh 2007; Hyvernat 2016; Iglesias 2020; Jamshidi 2021; Keh 2003; Keh 2016; Labib 2022; Liu 2012; Luce 1988; Lv 2017; Lyu 2022; Meduri 2007; Meduri 2009; Meduri 2022; Meijvis 2011; Menon 2017; Mirea 2014; Mohamed 2020; Mohamed 2023; Moskowitz 2020; Nagy 2013; Ngaosuwan 2018; Oppert 2005; Ram 2022; Rinaldi 2006; Sevransky 2021; Snijders 2010; Sprung 1984; Sprung 2008; Tagaro 2017; Tandan 2005; Tilouche 2019; Tomazini 2020; Tongyoo 2016; Torres 2015; Valoor 2009; VASSCSG 1987; Venkatesh 2018; Villar 2020; Wani 2020; Wittermans 2021; Yildiz 2002; Yildiz 2011).

Allocation concealment

We judged the method used for allocation concealment to be at low risk of bias in all but 23 trials. One trial assigned treatment using unsealed envelopes (Schumer 1976). In another trial, investigators enrolling participants at one of the two participating centres could have foreseen the upcoming assignment, as the local ethical committee refused to accept concealed allocation (Sprung 1984). For these two trials, we judged the risk of bias as being high in this domain. In 21 trials, the study authors did not report sufficient information to judge the method used for allocation concealment (CSG 1963; El Ghamrawy 2006; El-Nawawy 2017; Hu 2009; Huang 2014; Huh 2007; Jamshidi 2021; Jiang 2019; Kurungundla 2008; Li 2016; Liu 2012; Loisa 2007; Mirea 2014; Nafae 2013; Nagy 2013; Rezk 2013; Sabry 2011; Sai 2021; Slusher 1996; Sui 2013; Zhou 2015). We judged the risk of bias in this domain as being uncertain.

Blinding of participants and personnel

For 27 trials, we judged the method used for blinding participants and personnel as having a high risk of bias. In 22 studies, participants and personnel were not blinded to study interventions (Agarwal 2022; Angus 2020; Corral-Gudino 2021; Ghanei 2021; Horby 2021; Jamshidi 2021; Jiang 2019; Loisa 2007; McHardy 1972; Mirea 2014; Mohamed 2020; Mohamed 2023; Ngaosuwan 2018; Ram 2022; Rezk 2013; Rinaldi 2006; Sui 2013; Tilouche 2019; Tomazini 2020; Villar 2020; Wani 2020; Zhou 2015). In one study, only one centre used a placebo (Sprung 1984). Two trials that compared hydrocortisone versus hydrocortisone plus fludrocortisone did not blind the administration of fludrocortisone for technical reasons (Annane 2010; Labib 2022). Another trial compared the duration of hydrocortisone treatment (i.e. three days versus seven days) and it did not use a placebo (Huh 2007). In one trial, participants were blinded to the study interventions and not personnel (Chang 2020). We judged this trial as having a high risk of bias in this domain.

In 50 trials, the comparison of the effects of corticosteroid therapy and those of a placebo used appropriate blinding of participants and personnel (Aboab 2008; Annane 2002; Annane 2018; Arabi 2011; Birudaraju 2022; Blum 2015; Bollaert 1998; Bone 1987; Briegel 1999; Chawla 1999; Cicarelli 2007; Confalonieri 2005; CSG 1963; Dequin 2020; Dequin 2023; Doluee 2018; El-Nawawy 2017; Fernández-Serrano 2011; Gordon 2014; Gordon 2016; Hyvernat 2016; Iglesias 2020; Keh 2003; Keh 2016; Luce 1988; Lv 2017; Lyu 2022; Meduri 2007; Meduri 2009; Meduri 2022; Meijvis 2011; Menon 2017; Moskowitz 2020; Nafae 2013; Oppert 2005; Pierre de Oliveira 2023; Sabry 2011; Sai 2021; Sevransky 2021; Snijders 2010; Sprung 2008; Tagaro 2017; Tandan 2005; Tongyoo 2016; Torres 2015; VASSCSG 1987; Venkatesh 2018; Wittermans 2021; Yildiz 2002; Yildiz 2011).

Ten additional trials did not report the method used to ensure blinding (El Ghamrawy 2006; Hu 2009; Huang 2014; Kurungundla 2008; Li 2016; Liu 2012; Nagy 2013; Schumer 1976; Slusher 1996; Valoor 2009). We judged these trials as having an unclear risk of bias in this domain.

Blinding of outcomes assessors

In 27 trials, outcome assessors were not blinded to the study interventions (Agarwal 2022; Annane 2010; Angus 2020; Chang 2020; Corral-Gudino 2021; Ghanei 2021; Horby 2021; Huh 2007; Jamshidi 2021; Jiang 2019; Labib 2022; Loisa 2007;.McHardy 1972; Mirea 2014; Mohamed 2020; Mohamed 2023; Ngaosuwan 2018; Ram 2022; Rezk 2013; Rinaldi 2006; Sprung 1984; Sui 2013; Tilouche 2019; Tomazini 2020; Villar 2020; Wani 2020; Zhou 2015). We judged these trials to be at high risk of detection bias.

Ten trials did not report sufficient information about blinding of outcome assessors (El Ghamrawy 2006; Hu 2009; Huang 2014; Kurungundla 2008; Li 2016; Liu 2012; Nagy 2013; Schumer 1976; Slusher 1996; Valoor 2009). We judged these trials as having an unclear risk of bias in this domain.

For the remaining 50 trials, we deemed blinding of outcomes assessors to be appropriate (Aboab 2008; Annane 2002; Annane 2018; Arabi 2011; Birudaraju 2022; Blum 2015; Bollaert 1998; Bone 1987; Briegel 1999; Chawla 1999; Cicarelli 2007; Confalonieri 2005; CSG 1963; Dequin 2020; Dequin 2023; Doluee 2018; El-Nawawy 2017; Fernández-Serrano 2011; Gordon 2014; Gordon 2016; Hyvernat 2016; Iglesias 2020; Keh 2003; Keh 2016; Luce 1988; Lv 2017; Lyu 2022; Meduri 2007; Meduri 2009; Meduri 2022; Meijvis 2011; Menon 2017; Moskowitz 2020; Nafae 2013; Oppert 2005; Pierre de Oliveira 2023; Sabry 2011; Sai 2021; Sevransky 2021; Snijders 2010; Sprung 2008; Tagaro 2017; Tandan 2005; Tongyoo 2016; Torres 2015; VASSCSG 1987; Venkatesh 2018; Wittermans 2021; Yildiz 2002; Yildiz 2011). We judged these trials to be at low risk of bias in this domain.

Incomplete outcome data

We judged the risk of attrition bias as being low for 69 trials (Aboab 2008; Agarwal 2022; Angus 2020; Annane 2002; Annane 2010; Annane 2018; Birudaraju 2022; Blum 2015; Bollaert 1998; Bone 1987; Briegel 1999; Chang 2020; Chawla 1999; Confalonieri 2005; CSG 1963; Dequin 2020; Dequin 2023; Doluee 2018; El Ghamrawy 2006; El-Nawawy 2017; Gordon 2014; Gordon 2016; Horby 2021; Hu 2009; Huang 2014; Huh 2007; Hyvernat 2016; Iglesias 2020; Jamshidi 2021; Jiang 2019; Keh 2003; Keh 2016; Labib 2022; Lv 2017; Lyu 2022; McHardy 1972; Meduri 2007; Meduri 2009; Meduri 2022; Meijvis 2011; Menon 2017; Mirea 2014; Mohamed 2020; Mohamed 2023; Moskowitz 2020; Nagy 2013; Ngaosuwan 2018; Pierre de Oliveira 2023; Ram 2022; Rezk 2013; Rinaldi 2006; Sabry 2011; Schumer 1976; Sevransky 2021; Slusher 1996; Snijders 2010; Sprung 1984; Tagaro 2017; Tilouche 2019; Tomazini 2020; Tongyoo 2016; Torres 2015; Valoor 2009; VASSCSG 1987; Venkatesh 2018; Villar 2020; Wittermans 2021; Yildiz 2002; Yildiz 2011).

We judged four trials as being at high risk of bias in this domain (Corral-Gudino 2021; Fernández-Serrano 2011; Luce 1988; Oppert 2005). In one trial, due to a low recruitment rate, the expected sample size of 180 was not achieved, and only 64 patients were randomised (Corral-Gudino 2021). Five patients in the experimental arm did not receive the study drug and one patient in the usual care arm received methylprednisolone. In another trial, 21% of patients in the placebo arm and 18% in the active treatment arm were not included in the primary analysis (Fernández-Serrano 2011). In a third trial, 12 out of 87 randomly assigned participants were not analysed, and follow-up for these patients was not provided (Luce 1988). In the last study, seven randomly assigned participants (five in the corticosteroid group and two in the placebo group) were not analysed (Oppert 2005). For this study, contact with the primary author allowed us to obtain information on participants who had dropped out from the analysis. Four of these participants (two in the corticosteroid group and two in the placebo group) were discharged alive from the ICU and then were lost to follow-up. The three remaining participants (in the corticosteroid group) died: two before receiving hydrocortisone and the last on study day 17.

The risk of attrition bias was unclear for 14 trials (Arabi 2011; Cicarelli 2007; Ghanei 2021; Kurungundla 2008; Li 2016; Liu 2012; Loisa 2007; Nafae 2013; Sai 2021; Sprung 2008; Sui 2013; Tandan 2005; Wani 2020; Zhou 2015).

Selective reporting

For 55 trials, we could rule out selective reporting bias after contact with authors, full access to trial protocols or access to individual participant data (Aboab 2008; Agarwal 2022; Angus 2020; Annane 2002; Annane 2010; Annane 2018; Arabi 2011; Blum 2015; Bollaert 1998; Briegel 1999; Chang 2020; Chawla 1999; Confalonieri 2005; Corral-Gudino 2021; Dequin 2020; Dequin 2023; Doluee 2018; Fernández-Serrano 2011; Ghanei 2021; Gordon 2014; Gordon 2016; Horby 2021; Hyvernat 2016; Iglesias 2020; Jamshidi 2021; Keh 2003; Keh 2016; Lv 2017; Lyu 2022; Meduri 2007; Meduri 2009; Meduri 2022; Meijvis 2011; Menon 2017; Mirea 2014; Mohamed 2020; Mohamed 2023; Moskowitz 2020; Nagy 2013; Ngaosuwan 2018; Oppert 2005; Ram 2022; Rinaldi 2006; Sevransky 2021; Snijders 2010; Sprung 2008; Tagaro 2017; Tilouche 2019; Tomazini 2020; Tongyoo 2016; Torres 2015; Venkatesh 2018; Villar 2020; Wani 2020; Wittermans 2021). We judged these trials to be at low risk of bias in this domain.

In 32 trials, there was insufficient information to assess the risk of reporting bias (Birudaraju 2022; Bone 1987; Cicarelli 2007; CSG 1963; El Ghamrawy 2006; El-Nawawy 2017; Hu 2009; Huang 2014; Huh 2007; Jiang 2019; Kurungundla 2008; Labib 2022; Li 2016; Liu 2012; Loisa 2007; Luce 1988; McHardy 1972; Nafae 2013; Pierre de Oliveira 2023; Rezk 2013; Sabry 2011; Sai 2021; Schumer 1976; Slusher 1996; Sprung 1984; Sui 2013; Tandan 2005; Valoor 2009; VASSCSG 1987; Yildiz 2002; Yildiz 2011; Zhou 2015). We judged these trials as having an unclear risk of selective reporting bias.

Other potential sources of bias

We judged four trials as having a high risk of bias in this domain (Arabi 2011; Confalonieri 2005; Meduri 2007; Sprung 2008). One trial was stopped prematurely for futility after enrolment of only 75 out of 150 foreseen patients (Arabi 2011). Two trials were stopped for apparent benefit, yielding small-sized trials and some imbalances in baseline characteristics between placebo and intervention arms (Confalonieri 2005; Meduri 2007). The fourth trial stopped recruiting patients owing to loss of equipoise amongst investigators after only 500 of the 800 expected participants were recruited (Sprung 2008).

We judged 42 trials as having unclear risk of bias in this domain as there was insufficient information to draw a conclusion (Birudaraju 2022; Cicarelli 2007; Corral-Gudino 2021; CSG 1963; Doluee 2018; El Ghamrawy 2006; El-Nawawy 2017; Fernández-Serrano 2011; Horby 2021; Hu 2009; Huang 2014; Huh 2007; Jamshidi 2021; Jiang 2019; Kurungundla 2008; Li 2016; Liu 2012; Loisa 2007; Luce 1988; McHardy 1972; Meduri 2009; Meduri 2022; Mohamed 2023; Nafae 2013; Ngaosuwan 2018; Pierre de Oliveira 2023; Rezk 2013; Sabry 2011; Sai 2021; Schumer 1976; Slusher 1996; Snijders 2010; Sprung 1984; Sui 2013; Tandan 2005; Valoor 2009; VASSCSG 1987; Villar 2020; Wani 2020; Yildiz 2002; Yildiz 2011; Zhou 2015).

We judged the remaining trials as being at low risk of bias in this domain (Aboab 2008; Agarwal 2022; Angus 2020; Annane 2002; Annane 2010; Annane 2018; Blum 2015; Bollaert 1998; Bone 1987; Briegel 1999; Chang 2020; Chawla 1999; Dequin 2020; Dequin 2023; Ghanei 2021; Gordon 2014; Gordon 2016; Hyvernat 2016; Iglesias 2020; Keh 2003; Keh 2016; Labib 2022; Lv 2017; Lyu 2022; Meijvis 2011; Menon 2017; Mirea 2014; Mohamed 2020; Moskowitz 2020; Nagy 2013; Oppert 2005; Ram 2022; Rinaldi 2006; Sevransky 2021; Tagaro 2017; Tilouche 2019; Tomazini 2020; Tongyoo 2016; Torres 2015; Venkatesh 2018; Wittermans 2021).

Effects of interventions

See: **Summary of findings 1** Summary of findings table - Corticosteroids versus placebo or usual care; **Summary of findings 2** Summary of findings table - Continuous infusion versus bolus administration of corticosteroids

We did not pool the data from one trial that included both children and adults (CSG 1963). In this trial, hydrocortisone was given intravenously for only two days at low dose, followed by oral administration and rapid tapering off. One cross-over trial compared the effects of low-dose hydrocortisone to those of a placebo on haemodynamic and immune parameters (Keh 2003). This trial could not provide data for any of the review outcomes. One trial that compared two durations of hydrocortisone treatment (Huh 2007), and two trials that compared hydrocortisone versus the combination of hydrocortisone plus fludrocortisone (Annane 2010; Labib 2022), could not contribute to the meta-analysis. Indeed, the comparators in these trials did not correspond to the requirements for the comparison of corticosteroids versus placebo or usual care, nor for the comparison of continuous infusion versus intermittent bolus administration.

Corticosteroids versus placebo or usual care

We have summarised the main results in Summary of findings 1.

Primary outcome

28-day all-cause mortality

Data for 28-day mortality were available for 54 trials. In addition, we used data on 14-day mortality (two trials), hospital mortality (10 trials) or ICU mortality (two trials), or data on short-term mortality (four trials). Thus, we computed data from 72 trials that accounted for 22,915 participants. In the treated group, 2449 of 10,460 participants died by day 28, compared with 3207 of 12,455 participants in the control group. The heterogeneity in the results may be moderate (Chi^2 = 98.15, df = 70 (P = 0.01); I^2 = 29%). Compared to placebo or usual care, corticosteroids probably reduced 28-day mortality (risk ratio (RR) 0.89, 95% confidence interval (CI) 0.84 to 0.95; P = 0.0007; random-effects model; Analysis 1.1). We downgraded the certainty of evidence for this outcome from high to moderate for inconsistency.

A sensitivity analysis limited to trials judged to be at low risk of bias confirmed that corticosteroids probably reduced 28-day mortality (RR 0.89, 95% CI 0.81 to 0.99; P = 0.02; 25 studies, 11,125 participants; Analysis 1.2). The heterogeneity in the results might not be important (Chi^2 = 33.27, df = 24 (P = 0.10); I^2 = 28%). Another sensitivity analysis based on trials published after 2016 (i.e. after the 2016 consensus definition for Sepsis 3 (Singer 2016)) found a RR of 0.89, 95% CI 0.83 to 0.95; P = 0.0005; 28 trials, 17,023 participants; moderate-certainty evidence (Analysis 1.1). The heterogeneity in the results might not be important (Chi^2 = 32.33, df = 27 (P = 0.22); I^2 = 16%).

Subgroup analyses based on types of corticosteroids did not suggest that this may influence the response to treatment (test for subgroup differences: Chi^2 = 3.33, df = 5 (P = 0.65); I^2 = 0%; Analysis 1.3). Likewise, there was no evidence that dose/duration (long course of low dose versus short course of high dose) (test for subgroup differences: Chi^2 = 0.00, df = 1 (P = 0.97); I^2 = 0%; Analysis 1.4) or mode (continuous versus intermittent bolus) of corticosteroid administration (test for subgroup differences: Chi^2 = 1.17, df = 1 (P = 0.28); I^2 = 14.6%; Analysis 1.5) influences the response to treatment. Meta-regression analyses suggested that the lower the dose given on day one, the lower the RR for dying at 28 days (P = 0.0263; Figure 3), the lower the cumulative dose, the lower the RR for dying at 28 days (P =0.0227; Figure 4) and the longer the duration of treatment, the lower the RR for dying at 28 days (P = 0.041) (Figure 5). One trial of a large dose of corticosteroids was a statistical outlier and was excluded from the meta-regression analysis (Schumer 1976). Subgroup analysis based on modalities of drug termination (with versus without taper off) did not suggest that this may influence the response to treatment (test for subgroup differences: Chi^2 = 0.54, df = 1 (P = 0.46); I^2 = 0%; Analysis 1.6).

Figure 3. Figure represents the results from meta-regression of log of risk ratio of dying and the dose of corticosteroids given at day 1 and expressed as equivalent milligrams of hydrocortisone. Estimates from each study are represented by circles. Circle sizes depend on the precision of each estimate (the inverse of its within-study variance), which is the weight given to each study in the random-effects model. Meta-regression included 73 trials.
REML estimate of between-study variance: tau^2 = 5.1e-07
% residual variation due to heterogeneity: I^2 res = 0.00%
Proportion of between-study variance explained: Adj R^2 = 100% P = 0.0263

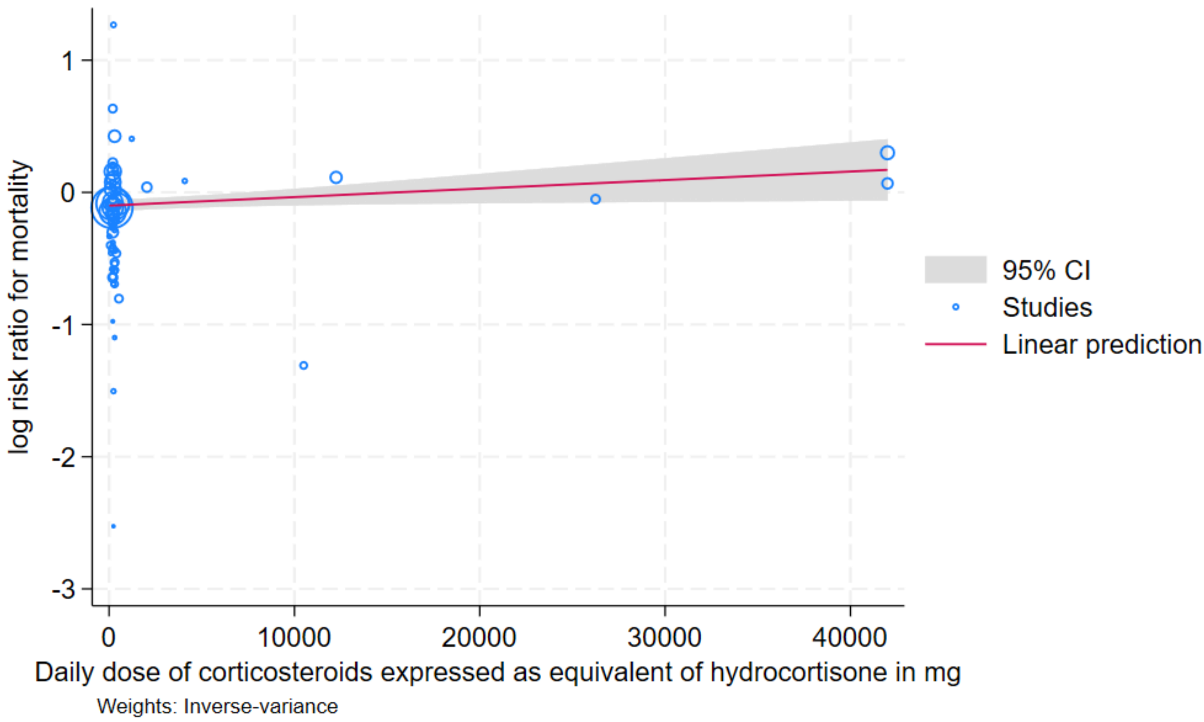


Figure 4. Figure represents results from meta-regression of log of risk ratio of dying and cumulated dose of corticosteroids expressed as equivalent milligrams of hydrocortisone. Estimates from each study are represented by circles. Circle sizes depend on the precision of each estimate (the inverse of its within-study variance), which is the weight given to each study in the random-effects model. Meta-regression included 70 trials. REML estimate of between-study variance: tau² = 1.4e-08
% residual variation due to heterogeneity: I² res = 0.00%
Proportion of between-study variance explained: Adj R² = 92.54% P = 0.0227

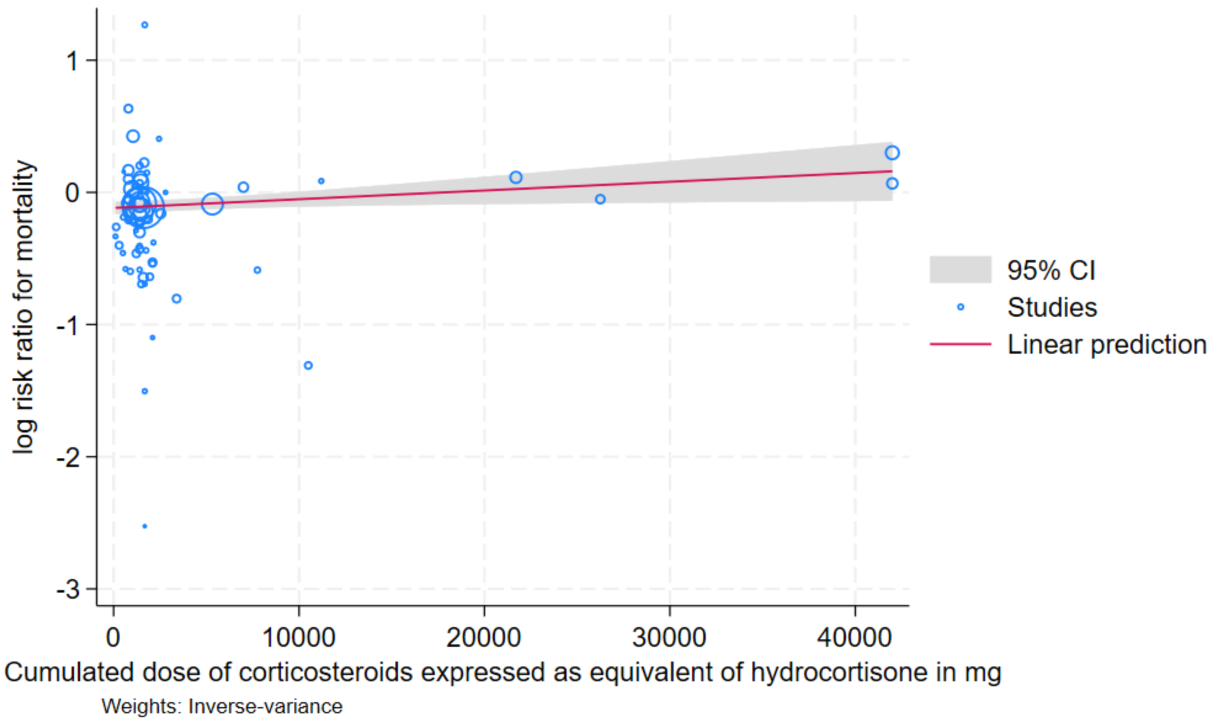
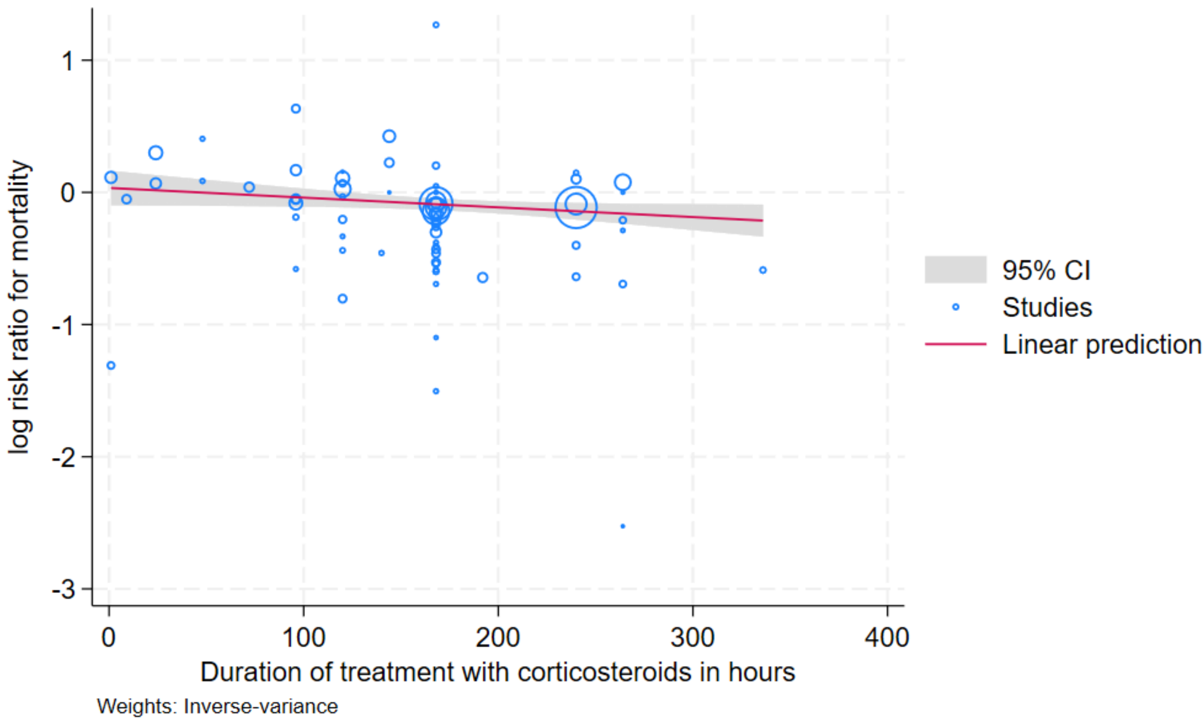


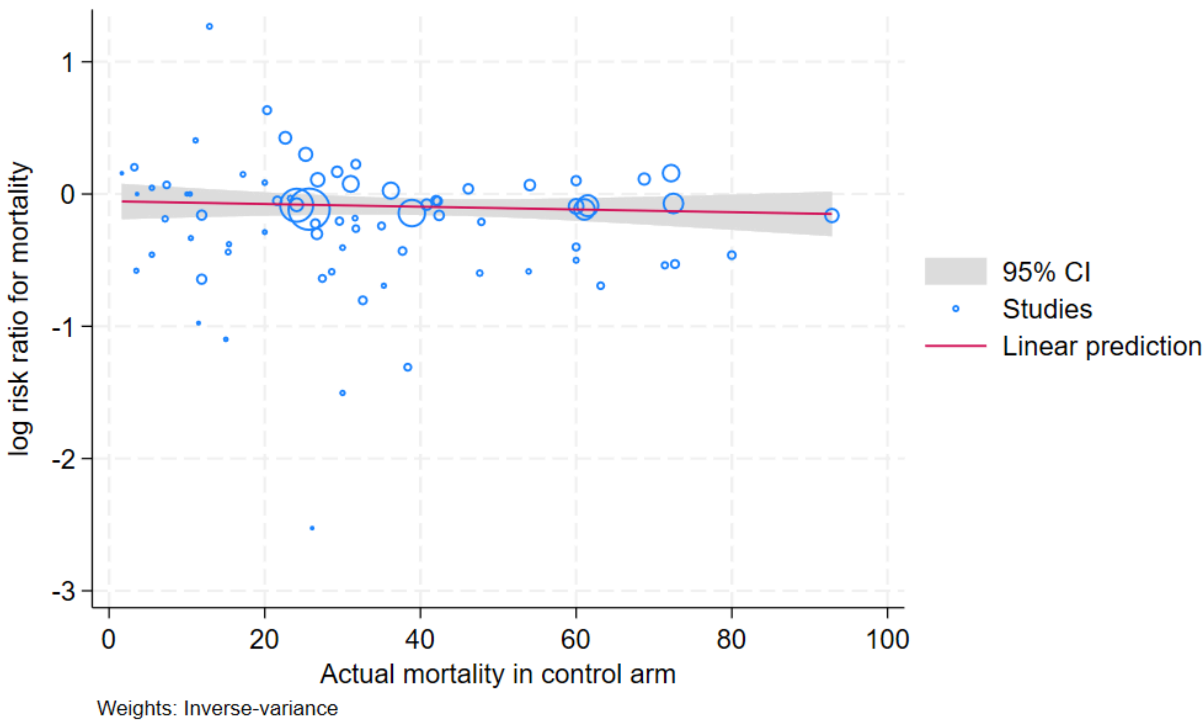
Figure 5. Figure represents results from meta-regression of log of risk ratio of dying and duration of corticotherapy expressed as equivalent milligrams of hydrocortisone. Estimates from each study are represented by circles. Circle sizes depend on the precision of each estimate (the inverse of its within-study variance), which is the weight given to each study in the random-effects model. Meta-regression included 70 trials. REML estimate of between-study variance: tau^2 = 1.6e-07
% residual variation due to heterogeneity: I^2 res = 0.00%
Proportion of between-study variance explained: Adj R^2 = 15.39% P = 0.0409



Subgroup analyses based on factors related to participants suggested no evidence of differences in response to treatment between children and adults (test for subgroup differences: Chi^2 = 0.34, df = 1 (P = 0.56); I^2 = 0%; Analysis 1.7).

Subgroup analysis based on targeted populations showed significant subgroup differences (test for subgroup differences: Chi^2 = 16.05, df = 4 (P = 0.003); I^2 = 75.1%; Analysis 1.8). In studies of heterogeneous populations of participants with sepsis, the RR for dying at 28 days was 1.08 (95% CI 0.90 to 1.28; P = 0.41; Chi^2 = 18.94, df = 13 (P = 0.12); I^2 = 31%; random-effects model; 14 studies, 1825 participants). In studies of only participants with septic shock, the RR for dying at 28 days was 0.93 (95% CI 0.86 to 1.00; P = 0.04; Chi^2 = 35.95, df = 28 (P = 0.14); I^2 = 22%; random-effects model; 29 studies, 8871 participants). In studies of participants with sepsis and ARDS, the RR was 0.59 (95% CI 0.42 to 0.83; P = 0.002; Chi^2 = 4.72, df = 4 (P = 0.32); I^2 = 15%; random-effects model; 5 studies, 496 participants). In studies of participants with sepsis and community-acquired pneumonia, the RR was 0.68 (95% CI 0.54 to 0.86; P = 0.001; Chi^2 = 13.07, df = 14 (P = 0.52); I^2 = 0%; random-effects model; 16 studies, 3818 participants). In studies of participants with COVID-19-related sepsis, the RR was 0.88 (95% CI 0.81 to 0.95; P = 0.002; Chi^2 = 3.93, df = 6 (P = 0.69); I^2 = 0%; random-effects model; 7 studies, 7764 participants). Subgroup analysis based on population suggests that the improved mortality with corticosteroids seen in other subsets of patients with sepsis may not be present in those with sepsis without shock. The credibility of this effect modification was rated as low to moderate. While the P value suggested that chance was an unlikely explanation (P = 0.003) and this subgroup effect was hypothesised a priori, this examination of the subgroup effect was based strictly on between-study comparisons and the meta-analysis examined numerous subgroups, increasing the possibility of chance findings. Meta-regression showed no evidence of interaction between the RR of dying at 28 days and severity of illness as assessed by crude mortality in the control arm (P = 0.5013; Figure 6). Subgroup analysis suggested no evidence of differences in response to treatment between participants with versus without critical illness associated with corticosteroid insufficiency (test for subgroup differences: Chi^2 = 0.20, df = 1 (P = 0.66); I^2 = 0%; Analysis 1.9). Subgroup analysis of participants with critical illness associated with corticosteroid insufficiency showed that heterogeneity in the results might not be important. Investigators reported 256 deaths amongst 535 participants in the treated group and 301 deaths amongst 566 in the placebo group. The RR for dying was 0.90 (95% CI 0.80 to 1.00; P = 0.06; Chi^2 = 8.74, df = 11 (P = 0.65); I^2 = 0%; random-effects model; 12 studies, 1101 participants; Analysis 1.9). Subgroup analysis of participants without critical illness associated with corticosteroid insufficiency showed that heterogeneity in the results might not be important. The RR for dying was 0.85 (95% CI 0.71 to 1.02; P = 0.06; Chi^2 = 8.21, df = 8 (P = 0.41); I^2 = 3%; random-effects model; 9 studies, 870 participants; Analysis 1.9).

Figure 6. Figure represents results from meta-regression of log of risk ratio of dying and actual mortality in the control group. Estimates from each study are represented by circles. Circle sizes depend on the precision of each estimate (the inverse of its within-study variance), which is the weight given to each study in the random-effects model. Meta-regression included 74 trials. REML estimate of between-study variance: tau² = 0.01369
% residual variation due to heterogeneity: I² res = 26.25%
Proportion of between-study variance explained: Adj R² = 0% P = 0.5013



Funnel plot analysis, including all trials, suggested some asymmetry (Figure 7). Contour-enhanced funnel plot analysis including trials of a long course of low-dose corticosteroids suggested small-study effects (Figure 8).

Figure 7. Funnel plot of comparison: 1 Steroids vs control, outcome: 1.1 28-day all-cause mortality. P = 0.0051 by regression-based Egger test for small-study effects
Random-effects model

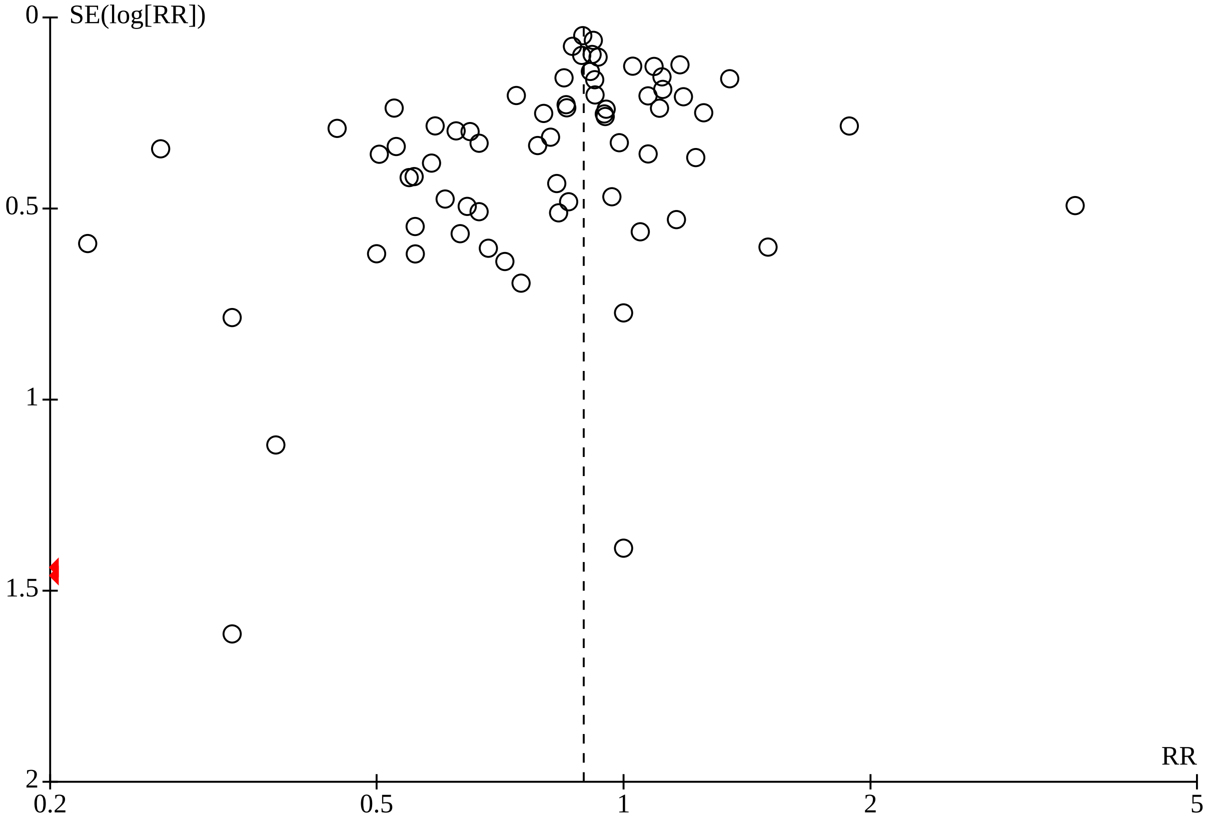
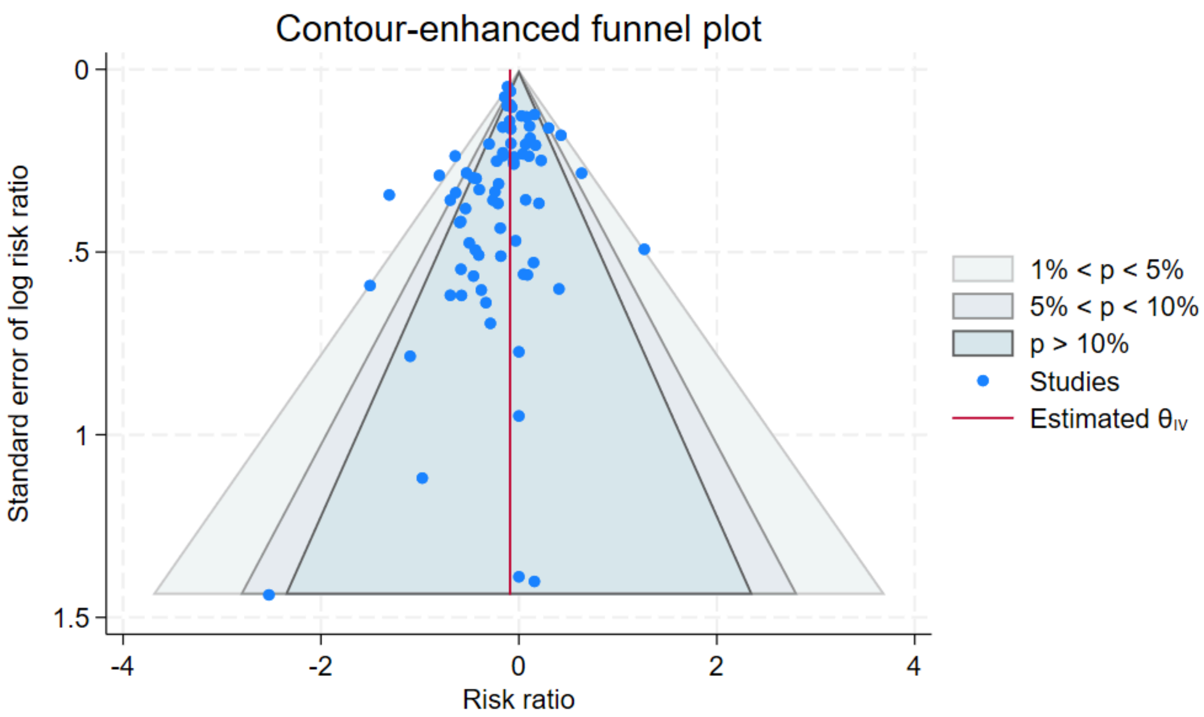


Figure 8. Contour-enhanced funnel plot. Log of risk ratio for 28-day mortality is plotted against its standard error.



In one trial comparing hydrocortisone alone versus hydrocortisone plus fludrocortisone, the hazard ratio of death was 0.94 (95% CI 0.73 to 1.21; Annane 2010).

Another trial compared hydrocortisone monotherapy versus hydrocortisone plus fludrocortisone versus usual care (Labib 2022). There were 9/22 (41%), 7/22 (32%) and 8/22 (36%) deaths in the control, hydrocortisone monotherapy and hydrocortisone plus fludrocortisone arms, respectively.

Secondary outcomes

Long-term mortality

We could extract data on mortality beyond three months for 12 trials. Compared to placebo or usual care, corticosteroids may result in little to no difference in long-term mortality (RR 0.97, 95% CI 0.91 to 1.03; P = 0.27; Chi^2 = 11.87, df = 11 (P = 0.37); I^2 = 7%; 12 studies, 8468 participants; Analysis 1.10). We downgraded the certainty of evidence to low owing to inconsistency in results and imprecision.

Hospital mortality

We could extract data on hospital mortality from 40 trials that accounted for 17,459 participants. A total of 2283 of the 7686 participants in the treated group, compared with 3247 of the 9773 in the control group, died in hospital. Heterogeneity in the results may be moderate (Chi^2 = 56.57, df = 39 (P = 0.03); I^2 = 31%). Compared to placebo or usual care, corticosteroids probably reduce in-hospital mortality (RR 0.90, 95% CI 0.84 to 0.97; P = 0.004; random-effects model; Analysis 1.11). We downgraded the certainty of evidence for this outcome from high to moderate for inconsistency.

90-day all-cause mortality

We could extract data on 90-day mortality from 13 trials. A total of 1244 of 4287 participants in the treated group and 1303 of 4073 in the control group died within 90 days. The heterogeneity in the results might not be important. Compared to placebo or usual care, corticosteroids may reduce 90-day mortality (RR 0.89, 95% CI 0.82 to 0.97; P = 0.010; Chi^2 =16.91, df = 12 (P = 0.15); I^2 = 29%; random-effects model; 13 studies, 8360 participants; Analysis 1.12). This was based on low-certainty evidence (lowered by one level for imprecision and inconsistency).

Intensive care unit (ICU) mortality

Data were available from 24 trials, accounting for 8866 participants. All of these trials investigated a long course of low-dose corticosteroids. A total of 1117 of 4441 participants in the treated group and 1235 of 4425 participants in the control group died in the ICU. The heterogeneity in the results might not be important (Chi^2 = 28.07, df = 23 (P = 0.21); I^2 = 18%). Compared to placebo or usual care, corticosteroids reduce ICU mortality (RR 0.90, 95% CI 0.83 to 0.98; P = 0.01; random-effects model; Analysis 1.13). This was based on high-certainty evidence.

Length of stay in the intensive care unit

For all participants

In 25 trials (8069 participants), compared to placebo or usual care, corticosteroids may reduce the ICU length of stay for all participants (mean difference (MD) -0.86 days, 95% CI -1.67 to -0.05; P = 0.04; random-effects model; Analysis 1.14), with moderate heterogeneity in the results (Chi^2 = 44.09, df = 24 (P = 0.007); I^2 = 46%). We judged the certainty of evidence for this outcome as low. We downgraded the level of certainty for imprecision and inconsistency.

For survivors only

We could extract data from 10 trials on 808 ICU survivors. Compared to placebo or usual care, corticosteroids may reduce the ICU length of stay for survivors (MD -2.19 days, 95% CI -3.93 to -0.46; P = 0.01; random-effects model; Analysis 1.15). The heterogeneity in the results might not be important (Chi^2 test = 8.63, df = 9 (P = 0.47); I^2 = 0%). We judged the certainty of evidence for this outcome as low. We downgraded certainty by one level for imprecision and inconsistency.

Length of hospital stay

For all participants

We could extract data on all participants from 31 trials (16,954 participants). The heterogeneity in the results was substantial (heterogeneity: Chi^2 = 89.01, df = 30 (P < 0.00001); I^2 = 66%). Compared to placebo or usual care, corticosteroids may reduce the length of hospital stay for all participants (MD -1.09 days, 95% CI -1.85 to -0.34; P = 0.005; random-effects model; Analysis 1.16). We judged the certainty of evidence for this outcome as low. We downgraded the level of certainty for imprecision and inconsistency.

For survivors only

We could extract data for hospital survivors from nine studies (710 participants). The heterogeneity in the results was moderate (I^2 = 43%). Compared to placebo or usual care, corticosteroids may not reduce the length of hospital stay for survivors (MD -4.11 days, 95% CI -8.50 to 0.28; P = 0.07; random-effects model; Analysis 1.17). We judged the certainty of evidence for this outcome as low. We downgraded certainty by one level for imprecision and inconsistency.

Number of participants with shock reversal (as defined by stable haemodynamic status ≥ 24 hours after withdrawal of vasopressor therapy) on day seven

We could extract data from 18 trials that accounted for 6938 participants. A total of 2711 of 3474 participants in the treated group and 2224 of 3464 in the control group had shock reversed on day seven. The heterogeneity in the results may be substantial (Chi^2 = 46.77, df = 17 (P = 0.0001); I^2 = 64%). Compared to placebo or usual care, corticosteroids probably increase the chance of having shock reversed on day seven (RR 1.22, 95% CI 1.13 to 1.33; P < 0.00001; random-effects model; Analysis 1.18). This was based on moderate-certainty evidence (lowered by one level for inconsistency).

Sensitivity analysis excluding the two trials that evaluated a short course of high-dose corticosteroids did not reduce heterogeneity in the results (Chi^2 = 34.55, df = 15 (P = 0.003); I^2 = 57%) (Bone 1987; Sprung 1984).

In one cross-over trial, hydrocortisone was given for three days at a dose of 240 mg per day (Keh 2003). Although this trial could not provide information on shock reversal on day seven, investigators showed that on day three fewer hydrocortisone-treated patients than placebo-treated patients required norepinephrine treatment (6/20 versus 14/20; P = 0.025).

Number of participants with shock reversal (as defined by stable haemodynamic status ≥ 24 hours after withdrawal of vasopressor therapy) on day 28

We could extract data from 16 trials, accounting for 7415 participants. A total of 2987 of 3701 participants in the treated group had shock reversed on day 28, as did 2844 of 3714 in the placebo group. The heterogeneity in the results might not be important (Chi^2 = 13.87, df = 15 (P = 0.54); I^2 = 0%). Compared to placebo or usual care, corticosteroids increase the chance of having shock reversed on day 28 (RR 1.05, 95% CI 1.03 to 1.07; P < 0.00001) (Analysis 1.19). This was based on high-certainty evidence.

Number of organs affected and severity of organ dysfunction on day seven, as measured by the sequential organ failure assessment (SOFA) score

Seventeen studies (3220 participants) reported the SOFA score seven days post-randomisation. Compared to placebo or usual care, corticosteroids reduce the SOFA score on day seven (MD -1.27, 95% CI -1.63 to -0.92; P < 0.00001; random-effects model). The heterogeneity in the results was moderate (Chi^2 = 25.21, df = 16 (P = 0.07); I^2 = 37%; Analysis 1.20). We downgraded the certainty of this outcome by one level from high to moderate for inconsistency.

Adverse events

Superinfection

We could extract data from 36 trials (7961 participants). A total of 722 of 4028 participants in the treated group and 717 of 3933 participants in the control group had an episode of nosocomial infection. The heterogeneity in the results might not be important (Chi^2 = 36.62, df = 32 (P = 0.26); I^2 = 13%). Compared to placebo or usual care, corticosteroids may result in little to no difference in the risk of superinfection (RR 0.96, 95% CI 0.86 to 1.07; P = 0.43; random-effects model; Analysis 1.21). We judged the certainty of evidence as low for this outcome (lowered for imprecision given large confidence intervals, and indirectness given the variability in how this outcome was defined across included trials).

Muscle weakness

The number of participants who presented with muscle weakness was reported for seven trials (6729 participants). The heterogeneity in the results was moderate (Chi^2 = 8.65, df = 6 (P = 0.19); I^2 = 31%). Compared to placebo or usual care, the effects of corticosteroids on the risk of muscle weakness are uncertain (RR 1.09, 95% CI 0.78 to 1.53; P = 0.61; random-effects model; Analysis 1.21). We judged the certainty of evidence as very low for this outcome. We rated it down for inconsistency, imprecision and indirectness (variability in how this outcome was defined across included studies).

Gastroduodenal bleeding

We could extract data from 32 trials (6978 participants). A total of 165 of 3529 participants in the treated group and 160 of 3449 in the control group had an episode of gastroduodenal bleeding. The heterogeneity in the results might not be important (I^2 = 0%). Compared to placebo or usual care, corticosteroids may not increase the risk of gastroduodenal bleeding (RR 1.00, 95% CI 0.82 to 1.23; P = 0.98; random-effects model; Analysis 1.21). We judged this to be based on low-certainty evidence (lowered due to imprecision by one level and to indirectness by one level given the variability in how this outcome was defined across included trials).

Hyperglycaemia

The number of participants who presented with hyperglycaemia was reported for 29 trials (10,933 participants). The heterogeneity in the results was substantial (Chi^2 = 98.76, df = 26 (P < 0.00001); I^2 = 74%). Compared to placebo or usual care, corticosteroids probably increase the risk of hyperglycaemia (RR 1.26, 95% CI 1.13 to 1.42; P < 0.0001; random-effects model; Analysis 1.21). This was based on moderate-certainty evidence (lowered by one level for inconsistency).

One trial comparing tight glucose control versus standard care found no benefit in normalising blood glucose levels amongst corticosteroid-treated septic shock participants (Annane 2010).

Hypernatraemia

The number of participants who presented with hypernatraemia was reported for 12 trials (6587 participants). The heterogeneity in the results was moderate (Chi^2 = 20.39, df = 11 (P = 0.04); I^2 = 46%). Compared to placebo or usual care, corticosteroids probably increase the risk of hypernatraemia (RR 1.44, 95% CI 1.08 to 1.93; P = 0.01; random-effects model; Analysis 1.21). This was based on moderate-certainty evidence (downgraded by one level for inconsistency).

Neuropsychiatric events

The number of participants who presented with neuropsychiatric events was reported for 10 trials (7926 participants). The heterogeneity in the results might not be important (Chi^2 = 10.98, df = 9 (P = 0.28); I^2 = 18%). Compared to placebo or usual care, corticosteroids may not increase the risk of neuropsychiatric events (RR 1.02, 95% CI 0.63 to 1.65; P = 0.95; random-effects model; Analysis 1.21). This was based on low-certainty evidence (lowered by one level for imprecision and inconsistency).

Stroke

The number of participants who presented with stroke was reported for eight trials (4742 participants). The heterogeneity in the results might not be important (Chi^2 = 5.30, df = 7 (P = 0.62); I^2 = 0%). Compared to placebo or usual care, the effects of corticosteroids on the risk of stroke are uncertain (RR 0.87, 95% CI 0.51 to 1.51; P = 0.63; random-effects model; Analysis 1.21). This was based on very low-certainty evidence (lowered for imprecision by one level and indirectness by two levels, given the variability in how this outcome was defined across included trials).

Cardiac events

The number of participants who presented with cardiac events was reported for 14 trials (12,488 participants). The heterogeneity in the results might not be important (Chi^2 = 9.59, df = 13 (P = 0.73); I^2 = 0%). Compared to placebo or usual care, the effects of corticosteroids on the risk of cardiac events are uncertain (RR 0.98, 95% CI 0.77 to 1.25; P = 0.87; random-effects model; Analysis 1.21). This was based on very low-certainty evidence (lowered by one level for imprecision and by two levels for indirectness, given the variability in how this outcome was defined across included trials).

Continuous infusion versus bolus administration of corticosteroids

We have summarised the main results in Summary of findings 2.

Primary outcome

28-day all-cause mortality

Data for 28-day mortality were available for three trials accounting for 310 participants. The heterogeneity in the results might not be important (Chi^2 = 1.94; df = 2 (P = 0.38); I^2 = 0%). A total of 73 of 159 participants in the continuous infusion group and 67 of 151 participants in the intermittent bolus group died at 28 days. Compared to bolus administration, the effects of corticosteroids on 28-day mortality are uncertain (RR 1.03, 95% CI 0.81 to 1.32; P = 0.79; random-effects model; Analysis 2.1). We downgraded the certainty of evidence to very low due to the high risk of bias in all except one trial and imprecision.

Secondary outcomes

Long-term mortality

Compared to bolus administration, the effects of corticosteroids on long-term mortality are uncertain (RR 1.36, 95% CI 1.02 to 1.81; 1 trial, 70 participants; Analysis 2.2) (Tilouche 2019). We downgraded the certainty of evidence to very low owing to the fact that data were available from only one small trial that was at high risk of performance and detection bias.

In-hospital mortality

We could extract data from three trials. Compared to bolus administration, the effects of corticosteroids on in-hospital mortality are uncertain (RR 0.92, 95% CI 0.71 to 1.19; P = 0.53; Chi^2 = 0.45, df = 2 (P = 0.80); I^2 = 0%; random-effects model; 3 trials, 352 participants; Analysis 2.3). We downgraded the certainty of evidence to very low owing to imprecision and risk of bias in one study.

90-day all-cause mortality

Data for 90-day mortality were available for only one trial (Hyvernat 2016). Compared to bolus administration, the effects of corticosteroids on 90-day mortality are uncertain (RR 0.87, 95% CI 0.61 to 1.22; Analysis 2.4). We judged this to be very low-certainty evidence (lowered by two levels for very high risk of bias and by one level for imprecision).

Intensive care unit mortality

We could extract data from four trials. The heterogeneity in the results might not be important (Chi^2 = 2.69, df = 3 (P = 0.44); I^2 = 0%). Compared to bolus administration, the effects of corticosteroids on ICU mortality are uncertain (RR 1.00, 95% CI 0.79 to 1.28; random-effects model; 358 participants; Analysis 2.5). This was based on very low-certainty evidence (lowered by two levels for very high risk of bias and by one level for imprecision).

Length of stay in the intensive care unit

For all participants

We could extract data about the length of stay in the ICU for four trials. The heterogeneity in the results was substantial (Chi^2 = 6.93, df = 3 (P = 0.07); I^2 = 57%). Compared to bolus administration, the effects of corticosteroids on ICU length of stay amongst all participants are uncertain (MD -0.56 days, 95% CI -3.44 to 2.32; P = 0.70; random-effects model; 422 participants; Analysis 2.6). We downgraded the certainty of evidence by one level for inconsistency and imprecision, and by two levels for high risk of bias in two studies, to very low.

For survivors only

No trial provided data for the length of ICU stay amongst survivors only.

Length of stay in the hospital

For all participants

We could extract data about the length of stay in the hospital from four trials. The heterogeneity in the results was substantial (Chi^2 = 6.75, df = 3 (P = 0.08); I^2 = 56%). Compared to bolus administration, the effects of corticosteroids on hospital length of stay are uncertain (MD -0.21 days, 95% CI -4.72 to 4.3; P = 0.93; random-effects model; 422 participants; Analysis 2.7). We downgraded the certainty of evidence by one level for inconsistency and imprecision, and by two levels for high risk of bias in two studies, to very low.

For survivors only

No trial provided data about length of hospital stay amongst survivors only.

Number of participants with shock reversal (as defined by stable haemodynamic status ≥ 24 hours after withdrawal of vasopressor therapy) on day seven

We could extract data about shock reversal from four trials (358 participants). The heterogeneity in the results was substantial (Chi^2 = 10.09; df = 3; P = 0.02; I^2 = 70%). A total of 104 of 183 participants in the continuous infusion group and 118 of 175 participants in the intermittent bolus group had shock reversal. Compared to bolus administration, the effects of corticosteroids on shock reversal on day seven are uncertain (RR 0.80, 95% CI 0.59 to 1.10; P = 0.17; random-effects model; Analysis 2.8). This was based on very low-certainty evidence (lowered by one level for inconsistency and imprecision, and by two levels for high risk of bias in two studies).

Number of participants with shock reversal (as defined by stable haemodynamic status ≥ 24 hours after withdrawal of vasopressor therapy) on day 28

Data were available from one trial (Tilouche 2019). Compared to bolus administration, the effects of corticosteroids on shock reversal on day 28 are uncertain (RR 0.78, 95% CI 0.45 to 1.34; 70 participants; Analysis 2.9). This was based on very low-certainty evidence (lowered by one level for inconsistency and imprecision, and by two levels for high risk of bias in two studies).

Number of organs affected and severity of organ dysfunction on day seven, as measured by the sequential organ failure assessment (SOFA) score

The SOFA score on day seven was available for three trials. The heterogeneity in the results might not be important (Chi^2 = 2.16, df = 2 (P = 0.34); I^2 = 7%). Compared to bolus administration, the effects of corticosteroids on the SOFA score are uncertain (MD 1.01, 95% CI -0.30 to 2.31; random-effects model; 260 participants; Analysis 2.10). This was based on very low-certainty evidence (lowered by two levels for high risk of bias and very severe imprecision).

Adverse events

Superinfection

Compared to bolus administration, the effects of corticosteroids on the risk of superinfection are uncertain (RR 1.12, 95% CI 0.37 to 3.33; P = 0.84; random-effects model; 2 trials, 193 participants; Analysis 2.11). The heterogeneity in the results was substantial (Chi^2 = 3.90, df = 1 (P = 0.05); I^2 = 74%). We downgraded the certainty of evidence to very low (lowered by one level for inconsistency and imprecision, and by two levels for high risk of bias in two studies).

Muscle weakness

Only one trial reported data about muscle weakness (Tilouche 2019). Compared to bolus administration, the effects of corticosteroids on the risk of muscle weakness are uncertain (RR 0.89, 95% CI 0.13 to 5.98; P = 0.91; 1 trial, 70 participants; Analysis 2.11). We downgraded the certainty of evidence to very low as this outcome was available from only one trial that was at high risk of performance and detection bias (lowered by one level for imprecision, and by two levels for high risk of bias in two studies).

Gastroduodenal bleeding

Compared to bolus administration, the effects of corticosteroids on the risk of gastroduodenal bleeding are uncertain (RR 0.79, 95% CI 0.10 to 6.37; P = 0.83; 2 trials, 193 participants; random-effects model; Analysis 2.11). The heterogeneity in the results was substantial (Chi^2 = 3.05, df = 1 (P = 0.08); I^2 = 67%). This was based on very low-certainty evidence (lowered by one level for inconsistency and imprecision, and by two levels for high risk of bias in two studies).

Hyperglycaemia

Compared to bolus administration, the effects of corticosteroids on the risk of hyperglycaemia are uncertain (RR 0.89, 95% CI 0.47 to 1.71; P = 0.74; 3 trials, 310 participants; random-effects model; Analysis 2.11). The heterogeneity in the results was considerable (Chi^2 = 9.77, df = 2 (P = 0.008); I^2 = 80%). This was based on very low-certainty evidence (lowered by one level for inconsistency and imprecision, and by two levels for high risk of bias in two studies).

Two trials reported mean blood glucose levels. One trial found slightly lower mean blood glucose levels in the continuous infusion group when compared to the intermittent boluses group (6.4 ± 0.7 mmol/L versus 6.2 ± 0.7; P = 0.04) (Loisa 2007). Another trial found no statistically significant difference between these groups (9 ± 2.5 mmol/L versus 8.5 ± 6.0; P = 0.55) (Hyvernat 2016). In addition, three trials found that managing hyperglycaemia required higher doses of insulin in continuous infusion versus bolus administration. The MDs for insulin dose were 12.92 (95% CI -15.43 to 41.27) (Hyvernat 2016), -5.00 (95% CI -28.50 to 18.50) (Loisa 2007) and 7.57 (95% CI 4.07 to 11.07) (Tilouche 2019).

Hypernatraemia

Compared to bolus administration, the effects of corticosteroids on the risk of hypernatraemia are uncertain (RR 0.74, 95% CI 0.34 to 1.61; P = 0.45; random-effects model; 2 trials, 187 participants; Analysis 2.11). The heterogeneity in the results was moderate (Chi^2 = 1.88, df = 1 (P = 0.17); I^2 = 47%). This was based on very low-certainty evidence (lowered by one level for inconsistency and imprecision, and by two levels for high risk of bias in two studies).

Neuropsychiatric events

Only one trial reported data about neuropsychiatric events (Tilouche 2019). Compared to bolus administration, the effects of corticosteroids on the risk of neuropsychiatric events are uncertain (RR 1.56, 95% CI 0.50 to 4.86; P = 0.44; 1 trial, 70 participants; Analysis 2.11). This was based on very low-certainty evidence (due to risk of bias, inconsistency and very severe imprecision).

Stroke

One trial reported that this outcome did not occur during patient follow-up (Tilouche 2019). The other trials did not report information about this outcome.

Cardiac events

One trial reported that this outcome did not occur during patient follow-up (Analysis 2.11) (Tilouche 2019). The other trials did not report information about this outcome.

DISCUSSION

Summary of main results

This review includes 87 trials (24,336 participants, individual study size from 21 to 3800). Most compared corticosteroids to no corticosteroids and were conducted in adults. The studies were located across the five continents, with about one-third being performed in Europe and four trials recruiting patients on multiple continents. Most studies used a parallel-group design and one-third were multicentric. Most of the trials included adults with sepsis or septic shock. Seventy-four studies contributed to the primary outcome of this review. We judged 25 as being at low risk of bias.

Compared to placebo or usual care, corticosteroids probably reduce all-cause mortality at 28 days, 90 days and in the hospital, and may result in little to no difference in mortality in the long term.

Compared to placebo or usual care, corticosteroids may reduce length of stay for all participants in the intensive care unit and in the hospital. The evidence is uncertain about the effect of corticosteroids on the risk of muscle weakness. Corticosteroids may not increase the risk of superinfection.

Compared to intermittent bolus administration, the effects of a continuous infusion of corticosteroids on all-cause mortality at 28 days, 90 days or in the long term, on the length of stay in the intensive care unit and in the hospital, and on the risk of muscle weakness and superinfection, are uncertain.

Overall completeness and applicability of evidence

We identified 72 trials addressing the question of the effects on mortality of corticosteroids versus a placebo or usual care for sepsis. These trials accounted for a large population. Most of the eligible trials contributed to the primary outcome of this review. Sensitivity analysis based on trials at low risk of bias confirmed the robustness of the results. In addition, four trials provided information on the direct comparison of continuous infusion versus intermittent bolus administration of corticosteroids.

Owing to the large number of trials, and to the large size of some individual trials, we could investigate all our prespecified outcomes. We found mild heterogeneity in the results for the primary outcome, which we explored through subgroup analyses and meta-regression analyses. Although subgroup analyses represent a between-study, not within-study, hypothesis, we thought its validity was acceptable according to the proposed criteria (Guyatt 2008b). First, we a priori defined the hypothesis for an interaction between practical modalities of administration and the effects of corticosteroids on mortality. Second, we conducted subgroup analyses based on only two factors (i.e. modalities of drug administration and population). Third, the findings show that the treatment effect was consistent in terms of 28-day, ICU, hospital, 90-day and long-term mortality (risk ratios 0.89, 0.90, 0.90, 0.89 and 0.96, respectively). Fourth, strong external evidence supports these results. Experimental and human studies have shown that a dose of 400 mg or less of hydrocortisone or equivalent can reverse the systemic inflammatory response, endothelial activation and coagulation disorders secondary to infection (Annane 2005; Heming 2018), thus arguing against the use of higher doses. Moreover, at these low doses, corticosteroids have been shown to improve rather than suppress innate immunity in patients with septic shock (Kaufman 2008). It is now established that sepsis results in a sustained pro-inflammatory state, arguing against a short course of treatment (Angus 2013; Kellum 2007; van der Poll 2017).

We noted some variation in the routine use of corticosteroids in sepsis. A survey of 542 US critical care physicians found that 83% do not routinely use corticosteroids in adults with sepsis, contrasting with 81% reporting common use of corticosteroids for septic shock (Bruno 2012). The findings of this review suggest that treatment effects on mortality may be greater in sepsis with versus without shock. Roughly one out of three respondents to the survey believed that corticosteroids reduce mortality in septic shock, whereas 27% did not and 45% were unsure. Hydrocortisone was the most common corticosteroid prescribed (93%), with a median dosage of 200 mg/day and administration via intermittent intravenous injection.

Comparison of corticosteroids to a placebo or usual care

We judged the certainty of the evidence for 28-day mortality as moderate rather than high. Indeed, the heterogeneity in the results may be moderate, in part related to differences in study populations, in the types of corticosteroids used and in the ways in which corticosteroids were given. Funnel plot analysis suggested some asymmetry and small-study effects (Figure 7).

In the evaluation of the effects of corticosteroids on long-term mortality, although the heterogeneity in the results might not be statistically important, populations and time points for the assessment of long-term mortality varied fairly between studies. There was also some imprecision in the results, as evidenced by large confidence intervals. Thus, we judged the certainty of the evidence for long-term mortality as being low.

There was moderate heterogeneity in the results when analysing in-hospital mortality, partly related to differences in the study populations and interventions. As a result, we downgraded the certainty of evidence from high to moderate.

We judged the certainty of the evidence for the length of stay in the ICU and in the hospital as being low. The heterogeneity in the results was substantial for both outcomes, and the upper limit of the confidence interval for the mean differences might not be clinically relevant.

We judged the certainty of evidence for the risk of superinfection as low because of imprecision and indirectness, given the variability in the definition of this outcome across included trials.

We rated the certainty of evidence for the risk of muscle weakness down to very low because of moderate heterogeneity in the results, large confidence intervals and variable definitions of this outcome across included studies.

Comparison of continuous infusion to intermittent bolus administration of corticosteroids

We judged the evidence to be uncertain about the effects of continuous intravenous infusion of corticosteroids on 28-day mortality, long-term mortality, in-hospital mortality, the length of stay in the ICU and in hospital, and the risk of superinfection and muscle weakness. Indeed, the evidence was generated from very few small-sized trials, with large confidence intervals, and substantial to considerable heterogeneity.

Potential biases in the review process

For this review, we performed a comprehensive search of the literature with no restrictions on language. The fact that 12 studies have not yet been incorporated may be a source of potential bias. However, these studies are unlikely to change the results as most are comparing different regimes of corticosteroids to each other. Slight asymmetry in the funnel plot for the primary outcome of this review may suggest some publication bias. However, potential sources of an asymmetrical funnel plot also include selection biases, the poor methodological quality of smaller studies, true heterogeneity, artefacts and chance (Egger 1997). Visual inspection of the funnel plot and a regression-based Egger test suggest a small-study effect (i.e. amongst small studies, the positive ones are more likely to be published). Nevertheless, our thorough search strategy and the need to register studies in public clinical trial registries may have decreased the risk of missing any randomised controlled trials. True heterogeneity seems to be a more plausible explanation for the observed asymmetrical funnel plot. Indeed, the effects of corticosteroids on mortality may be proportionate to the basal risk of death, and the two large trials that did not find survival benefit (Sprung 2008 and Venkatesh 2018) included patients at lower risk of death than was seen in the two large trials that found a survival benefit (Annane 2002 and Annane 2018). Finally, the asymmetrical funnel plot may have been due to chance.

One trial used a cross-over design (Keh 2003), and we could obtain none of the prespecified outcomes for this review. This trial concluded that prolonged treatment with a low dose of hydrocortisone improved haemodynamic and immune outcomes. Another trial compared three days versus seven days of hydrocortisone therapy and provided no evidence of differences in outcomes between patients treated for three days or seven days (Huh 2007). However, this trial had some limitations, including lack of blinding and small sample size. We considered that pooling the results of the remaining trials in a meta-analysis was acceptable.

Three trials were published only as an abstract (Chawla 1999; Mirea 2014; Tandan 2005). Nevertheless, the primary investigators for these studies provided sufficient unpublished data for us to compute the primary outcome and several secondary outcomes for this review, allowing us to include these trials in the meta-analysis. Both published and unpublished data were available for 33 trials (Appendix 6), and the primary author for each trial validated the data extraction form. In four studies, contact with the primary investigator yielded no additional data (Luce 1988; Meijvis 2011; Rezk 2013; Snijders 2010).

We chose to convert outcome measures that correspond to censored data into dichotomous variables; that is, the proportion of participants with a particular event after one week and after four weeks, or at ICU or hospital discharge.

Some of the review authors were authors on included trials, i.e. DA (Aboab 2008; Angus 2020; Annane 2002; Annane 2010; Annane 2018; Dequin 2020; Dequin 2023; Sprung 2008), JB (Briegel 1999; Keh 2016; Sprung 2008), EB (Annane 2002; Annane 2018), PEB (Annane 2002; Bollaert 1998), DK (Keh 2003; Keh 2016; Sprung 2008) and YK (Chawla 1999). The review authors did not evaluate the risk of bias in studies of which they were authors and did not extract data from their studies.

Agreements and disagreements with other studies or reviews

Findings in this review that a short course of high-dose corticosteroids provides no benefit for patients with sepsis are in line with reports from previous systematic reviews (Cronin 1995; Lefering 1995), as well as with current international guidelines (Annane 2017b; Chaudhuri 2024; Evans 2021; Rhodes 2017).

We found scarce data that could not allow conclusions on the effects of corticosteroids in children with sepsis, in keeping with a recent systematic review (Menon 2013) and recent guidelines (Chaudhuri 2024; Weiss 2020). Nevertheless, we found no evidence of a difference in response to corticosteroids between children and adults.

The beneficial effects of corticosteroids on shock reversal in patients with septic shock are consistent across recent systematic reviews (Allen 2018; Fang 2018; Kalil 2011; Moran 2010; Ni 2018; Pirracchio 2023; Pitre 2023; Rochwerg 2018; Rygard 2018; Sherwin 2012). The survival benefit derived from corticosteroids for patients with sepsis was suggested by some previous authors (Allen 2018; Fang 2018; Moran 2010; Ni 2018; Pitre 2023; Rochwerg 2018), but not by others (Table 1) (Kalil 2011; Rygard 2018; Sherwin 2012). A recent patient-level meta-analysis did not find survival benefit from hydrocortisone therapy in adults with septic shock (Pirracchio 2023). Nevertheless, this current systematic review included trials that were not included in previous systematic reviews, as they were published only recently or were published in a non-English language. Also, the current review has no restriction in terms of patient characteristics, including all ages, all ranges of illness severity, all causes of sepsis and all types of corticosteroids, including combination with vitamins. The current review includes non-published information for a large number of trials after contact was made with the original study authors, resulting in the inclusion of qualitatively and quantitatively better data than were provided previously.

A network meta-analysis suggested that hydrocortisone was more likely than methylprednisolone to achieve shock reversal (Gibbison 2017). A recent patient-level and network meta-analysis suggested that the effects on mortality may be greater with hydrocortisone plus fludrocortisone (Pirracchio 2023). Likewise, a trial-level, Bayesian network meta-analysis has compared the effects of hydrocortisone plus fludrocortisone versus hydrocortisone monotherapy (Teja 2024). All-cause mortality at last follow-up was lowest with fludrocortisone plus hydrocortisone (RR 0.85, 95% credible interval (CrI) 0.72 to 0.99, 98.3% probability of superiority, moderate-certainty evidence), followed by hydrocortisone alone (RR 0.97, 95% CrI 0.87 to 1.07, 73.1% probability of superiority, low-certainty evidence). Fludrocortisone plus hydrocortisone was associated with a 12% lower risk of all-cause mortality compared to hydrocortisone alone (RR 0.88, 95% CrI 0.74 to 1.03, 94.2% probability of superiority, moderate-certainty evidence). A large retrospective study has compared the effectiveness, in real life, of adding fludrocortisone to hydrocortisone versus hydrocortisone monotherapy among patients with septic shock using target trial emulation (Bosch 2023). In this study, the primary composite outcome of death in hospital or discharge to hospice occurred among 1076/2280 (47.2%) patients treated with hydrocortisone-fludrocortisone versus 43,669/85,995 (50.8%) treated with hydrocortisone alone (adjusted absolute risk difference -3.7%, 95% CI -4.2% to -3.1%). In the current review we found no evidence that the type of corticosteroid, namely hydrocortisone alone, hydrocortisone plus fludrocortisone, hydrocortisone plus vitamin C and thiamine, prednisone/prednisolone, methylprednisolone or dexamethasone, did influence treatment effects on 28-day mortality.

AUTHORS' CONCLUSIONS

Implications for practice

Moderate-certainty evidence indicates that corticosteroids probably reduce 28-day and hospital mortality amongst patients with sepsis. Corticosteroids may reduce intensive care unit (ICU) and hospital length of stay (low-certainty evidence). There may be little or no difference in the risk of superinfection or the risk of muscle weakness (very low-certainty evidence). Corticosteroids probably increase the risk of hyperglycaemia and hypernatraemia (moderate-certainty evidence). The effects of continuous versus intermittent bolus administration of corticosteroids are uncertain.

The 12 studies in 'Studies awaiting classification' may not alter the conclusions of the review once assessed.

Implications for research

The criteria for critical illness-related corticosteroid insufficiency in septic shock remain to be defined.

Subgroup analyses suggest that additional studies are needed to address the following topics related to the use of corticosteroids in patients with sepsis.

- The role of a long course of low-dose corticosteroids in the treatment of septic shock in children.
- The role of a long course of low-dose corticosteroids in the treatment of patients with sepsis without shock, or with a mild form of septic shock, and patients with acute respiratory distress syndrome (ARDS).
- The role of mineralocorticoid replacement.
- Optimal timing of initiation of treatment.
- Optimal dose and duration of hydrocortisone (or equivalent).
- Optimal modality to administer treatment that is continuous versus intermittent bolus.
- Optimal modality to stop treatment with or without taper off.
- The role of a long course of low-dose corticosteroids for the treatment of sepsis caused by different types of infections.

Future studies should also address the following.

- The effects of corticosteroids on patient-reported outcomes measured in the long term.
- The risk factors associated with the adverse effects of corticosteroids, such as superinfection and muscle weakness.

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* Indicates the major publication for the study

CHARACTERISTICS OF STUDIES

Characteristics of included studies *[ordered by study ID]*

Aboab 2008

Study characteristics	
Methods	Randomised controlled trial with 2 parallel groups 1 centre Dates study was conducted: not reported
Participants	Adults (n = 23) with septic shock and adrenal insufficiency as defined by a cortisol response to 250 µg synacthene (delta cortisol) ≤ 9 µg/dL

Aboab 2008 *(Continued)*

	Setting: intensive care unit Study location: France
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) plus fludrocortisone (50 µg taken orally every 24 hours for 7 days) • Respective placebos
Outcomes	• Short-term improvement in autonomic failure as assessed by changes in the components of spectral analysis of cardiac and vascular signal variability between baseline and day 3 • In-hospital mortality
Identification	
Notes	We contacted the study authors and they could not provide missing data. Funding source: authors' institution (public source) Data were extracted by BR Risk of bias was assessed by EB and BR

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Centralised
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of placebos
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of placebos
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol excluding reporting bias
Other bias	Low risk	Full access to data excluding selection bias

Agarwal 2022

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: parallel groups

Agarwal 2022 *(Continued)*

	Number of sites: 1 Dates study was conducted: May 2020 to February 2021
Participants	Geriatric patients (age > 60 years) admitted to the intensive care unit with septic shock, or who developed septic shock during their hospital stay Setting: intensive care unit Study location: India
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours, during vasopressor therapy, when off vasopressors, hydrocortisone was tapered off as 100 mg per day for 3 days then 50 mg per day for 3 days), cumulated total dose: unclear. Duration: during vasopressor administration and then taper: when off vasopressors 100 mg/d x 3 d then 50 mg/d x 3 d then discontinued: 4 bolus per day • Usual care
Outcomes	**Primary outcome** • 28-day all-cause mortality **Secondary outcomes** • Length of stay in the intensive care unit • Duration of vasopressor requirement • Need for mechanical ventilation • Incidence of adverse events
Identification	Sponsorship source: not specified Comments: N/A Author's name: Mayank Agarwal Institution: India Institute of Medical Sciences Email: N/A Address: India Institute of Medical Sciences, Rishikesh, Uttarakhand
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Judgement comment: Open access computer-based software used for random allocation.
Allocation concealment (selection bias)	Low risk	Judgement comment: Computer program used although no details of exactly how allocation concealment was preserved, but no convincing data to suggest this was breached.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: Mentions that this is a single-blind study, however it is unclear what methods (if any) were used to facilitate this. Unclear if medications were delivered unmarked/covered etc. Seems possible that participants could discern what arm of the trial they were in based on when hydrocortisone or "study drug" was administered.
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: Outcomes assessment was not blinded to allocated interventions.
Incomplete outcome data (attrition bias)	Low risk	Judgement comment: No evidence of post-randomisation exclusions/attrition.

Agarwal 2022 *(Continued)*

All outcomes		
Selective reporting (reporting bias)	Low risk	Judgement comment: No protocol available but all outcomes presented in the methods are presented appropriately in adequate detail. The outcomes reported are typical of this study.
Other bias	Low risk	Judgement comment: No other sources of bias.

Angus 2020

Study characteristics	
Methods	Study design: platform, randomised controlled trial Study grouping: parallel groups Dates study was conducted: 9 March and 17 June 2020
Participants	Adults (> 18 years or older) with presumed or confirmed SARS-CoV-2 infection and admitted to an intensive care unit (ICU) for provision of respiratory or cardiovascular organ support Setting: intensive care unit Location: multinational over the 5 continents
Interventions	• Hydrocortisone (fixed 50 mg intravenous bolus every 6 hours for 7 days) • Hydrocortisone (fixed 100 mg intravenous bolus every 6 hours for 7 days) • Hydrocortisone (50 mg intravenous bolus every 6 hours while in shock for up to 28 days) • Usual care
Outcomes	**Primary outcome:** • Respiratory and cardiovascular organ support-free days up to day 21 (an ordinal endpoint with death within the hospital as the worst outcome (labelled –1), then the length of time free of both respiratory and cardiovascular organ support, such that the best outcome would be 21 organ support-free days). **Secondary outcomes** • In-hospital mortality • ICU and hospital length of stay • Respiratory support–free days • Cardiovascular organ support–free days • A composite outcome of progression to invasive mechanical ventilation, extracorporeal membrane oxygenation (ECMO) or death among those not ventilated at baseline • WHO ordinal scale (range, 0 to 8, where 0 = no illness, 1 to 7 = increasing level of care and 8 = death) assessed at day 14
Identification	Sponsorship source: Platform for European Preparedness Against (Re-) emerging Epidemics (PREPARE) consortium by the European Union, FP7-HEALTH-2013-INNOVATION-1 (grant 602525), the Australian National Health and Medical Research Council (grant APP1101719), the New Zealand Health Research Council (grant 16/631), the Canadian Institute of Health Research Strategy for Patient-Oriented Research Innovative Clinical Trials Program (grant 158584), the UK National Institute for Health Research (NIHR) and the NIHR Imperial Biomedical Research Centre, the Health Research Board of Ireland (grant CTN 2014-012), the UPMC Learning While Doing Program, the Breast Cancer Research Foundation, the French Ministry of Health (grant PHRC-20-0147), and the Minderoo Foundation Comments: no Author's name: Derek C. Angus

Angus 2020 *(Continued)*

	Institution: Department of Critical Care Medicine, University of Pittsburgh Email: angusdc@upmc.edu Address: Department of Critical Care Medicine, University of Pittsburgh, 3550 Terrace St, 614 Scaife Hall, Pittsburgh, PA 15261
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated list
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	High risk	Open-label
Blinding of outcome assessment (detection bias) All outcomes	High risk	Open-label
Incomplete outcome data (attrition bias) All outcomes	Low risk	No evidence of loss to follow-up or withdrawal of patients.
Selective reporting (reporting bias)	Low risk	Judgement comment: Reported outcomes matched protocol information.
Other bias	Low risk	Judgement comment: No evidence of other sources of bias.

Annane 2002

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups 19 centres Dates study was conducted: from October 1995 to February 1999
Participants	Adults (n = 300) with vasopressor- and ventilator-dependent septic shock Stratification according to cortisol response to 250 µg synacthene for non-responders (delta cortisol ≤ 9 µg/dL) and responders (> 9 µg/dL) Setting: intensive care unit Study location: France
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) plus fludrocortisone (50 µg taken orally every 24 hours for 7 days)

Annane 2002 *(Continued)*

	• Respective placebos Treatments had to be initiated within 8 hours of shock onset.
Outcomes	**Primary** • 28-day mortality in non-responders **Secondary** • 28-day mortality in responders and in all participants • Intensive care unit (ICU) mortality rate • Hospital mortality rate • 1-year mortality rate • Shock reversal • Organ system failure-free days • Length of stay in ICU and at hospital • Safety
Identification	
Notes	We contacted the study authors and obtained access to individual patient data Funding source: French Ministry of Health Data were extracted by JB. Risk of bias was assessed by RP and BR.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatments or the placebos in an indistinguishable form.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatments or the placebos in an indistinguishable form.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol, excluding reporting bias.
Other bias	Low risk	Full access to data, excluding selection bias.

Annane 2010

Study characteristics	
Methods	Study design: randomised controlled trial with 2 × 2 factorial design 11 centres Dates study was conducted: from July 2014 to October 2014
Participants	Adults (n = 509) with vasopressor-dependent septic shock Subgroups based on adrenal status assessed by a 250 µg ACTH test Setting: intensive care unit Study location: France
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) plus fludrocortisone (50 µg taken orally every 24 hours for 7 days) and intravenous insulin to maintain blood glucose between 80 and 110 mg/dL • Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) and intravenous insulin to maintain blood glucose between 80 and 110 mg/dL • Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) plus fludrocortisone (50 µg taken orally every 24 hours for 7 days) and conventional control of blood glucose levels • Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) and conventional control of blood glucose levels Treatments had to be initiated within 24 hours of shock onset.
Outcomes	**Primary** • Hospital mortality in non-responders **Secondary** • Mortality rates at 28 days, 90 days and 180 days and at ICU discharge • Vasopressor-free days • Organ failure-free days • ICU and hospital length of stay • Safety
Identification	
Notes	We contacted the study authors and obtained access to individual patient data. Funding source: French Ministry of Health Data were extracted by EB. Risk of bias was assessed by EB and BR.

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme

Annane 2010 *(Continued)*

Allocation concealment (selection bias)	Low risk	Centralised randomisation through a secured website
Blinding of participants and personnel (performance bias) All outcomes	High risk	Open-label
Blinding of outcome assessment (detection bias) All outcomes	High risk	Open-label
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol, excluding reporting bias.
Other bias	Low risk	Full access to data excluding selection bias.

Annane 2018

Study characteristics	
Methods	Study design: randomised controlled trial with 2 × 2 factorial design 34 centres Dates study was conducted: from July 2014 to October 2014
Participants	Adults (n = 1241) with vasopressor-dependent septic shock Subgroups based on adrenal status assessed by a 250 µg ACTH test Setting: intensive care unit Study location: France
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) plus fludrocortisone (50 µg taken orally every 24 hours for 7 days) and drotrecogin alfa (activated) at a dose of 24 µg/24 h • Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) plus fludrocortisone (50 µg taken orally every 24 hours for 7 days) and placebo of drotrecogin alfa (activated) • Placebo of hydrocortisone plus placebo of fludrocortisone (50 µg taken orally every 24 hours for 7 days) and drotrecogin alfa (activated) at a dose of 24 µg/24 h • Placebo of hydrocortisone plus placebo of fludrocortisone (50 µg taken orally every 24 hours for 7 days) and placebo of drotrecogin alfa (activated) Treatments had to be initiated within 24 hours of shock onset.
Outcomes	**Primary** • 90-day all-cause mortality **Secondary** • All-cause mortality at ICU discharge, hospital discharge, day 28 and day 180 • Percentage of participants from whom care was withheld or withdrawn

Annane 2018 *(Continued)*

	• Percentage of participants weaned from vasopressors at day 28 and at day 90 • Time to weaning from vasopressors • Number of days that participants were alive and free of vasopressors (vasopressor-free days) up to day 28 and day 90 (patients who died before day 28 or day 90 were assigned zero free days) • Percentage of participants weaned from mechanical ventilation at day 28 and at day 90 • Time to weaning from mechanical ventilation - ventilator-free days up to day 28 and day 90 • Percentage of participants with a total SOFA score < 6 (organ failure–free) at day 28 and at day 90 • Time to reaching a SOFA score < 6 organ failure-free days up to day 28 and day 90 • Percentage of participants discharged from the ICU and hospital up to day 28 and day 90 • Time to discharge from ICU and hospital and ICU-free and hospital-free days up to day 28 and day 90 **Safety outcomes included** • Superinfection up to day 180 • Gastrointestinal bleeding up to day 28 • Episodes of hyperglycaemia up to day 7 • Neurological sequelae (cognitive impairment and muscle weakness) at time of ICU and hospital discharge, at day 90 and at day 18
Identification	
Notes	We contacted the study authors and obtained access to individual patient data. Funding source: French Ministry of Health Data were extracted by BR. Risk of bias was assessed by BR and RP.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatments or the placebos in an indistinguishable form.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatments or the placebos in an indistinguishable form.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol, excluding reporting bias
Other bias	Low risk	Full access to data, excluding selection bias

Arabi 2011

Study characteristics	
Methods	Study design: randomised controlled trial 1 centre Dates study was conducted: from April 2004 to October 2007
Participants	Adults (n = 75) with liver cirrhosis and septic shock Subgroups based on adrenal status assessed by a 250 µg ACTH test Setting: intensive care unit Study location: Saudi Arabia
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours until shock resolution, then treatment tapered off by 1 mL every 2 days until discontinuation) • Placebo (normal saline)
Outcomes	**Primary** • 28-day all-cause mortality **Secondary** • ICU and hospital mortality • Shock reversal • Mechanical ventilation-free days • Renal replacement therapy-free days • Length of stay • SOFA score at day 7 • Adverse events Outcomes were also analysed in relation to adrenal insufficiency.
Identification	
Notes	We contacted the authors and obtained access to individual patient data. Funding source: King Abdulaziz City for Science and Technology

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random numbers
Allocation concealment (selection bias)	Low risk	Use of sealed envelopes by pharmacists
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatment or the placebo in an indistinguishable form

Arabi 2011 *(Continued)*

Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatment or the placebo in an indistinguishable form
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unexplained discrepancy between reported K-M curves and number of deaths at 28 days in placebo arm
Selective reporting (reporting bias)	Low risk	Access to unpublished data
Other bias	High risk	Trial terminated prematurely after enrolment of 75 participants; planned sample size was 150

Birudaraju 2022

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: parallel groups Centre: 1
Participants	Adults (age of 18 years or more) who meet the criteria for severe sepsis or septic shock Setting: intensive care unit Study location: USA
Interventions	• Methylprenisolone (20 mg intravenously every 8 h for 7 days) • Placebo
Outcomes	**Primary outcome** • Not mentioned **Secondary outcomes** • APACHE III score measured at 14 days • 14-day mortality • 28-day mortality • Serum albumin levels on day 14 • Length of stay in the intensive care unit • Duration of dependency on vasopressor • Duration of dependency on mechanical ventilation
Identification	Country: USA
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**

Birudaraju 2022 *(Continued)*

Random sequence generation (selection bias)	Low risk	Judgement comment: computer-generated
Allocation concealment (selection bias)	Low risk	Judgement comment: allocation concealed by pharmacists
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Judgement comment: pharmacists masked treatments and all other stakeholders remained blinded to treatments
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement comment: all outcomes were assessed by blinded assessors
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: no exclusions, no loss to follow-up, no attrition
Selective reporting (reporting bias)	Unclear risk	Judgement comment: insufficient information
Other bias	Unclear risk	Judgement comment: insufficient information

Blum 2015

Study characteristics	
Methods	Study design: multi-centre, randomised, placebo-controlled study Study grouping: 2 parallel groups Dates study was conducted: from December 2009 to May 2014
Participants	Adults (n = 800) patients hospitalised with community-acquired pneumonia Subgroups based on adrenal status assessed by a 250 µg ACTH test Setting: emergency and medical wards Study location: Switzerland
Interventions	• Prednisone 50 mg per day for 7 days • Placebo
Outcomes	**Primary** • Clinical stability **Secondary** • All-cause mortality within 30 and 180 days post randomisation • ICU admission and length of stay • Duration of antibiotic treatment • Disease activity scores • Adverse events

Blum 2015 *(Continued)*

Identification	
Notes	We contacted the study authors and obtained access to the trial protocol and some missing data. Funding source: Swiss National Science Foundation, Viollier AG, Nora van Meeuwen Haefliger Stiftung, Julia und Gottfried Bangerter-Rhyner Stiftung

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation list
Allocation concealment (selection bias)	Low risk	Randomisation was centralised Generator and executor of randomisation were separated Variable block sizes of 4 to 6
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatments or the placebos in an indistinguishable form
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatments or the placebos in an indistinguishable form
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Outcomes reported matched outcomes planned in trial protocol, which was publicly released before the end of the recruitment period.
Other bias	Low risk	Contact with trial authors; access to trial protocol but could not identify areas of bias

Bollaert 1998

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups 2 centres Dates the study was conducted: not reported
Participants	Adults (n = 41) with vasopressor- and ventilator-dependent septic shock Stratification according to cortisol response to 250 µg synacthene for non-responders (delta cortisol ≤ 6 µg/dL) and responders (> 6 µg/dL) Setting: intensive care unit

Bollaert 1998 *(Continued)*

	Study location: France
Interventions	• Hydrocortisone (100 mg intravenous bolus every 8 hours for 5 days, then tapered over 6 days) • Placebo Treatments had to be initiated after 48 hours or longer from shock onset.
Outcomes	**Primary** • Shock reversal **Secondary** • 28-day mortality • ICU mortality • Hospital mortality • Improvement in haemodynamics • Length of stay in ICU and at hospital • Safety
Identification	
Notes	We contacted the study authors and obtained access to individual patient data. Funding source: Nancy University Hospital Data were extracted by DA. Risk of bias was assessed by DA and EB.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Randomisation list kept confidential by the pharmacist
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Pharmacist masked study treatment and all other stakeholders remained blinded.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Pharmacist masked study treatment and all other stakeholders remained blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol, excluding reporting bias
Other bias	Low risk	Full access to data, excluding selection bias

Bone 1987

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups 19 centres Dates study was conducted: from November 1982 to December 1985
Participants	Adults (n = 382) with sepsis (n = 234) or septic shock (n = 148) Setting: intensive care unit Study location: USA
Interventions	• Methylprednisolone (30 mg/kg 20-minute intravenous infusion, every 6 hours for 24 hours) • Placebo Treatments had to be initiated 2 hours from time entry criteria were met
Outcomes	**Primary** • 14-day development of shock for sepsis • Shock reversal for septic shock • 14-day death and safety
Identification	
Notes	We did not contact the study authors. Funding source: Upjohn Company

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatments or the placebos in an indistinguishable form
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatments or the placebos in an indistinguishable form
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none

Bone 1987 *(Continued)*

Selective reporting (reporting bias)	Unclear risk	No access to study protocol to exclude reporting bias
Other bias	Low risk	No access to full data including screening log to exclude selection bias

Briegel 1999

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups 1 centre Dates study was conducted: from November 1993 to September 1996
Participants	Adults (n = 40) with vasopressor- and ventilator-dependent septic shock Setting: intensive care unit Study location: Germany
Interventions	• Hydrocortisone (100 mg 30-minute intravenous infusion followed by 0.18 mg/kg/h continuous infusion until shock reversal, then tapered off) • Placebo Treatments had to be initiated within 72 hours from shock onset.
Outcomes	**Primary** • Shock reversal **Secondary** • 28-day mortality • ICU mortality • Hospital mortality • Improvement in haemodynamics • Organ system failure (SOFA at day 7) • Length of stay in ICU • Safety
Identification	
Notes	We contacted the study authors and obtained access to individual patient data. Funding source: Friedrich-Baur-Stiftung, Germany, by departmental funds Data were extracted by DA. Risk of bias was assessed by DA and EB.

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**

Briegel 1999 *(Continued)*

Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Adequate randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	The study drugs were prepared by research assistants who were involved in neither the study nor the clinical care of the patients. All other stakeholders remained blinded.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	The study drugs were prepared by research assistants who were involved in neither the study nor the clinical care of the patients. All other stakeholders remained blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol, excluding reporting bias
Other bias	Low risk	Access to full data including screening log

Chang 2020

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: parallel groups Number of sites: 1 Dates study was conducted: 25 September 2017 to 7 January 2019
Participants	Adults (age of 18 years old or more, n = 80) who met the diagnostic criteria for sepsis-3 and serum levels of Procalcitonin ≥ 2 ng/mL when entering the intensive care unit Setting: intensive care unit Study location: China
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) + thiamine (200 mg intravenous infusion every 12 hours for 4 days) + vitamin C (1500 mg intravenous infusion every 6 hours for 4 days) • Placebos
Outcomes	**Primary outcome** • 28-day all-cause mortality **Secondary outcomes** • Duration of dependency on vasopressor • Length of stay in the intensive care unit • Change in SOFA (DSOFA) score within 72 h after experimental intervention • Procalcitonin clearance rate within 72 h after experimental intervention

Chang 2020 *(Continued)*

Identification	Sponsorship source: National Natural Science Foundation of China [Grant 81971859], the Science and Technology Planning Project of Guangdong Province [Grant 2014A020212203], and the Clinical Research Startup Program of Southern Medical University by the High-level University Construction Funding of Guangdong Provincial Department of Education [Grant LC2019ZD014] Comments: no Author's name: Zhanguo Liu Institution: Zhujiang Hospital Email: zhguoliu@163. com Address: Department of Critical Care Medicine, Zhujiang Hospital, Southern Medical University, 253 Gongye Rd, Guangzhou 510282, China
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement comment: No concerns. A random sequence with a block of 10 was generated using a random number table.
Allocation concealment (selection bias)	Low risk	Judgement comment: No concerns. Each random group card was marked with a patient's enrolment group and sealed into an opaque envelope until each patient had successfully recruited.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: Single-blind with NS placebo and mentioned patients and families were unaware of treatment allocation. However, no mention that other personnel were blinded to treatment allocation and also Figure 1 mentions that 28 patients did not get any placebo at all; therefore may have been able to ascertain treatment allocation. Additionally, there is evidence that lack of blinding of clinicians impacted this, given concerns that the extra volume of placebo could be detrimental and, as such, placebo not given.
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: Single-blind study, so treatment allocation likely known to outcome assessors and researchers.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: ITT used and no significant exclusions or attrition post-randomisation
Selective reporting (reporting bias)	Low risk	Judgement comment: All pre-specified outcomes in the protocol were present in the trial and reported in adequate detail.
Other bias	Low risk	Judgement comment: No evidence of other sources of bias

Chawla 1999

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups 1 centre Dates study was conducted: not reported

Chawla 1999 *(Continued)*

Participants	Adults (n = 44) with vasopressor-dependent septic shock Setting: intensive care unit Study location: USA
Interventions	• Hydrocortisone (100 mg intravenous bolus every 8 hours for 3 days, then tapered over 4 days) • Placebo Treatments had to be initiated after 72 hours or longer from shock onset.
Outcomes	**Primary** • Shock reversal **Secondary** • 28-day mortality • Hospital mortality • Improvement in haemodynamics • Length of stay in ICU • Safety
Identification	
Notes	We contacted the study authors and obtained access to the trial protocol and some missing data. Funding source: not declared Data were extracted by DA. Risk of bias was assessed by DA and EB.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	The randomisation list was kept confidential by the pharmacist.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Pharmacist masked study treatments and all other stakeholders remained blinded.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Pharmacist masked study treatments and all other stakeholders remained blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol excluding reporting bias

Chawla 1999 *(Continued)*

Other bias	Low risk	Access to full data including screening log

Cicarelli 2007

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups 1 centre Dates study was conducted: from November 2004 through December 2005
Participants	Adults (n = 29) with vasopressor-dependent septic shock Setting: intensive care unit Study location: Brazil
Interventions	• Dexamethasone (0.2 mg/kg intravenous, 3 doses at intervals of 36 hours) • Placebo (normal saline)
Outcomes	**Primary outcome** • Not mentioned **Secondary outcomes** • Duration of vasopressor support (SOFA score for cardiovascular system ≥ 2) • Duration of mechanical ventilation • 28-day mortality
Identification	
Notes	We contacted the study authors and obtained information on study design but no additional unpublished data. Funding source: not declared

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Randomisation list kept confidential by the pharmacist
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of syringe prepared by the pharmacist containing either the active treatments or the placebos in an indistinguishable form.
Blinding of outcome assessment (detection bias)	Low risk	Use of syringe prepared by the pharmacist containing either the active treatments or the placebos in an indistinguishable form.

Cicarelli 2007 *(Continued)*

All outcomes		
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Lost to follow-up: none; 3 participants were withdrawn after next of kin refused to consent.
Selective reporting (reporting bias)	Unclear risk	No access to study protocol to rule out reporting bias
Other bias	Unclear risk	No access to data to rule out selection bias

Confalonieri 2005

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 6 centres Dates study was conducted: from July 2000 through March 2003
Participants	Adults (n = 46) with severe community-acquired pneumonia Setting: intensive care unit Study location: Italy
Interventions	• Hydrocortisone (200 mg intravenous loading bolus followed by a continuous infusion at a rate of 10 mg/h for 7 days, then tapered over 4 days) • Placebo
Outcomes	**Primary** • Improvement in PaO_2:FiO_2 and in multiple organ dysfunction syndrome score by study day 8 **Secondary** • Duration of mechanical ventilation • Length of stay • 60-day mortality • ICU mortality • Hospital mortality • Safety
Identification	
Notes	We contacted the study authors and obtained access to individual patient data. Funding source: Assisi Foundation of Memphis

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**

Confalonieri 2005 *(Continued)*

Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of syringe prepared by the pharmacist containing either the active treatments or the placebos in an indistinguishable form.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of syringe prepared by the pharmacist containing either the active treatments or the placebos in an indistinguishable form.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: 2 at 60 days after randomisation, all in the placebo group
Selective reporting (reporting bias)	Low risk	Access to full data including screening log
Other bias	High risk	Study was stopped prematurely for apparent benefit; no sample size was defined a priori, but study authors used the triangular test as a stopping rule, analysing the primary outcome after each 20 participants.

Corral-Gudino 2021

Study characteristics	
Methods	Study design: randomised, open-label, controlled Study grouping: 2-arm, parallel-group trial Number of sites: 5 centres
Participants	Adults (age of 18 years or more, n = 180) with laboratory-confirmed diagnosis of SARS-CoV-2 infection, symptom duration of at least 7 days, radiological evidence of lung disease on chest X-ray or computed tomography (CT) scan, moderate to severe disease with abnormal gas exchange: (a) partial pressure of oxygen/fraction of inspired oxygen < 300, (b) arterial oxygen saturation/fraction of inspired oxygen < 400 or (c) at least two criteria of the BRESCIA-COVID Respiratory Severity Scale (BCRSS), and evidence of a systemic inflammatory response with any of the following: serum C-reactive protein > 15 mg/dL, D-dimer > 800 ng/mL, ferritin > 1000 mg/dL, or interleukin-6 levels > 20 pg/mL (the available biochemical tests in every hospital depended on local protocols) Setting: hospital, intensive care unit and general wards Study location: Spain
Interventions	• Methylprednisolone (40 mg intravenously twice a day for days and then 20 mg twice a day for 3 more days) • Usual care
Outcomes	**Primary outcome measure** • Composite endpoint that included in-hospital all-cause mortality, escalation to ICU admission, or progression of respiratory insufficiency that required non-invasive ventilation

Corral-Gudino 2021 *(Continued)*

	Secondary outcomes measures
	
	• Individual components of the composite endpoint
	  ○ In-hospital all-cause mortality
	  ○ Escalation to ICU admission
	  ○ Progression of respiratory insufficiency that required non-invasive ventilation
	• Variation of the laboratory biomarkers between baseline and 6 days after inclusion (time window 4 to 8 days)
	• Adverse events
	  ○ Hyperglycaemia (> 180 mg/dL based on the fasting blood glucose test) in the first 6 days after being included in the study (period of methylprednisolone administration)
	  ○ Nosocomial infection (infection confirmed with positive cultures that was not present at the time trial inclusion but occurring within 3 to 28 days after first dose of methylprednisolone)

Identification	Contact information: Corral-Gudino- Hospital Rio Hortega, Servicio de Medicina Interna, Universidad de Valladolid, Valladolid, Spain; lcorral@saludcastillayleon.es No information about funding sources
Notes	Spain

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	The randomisation list was automatically computed from medical record number.
Allocation concealment (selection bias)	Low risk	They use a concealed algorithm for allocation
Blinding of participants and personnel (performance bias) All outcomes	High risk	This was an open-label trial.
Blinding of outcome assessment (detection bias) All outcomes	High risk	This was an open-label trial.
Incomplete outcome data (attrition bias) All outcomes	High risk	Due to a low recruitment rate, the expected sample size of 180 was not achieved, and only 64 patients were randomised. Five patients in the experimental arm did not receive the study drug and one patient in the usual care arm did receive methylprednisolone.
Selective reporting (reporting bias)	Low risk	Reported outcomes are consistent with trial protocol
Other bias	Unclear risk	The trial was terminated early due to a low recruitment rate. Before randomisation, investigators' preference to use or not corticosteroids was checked, which might have biased the selection process.

CSG 1963

Study characteristics

CSG 1963 *(Continued)*

Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 5 centres
Participants	Adults (n = 194) and children (n = 135) with vasopressor-dependent septic shock Setting: intensive care unit Study location: USA
Interventions	• Hydrocortisone (intravenous infusion of 300 mg for 24 hours, then 250 mg for 24 hours, followed by 200 mg orally on day 3, then tapered off in steps of 50 mg per day, i.e. total duration of treatment 6 days) • Placebo
Outcomes	**Primary** • Hospital mortality **Secondary** • Safety
Identification	
Notes	We did not contact the study authors. Funding source: not declared

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not given
Allocation concealment (selection bias)	Unclear risk	Not given
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of syringe prepared by the pharmacist containing either the active treatments or the placebos in an indistinguishable form.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of syringe prepared by the pharmacist containing either the active treatments or the placebos in an indistinguishable form.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Unclear risk	No access to study protocol to exclude reporting bias
Other bias	Unclear risk	No access to data to exclude selection bias

Dequin 2020

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: parallel groups Number of sites: 9
Participants	Adults (age of 18 years or more, n = 149) with suspected or proven SARS-CoV-2 infection and acute respiratory failure requiring admission to the intensive care unit Setting: intensive care unit Study location: France
Interventions	• Hydrocortisone (200 mg per day as a continuous intravenous infusion for 7 days, and then decreased to 100 mg per day for 4 days and 50 mg per day for 3 days, for a total of 14 days. If the patient's respiratory and general status had sufficiently improved by day 4, a short treatment regimen was used, i.e. 200 mg per day for 4 days, followed by100 mg per day for 2 days and then 50 mg per day for the next 2 days, for a total of 8 days) • Placebo

Outcomes	**Primary outcome**
	
	• Treatment failure on day 21, defined as death or persistent dependency on mechanical ventilation or high-flow oxygen therapy
	
	Secondary outcomes
	
	• Need for respiratory support
	  ◦ Intubation and invasive mechanical ventilation
	  ◦ Prone positioning
	  ◦ ECMO
	• The PaO2:FIO2 ratio recorded daily from day 1 to day 7 and then on days 14 and 21
	• The proportion of patients with and the number of episodes of nosocomial infections recorded during the ICU stay
	
	Post hoc outcomes
	
	• Death on day 21
	• Status on day 21: determined using a 5-item scale:
	  ◦ Death
	  ◦ Presence in the ICU
	  ◦ On mechanical ventilation
	  ◦ High-flow or low-flow oxygen therapy
	  ◦ ICU discharge

Identification	Author's name: Pierre-François Dequin Email: dequin@univ-tours.fr
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement

Dequin 2020 *(Continued)*

Random sequence generation (selection bias)	Low risk	Computer-generated list
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Indiscernible active and placebo
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Indiscernible active and placebo
Incomplete outcome data (attrition bias) All outcomes	Low risk	There were no losses to follow-up.
Selective reporting (reporting bias)	Low risk	Judgement comment: All reported outcomes matched protocol information.
Other bias	Low risk	No evidence of other sources of bias

Dequin 2023

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: parallel groups Number of sites: 31 Dates study was conducted: from 28 October 2015 to 11 March 2020
Participants	Adult patients (≥ 18 years of age, n = 795) admitted to the intensive care unit for severe community-acquired pneumonia. The severity of pneumonia was defined by the presence of at least one of four criteria: • The initiation of mechanical ventilation (invasive or non-invasive) with a positive end-expiratory pressure level of at least 5 cm of water • The initiation of the administration of oxygen through a high-flow nasal cannula with a ratio of the partial pressure of arterial oxygen to the inspired fraction of oxygen of less than 300, with a Fio2 of 50% or more; for patients wearing a nonrebreathing mask, an estimated PaO2:FiO2 ratio of less than 300, according to prespecified charts • A score of more than 130 on the Pulmonary Severity Index, which classifies patients with community-acquired pneumonia into 5 groups according to increasing severity, with a score of more than 130 defining group V, which is associated with the highest mortality Setting: intensive care unit Study location: France
Interventions	• Hydrocortisone (200 mg as continuous intravenous infusion per day during the first 4 days. On the fourth day, the medical team used predefined criteria to decide whether to administer hydrocortisone for a total of 8 or 14 days, depending on whether the patient's condition had improved. Regardless of

Dequin 2023 *(Continued)*

	the duration of treatment, the dose of hydrocortisone was gradually tapered according to a prespecified plan. In all cases, treatment was discontinued at the time of discharge from the ICU) • Placebo
Outcomes	**Primary outcome** • 28-day all-cause mortality **Secondary outcomes** • 90-day all-cause mortality • The length of stay in the intensive care unit • Number of intubation and invasive mechanical ventilation among patients free of respiratory support at baseline • Number of intubation and invasive mechanical ventilation among patients on non-invasive mechanical ventilation at baseline • The initiation of vasopressor therapy by day 28 • The number of ventilator-free days • The number of vasopressor-free days • The change in the PaO_2:FiO_2 ratio by day 7 • The change by day 7 in the score on the Sequential Organ Failure Assessment • Quality of life by day 90, as measured on the 36-Item Short-Form Health Survey
Identification	Sponsorship source: French Ministry of Health (Programme Hospitalier de Recherche Clinique 2014) CHU Tours Comments: no Author's name: PF Dequin Institution: University of Tours Email: dequin@univ-tours.fr Address: Service de Médecine Intensive–Réanimation, Hôpital Bretonneau, 37 044 Tours Cedex 9, France
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated list
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Indiscernible active drug and placebo
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Indiscernible active drug and placebo
Incomplete outcome data (attrition bias) All outcomes	Low risk	No evidence of attrition bias

Dequin 2023 *(Continued)*

Selective reporting (reporting bias)	Low risk	Reported outcomes matched those listed in the protocol.
Other bias	Low risk	Judgement comment: No evidence of other sources of bias.

Doluee 2018

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 centre Dates study was conducted: from August 2014 to April 2015
Participants	Adults (age 18 years or more, n = 160) that did not respond to vasopressor therapy for longer than 60 minutes Subgroup based on adrenal status as assessed by 250 µg ACTH given intramuscularly Setting: intensive care unit Study location: Iran
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) • Placebo
Outcomes	**Primary** • 28-day all-cause mortality **Secondary** • Shock termination as defined by full withdrawal of vasopressor for at least 6 consecutive hours
Identification	
Notes	Study authors contacted but did not reply Funding source: Mashhad University of Medical Sciences

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	No information was provided on the method for generating the randomisation list.
Allocation concealment (selection bias)	Low risk	Sealed envelopes were used.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of a placebo (saline in the same volume and indistinguishable from active treatment)

Doluee 2018 *(Continued)*

Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of a placebo (saline in the same volume and indistinguishable from active treatment)
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Reported outcomes in the publication matched those planned in the protocol as recorded in the Iranian Registry of Clinical Trials (https://en.irct.ir/trial/12131?revision=12131; accessed 7 April 2019)
Other bias	Unclear risk	No access to data to exclude selection bias

El Ghamrawy 2006

Study characteristics	
Methods	Study design: randomised trial Study grouping: 2 parallel groups Single centre Dates study was conducted: not reported
Participants	Adults (n = 34) with severe community-acquired pneumonia Setting: intensive care unit Study location: Saudi Arabia
Interventions	• Hydrocortisone 200 mg intravenous bolus followed by infusion at 10 mg/h for 7 days • Normal saline
Outcomes	• In-hospital mortality • Serious adverse events
Identification	
Notes	Study authors contacted but did not reply Funding source: not declared

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Insufficient information reported
Allocation concealment (selection bias)	Unclear risk	Insufficient information reported
Blinding of participants and personnel (performance bias)	Unclear risk	Insufficient information reported

El Ghamrawy 2006 *(Continued)*

All outcomes		
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Insufficient information reported
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Unclear risk	Insufficient information reported
Other bias	Unclear risk	Insufficient information reported

El-Nawawy 2017

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 centre Dates the study was conducted: not reported
Participants	Children (n = 96) aged 1 month to 4 years with septic shock Subgroups based on adrenal status as assessed by a 250 µg ACTH test Setting: intensive care unit Study location: Egypt
Interventions	• Standard management without any study-specific intervention (n = 32) • ACTH test, diagnosed positive when maximal cortisol increment < 9 µg/dL (n = 32) • Hydrocortisone continuous infusion of 50 mg/m²/24 h for 5 days, then weaned off over 5 days (n = 32)
Outcomes	**Primary outcome** • Not mentioned **Secondary outcomes** • Paediatric logistic organ dysfunction (PELOD) score • Shock reversal time • Length of stay in the paediatric intensive care unit • Survival time • Safety
Identification	
Notes	Study authors contacted but did not reply. Funding source: not declared

El-Nawawy 2017 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No information was provided on the method used for generating the randomisation list.
Allocation concealment (selection bias)	Unclear risk	No information was provided on the method used for randomisation and allocation concealment.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	The physicians, nursing team, data collectors and statistician were blinded to the use of study interventions.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	The physicians, nursing team, data collectors and statistician were blinded to the use of study interventions.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: yes
Selective reporting (reporting bias)	Unclear risk	No access to study protocol to exclude reporting bias
Other bias	Unclear risk	No access to data to exclude selection bias

Fernández-Serrano 2011

Study characteristics	
Methods	Study design: randomised trial Study grouping: 2 parallel groups Number of sites: 1 centre Dates the study was conducted: not reported
Participants	Adults (n = 56) with severe community-acquired pneumonia without shock and spontaneously breathing Setting: respiratory medicine department Study location: Spain
Interventions	• Methylprednisolone as an intravenous bolus of 200 mg administered 30 minutes before the start of antibiotic treatment, followed by 29 mg every 6 hours for 3 days, then 20 mg every 12 hours for 3 days, and finally 20 mg/d for another 3 days • Placebo
Outcomes	**Primary outcome** • Need for mechanical ventilation **Secondary outcomes**

Fernández-Serrano 2011 *(Continued)*

	• Time to resolve morbidity score • Time course of PaO_2/FiO_2 • Time for radiological clearance • ICU length of stay • Hospital length of stay • Short-term mortality • Time course of serum levels of various inflammatory biomarkers • Adverse events
Identification	
Notes	Study authors contacted but did not reply. Funding source: Fondo de Investigaciones Sanitarias (FIS) and partial funding from ISCIII RTIC (Red Respira)

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation list
Allocation concealment (selection bias)	Low risk	The generator and executor of randomisation were separated.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatments or the placebos in an indistinguishable form
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatments or the placebos in an indistinguishable form
Incomplete outcome data (attrition bias) All outcomes	High risk	> 10% of participants not analysed
Selective reporting (reporting bias)	Low risk	Outcomes reported in the publication matched those planned in the protocol as released on the ISRCTN registry.
Other bias	Unclear risk	Insufficient information

Ghanei 2021

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: parallel groups Number of sites: 6 Dates study was conducted: 13 April to 9 August 2020
Participants	Prednisolone: Adults: 116

Ghanei 2021 *(Continued)*

	Children: 0 Sepsis: N/A Septic shock: N/A CAP: N/A ARDS: N/A Male/Female: 57/59 Adrenal insufficiency/CIRCI: N/A COVID-19: 116 Usual care: Adults: 110 Children: 0 Sepsis: N/A Septic shock: N/A CAP: N/A ARDS: N/A Male/female: 55/55 Adrenal insufficiency/CIRCI: N/A COVID-19: 110 Overall: Adults: 336 Children: 0 Sepsis: N/A Septic shock: N/A CAP: N/A ARDS: N/A Male/female: 173/163 Adrenal insufficiency/CIRCI: N/A COVID-19: 336 Inclusion criteria: hospitalised patients, 16 years of age or older, were enrolled in the trial if they had a positive polymerase chain reaction (PCR) assay and oxygen saturation (SpO2) less than 94% Exclusion criteria: history of receiving any medications (i.e. immunosuppressive drugs, systemic steroids, chemotherapy drugs, hydroxychloroquine, lopinavir/ritonavir, ribavirin and oseltamivir) for COVID-19 in the last month. Patients with uncontrolled diabetes or asthma were excluded from this trial. Patients with other comorbidities were not excluded from the study if they had received the drug for their condition. Other exclusion criteria included chronic renal or liver disease, gastrointestinal haemorrhage, untreated bacterial infection, pregnancy or breast-feeding, and QT interval ≥ 500 ms. COVID-19 patients who were ill within less than 48 h were also excluded. Group differences: no evidence for clinically relevant difference in baseline data ACTH test: no
Interventions	Prednisolone: Dose per day: 25 mg Cumulated total dose: prednisolone 125 mg, azithromycin 1500 mg, naproxen 2500 mg, HCQ 400 mg, Pantoloc 200 mg Duration: 5 d (up to 10) Taper: Y (prednisolone was tapered to 5 mg per week after discharge) Continuous: no Usual care: Doses per day: HCQ 400 mg x 1, azithromycin 500 mg x 1 then 250 mg daily x 4 d, naproxen 250 mg twice a day x 5 d, Pantoloc 40 mg daily Cumulated total dose: azithromycin 1500 mg, naproxen 2500 mg, HCQ 400 mg, Pantoloc 200 mg Duration: 5 d (up to 10) Taper: N Continuous: N
Outcomes	28-day all-cause mortality

Ghanei 2021 *(Continued)*

Outcome type: dichotomous outcome
Reported as: number of participants with event (n, N)
90-day mortality
Outcome type: dichotomous outcome
Reported as: number of participants with event (n, N)
Long-term mortality
Outcome type: dichotomous outcome
Reported as: number of participants with event (n, N)
ICU mortality
Outcome type: dichotomous outcome
Reported as: number of participants with event (n, N)
Hospital mortality
Outcome type: dichotomous outcome
Reported as: number of participants with event (n, N)
Number of patients with shock reversal at day 7
Outcome type: dichotomous outcome
Reported as: number of participants with event (n, N)
Number of patients with shock reversal at day 28
Outcome type: dichotomous outcome
Reported as: number of participants with event (n, N)
SOFA score at day 7
Outcome type: continuous outcome
Reported as: standard deviation (mean, SD, N)
Length of ICU stay in all patients
Outcome type: continuous outcome
Reported as: standard deviation (mean, SD, N)
Length of ICU stay in survivors
Outcome type: continuous outcome
Reported as: standard deviation (mean, SD, N)
Length of hospital stay in all patients
Outcome type: continuous outcome
Reported as: standard deviation (mean, SD, N)
Length of hospital stay in survivors
Outcome type: continuous outcome
Reported as: standard deviation (mean, SD, N)
Number of participants with at least one AE
Outcome type: dichotomous outcome
Reported as: number of participants with event (n, N)
Number of patients with superinfection
Outcome type: dichotomous outcome
Reported as: number of participants with event (n, N)
Number of patients with at least one hyperglycaemia event
Outcome type: dichotomous outcome
Reported as: number of participants with event (n, N)
Number of patients with hypernatraemia
Outcome type: dichotomous outcome
Reported as: number of participants with event (n, N)
Number of patients with muscle weakness
Outcome type: dichotomous outcome
Reported as: number of participants with event (n, N)
Number of patients with neuropsychiatric events
Outcome type: dichotomous outcome
Reported as: number of participants with event (n, N)
Number of patients with stroke
Outcome type: dichotomous outcome
Reported as: number of participants with event (n, N)
Number of patients with cardiac event
Outcome type: dichotomous outcome
Reported as: number of participants with event (n, N)
28-day mortality in subgroup 1
Outcome type: dichotomous outcome

Ghanei 2021 *(Continued)*

	Reported as: number of participants with event (n, N) 28-day mortality in subgroup 2 Outcome type: dichotomous outcome Reported as: number of participants with event (n, N) Number of patients with gastrointestinal bleeding Outcome type: dichotomous outcome Reported as: number of participants with event (n, N)
Identification	Sponsorship source: Chemical Injuries Center, Systems Biology and Poisoning Institute, Baqiyatal lah University of Medical Sciences Country: Iran Setting: hospital ward Comments: no Author's name: Mostafa Ghanei Institution: Chemical Injuries Center, Systems Biology and Poisoning Institute, Baqiyatal lah University of Medical Sciences Email: mghaneister@gmail.com Address: Chemical Injuries Center, Systems Biology and Poisoning Institute, Baqiyatallah University of Medical Sciences, MollaSadra St., Tehran, Iran
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Judgement comment: Mentioned stratified block randomisation but unclear how this was done
Allocation concealment (selection bias)	Low risk	Judgement comment: Allocation concealment was ensured by the use of sealed envelopes, and variable block size.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: No blinding was performed - open-label trial
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: No blinding was performed - open-label trial
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Judgement comment: Modified ITT (mITT) performed, and 16 total participants were excluded from the mITT analysis -- it is unclear if this was balanced or not between groups, which may have suggested potential bias from these exclusions.
Selective reporting (reporting bias)	Low risk	Judgement comment: Reported outcomes matched protocol information
Other bias	Low risk	Judgement comment: No evidence of other sources of bias

Gordon 2014

Study characteristics

Methods	Study design: randomised controlled trial

Gordon 2014 *(Continued)*

	Study grouping: 2 parallel groups Number of sites: 4 centres The dates the study was conducted: from October 2010 through March 2012
Participants	Adults (n = 61) with septic shock on a maximal dose of vasopressin up to 0.06 U/min Setting: intensive care unit Study location: UK
Interventions	• Hydrocortisone phosphate (50 mg IV bolus 6-hourly for 5 days, 12-hourly for 3 days, then once daily for 3 days) • Placebo (0.5 mL of 0.9% saline)
Outcomes	**Primary outcome** • Difference in plasma vasopressin concentrations between treatment groups **Secondary outcomes** • Difference in vasopressin requirements • 28-day mortality • ICU mortality • Hospital mortality • Organ failure-free days to 28 days post randomisation • Shock reversal • Length of stay in ICU and at hospital • Safety
Identification	
Notes	We contacted the study authors and obtained access to individual patient data. Funding source: National Institute for Health Research

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random numbers prepared by an independent statistician
Allocation concealment (selection bias)	Low risk	Randomisation done via an online system
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatments or the placebos in an indistinguishable form
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatments or the placebos in an indistinguishable form
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none

Gordon 2014 *(Continued)*

Selective reporting (reporting bias)	Low risk	Reported information matched published statistical plan
Other bias	Low risk	Accessed unpublished information to exclude other risk of bias

Gordon 2016

Study characteristics	
Methods	Study design: randomised controlled trial with 2 × 2 factorial design Number of sites: 18 centres Dates study was conducted: from February 2013 through May 2015
Participants	Adults (n = 409) with septic shock requiring vasopressors despite fluid resuscitation within a maximum of 6 hours after onset of shock Setting: intensive care unit Study location: UK
Interventions	• Vasopressin (titrated up to 0.06 U/min) and hydrocortisone intravenous bolus every 6 hours for 5 days, every 12 hours for 3 days, and then once daily for 3 days (n = 101) • Vasopressin (titrated up to 0.06 U/min) and placebo of hydrocortisone (n = 104) • Norepinephrine (titrated up to 12 µg/min) and hydrocortisone intravenous bolus every 6 hours for 5 days, every 12 hours for 3 days, and then once daily for 3 days (n = 101) • Norepinephrine (titrated up to 12 µg/min) and placebo (n = 103) Hydrocortisone or its placebo started only when the maximum dose of vasopressin or norepinephrine was reached to maintain the target mean arterial pressure of 65 to 75 mmHg
Outcomes	**Primary outcomes** • Kidney failure-free days (i.e. number of days alive and free of kidney failure), defined by the Acute Kidney Injury Network (AKIN) group stage 3 definition during the 28 days after randomisation, with no additional penalty for death **Secondary outcomes** Rate and duration of renal replacement therapy • Length of kidney failure in survivors and non-survivors • 28-day, ICU and hospital mortality rates • Organ failure-free days in the first 28 days, assessed via the SOFA score
Identification	
Notes	We contacted the study authors and obtained access to individual patient data. Funding source: National Institute for Health Research

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation list

Gordon 2016 *(Continued)*

Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatments or the placebos in an indistinguishable form
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of numbered treatment batches containing either the active treatments or the placebos in an indistinguishable form
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol excluding reporting bias
Other bias	Low risk	Full access to data excluding selection bias

Horby 2021

Study characteristics	
Methods	Study design: platform randomised controlled trial Study grouping: parallel groups Number of sites: 176 Dates study was conducted: March 19 to 8 June 2020
Participants	Adults (age 18 years or more, n = 6425) with clinically suspected or laboratory-confirmed SARS-CoV-2 infection and no medical history that might, in the opinion of the attending clinician, put patients at substantial risk if they were to participate in the trial. Initially, recruitment was limited to patients who were at least 18 years of age, but the age limit was removed starting on 9 May. Setting: hospital general wards and intensive care units Study location: UK
Interventions	• Dexamethasone (6 mg per day orally or intravenously for up to 10 days, or until hospital discharge if sooner) • Usual care
Outcomes	**Primary outcome** • 28-day all-cause mortality **Secondary outcomes** • The time until discharge from the hospital • Amongst patients not receiving invasive mechanical ventilation at the time of randomisation, subsequent receipt of invasive mechanical ventilation (including extracorporeal membrane oxygenation) or death • Cause-specific mortality

Horby 2021 *(Continued)*

	• Receipt of renal dialysis or haemofiltration • Major cardiac arrhythmia (recorded in a subgroup) • Receipt and duration of ventilation. Amongst those receiving invasive mechanical ventilation at the time of randomisation, the outcome of successful cessation of invasive mechanical ventilation was defined as cessation within (and survival to) 28 days.
Identification	Sponsorship source: Nuffield Department of Population Health at the University of Oxford NIHR Clinical Research Network Comments: no Author's name: Drs. Horby and Landray Institution: Oxford University Email: recoverytrial@ndph.ox.ac.uk Address: RECOVERY Central Coordinating Office, Richard Doll Bldg., Old Road Campus, Roosevelt Dr., Oxford OX3 7LF, United Kingdom
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement comment: Randomised using a central web-based randomisation service
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: Open-label trial
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: Open-label trial
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: No significant exclusions or attrition and appropriate analyses used
Selective reporting (reporting bias)	Low risk	Judgement comment: Pre-specified outcomes in trial protocol reported appropriately in sufficient detail
Other bias	Unclear risk	Judgement comment: No other significant concerns of bias. It is unclear whether the pandemic context, high flow of patients, high workload of healthcare staff, and availability of treatment and staff might have biased the study.

Hu 2009

Study characteristics

Methods	Study design: randomised controlled trial Number of sites: 1 centre

Hu 2009 *(Continued)*

	Dates the study was conducted: not reported
Participants	Adults (n = 77) with septic shock Setting: intensive care unit Study location: China
Interventions	• Hydrocortisone (50 mg intravenous bolus 6-hourly for 7 days, then 50 mg 8-hourly for 3 days, then 50 mg 12-hourly for 2 days and 50 mg once daily for 2 days) • Control group: no mention of placebo
Outcomes	**Primary outcome** • Time on norepinephrine and lactate clearance **Secondary outcome** • ICU mortality • ICU length of stay • Shock reversal
Identification	
Notes	We contacted the study authors, who did not reply. Funding source: not declared

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not stated in the manuscript
Allocation concealment (selection bias)	Unclear risk	Not stated in the manuscript
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	No information is provided about blinding
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	No information is provided about blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Unclear risk	No information
Other bias	Unclear risk	No information

Huang 2014

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 3 parallel groups Number of sites: 1 centre Dates the study was conducted: not reported
Participants	Adults (age 18 years or more, n = 60) with sepsis Setting: intensive care unit Study location: China
Interventions	• Hydrocortisone 200 mg as continuous infusion for 7 days • Chinese herb (Sini decoction) 100 mL per day enterally for 7 days • Standard therapy
Outcomes	**Primary outcome** • Hypothalamic-pituitary-adrenal function assessed by levels of cortisol and ACTH and cortisol increment post ACTH injection at day 3 and day 14 **Secondary outcomes** • 3-day shock reversal • 28-day all-cause mortality
Identification	
Notes	We contacted the study authors, who did not reply. Funding source: not declared

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Unclear risk	No information
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	No information on blinding
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	No information on blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none

Huang 2014 *(Continued)*

Selective reporting (reporting bias)	Unclear risk	No information
Other bias	Unclear risk	No information

Huh 2007

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 centre Dates study was conducted: from July 2005 through June 2006
Participants	Adults (age 18 years or more, n = 82) with septic shock and adrenal insufficiency Setting: intensive care unit Study location: South Korea
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) • Hydrocortisone (50 mg intravenous bolus every 6 hours for 3 days)
Outcomes	**Primary outcome** • 28-day mortality **Secondary outcomes** • Shock reversal • Duration of mechanical ventilation • Length of stay • Safety
Identification	
Notes	We contacted the study authors, who did not reply. Funding source: not declared

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Unclear risk	Not given
Blinding of participants and personnel (performance bias) All outcomes	High risk	Open-label trial

Huh 2007 *(Continued)*

Blinding of outcome assessment (detection bias) All outcomes	High risk	Open-label trial
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Unclear risk	No explicit information on plan analysis
Other bias	Unclear risk	No information

Hyvernat 2016

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 4 centres Dates study was conducted: from November 2008 through June 2010
Participants	Adults (18 years or more, n = 120) with septic shock Subgroups based on adrenal status as assessed by a 250 µg ACTH test Setting: intensive care unit Study location: France
Interventions	• Hydrocortisone 50 mg intravenous bolus every 6 hours for 5 days without tapering + continuous infusion of a placebo for 5 days • Hydrocortisone as an initial bolus of 100 mg followed by continuous infusion of 300 mg per day for 5 days without tapering + intravenous bolus of placebo every 6 hours for 7 days
Outcomes	**Primary outcome** • 28-day all-cause mortality **Secondary outcomes** • Survival time • Days free of vasopressor (with zero days assigned to participants dying before being weaned off) • Days free of mechanical ventilation (with zero days assigned to participants dying before being weaned off) • Days free of renal replacement therapy (with zero days assigned to participants dying before being weaned off) • Safety
Identification	
Notes	We contacted the study authors and obtained access to individual patient data. Funding source: Association Niçoise de Réanimation Médicale

Hyvernat 2016 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of placebo: saline serum supplied as 10 mL ampoules, indistinguishable from active treatment
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Use of placebo: saline serum supplied as 10 mL ampoules, indistinguishable from active treatment
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol excluding reporting bias
Other bias	Low risk	No evidence of other bias

Iglesias 2020

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: parallel groups Number of sites: 2 Dates study was conducted: 14 February 2018 and 29 April 2019
Participants	Adults (age of 18 years or more, n = 137) with a primary diagnosis of sepsis or septic shock according to the 2016 Surviving Sepsis Campaign definitions, within 12 hours of admission to the ICU and compliance with the 3-hour sepsis bundle Setting: intensive care unit Study location: USA
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 4 days or ICU discharge) + ascorbic acid (1500 mg intravenous infusion every 6 hours 4 days or ICU discharge) + thiamine (200 mg intravenous infusion every 12 hours 4 days or ICU discharge) • Placebo
Outcomes	**Primary outcomes** • Resolution of shock defined as the time from starting blinded study medications to discontinuation of all vasopressor support

Iglesias 2020 *(Continued)*

	• Change in SOFA score defined as the initial SOFA score minus the day 4 SOFA score **Secondary outcomes** • ICU mortality • Hospital mortality • Procalcitonin clearance • The length of stay in the hospital • The length of stay in ICU • Ventilator-free days
Identification	Sponsorship source: Community Medical Centre, Toms River, NJ Comments: no Author's name: Jose Iglesia Institution: Department of Critical Care, Department of Nephrology, Community Medical Centre Email: iglesias23@gmail.com Address: Department of Critical Care, Department of Nephrology, Community Medical Centre, 99 NJ-37, Toms River, NJ 08755;
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement comment: Randomised using a random number table provided to pharmacists
Allocation concealment (selection bias)	Low risk	Judgement comment: Randomisation kept in password-protected file and only available to site pharmacists. Performed in blocks of 2.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Judgement comment: Adequate blinding of participants and personnel using adequate matching saline placebo
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement comment: Outcome assessors were adequately blinded to participant treatment allocation throughout enrolment and outcomes met
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: ITT employed and minimal exclusions post-randomisation with no significant concerns about incomplete outcome data (only 3 patients excluded post-randomisation with explanation). No other exclusions or loss to follow-up.
Selective reporting (reporting bias)	Low risk	Judgement comment: No clear previously published trial protocol available. When looking back on the archived clinicaltrials.gov registrations for this study prior to enrolment (2018-02-02) the primary outcome is different (listed as hospital mortality). However, despite the change in primary outcome, essentially all previously specified outcomes are reported in the parent trial adequately.
Other bias	Low risk	Judgement comment: No evidence of other sources of bias

Jamshidi 2021

Study characteristics

Jamshidi 2021 *(Continued)*

Methods	Study design: randomised controlled trial Study grouping: parallel groups Number of sites: 1 Dates study was conducted: May and November 2018
Participants	Adults (age 18 years or more, n = 58) with septic shock (mean systolic blood pressure lower than 65 mmHg), on invasive mechanical ventilation, and admitted to the intensive care unit Setting: intensive care unit Study location: Iran
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 3 days) + ascorbic acid (1500 mg intravenous infusion every 6 hours for 3 days) + thiamine (600 mg intravenous infusion every 12 hours for 3 days) • Usual care
Outcomes	**Primary outcome** • Not mentioned **Secondary outcomes** • The change in serum levels of procalcitonin, lactate and leukocyte count after 72 h of treatment • In-hospital mortality • The change in the sequential organ failure assessment (SOFA) score
Identification	Sponsorship source: Zanjan University of Medical Sciences for funding and supporting the study Comments: none Author's name: Mohammad Reza Jamshidi Institution: Department of Anesthesiology and Critical Care Medicine, Ayatollah Mousavi Hospital, Zanjan University of Medical Sciences, Zanjan, Iran Email: mohammadrezazeraati@zums.ac.ir Address: r. Mohammad Reza Zeraati, Department of Anesthesiology and Critical Care Medicine, Ayatollah Mousavi Hospital, Zanjan University of Medical Sciences, Zanjan, Iran
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated list
Allocation concealment (selection bias)	Unclear risk	No information on how allocation was concealed
Blinding of participants and personnel (performance bias) All outcomes	High risk	Open-label trial
Blinding of outcome assessment (detection bias)	High risk	Open-label trial

Jamshidi 2021 *(Continued)*

All outcomes		
Incomplete outcome data (attrition bias) All outcomes	Low risk	No evidence of attrition bias
Selective reporting (reporting bias)	Low risk	Reported outcomes matched those listed in methods and protocol
Other bias	Unclear risk	Judgement comment: The cohort under study with multiple trauma as the main reason for developing septic shock differs from other reported trials in the field.

Jiang 2019

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 3 parallel groups Number of sites: 1 Dates study was conducted: June 2013 to August 2015
Participants	Adults (age 18 years or more, n = 41) with septic shock as per Sepsis 3 definition Setting: intensive care unit Study location: China
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) administered within 6 hours after the occurrence of septic shock • Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) administered within 12 to 25 hours • Usual care (routine anti-infection, fluid resuscitation, nutritional support and organ protection. No clear placebo).

Outcomes	**Primary outcome**
	• Not mentioned
	Secondary outcomes
	• Serum levels of immune biomarkers at baseline, days 1, 3 and 7, including:
	  ◦ Interleukin 6
	  ◦ Interleukin 10
	  ◦ The expression of serum human leucocyte antigen DR locus (HLA-DR)
	  ◦ The apoptosis rate of peripheral blood CD4+ cells.
	• Occurrence of complications, at any time during hospital stay, including:
	  ◦ Shock recurrence
	  ◦ Arrhythmia
	  ◦ Gastrointestinal bleeding
	  ◦ Blood glucose ≥ 10 mmol/L
	  ◦ Fungal infections

Identification	Sponsorship source: Science and Technology support project of Hebei province, China (No: 14277744D) Author's name: Huiyu Tian

Jiang 2019 *(Continued)*

	Institution: The First Hospital of Hebei Medical University Email: hi2753@163.com Address: Department of Intensive Care Unit, The First Hospital of Hebei Medical University, 89 Donggang Road, Shijiazhuang 050031, China
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Judgement comment: Insufficient data to assess how the random list has been generated
Allocation concealment (selection bias)	Unclear risk	Judgement comment: There are no details provided on how or if allocation concealment was achieved.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: There is no evidence to suggest that blinding of treatment allocation was performed.
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: There is no evidence to suggest that blinding of treatment allocation was performed.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: No evidence of incomplete outcome data for included participants or attrition.
Selective reporting (reporting bias)	Unclear risk	Judgement comment: There was no trial protocol available but within the limitations of this there was no clear evidence of selective outcome reporting.
Other bias	Unclear risk	Judgement comment: Insufficient data

Keh 2003

Study characteristics

Methods	Study design: randomised controlled trial with cross-over design Number of sites: 1 centre Dates study was conducted: from March 1997 through September 2000
Participants	Adults (age 18 years or more, n = 40) with vasopressor-dependent septic shock Setting: intensive care unit Study location: Germany
Interventions	• Hydrocortisone (100 mg 30-minute intravenous infusion followed by 10 mg/h continuous infusion for 3 days) • Placebo All participants received hydrocortisone for 3 days, preceded or followed by placebo for 3 days

Keh 2003 *(Continued)*

Outcomes	**Primary** • Immune response **Secondary** • Improvement in haemodynamics and organ system failure • Safety
Identification	
Notes	We contacted the study authors and obtained access to individual patient data. Funding sources: Deutsche Forschungsgemeinschaft (expenses for analytes) and Pharmacia/Upjohn (expenses for patients' insurance) Data were extracted by DA. The risk of bias was assessed by DA and BR.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Randomisation list kept confidential by the pharmacist
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Pharmacist masked study drugs and all other stakeholders remained blinded.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Pharmacist masked study drugs and all other stakeholders remained blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol
Other bias	Low risk	Full access to data including screening log

Keh 2016

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of study sites: 34 centres

Keh 2016 *(Continued)*

	Dates study was conducted: from January 2009 to August 2013
Participants	Adults (n = 380) with severe sepsis who were not in septic shock Subgroups based on adrenal status as assessed by the 250 µg ACTH test Setting: intensive care unit Study location: Germany
Interventions	• Hydrocortisone (continuous infusion of 200 mg per day for 5 days followed by dose tapering until day 11) • Placebo
Outcomes	**Primary** • Proportion of participants who developed septic shock within 14 days **Secondary** • Time until septic shock • Mortality in the intensive care unit or hospital • Survival up to 180 days • Safety • Secondary infection • Mechanical ventilation weaning failure • Muscle weakness • Hyperglycaemia (blood glucose level > 150 mg/dL)
Identification	
Notes	We contacted the study authors and obtained access to individual patient data. Funding sources: Charité-Universitätsmedizin Berlin and grant from the German Federal Ministry of Education and Research Data were extracted by DA. The risk of bias was assessed by DA and BR.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	The study medication (hydrocortisone and placebo) was produced and released by BAG Health Care GmbH. The medication was delivered in boxes, each containing 17 brown glass vials for 1 patient. Each vial contained 100 mg of lyophilised hydrocortisone hydrogen succinate or the same amount of lyophilised mannitol as placebo, which was indistinguishable from hydrocortisone.
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	The study medication (hydrocortisone and placebo) was produced and released by BAG Health Care GmbH. The medication was delivered in boxes, each containing 17 brown glass vials for 1 patient. Each vial contained 100

Keh 2016 *(Continued)*

		mg of lyophilised hydrocortisone hydrogen succinate or the same amount of lyophilised mannitol as placebo, which was indistinguishable from hydrocortisone.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol, excluding reporting bias
Other bias	Low risk	Full access to data, excluding selection bias

Kurungundla 2008

Study characteristics	
Methods	Study design: randomised trial Study grouping: 2 parallel groups Number of sites: 1 centre Dates the study was conducted: unknown
Participants	Adults (n = 21) with sepsis Subgroups based on adrenal status as assessed by the 250 µg ACTH test Setting: intensive care unit Study location: USA
Interventions	• Corticosteroids at stress dose (molecule, exact dose and duration not reported) • Placebo
Outcomes	• Short-term mortality • ICU length of stay
Identification	
Notes	We contacted the study authors, who did not reply. Funding source: not declared

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Not reported
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	Unclear risk	There is insufficient information about blinding

Kurungundla 2008 *(Continued)*

All outcomes		
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	There is insufficient information about blinding
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Not reported
Selective reporting (reporting bias)	Unclear risk	Not reported
Other bias	Unclear risk	Not reported

Labib 2022

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 3 parallel groups Number of sites: 1 centre The study was conducted between September 2018 and September 2020
Participants	Adults (aged between 18 and 80 years, n = 66) with septic shock defined by clinical evidence of infection within the previous 72 h of ICU admission or Sepsis-Related Organ Failure Assessment (SOFA) score of 3 or 4 (on a scale of 0 to 4 for each of 6 organ systems) for at least 2 organs and at least 6 h or vasopressors therapy (norepinephrine, epinephrine or any other vasopressors at a dose of ≥ 0.25 μg/kg/min) for at least 6 h to maintain a systolic blood pressure of at least 90 mmHg or mean blood pressure of at least 65 mmHg Setting: intensive care unit Study location: Egypt
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) • Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) + fludrocortisone (50 μg once per day via a nasogastric tube for 7 days) • Usual care
Outcomes	**Primary outcome** • Mortality rate due to septic shock **Secondary outcomes** Improvement of the patient in the form of maintaining MAP ≥ 65 mmHg without vasopressors or death of patient The time of weaning from vasopressors The SOFA score as measured daily during the stay in the intensive care unit The length of stay in the intensive care unit Complications that may be related to corticosteroid treatment

Labib 2022 *(Continued)*

Identification		
Notes		

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement comment: Sequence was generated by a computer
Allocation concealment (selection bias)	Low risk	Judgement comment: Use of sealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: Only patients were blinded to treatment. Investigators, healthcare staff, statisticians and sponsors were not blinded.
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: No blinding of outcome assessors
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: No evidence of attrition bias
Selective reporting (reporting bias)	Unclear risk	Judgement comment: Protocol not available. Insufficient information reported at NCT04492280 clinicaltrial.gov
Other bias	Low risk	Judgement comment: No evidence of other bias

Li 2016

Study characteristics

Methods	Study design: randomised trial Study grouping: 2 parallel groups Number of sites: 1 centre Dates study was conducted: from May 2014 to February 2016
Participants	Adults (age 18 years old or more, n = 58) with community-acquired pneumonia and septic shock Setting: intensive care unit Study location: China
Interventions	• Methylprednisolone 80 mg intravenously once a day for 7 days • Standard therapy
Outcomes	• 28-day mortality • Hospital length of stay • Time on mechanical ventilation

Li 2016 *(Continued)*

	• Time on vasopressor • Time course of PaO_2/FiO_2 and C-reactive protein levels • Adverse events
Identification	
Notes	We contacted the study authors, who did not reply. Funding source: not declared

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not reported
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	There is insufficient information about blinding
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	There is insufficient information about blinding
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Insufficient information reported
Selective reporting (reporting bias)	Unclear risk	No access to trial protocol
Other bias	Unclear risk	Insufficient information reported

Liu 2012

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: parallel groups Number of sites: 1 centre Dates the study was conducted: not reported
Participants	Adults (age 18 years or more, n = 26) with ARDS and sepsis, including septic shock (n = 12), and with critical illness-associated corticosteroid insufficiency Setting: intensive care unit Study location: China

Liu 2012 *(Continued)*

Interventions	• Hydrocortisone (100 mg intravenous bolus 8-hourly for 7 consecutive days) • Placebo (normal saline)
Outcomes	**Primary** • Unclear **Secondary** • 28-day mortality • Prevalence of shock within 28 days • SOFA score (information for SOFA score on day 7 not available) • ICU length of stay • Safety
Identification	
Notes	We contacted the study authors and obtained access to individual patient data. Funding source: not declared

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Unclear risk	No explicit information in the manuscript
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	The authors stated they used normal saline as a placebo but no information is provided on the blinding process.
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	The authors stated they used normal saline as a placebo but no information is provided on the blinding process.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No explicit information in the manuscript
Selective reporting (reporting bias)	Unclear risk	No information
Other bias	Unclear risk	No information

Loisa 2007

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups

Loisa 2007 *(Continued)*

	Number of sites: 1 centre Dates study was conducted: from July 2005 through April 2006
Participants	Adults (age 18 years or more, n = 48) with septic shock Setting: intensive care unit Study location: Finland
Interventions	• Hydrocortisone intravenous bolus of 50 mg every 6 hours for 5 days • Hydrocortisone as a continuous infusion of 200 mg per day for 5 days
Outcomes	**Primary** • Difference in mean blood glucose levels between study groups and occurrence of hyperglycaemic and hypoglycaemic episodes **Secondary** • Reversal of shock • Extent of nursing workload required to maintain strict normoglycaemia
Identification	
Notes	We contacted the study authors, who did not reply. Funding source: Medical Research Fund of Tampere University Hospital and Medical Research Fund of Päijät-Häme Central Hospital, Lahti

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Not stated by the study authors
Allocation concealment (selection bias)	Unclear risk	Not stated by the study authors
Blinding of participants and personnel (performance bias) All outcomes	High risk	Open-label trial
Blinding of outcome assessment (detection bias) All outcomes	High risk	Open-label trial
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Not stated by study authors
Selective reporting (reporting bias)	Unclear risk	Not stated by study authors
Other bias	Unclear risk	Not stated by study authors

Luce 1988

Study characteristics	
Methods	Study design: randomised controlled trial Number of sites: 1 centre Dates the study was conducted: unknown
Participants	Adults (age 18 years or more, n = 75) with sepsis and septic shock Setting: intensive care unit Study location: USA
Interventions	• Methylprednisolone (30 mg/kg 15-minute intravenous infusion every 6 hours for 24 hours) • Placebo
Outcomes	**Primary** • Prevention of ARDS **Secondary** • Hospital mortality
Identification	
Notes	We contacted the study authors and their data were no longer available. Funding source: National Heart, Lung and Blood Institute, and Upjohn Company

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Randomisation list kept confidential by the pharmacist
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Pharmacist masked the study drugs and all other stakeholders remained blinided
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Pharmacist masked the study drugs and all other stakeholders remained blinided
Incomplete outcome data (attrition bias) All outcomes	High risk	12 out of 87 randomly assigned participants were not analysed, and follow-up was not provided.
Selective reporting (reporting bias)	Unclear risk	No access to study protocol
Other bias	Unclear risk	No access to data to exclude selection bias

Lv 2017

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 centre Dates study was conducted: from September 2015 to September 2016
Participants	Adults (age 60 years or more, n = 118) with septic shock that has developed within 6 hours before randomisation Setting: intensive care unit Study location: China
Interventions	• Hydrocortisone (200 mg per day as a continuous infusion for 6 days, and then tapered off. Once all vasopressors were discontinued, the taper protocol was initiated: half dose for 3 days, then quarter dose for 3 days, then stopped). • Placebo (vials containing normal saline as placebo were identical to those containing hydrocortisone. Placebo administration procedures were similar).
Outcomes	**Primary** • 28-day all-cause mortality **Secondary** • Reversal of shock • In-hospital mortality • Length of stay at ICU and at hospital
Identification	
Notes	We contacted the study authors and obtained access to individual patient data. Funding source: not declared

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Vials containing normal saline as placebo were identical to those containing hydrocortisone. Placebo administration procedures were similar.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Vials containing normal saline as placebo were identical to those containing hydrocortisone. Placebo administration procedures were similar.

Lv 2017 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol excluding reporting bias
Other bias	Low risk	Full access to data excluding selection bias

Lyu 2022

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 Dates study was conducted: February 2019 through June 2021
Participants	Adults (age 18 years old or older, n = 426) with diagnosis of septic shock within 12 h Setting: intensive care unit Study location: China
Interventions	• Hydrocortisone (200 mg per day as a continuous infusion for 5 days or until ICU discharge) + vitamin C (2 g intravenous infusion every 6 h for 5 days or until ICU discharge) + thiamine (200 mg intravenous infusion every 12 h for 5 days or until ICU discharge) • Placebo (normal saline)
Outcomes	**Primary outcome** • 90-day all-cause mortality **Secondary outcomes** • 28-day all-cause mortality • Mortality rates at intensive care unit discharge • Mortality rates at hospital discharge • Shock reversal rate • Time to shock reversal • 72 h delta SOFA score • ICU-free days • Vasopressor-free days up to day 28 (patients who died before day 28 were assigned zero free days) • Ventilator support-free days up to day 28 (patients who died before day 28 were assigned zero free days) • Length of stay in the intensive care unit • Length of stay in the hospital
Identification	Sponsorship source: Department of Critical Care Medicine, Northern Jiangsu People's Hospital; Clinical Medical College, Yangzhou University, Yangzhou, Jiangsu, People's Republic of China Comments: no Author's name: Xiao-Hua Gu Institution: Northern Jiangsu People's Hospital

Lyu 2022 *(Continued)*

	Email: njguxiaohua@163.com Address: Department of Critical Care Medicine, Northern Jiangsu People's Hospital; Clinical Medical College, Yangzhou University, Yangzhou, Jiangsu, People's Republic of China
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement comment: Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Judgement comment: Computer program used and patients, investigators, clinical staff and research staff remained blinded to the allocated therapy
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Judgement comment: Adequate placebo used and trial regimen/placebo was ensured with identical masked bags
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement comment: Outcome assessors were blinded to allocated therapy. Only exception was designated nurses, who were aware of treatment allocation; however, they were not involved in clinical care or outcome evaluation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: There were no significant exclusions in the ITT analysis and no attrition.
Selective reporting (reporting bias)	Low risk	Judgement comment: No concerns. All pre-specified outcome data both in methods and clinicaltrials.gov registry are reported in sufficient detail.
Other bias	Low risk	Judgement comment: No evidence of other sources of bias.

McHardy 1972

Study characteristics

Methods	Study design: randomised trial on 4 parallel groups Number of sites: 1 centre The study was conducted in January 1966 and continued until June 1970.
Participants	Children and adults (age 12 years or more, n = 126) with severe community-acquired pneumonia Setting: respiratory wards Study location: Scotland
Interventions	• Ampicillin 1 g per day orally for 7 to 14 days according to clinical response • Ampicillin 1 g per day orally for 7 to 14 days according to clinical response, plus prednisolone 5 mg orally every 6 hours for 7 days • Ampicillin 2 g per day orally for 7 to 14 days according to clinical response • Ampicillin 2 g per day orally for 7 to 14 days according to clinical response, plus prednisolone 5 mg orally every 6 hours for 7 days

McHardy 1972 *(Continued)*

Outcomes	**Primary outcome** • Not mentioned **Secondary outcomes** • In-hospital death • Duration of antibiotic therapy • Need to change antibiotic therapy • Time to resolve fever • Time for pathogen clearance • Time for radiological clearance
Identification	
Notes	We did not contact the study authors. Funding source: anonymous gift to the Department of Respiratory Diseases, University of Edinburgh

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No information
Allocation concealment (selection bias)	Low risk	Sealed, opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Open-label trial
Blinding of outcome assessment (detection bias) All outcomes	High risk	Open-label trial
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Unclear risk	No information
Other bias	Unclear risk	No information

Meduri 2007

Study characteristics

Methods	Study design: randomised controlled trial (2:1 scheme) Number of sites: 5 centres Dates the study was conducted: from April 1997 through April 2002

Meduri 2007 *(Continued)*

Participants	Adults (age 18 years or more, n = 91) with early ARDS (≤ 72 hours from diagnosis of ARDS); 61 (67%) had sepsis or septic shock, and the primary author provided separate data for these participants Stratification according to cortisol response to 250 µg synacthene for non-responders (delta cortisol ≤ 9 µg/dL) and responders (> 9 µg/dL) Setting: intensive care unit Study location: USA
Interventions	• Methylprednisolone loading dose of 1 mg/kg followed by continuous infusion of 1 mg/kg/d from day 1 to day 14, then 0.5 mg/kg/d from day 15 to day 21, then 0.25 mg/kg/d from day 22 to day 25, then 0.125 mg/kg/d from day 26 to day 28. If a participant was extubated before day 14, he/she is advanced to day 15 of drug therapy. Treatment was given intravenously until internal intake was restored, then it was given as a single oral dose. • Placebo
Outcomes	**Primary** • Improvement in Lung Injury Score (LIS) on day 7. This improvement was defined as a reduction in score ≥ 1 point and a day 7 score ≤ 2 (if randomisation LIS score < 3) or ≤ 2.5 (if randomisation LIS score < 3). **Secondary** • Mechanical ventilation-free days • Multiple organ dysfunction (MOD) score on study day 7 • 28-day mortality • ICU mortality • Hospital mortality • Length of stay in ICU and at hospital • C-reactive protein levels on study day 7 • Safety
Identification	
Notes	If a participant failed to improve on LIS between day 7 and day 9, he/she received open-label methyl-prednisolone at 2 mg/kg/d for non-resolving ARDS. We contacted the study authors, who provided separate data for patients with sepsis. Funding source: Baptist Memorial Health Care Foundation and the Assisi Foundation of Memphis

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Pharmacists masked the study drugs and all other stakeholders remained blinded.
Blinding of outcome assessment (detection bias)	Low risk	Pharmacists masked the study drugs and all other stakeholders remained blinded.

Meduri 2007 *(Continued)*

All outcomes		
Incomplete outcome data (attrition bias) All outcomes	Low risk	Full access to data, excluding any attrition bias
Selective reporting (reporting bias)	Low risk	Full access to data including screening log - reported outcomes matched those planned in protocol.
Other bias	High risk	Study was stopped prematurely for efficacy.

Meduri 2009

Study characteristics	
Methods	Study design: randomised controlled trial with 2 parallel groups, stratified according to absence of shock and a multiple organ dysfunction score (MODS) < 3 (strata A), or presence of shock and MODS ≥ 3 (strata B) Number of sites: 1 centre Dates the study was conducted: not reported
Participants	Adults (age 18 years or more, n = 80) with sepsis with or without septic shock Setting: intensive care unit Study location: USA
Interventions	• Hydrocortisone as a bolus of 300 mg followed by continuous infusion at 10 mg per hour for 7 days without tapering • Placebo
Outcomes	**Primary** • In strata A: improvement in MODS by day 7 • In strata B: resolution of shock by day 7 **Secondary** • Mortality on day 7 and on day 28 • Time to wean off vasopressors • New onset of shock • New onset of ARDS • MODS on day 7
Identification	
Notes	The study was published as an abstract only. We contacted the study authors and obtained access to individual patient data. Funding source: Baptist Memorial Health Care Foundation and Assisi Foundation of Memphis

Risk of bias

Bias	Authors' judgement	Support for judgement

Meduri 2009 *(Continued)*

Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Pharmacists masked the study drugs and all other stakeholders remained blinded.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Pharmacists masked the study drugs and all other stakeholders remained blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Full access to data, excluding any attrition bias
Selective reporting (reporting bias)	Low risk	Study was stopped prematurely for efficacy
Other bias	Unclear risk	Full access to data including screening log Some imbalance in numbers allocated to hydrocortisone versus placebo

Meduri 2022

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 42 centres Dates study was conducted: 1 January 2012 to 31 August 2016
Participants	Adults (age 18 years or more, n = 584, male/female: 562/21) with a clinical diagnosis of severe CAP/HCAP (HCAP = from nursing facility/rehab/LTC, CAP = admitted from home), enroled within 72-96h of hospital presentation (additional 24h in patients not yet meeting severity criteria), admitted to ICU or intermediate care unit, having at least 1 major, or 3 minor, modified ATS/IDSA criteria for severe pneumonia (major = IMV or IMV, vasopressors for shock, arterial pH 30, hypotension requiring aggressive IV fluids, uraemia, PF400, multilobar infiltrates) Setting: intensive care unit or intermediate care unit Study location: USA
Interventions	• Methylprednisolone (loading intravenous dose 40 mg, 40 mg for 7 days, 20 mg for 7 days, 12 mg for 3 days, 4 mg for 3 days for a total of 21 days) • Placebo (saline or enteral capsules when discharged from ICU)
Outcomes	**Primary outcome** • 60-day all-cause mortality **Secondary outcomes**

Meduri 2022 *(Continued)*

	• 28-day all-cause mortality
	• 180-day all-cause mortality
	• 365-day all-cause mortality
	• In-hospital mortality
	• Development of vasopressor-dependent septic shock
	• Development of ARDS
	• Number of multiple organ dysfunction syndrome (MODS)-free days to day 8
	• Number of MV-free days up to days 8 and 28
	• Length of stay in the intensive care unit
	• Length of stay in the hospital
	• Adverse events
	• Post-discharge
	  ◦ Cardiovascular complications within 180 days of randomisation
	  ◦ Quality of life and functional status at days 28, 60 and 180
	  ◦ Number and causes of rehospitalisation at VA hospitals within 1 year
	  ◦ Serious adverse events

Identification	Sponsorship source: VA Cooperative Studies Program Comments: no Author's name: G. Umberto Meduri Institution: Memphis VA Medical Center, Memphis, USA Email: gmeduri@uthsc.edu Address: 1030 Jefferson Ave, Memphis, TN 38104, United States
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement comment: Trial protocol mentioned a centralised computer system (VA cooperate studies programs' eDC) was used.
Allocation concealment (selection bias)	Low risk	Judgement comment: Centralised computer system (VA cooperate studies programs' eDC) was used and sealed, opaque envelopes were used per supplemental protocol.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Judgement comment: Double-blind with placebo, including IV NS or enteral capsules. No significant concerns.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement comment: Double-blind with placebo including IV NS or enteral capsules. No significant concerns.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: The number of participants with missing outcome data is small across trial outcomes and seems fairly balanced between both arms. Also, sensitivity analysis was performed on the 21 participants who missed primary outcomes due to early withdrawal.
Selective reporting (reporting bias)	Low risk	Judgement comment: No evidence of selective reporting of multiple eligible outcome measurements and the analysis is overall in keeping with the pre-specified protocol.
Other bias	Unclear risk	Judgement comment: External validity in question given this trial featured almost exclusively males. Additionally, there is no clear definition of sepsis or

Meduri 2022 *(Continued)*

		septic shock, and we are unable to find a number of patients specifically with sepsis in each arm. The trial stopped early, but no extremes of effect were seen.

Meijvis 2011

Study characteristics	
Methods	Study design: randomised controlled trial with 2 parallel groups Number of sites: 2 centres Dates the study was conducted: from November 2007 to September 2010
Participants	Adults (n = 304) with confirmed community-acquired pneumonia Setting: emergency departments and medical wards Study location: The Netherlands
Interventions	• Dexamethasone (5 mg intravenous bolus once a day for 4 days) • Placebo (normal saline)
Outcomes	**Primary** • Length of hospital stay **Secondary** • 30-day mortality • Hospital mortality • Duration of treatment with intravenous antibiotics • Admission to ICU • Inflammation markers and health performance • Lung function • Safety
Identification	
Notes	We contacted the study authors and obtained separate data for patients with sepsis. Funding source: no funding source identified for this study

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Randomisation list kept confidential by the pharmacist
Blinding of participants and personnel (performance bias)	Low risk	Use of pre-numbered boxes containing four ampoules (identical appearance of dexamethasone and placebo) for intravenous administration. Patients, investigators and data assessors were masked to treatment allocation.

Meijvis 2011 *(Continued)*

All outcomes		
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of pre-numbered boxes containing four ampoules (identical appearance of dexamethasone and placebo) for intravenous administration. Patients, investigators and data assessors were masked to treatment allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	All outcomes reported in the study protocol are reported in the final analysis.
Other bias	Low risk	Full access to study protocol

Menon 2017

Study characteristics	
Methods	Study design: randomised controlled trial with 2 parallel groups Number of sites: 7 centres Dates study was conducted: from July 2014 through March 2016
Participants	Children (n = 57) newborn to 17 years old inclusive with suspected septic shock on vasopressor therapy from 1 to 6 hours before randomisation Setting: intensive care unit Study location: Canada
Interventions	• Hydrocortisone initial IV bolus of 2 mg/kg followed by 1 mg/kg of hydrocortisone every 6 hours until the patient meets stability criteria (defined as no increase in vasoactive infusions and no administration of a fluid bolus) for at least 12 hours • Placebo
Outcomes	**Primary** • Patient accrual rate **Secondary** • Agreement between clinical diagnosis of septic shock with either the International Paediatric Sepsis Consensus Conference definition of septic shock (24) or the International Classification of Diseases, 10th edition (ICD-10), diagnostic codes • Number of eligible patients who were randomised within 6 hours from identification • Rate of protocol adherence (percentage of study drug doses correctly administered) • Frequency of corticosteroid use outside the protocol (pre-screening and post-randomization) • PICU length of stay • Hospital length of stay • Time on vasopressors • Duration of mechanical ventilation
Identification	
Notes	We did not contact the study authors.

Menon 2017 *(Continued)*

	Funding source: Canadian Institutes of Health Research (CIHR)

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Centralised randomisation through a secured website
Blinding of participants and personnel (performance bias) All outcomes	Low risk	The active drug and placebo (hydrocortisone and normal saline) were identical in appearance, volume and smell and all study personnel, members of the bedside health care team and patients/families were blinded to the study group assignment.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	The active drug and placebo (hydrocortisone and normal saline) were identical in appearance, volume and smell and all study personnel, members of the bedside health care team and patients/families were blinded to the study group assignment.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol, excluding reporting bias
Other bias	Low risk	Access to study protocol with no evidence of other bias

Mirea 2014

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: 3 parallel groups Number of sites: 1 centre Dates the study was conducted: not reported
Participants	Adults (n = 171) with septic shock Setting: intensive care unit Study location: Romania
Interventions	• Hydrocortisone 50 mg intravenous bolus every 6 hours for a maximum of 7 days • Hydrocortisone 200 mg per day as a continuous infusion for a maximum of 7 days • Usual care
Outcomes	**Primary** • Mean serum sodium values over 7 days **Secondary**

Mirea 2014 *(Continued)*

	• Number of hyperglycaemia episodes, number of severe hyperglycaemia episodes (> 150 mg/dL) during 7-day study period • Short-term mortality
Identification	
Notes	We contacted the study authors and obtained access to individual patient data. Funding source: not declared

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation list
Allocation concealment (selection bias)	Unclear risk	No information provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Open-label trial
Blinding of outcome assessment (detection bias) All outcomes	High risk	Open-label trial
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol and to raw data of individual patients, excluding reporting bias
Other bias	Low risk	Full access to raw data of individual patients including screening log

Mohamed 2020

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 Dates the study was conducted: June 2018 to August 2019
Participants	Adults (age 18 years or more, n = 86) with septic shock (Surviving Sepsis Campaign 2016 guidelines) enrolled within 6 hours of initiation of inotropic support Setting: intensive care unit Study location: India

Mohamed 2020 *(Continued)*

Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 4 days) + vitamin C (1500 mg intravenous infusion every 6 hours for 4 days) + thiamine (200 mg intravenous infusion every 12 hours for 4 days) • Usual care
Outcomes	**Primary outcome** • All-cause mortality **Secondary outcomes** • Time to shock reversal defined as the amount of time (in hours) required to withdraw all vasoactive medication supports from recruitment in study • Change in SOFA score over 72 hours • Need for mechanical ventilation • New onset of AKI • Length of stay in the intensive care unit • Length of stay in the hospital
Identification	Sponsorship source: Indian Society of Critical Care Medicine and Dr. Subhash Todi Comments: no Author's name: Dipu T Sathyapalan Institution: Amrita Institute of Medical Sciences Email: diputsmck@gmail.com Address: Department of General Medicine and Division of Infectious Diseases, Amrita Institute of Medical Sciences, Kochi, Kerala, India
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement comment: Online computer system used
Allocation concealment (selection bias)	Low risk	Judgement comment: Computer system used and random permuted block sizes of length 6
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: Open-label trial so participants and personnel are aware of treatment allocation.
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: Open-label trial so outcome assessors and research group are aware of treatment allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: Only 2 participants were excluded in the routine care arm for receiving Vit C by error. Small number so unlikely to have introduced bias.
Selective reporting (reporting bias)	Low risk	Judgement comment: Reported outcomes matched protocol information
Other bias	Low risk	Judgement comment: No evidence of other sources of bias

Mohamed 2023

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 4 centres Dates study was conducted: March 2018 and August 2019
Participants	Adults (age 18 years or more, n = 106) with suspected or documented infection who met the criteria for septic shock requiring norepinephrine at a dose of more than or equal to 0.1 µg/kg/min were screened for inclusion. The diagnosis of septic shock was based on persistent hypotension requiring vasopressor therapy to maintain a MAP of more than or equal to 65 mmHg and having a serum lactate level of more than 2 mmol/L, despite adequate volume resuscitation. Setting: intensive care unit Study location: Qatar
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) + thiamine (200 mg intravenous infusion every 12 hours for 4 days) + vitamin C (1500 mg intravenous infusion every 6 hours for 4 days) • Usual care

Outcomes	**Primary outcome**
	• In-hospital mortality evaluated at hospital discharge or at 60 days, whichever came first
	Secondary outcomes
	• Time to death
	• Clinical evidence of organ dysfunction (SOFA scores at 72 h)
	• Length of stay in the intensive care unit
	• Length of stay in the hospital
	• Duration of vasopressor therapy
	• Need for renal replacement therapy for acute kidney injury
	• Need for extracorporeal membrane oxygenation
	• Safety outcomes included
	  ○ Incidence of nephrolithiasis
	  ○ Secondary infections
	  ○ Mechanical ventilator weaning failure
	  ○ Hypernatraemia
	  ○ Hypokalemia
	  ○ Haemolysis
	  ○ Gastrointestinal bleeding

Identification	Sponsorship source: Hamad Medical Corporation Medical Research Centre Comments: no Author's name: Mohamed O Saad Institution: Al Wakra Hospital, Hamad Medical Corporation Email: msaad4@hamad.qa Address: Department of Pharmacy, Al Wakra Hospital, Hamad Medical Corporation, 82228 Doha, Qatar
Notes	
Risk of bias	

Mohamed 2023 *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement comment: Web-based randomisation system - 1:1 ratio with stratification by study site
Allocation concealment (selection bias)	Low risk	Judgement comment: Web-based tool and permuted block randomisation with variable block sizes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: Open-label trial without blinding of participants or study personnel
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: Open-label trial without blinding of outcome assessors
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: ITT performed without any evidence of dropouts or attrition
Selective reporting (reporting bias)	Low risk	Judgement comment: No pre-published protocol is clearly available, but the reported outcomes are typical for the study, and published in sufficient detail and match what was mentioned in the methods.
Other bias	Unclear risk	Judgement comment: The trial was suspended during the COVID pandemic and then terminated early due to lack of funding and possibly futility. A high proportion of patients in the control arm received hydrocortisone (> 65%), possibly diluting treatment effect.

Moskowitz 2020

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 14 Dates the study was conducted: 9 February 2018 to 26 November 2019
Participants	Adults (age 18 years or more, n = 200) with suspected (cultures drawn and antibiotic given) or confirmed (via culture results) infection, receiving (continuous infusion) vasopressor (norepinephrine, phenylephrine, epinephrine, dopamine, angiotensin II, or vasopressin), hypotension related primarily to sepsis as opposed to another cause of hypotension (e.g. bleed, cardiogenic shock) Setting: intensive care unit Study: USA
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) + thiamine (200 mg intravenous infusion every 12 hours for 4 days) + vitamin C (1500 mg intravenous infusion every 6 hours for 4 days) • Placebo (normal saline administered in equal volume and technique to trial drug)
Outcomes	**Primary outcome** • Absolute change in the SOFA score from enrolment to 72 hours after enrolment **Secondary outcomes**

Moskowitz 2020 *(Continued)*

	• 30-day all-cause mortality • All-cause mortality at ICU discharge • All-cause mortality at hospital discharge • Renal failure during the index ICU which is a composite outcome of all-cause death or Kidney Disease Improving Global Outcomes (KDIGO) stage 3 acute renal failure • 72 hours change in each individual component of the SOFA score • Number of ICU-free days in the first 28 days following enrolment • Hospital disposition in patients who survived to hospital discharge • Number of shock-free days over the first 7 days after enrolment • Number of ventilator-free days over the first 7 days after enrolment • Incidence of delirium as measured by the Confusion Assessment Method (CAM)-ICU11 on study day 3
Identification	Sponsorship source: Open Philanthropy (nonprofit foundation) and NIH Author's name: Michael W. Donnino Institution: Harvard Medical School Email: mdonnino@bidmc.harvard.edu Address: Beth Israel Deaconess Medical Center, West Clinical Campus 2nd Floorl, One Deaconess Road, Boston MA 02215
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement comment: Computer-generated
Allocation concealment (selection bias)	Low risk	Judgement comment: Use of an online-based system for randomisation and variable block sizes
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Judgement comment: Double-blind trial with appropriate NS placebo
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement comment: Double-blind trial. Investigators and clinical team investigators were unaware of the allocation. Only a pharmacist at the study site was aware and was independent. There were no reported cases of clinical teams requesting unblinding for drug safety concerns.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: It seems a modified ITT was performed as 2 + 3 participants were excluded for not receiving at least 1 dose of the trial drug or placebo respectively; however, this was balanced and represented very few patients. Otherwise, no significant exclusions or attrition.
Selective reporting (reporting bias)	Low risk	Judgement comment: Notably, ICU and hospital length of stay were identified as a pre-specified outcome but not reported anywhere. However, the impact of this is unclear and likely minimal. All other pre-specified outcomes in the initial published trial protocol were adequately reported in sufficient detail in the parent trial or accompanying supplement.
Other bias	Low risk	Judgement comment: No evidence of other sources of bias

Nafae 2013

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups (3:1) Number of sites: 1 centre Dates the study was conducted: not reported
Participants	Adults (age 18 years or more, n = 80) with severe community-acquired pneumonia Setting: chest department, respiratory, intensive care unit, general medicine department and general medicine intensive care unit Study location: Egypt
Interventions	• Hydrocortisone (200 mg intravenous bolus followed by infusion at 10 mg/h for 7 days) • Placebo (sterile normal saline in same volume and indistinguishable from active treatment)
Outcomes	**Primary outcome** • Not mentioned **Secondary outcomes** • In-hospital mortality • Serious adverse events
Identification	
Notes	We contacted the study authors, who did not reply. Funding source: not declared

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Insufficient information reported
Allocation concealment (selection bias)	Unclear risk	Insufficient information reported
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of placebo: sterile normal saline in a volume equal to the study drug
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of placebo: sterile normal saline in a volume equal to the study drug
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Insufficient information reported

Nafae 2013 *(Continued)*

Selective reporting (reporting bias)	Unclear risk	Insufficient information reported
Other bias	Unclear risk	Insufficient information reported

Nagy 2013

Study characteristics	
Methods	Study design: randomised controlled trial performed Study grouping: 2 parallel groups Number of sites: 1 centre Dates the study was conducted: from June 2007 through September 2009
Participants	Children (n = 59) with severe community-acquired pneumonia Setting: Department of Paediatric Medical Health Study location: Hungary
Interventions	• Intravenous methylprednisolone 20 mg (0.5 to 2.0 mg/kg body weight in different ages) for 5 days • Intravenous 5% dextrose for 5 days
Outcomes	**Primary outcome** • Clinical improvement at 7 days (defined as decrease in coughing and respiratory distress, ≤ 4 days to defervescence, i.e. temperature < 37.58 °C, improved levels of white blood cell count and CRP) **Secondary outcomes** • Length of hospital stay • Complications including worsening of radiological findings such as pleural effusion, abscess formation, empyema and pneumothorax
Identification	
Notes	We did not contact the study authors. Funding source: New Hungary Development Plan, co-financed by the European Social Fund and the European Regional Development Fund

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The sequence of random numbers for parallel groups was computer-generated without block randomisation.
Allocation concealment (selection bias)	Unclear risk	No information is given on how people were blinded to the randomisation list.
Blinding of participants and personnel (performance bias)	Unclear risk	It is stated that they used a placebo, and that this was an open-label, unblinded trial.

Nagy 2013 *(Continued)*

All outcomes		
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	It is stated that they used a placebo, and that this was an open-label, unblinded trial.
Incomplete outcome data (attrition bias) All outcomes	Low risk	None lost to follow-up
Selective reporting (reporting bias)	Low risk	Reported outcomes matched outcomes released on the EudraCT registry.
Other bias	Low risk	There is no evidence of other sources of bias. The trial was completed with inclusion of the planned number of participants.

Ngaosuwan 2018

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 centre Dates the study was conducted: from October 2014 to October 2016
Participants	Adults (age 18 years or more, n = 80) with septic shock defined by the Surviving Sepsis Campaign Guidelines Committee Setting: intensive care unit and intermediate care unit Study location: Thailand
Interventions	• Hydrocortisone (100 mg per day, 24-hour intravenous hydrocortisone sodium succinate, stepwise de-escalated into half of prior dosage, then discontinued after 7-day completion) • Hydrocortisone (200 mg per day, 24-hour intravenous hydrocortisone sodium succinate, stepwise de-escalated into half of prior dosage, then discontinued after 7-day completion)
Outcomes	**Primary outcome** • Occurrence of hyperglycaemia after hydrocortisone initiation (defined as peak blood glucose greater than or equal to 180 mg/dL), which would be assessed every 6 hours up to the 8th day **Secondary outcomes** • 28-day survival • 90-day survival • Increment in blood glucose from baseline to peak blood glucose during 4 separate periods (in the first 3 days, day 4 to 5, day 6 to 7, and day 8) • Time to shock reversal (defined as discontinuation of vasopressors for longer than 24 hours) • Superinfection • Gastrointestinal bleeding
Identification	

Ngaosuwan 2018 *(Continued)*

Notes	We contacted the study authors, who did not provide additional information. Funding source: Srinakharinwirot University

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Randomisation list kept confidential by pharmacist
Blinding of participants and personnel (performance bias) All outcomes	High risk	This was an open-label trial.
Blinding of outcome assessment (detection bias) All outcomes	High risk	This was an open-label trial.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Reported outcomes in publication matched planned outcomes as released on clinicaltrial.gov registry.
Other bias	Unclear risk	The study authors did not reply, and we could not access the full protocol.

Oppert 2005

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 centre Dates the study was conducted: not reported
Participants	Adults (age 18 years or more, n = 40) with vasopressor-dependent septic shock and a cardiac index of ≥ 3.5 L/min/m^2 Subgroups based on adrenal status as assessed by the 250 µg ACTH test Setting: intensive care unit Study location: Germany
Interventions	• Hydrocortisone (50 mg intravenous bolus followed by 0.18 mg/kg/h continuous infusion up to cessation of vasopressor for ≥ 1 hour, reduced to a dose of 0.02 mg/kg/h for 24 hours, then reduced by 0.02 mg/kg/h every day) • Placebo

Oppert 2005 *(Continued)*

Outcomes	**Primary outcome** • Time to cessation of vasopressor support **Secondary outcomes** • Cytokine response • 28-day survival • Sequential organ failure assessment (SOFA) score
Identification	
Notes	We contacted the study authors and obtained details of randomisation and blinding procedures, along with additional information about mortality, for outcomes of participants randomised and not analysed, for shock reversal and for adverse events. Funding source: not declared

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Randomisation list kept confidential by the pharmacist
Blinding of participants and personnel (performance bias) All outcomes	Low risk	The pharmacist masked the study drugs and all other stakeholders remained blinded.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	The pharmacist masked the study drugs and all other stakeholders remained blinded.
Incomplete outcome data (attrition bias) All outcomes	High risk	7 of 48 randomly assigned participants were not analysed: 5 in the corticosteroid group and 2 in the placebo group. 4 of these 7 participants were lost to follow-up, and 3 died (all in the steroid group).
Selective reporting (reporting bias)	Low risk	Access to study protocol, excluding reporting bias
Other bias	Low risk	Full access to data including screening log

Pierre de Oliveira 2023

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 Dates the study was conducted: from 7 July 2020 to March 2021

Pierre de Oliveira 2023 *(Continued)*

Participants	Adults (age 18 years or more, n = 148) with COVID-19 on invasive mechanical ventilation within 24 hours of meeting criteria of moderate or severe acute respiratory distress syndrome (ARDS) defined as the ratio of arterial partial pressure of oxygen to fraction of inspired oxygen (PaO2:FIO2) of 200 or less, with positive RT-PCR test for COVID-19 at the time of inclusion. COVID-19; mechanical ventilation within 24 hours; Acute Respiratory Distress Syndrome Setting: intensive care unit Study location: Brazil
Interventions	• Methylprednisolone (initial dose of 1 mg/kg/day for 5 days, followed by 0.5 mg/kg/day for 3 days and 0.25 mg/kg/day for 2 days) • Placebo (normal saline)
Outcomes	**Primary outcome** • 28-day all-cause mortality **Secondary outcomes** • Length of stay in the intensive care unit • Length of stay in the hospital • In-hospital mortality • Duration of mechanical ventilation • Number of extubations • Number of tracheostomies • Ventilator-associated pneumonia • Bloodstream infections
Identification	Sponsorship source: Instituto de Assistência Médica ao Servidor Público Estadual de São Paulo Comments: no Author's name: Ellen Pierre Oliveira Institution: Instituto de Assistência Médica ao Servidor Público Estadual de São Paulo Email: lenna7772003@yahoo.com Address: R Pedro de Toledo, 1800, São Paulo / Brazil, 04029-000
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Judgement comment: Trial pharmacist performed randomisation but unclear how this was done
Allocation concealment (selection bias)	Low risk	Judgement comment: Only the pharmacist was aware of the allocation concealment
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Judgement comment: Double-blind trial with placebo stated to be indistinguishable from trial drug
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement comment: Presumed blinding of outcome assessors as was run as a double-blind, placebo-controlled trial

Pierre de Oliveira 2023 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: Per text, 3 patients total were excluded after randomisation, and 33 out of the initially screened patients did not consent.
Selective reporting (reporting bias)	Unclear risk	Judgement comment: Unclear if there was selective reporting. No trial protocol available.
Other bias	Unclear risk	Judgement comment: Only available in essentially abstract form on a Brazil-based trial registry.

Ram 2022

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 centre The study was conducted between June 2021 and May 2022
Participants	Adults (age 18 years or more, n = 140) with norepinephrine-dependent septic shock Setting: intensive care unit Study location: India
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours; the duration of treatment was not stated) • Hydrocortisone (200 mg per day as a continuous infusion; the duration of treatment was not stated)
Outcomes	**Primary outcome** • Mean blood glucose (it was not stated when and how this outcome was assessed **Secondary outcomes** • Incidence of hypoglycaemia • Incidence of hyperglycaemia • Mortality rates • Duration of vasopressor therapy • Length of stay in the hospital • Length of stay in the ICU • Glycaemic variability (estimated as the glucose coefficient of variation, standard deviation/mean × 100)
Identification	Country: India
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated

Ram 2022 *(Continued)*

Allocation concealment (selection bias)	Low risk	Use of online system for group allocation
Blinding of participants and personnel (performance bias) All outcomes	High risk	This was an open-label trial.
Blinding of outcome assessment (detection bias) All outcomes	High risk	This was an open-label trial.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	The reported outcomes matched the protocol list.
Other bias	Low risk	No evidence of other sources of bias

Rezk 2013

Study characteristics	
Methods	Study design: randomised controlled trial (2:1 scheme) Study grouping: with 2 parallel groups Number of sites: 1 centre Dates the study was conducted: from October 2011 through October 2012
Participants	Adults (n = 27) with ARDS and hospital- or community-acquired pneumonia Setting: intensive care unit Study location: Egypt
Interventions	• Methylprednisolone (loading dose of 1 mg/kg followed by infusion of 1 mg/kg/d from day 1 to day 14, 0.5 mg/kg/d from day 15 to day 21, 0.25 mg/kg/d from day 22 to day 25, and 0.125 mg/kg/d from day 26 to day 28) • Usual care
Outcomes	**Primary** • Not mentioned **Secondary** • Short-term mortality (time point unclear) • Time on mechanical ventilation • Vital signs • Safety
Identification	

Rezk 2013 *(Continued)*

Notes	We contacted the study authors, who did not provide additional information. Funding source: not declared

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No explicit information in the manuscript
Allocation concealment (selection bias)	Unclear risk	No explicit information in the manuscript
Blinding of participants and personnel (performance bias) All outcomes	High risk	This was an open-label trial.
Blinding of outcome assessment (detection bias) All outcomes	High risk	This was an open-label trial.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Unclear risk	No information
Other bias	Unclear risk	No information

Rinaldi 2006

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 centre Dates the study was conducted: not reported
Participants	Adults (age 18 years or more, n = 40) with severe sepsis and not receiving vasopressor support Setting: intensive care unit Study location: Italy
Interventions	• Hydrocortisone (300 mg per day as a continuous infusion for 6 days, then tapered off) • Usual care
Outcomes	**Primary** • Not explicitly stated **Secondary**

Rinaldi 2006 *(Continued)*

	• Markers of inflammation: microalbuminuria-to-creatinine ratio, serum levels of C-reactive protein and procalcitonin • Duration of mechanical ventilation • Sequential organ failure assessment (SOFA) score • Length of stay in the intensive care unit • In-hospital mortality
Identification	
Notes	We contacted the study authors and obtained details of randomisation and blinding procedures, along with additional information about mortality, for outcomes of participants randomised and not analysed, and for adverse events. Funding source: not declared

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation list
Allocation concealment (selection bias)	Low risk	Sealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	This was an open-label trial.
Blinding of outcome assessment (detection bias) All outcomes	High risk	This was an open-label trial.
Incomplete outcome data (attrition bias) All outcomes	Low risk	12 of 52 participants dropped out of the study: 6 in the control group and 6 in the corticosteroid group; contact with the primary author permitted completion of follow-up for all 12 participants.
Selective reporting (reporting bias)	Low risk	Access to study protocol, excluding any reporting bias
Other bias	Low risk	Full access to data including screening log

Sabry 2011

Study characteristics

Methods	Study design: randomised controlled trial Number of sites: 3 centres Dates the study was conducted: from July 2010 through January 2011
Participants	Adults (age 18 years or more, n = 80) admitted to the ICU with community-acquired pneumonia and sepsis Setting: intensive care unit

Sabry 2011 *(Continued)*

	Study location: Egypt
Interventions	• Hydrocortisone (intravenous loading dose of 200 mg over 30 minutes, followed by 300 mg in 500 mL 0.9% saline at a rate of 12.5 mg/h) for 7 days • Placebo (normal saline)
Outcomes	**Primary outcomes** • Improvement in PaO_2:FiO_2 (PaO_2:FiO_2 > 300 or ≥ 100 increase from study entry) • SOFA score by day 8 • Development of delayed septic shock **Secondary** • Survival by day 8 • ICU mortality rate • Number of patients on mechanical ventilation by day 8 • Time to wean off mechanical ventilation up to day 8 • Number of patients with PaO2:FIO2 ≥ 300 • Mean chest radiograph score • Number of patients with improvement in chest radiograph score from day 1 to 8 • Number of patients with MODS by day 8 • Number of patients with New ARDS by day 8 • Mean CRP by day 8 (mg/dL) (range)
Identification	
Notes	We contacted the study authors, who did not provide additional information. Funding source: authors' institution (public)

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No information in the manuscript
Allocation concealment (selection bias)	Unclear risk	No information in the manuscript
Blinding of participants and personnel (performance bias) All outcomes	Low risk	The pharmacist masked the study drugs and all other stakeholders remained blinded.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	The pharmacist masked the study drugs and all other stakeholders remained blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Unclear risk	No information in the manuscript

Sabry 2011 *(Continued)*

Other bias	Unclear risk	No information in the manuscript

Sai 2021

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: parallel groups Number of sites: 1 Dates study was conducted: N/A
Participants	Adults (age 18 years or more, n = 150) with severe sepsis and septic shock Setting: intensive care unit Study location: India
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) + fludrocortisone (50 µg once a day via a nasogastric tube for 7 days) • Placebo (described as indiscernible from active treatments)
Outcomes	**Primary outcome** • 28-day all-cause mortality **Secondary outcomes** • In-hospital mortality • ICU mortality
Identification	Sponsorship source: unclear Comments: no Author's name: Sai TA Institution: N/A Email: N/A Address: N/A
Notes	Available as an abstract only

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Judgement comment: There is insufficient information to assess how the randomisation sequence was generated.
Allocation concealment (selection bias)	Unclear risk	Judgement comment: There is insufficient information to assess how the randomisation was concealed. It is said that active treatment and placebo were indistinguishable.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Judgement comment: It is said that active treatment and placebo were indistinguishable.

Sai 2021 *(Continued)*

Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement comment: It is said that active treatment and placebo were indistinguishable.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Judgement comment: There is insufficient information to assess this risk of bias.
Selective reporting (reporting bias)	Unclear risk	Judgement comment: There is insufficient information to assess this risk of bias.
Other bias	Unclear risk	Judgement comment: There is insufficient information to assess this risk of bias.

Schumer 1976

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 3 parallel groups Number of sites: 1 centre Dates the study was conducted: from 1967 through 1975
Participants	Adults (n = 172) with septic shock with positive blood culture Setting: surgical department Study location: USA
Interventions	• Dexamethasone (3 mg/kg as a single intravenous bolus) • Methylprednisolone (30 mg/kg as a single intravenous bolus) • Placebo Treatments might have been repeated once after 4 hours and had to be initiated at the time of diagnosis.
Outcomes	**Primary** • Hospital mortality **Secondary** • Complication rates
Identification	
Notes	We did not contact the study authors. Funding source: not declared

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**

Schumer 1976 *(Continued)*

Random sequence generation (selection bias)	High risk	Randomised card system
Allocation concealment (selection bias)	High risk	Unsealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	There is insufficient information about blinding.
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	There is insufficient information about blinding.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Unclear risk	No access to study protocol
Other bias	Unclear risk	No data to exclude selection bias

Sevransky 2021

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 43 The study was conducted between August 2018 and July 2019.
Participants	Adults (age 18 years or more, n = 501) with suspected or confirmed infection as evidenced by ordering of blood cultures and administration of at least one antimicrobial agent, anticipated or confirmed intensive care unit admission, acute respiratory and/or cardiovascular organ dysfunction attributed to sepsis Setting: intensive care unit Study location: USA
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours up to 96 h, death or ICU discharge) + vitamin C (1500 mg every 6 hours up to 96 h, death or ICU discharge) + thiamine (200 mg every 12 hours up to 96 h, death or ICU discharge) • Placebos
Outcomes	**Primary outcome** • Ventilator-and vasopressor-free days (VVFDs) in the first 30 days following randomisation (day 0) **Secondary outcomes** • 30-day all-cause mortality • ICU mortality

Sevransky 2021 *(Continued)*

	• Length of stay in the intensive care unit • Length of stay in the hospital • ICU delirium- and coma-free days • Kidney replacement therapy-free days at day 30 • Change between pre-randomisation and day 4 in Sequential Organ Failure Assessment (SOFA) score (range, 0 (best) to 28 (worst)) • Safety endpoints included: ◦ Prespecified potentially associated adverse events of nephrolithiasis, haemolysis, hypersensitivity reactions and injection site reactions • At discharge or at 30 days, whichever occurred sooner, participants were invited to participate in a 180-day telephone follow-up. For patients who died between discharge and 180 days, date of death was obtained at this follow-up.
Identification	Sponsorship source: Marcus Foundation to Emory University Nova Biomedical Inc provided support to 6 sites SCCM Discovery, a critical care research infrastructure programme, facilitated the design and conduct of the study, provided administrative support to conducting the study, and was involved in critical manuscript revision. Comments: no Author's name: Jonathan E. Sevransky, Institution: Emory University Hospital Email: jsevran@emory.edu Address: Emory University Hospital, Room D226, Atlanta, GA 30322
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement comment: Randomisation done using computer software
Allocation concealment (selection bias)	Low risk	Judgement comment: Computer software used in EDC portal to ensure allocation concealment
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Judgement comment: Participants/personnel were blinded to treatment allocation. Only trial pharmacists were unblinded. Otherwise, kits are sufficiently blinded.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement comment: Outcome assessors were blinded to treatment allocation. Only DSMB members were unblind to treatment allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: No significant exclusions or attrition. ITT performed. The protocol deviations appear balanced and do not appear to affect outcome data assessment or dropout.
Selective reporting (reporting bias)	Low risk	Judgement comment: When comparing trial protocol to reported results, there was no evidence of selective outcome reporting. All pre-specified outcomes were adequately reported.
Other bias	Low risk	Judgement comment: No evidence of other sources of bias.

Slusher 1996

Study characteristics	
Methods	Study design: randomised controlled trial Number of sites: 2 centres Dates the study was conducted: from September 1991 through October 1992
Participants	African children (1 to 16 years of age; n = 72) with sepsis or septic shock Setting: 2 rural hospitals in Africa Study location: USA, Kenya and Nigeria
Interventions	• Dexamethasone (0.20 mg/kg intravenous bolus every 8 hours for 2 days) • Placebo (0.45% sodium chloride solution) Treatments had to be initiated 5 to 10 minutes before the first dose of antibiotics.
Outcomes	**Primary outcome** • Not mentioned **Secondary outcomes** • Survival to hospital discharge • Haemodynamic stability at 48 hours • Length of stay in the hospital • Complication at discharge (unclear)
Identification	
Notes	We did not contact the study authors. Funding source: Roche Pharmaceuticals donated ceftriaxone and provided partial financial support for the study.

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Not given
Allocation concealment (selection bias)	Unclear risk	Unclear; not reported
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	There is insufficient information about blinding.
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	There is insufficient information about blinding.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none

Slusher 1996 *(Continued)*

Selective reporting (reporting bias)	Unclear risk	No access to study protocol
Other bias	Unclear risk	No data to exclude selection bias

Snijders 2010

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 centre Dates the study was conducted: from August 2005 through July 2008
Participants	Adults (n = 213) with severe community-acquired pneumonia Setting: medical wards Study location: The Netherlands
Interventions	• Prednisolone (40 mg intravenously once per day for 7 days) • Placebo
Outcomes	**Primary outcome** • Day 7 and day 30 rates of treatment failure, defined by persistence or progression of all signs and symptoms that developed during an acute disease episode after randomisation, or development of new pulmonary or extrapulmonary respiratory tract infection, or deterioration of chest radiography after randomisation or death due to pneumonia, or inability to complete the study due to adverse events **Secondary outcomes** • Time for clinical stability • Length of hospital stay • 30-day mortality • Inflammatory markers • Safety
Identification	
Notes	We contacted the study authors but did not obtain additional information. Funding source: unrestricted grant by AstraZeneca

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Randomisation list kept confidential by the pharmacist

Snijders 2010 *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	Low risk	Pharmacist masked study drugs and all other stakeholders remained blinded.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Pharmacist masked study drugs and all other stakeholders remained blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	All outcomes reported in the study protocol are reported in the final analysis.
Other bias	Unclear risk	No access to full protocol

Sprung 1984

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 3 parallel groups Number of sites: 2 centres Dates the study was conducted: from August 1979 to February 1982
Participants	Adults (age 18 years or more, n = 59) with vasopressor-dependent septic shock Setting: intensive care unit Study location: USA
Interventions	• Dexamethasone (6 mg/kg as a single intravenous 10- to 15-minute infusion) • Methylprednisolone (30 mg/kg as a single intravenous 10- to 15-minute infusion) • No treatment • Placebo Treatments might have been repeated once after 4 hours if shock persisted, and they had to be initiated at the time of diagnosis.
Outcomes	**Primary** • Hospital mortality • Shock reversal **Secondary** • Complications of septic shock • Safety
Identification	
Notes	We contacted the study authors and obtained additional data (i.e. 28-day all-cause mortality).

Sprung 1984 *(Continued)*

	Funding source: Veterans Administration and Upjohn Company

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	High risk	At one centre, it is not clear how the randomisation list was kept confidential.
Blinding of participants and personnel (performance bias) All outcomes	High risk	On one site, the trial was open-label.
Blinding of outcome assessment (detection bias) All outcomes	High risk	In one site, the trial was open-label.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Unclear risk	No access to study protocol
Other bias	Unclear risk	No data to exclude selection bias

Sprung 2008

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 52 centres The study was conducted from March 2002 to November 2005
Participants	Adults (n = 499) with septic shock Subgroups based on adrenal status as assessed by the 250 µg ACTH test Setting: intensive care unit Study locations: Europe and Israel
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 5 days, then 50 mg every 12 hours for 3 days, then 50 mg once a day for 3 days) • Placebo
Outcomes	**Primary** • 28-day mortality in non-responders

Sprung 2008 *(Continued)*

	Secondary • 28-day mortality in responders and in all participants • ICU mortality rate • Hospital mortality rate • 1-year mortality rate • Shock reversal • Organ system failure-free days • Safety
Identification	
Notes	We contacted the study authors and obtained access to individual patient data. Funding sources: contract (QLK2-CT-2000-00589) from the European Commission, the European Society of Intensive Care Medicine, the European Critical Care Research Network, the International Sepsis Forum, and the Gorham Foundation. Roche Diagnostics provided the Elecsys cortisol immunoassay Data were extracted by EB. The risk of bias was assessed by EB and BR.

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of pre-numbered boxes containing hydrocortisone or its placebo in an indistinguishable form
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of pre-numbered boxes containing hydrocortisone or its placebo in an indistinguishable form
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Lost to follow-up: none; 1 participant withdrew his consent Data for serious adverse events were reported for only 466 of 499 participants, and analysis of these outcomes was performed per-protocol, not by intention-to-treat.
Selective reporting (reporting bias)	Low risk	Access to study protocol and to raw data of individual participants to confirm the absence of reporting bias
Other bias	High risk	Only 500 participants included; the expected sample size was 800 participants.

Sui 2013

Study characteristics	
Methods	Study design: randomised controlled trial

Sui 2013 *(Continued)*

	Study grouping: 2 parallel groups Number of sites: 1 centre Dates the study was conducted: not reported
Participants	Adults (age 18 years or more, n = 60) with severe community-acquired pneumonia Setting: medical wards Study location: China
Interventions	• Methylprednisolone (80 mg intravenously for 3 days, then 16 mg per day for 4 days) • Usual care
Outcomes	**Primary outcome** • Not mentioned **Secondary outcomes** • 7-day clinical efficacy • Time course of C-reactive protein levels in plasma • Hospital length of stay • Reinfection rate • Serious adverse events
Identification	
Notes	We contacted the study authors, who did not reply. Funding source: not declared

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Insufficient information reported
Allocation concealment (selection bias)	Unclear risk	Insufficient information reported
Blinding of participants and personnel (performance bias) All outcomes	High risk	This was an open-label trial.
Blinding of outcome assessment (detection bias) All outcomes	High risk	This was an open-label trial.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Insufficient information reported
Selective reporting (reporting bias)	Unclear risk	Insufficient information reported

Sui 2013 *(Continued)*

Other bias	Unclear risk	Insufficient information reported

Tagaro 2017

Study characteristics	
Methods	Study design: randomised trial performed Study grouping: 2 parallel groups Number of sites: 9 centres The study was conducted from 1 January 2011 to 31 May 2015.
Participants	Children (aged from 1 month to 14 years, n = 60) with community-acquired pneumonia and pleural effusion Setting: intensive care unit Study location: Spain
Interventions	• Intravenous bolus of dexamethasone, 0.25 mg/kg every 6 hours for 48 hours • Intravenous bolus of placebo (0.9% sodium chloride) every 6 hours for 48 hours
Outcomes	**Primary outcome** • Time to recovery, measured in hours. Recovery criteria were ambient oxygen saturation (SaO_2) > 92%, and temperature < 37 °C, no respiratory distress, end of invasive procedures, pneumonia in resolution and oral feeding. Time to recovery was calculated from the first dose of trial treatment until the time when all recovery criteria were fulfilled; for fever, this was the first hour of absence of fever. **Secondary outcomes** • Complications of disease from time of hospitalisation to day 30 after discharge, including all-cause mortality, pneumothorax, necrotising pneumonia and initial uncomplicated effusion eventually needing drainage or surgery • Progression of simple effusion to complicated effusion requiring chest drainage • Decreased CRP level • Decreased effusion during days 1 to 3 • Adverse events including hyperglycaemia (up to study day 3), gastroduodenal bleeding, anaemia, oropharyngeal candidiasis, allergic reaction, rash
Identification	
Notes	Source of funding: Spanish Ministry of Health and Sociedad de Pediatría de Madrid y Castilla La Mancha. Kern Pharma, Inc, Barcelona, Spain, supplied the drugs (dexamethasone and saline) and randomisation. We did not contact the trial authors.

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	A computer-generated randomisation scheme performed by the manufacturer of the study drug (Kern Pharma, Barcelona, Spain)
Allocation concealment (selection bias)	Low risk	Use of pre-packaged boxes that were provided by the manufacturer of the study drug (Kern Pharma, Barcelona, Spain). The boxes were numbered con-

Tagaro 2017 *(Continued)*

		secutively for each hospital, severity stratum and patient, according to the randomisation scheme. They contained 15 transparent ampoules of dexamethasone or placebo, which were indiscernible from each other.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of pre-packaged boxes that were provided by the manufacturer of the study drug (Kern Pharma, Barcelona, Spain). The boxes were numbered consecutively for each hospital, severity stratum and patient, according to the randomisation scheme. They contained 15 transparent ampoules of dexamethasone or placebo, which were indiscernible from each other.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of pre-packaged boxes that were provided by the manufacturer of the study drug (Kern Pharma, Barcelona, Spain). The boxes were numbered consecutively for each hospital, severity stratum and patient, according to the randomisation scheme. They contained 15 transparent ampoules of dexamethasone or placebo, which were indiscernible from each other.
Incomplete outcome data (attrition bias) All outcomes	Low risk	None lost to follow-up
Selective reporting (reporting bias)	Low risk	No difference between planned outcomes as released on ClinicalTrial.gov registry and reported outcomes in the main publication
Other bias	Low risk	No evidence for other bias - trial was completed with planned sample size

Tandan 2005

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 centre Dates the study was conducted: not reported
Participants	Adults (age 18 years or more, n = 28) with septic shock and adrenal insufficiency Setting: intensive care unit Study location: India
Interventions	• Hydrocortisone (stated low dose but actual dose and duration not reported) • Placebo
Outcomes	**Primary** • 28-day mortality or survival until hospital discharge **Secondary** • Shock reversal • Improvement in APACHE II score • Safety
Identification	

Tandan 2005 *(Continued)*

Notes	We contacted the study authors and obtained information on study design but no additional data. Funding source: not declared

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Randomisation list kept confidential by the local pharmacist
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Pharmacist masked study drugs and all other stakeholders remained blinded.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Pharmacist masked study drugs and all other stakeholders remained blinded.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Lost to follow-up: unknown; data reported only as an abstract with minimal information to allow accurate assessment of bias
Selective reporting (reporting bias)	Unclear risk	No access to study protocol; data reported only as an abstract with minimal information to allow accurate assessment of bias
Other bias	Unclear risk	Data reported only as an abstract with minimal information to allow accurate assessment of bias

Tilouche 2019

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 centre The study was conducted from April 2013 through June 2016.
Participants	Adults (n = 70) with septic shock Setting: intensive care unit Study location: Tunisia
Interventions	• Hydrocortisone (200 mg per day by continuous intravenous infusion for 7 days) • Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days)
Outcomes	**Primary** • Shock reversal on day 7 (defined as non need for vasopressors for 24 consecutive hours)

Tilouche 2019 *(Continued)*

	Secondary • 28-day all-cause mortality • Number of days under vasopressors • Vasopressor free-days • ICU and hospital length of stay • Duration of mechanical ventilation • Occurrence of superinfection • New episode of septic shock • Episodes of hyperglycaemia • Dose of insulin administered • Occurrence of side effects of corticosteroids
Identification	
Notes	We contacted the study authors and obtained additional information and unpublished data. Funding source: not declared

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation list
Allocation concealment (selection bias)	Low risk	Opaque, sealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	This was an open-label trial.
Blinding of outcome assessment (detection bias) All outcomes	High risk	This was an open-label trial.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Reported outcomes matched those mentioned in the trial protocol.
Other bias	Low risk	No evidence of other risks of bias

Tomazini 2020

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: parallel groups Number of sites: 41

Tomazini 2020 *(Continued)*

	The study was conducted from 17 April to 23 June 2020.
Participants	Adults (age 18 years or more, n = 299) with probable or confirmed infection by SARS-CoV2 3, intubated and mechanically ventilated, with onset of moderate or severe ARDS according to Berlin criteria, in less than 48 h before randomisation. Setting: intensive care unit Study location: Brazil
Interventions	• Dexamethasone (20 mg intravenous bolus once daily for 5 days, followed by 10 mg intravenously once daily for an additional 5 days or until ICU discharge, whichever occurred first • Usual care

Outcomes	**Primary outcome**
	
	• Ventilator-free days during the first 28 days, defined as the number of days alive and free from mechanical ventilation for at least 48 consecutive hours
	
	Secondary outcomes
	
	• 28-day all-cause mortality
	• Clinical status at day 15 using the 6-point ordinal scale adapted from the World Health Organization R&D Blueprint expert group
	• ICU-free days during the first 28 days
	• Mechanical ventilation duration at 28 days
	• Sequential Organ Failure Assessment (SOFA) scores at 48 hours, 72 hours and 7 days.
	• Post hoc analyses
	  ○ Components of ventilator-free days during the first 28 days
	  ○ Cumulative proportions of the 6-point ordinal scale at 15 days, and the outcome of discharge from hospital alive within 28 days

Identification	Sponsorship source: Sponsor: Hospital Sirio-Libanes Funded and supported by: Coalition COVID-19 Brazil and Laboratórios Farmacêuticos Comments: no Author's name: Luciano C. P. Azevedo Institution: Hospital Sirio-Libanes, Rua Prof Daher Cutait, 69, 01308-060, São Paulo, Brazil Email: luciano.azevedo@hsl.org.br Address: Rua Dona Adma Jafet, 91 - Bela Vista, São Paulo - SP, 01308-050, Brazil
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement comment: Performed via an online web-based system using computer-generated random numbers in blocks of 2 and 4
Allocation concealment (selection bias)	Low risk	Judgement comment: Allocation was performed on a computer system and not disclosed to investigators until all information regarding enrolment was documented
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: There was no blinding of participants and research personnel to assigned treatment.

Tomazini 2020 *(Continued)*

Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: There was no blinding of treatment assignment to individuals who assessed outcomes.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: No significant concerns. No loss to follow-up on primary outcome or mortality.
Selective reporting (reporting bias)	Low risk	Judgement comment: No significant concerns. Pre-specified outcomes in trial protocol are reported in sufficient detail. No major changes to protocol or outcomes identified.
Other bias	Low risk	Judgement comment: The use of steroids in the control arm was allowed for typical ICU indications including bronchospasm and septic shock, thus possibly diluting the treatment effect.

Tongyoo 2016

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 centre The study was conducted from December 2010 through December 2014.
Participants	Adults (age 18 years or more, n = 197) with septic shock and ARDS Setting: intensive care unit Study location: Thailand
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours for 7 days) • Placebo (normal saline)
Outcomes	**Primary outcome** • 28-day all-cause mortality **Secondary outcomes** • Patients alive without organ support (ventilator, renal replacement therapy and vasopressors) on day 28 • MV-free days up to day 28 (free days to day 28 for patients who died before or on study day 28 was set to 0) • Vasopressor (dopamine, norepinephrine or adrenaline)-free days up to day 28 (free days to day 28 for patients who died before or on study day 28 was set to 0) • 60-day mortality • Return to MV after initial successful (off or at least 48 h) extubation was included in the total duration of MV until study day 28
Identification	
Notes	We contacted the study authors and obtained access to individual patient data.

Tongyoo 2016 *(Continued)*

	Funding source: Siriraj critical care research funding

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	A research nurse not otherwise involved in the study prepared both the study drug and placebo. The attending physicians, nursing care teams, research investigators and participants and their family members were blinded to treatment allocation.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	A research nurse not otherwise involved in the study prepared both the study drug and placebo. The attending physicians, nursing care teams, research investigators, and participants and their family members were blinded to treatment allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol, excluding reporting bias
Other bias	Low risk	Full access to data, excluding selection bias

Torres 2015

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 3 centres The study was conducted from June 2004 through February 2012.
Participants	Adults (age 18 years or more, n = 61) with both severe CAP and high inflammatory response, defined as levels of C-reactive protein > 15 mg/dL on admission Setting: intensive care unit Study location: Spain
Interventions	• Methylprednisolone (intravenous bolus of 0.5 mg/kg/12 h for 5 days started within 36 hours of hospital admission) • Placebo (normal saline)
Outcomes	**Primary outcome** • Rate of treatment failure, which includes early and/or late treatment failure. Early treatment failure was defined as clinical deterioration within 72 hours of treatment, as indicated by development of

Torres 2015 *(Continued)*

	shock or need for invasive mechanical ventilation not present at baseline, or death. Late treatment failure was defined as radiographic progression (increase ≥ 50% of pulmonary infiltrates compared with baseline), persistence of severe respiratory failure (PaO_2/FiO_2 < 200, with respiratory rate ≥ 30 min^{-1} in non-intubated participants), development of shock or need for invasive mechanical ventilation not present at baseline or death between 72 and 120 hours after treatment initiation
	
	Secondary
	
	• Time to clinical stability
	• Length of stay in the intensive care unit
	• Length of stay in the hospital
	• In-hospital mortality
	• Inflammatory markers
	• Adverse events during hospital stay included:
	  ○ Hyperglycaemia
	  ○ Superinfection
	  ○ Gastrointestinal bleeding
	  ○ Delirium
	  ○ Acute kidney injury
	  ○ Acute hepatic failure

Identification	
Notes	We contacted the study authors and obtained details about randomisation and blinding procedures, along with additional information about mortality, shock reversal, SOFA, length of stay and adverse events. Funding source: SEPAR, SOCAP, FUCAP, SGR-2011, Fondo de Investigación Sanitaria, IDIBAPS, CIBERES

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Randomisation list kept confidential by the pharmacist
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of pre-numbered boxes containing dosing units with identical appearance for methylprednisolone and placebo. Patients, investigators and data assessors were blinded to treatment allocation.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of pre-numbered boxes containing dosing units with identical appearance for methylprednisolone and placebo. Patients, investigators and data assessors were blinded to treatment allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to full protocol and unpublished information
Other bias	Low risk	Access to full protocol and unpublished information

Valoor 2009

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 centre Dates the study was conducted: not reported
Participants	Children (aged 2 months to 12 years, n = 38) with sepsis or septic shock, unresponsive to fluid therapy alone Setting: intensive care unit Study location: India
Interventions	• Hydrocortisone (intravenous dose of 5 mg/kg per day in 4 divided doses followed by half the dose for a total duration of 7 days) • Placebo (normal saline)
Outcomes	**Primary** • Time to shock reversal **Secondary** • Vasopressor doses • Mortality (unclear time point) • Safety
Identification	
Notes	We did not contact the study authors. Funding source: not declared

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	A computer-generated random number list
Allocation concealment (selection bias)	Low risk	Use of sealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	A faculty member not directly involved with data collection, analysis or interpretation kept the information about study drugs. It is unclear who masked the study drugs and who was blinded to treatment allocation.
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	A faculty member not directly involved with data collection, analysis or interpretation kept the information about study drugs. It is unclear who masked the study drugs and who was blinded to treatment allocation.
Incomplete outcome data (attrition bias)	Low risk	Lost to follow-up: none

Valoor 2009 *(Continued)*

All outcomes		
Selective reporting (reporting bias)	Unclear risk	No information
Other bias	Unclear risk	No information

VASSCSG 1987

Study characteristics	
Methods	Study design: randomised controlled trial Number of sites: 10 centres Dates the study was conducted: not reported
Participants	Adults (age 18 years or more, n = 223) with sepsis or septic shock (n = 100) Setting: intensive care unit Study location: USA
Interventions	• Methylprednisolone (30 mg/kg as a single intravenous 10- to 15-minute infusion, followed by a constant infusion of 5 mg/kg/h for 9 hours) • Placebo Treatment had to be initiated within 2 hours.
Outcomes	**Primary** • 14-day mortality **Secondary** • Complications
Identification	
Notes	We did not contact the study authors. Funding source: Veterans Administration Cooperative Studies Program, Medical Research Service, Veterans Administration Central Office, Washington, DC

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of pre-numbered boxes containing active treatment or its placebo in indistinguishable form.

VASSCSG 1987 *(Continued)*

Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of pre-numbered boxes containing active treatment or its placebo in indistinguishable form.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Unclear risk	No access to study protocol
Other bias	Unclear risk	No data to exclude selection bias

Venkatesh 2018

Study characteristics	
Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 69 centres The study was conducted from March 2013 to April 2017.
Participants	Adults (n = 3800) with vasopressor- and ventilator-dependent septic shock Setting: intensive care unit Study locations: Australia, Denmark, New Zealand, Saudi Arabia, UK
Interventions	• Hydrocortisone (200 mg continuous intravenous infusion over a period of 24 hours for a maximum of 7 days or until ICU discharge or death, whichever occurred first) • Placebo
Outcomes	**Primary** • 90-day all-cause mortality **Secondary** • 28-day all-cause mortality • Time to resolve shock • Recurrence of shock • Frequency and duration of mechanical ventilation • Frequency and duration of treatment with renal replacement therapy • Organ system failure-free days • Length of stay in ICU and at hospital • Safety
Identification	
Notes	We contacted the study authors and did not obtain additional data. Funding source: National Health and Medical Research Council of Australia; Pfizer (which supplied hydrocortisone); Radpharm Scientific (which supplied placebo)

Venkatesh 2018 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Centralised randomisation
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Blinding regarding the trial regimen was ensured by the supply of hydrocortisone and placebo in identical, masked vials. The integrity of the trial-group assignment was confirmed by an independent person who assessed a random sample of hydrocortisone and placebo packs from 10% of the trial population.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Blinding regarding the trial regimen was ensured by the supply of hydrocortisone and placebo in identical, masked vials. The integrity of the trial-group assignment was confirmed by an independent person who assessed a random sample of hydrocortisone and placebo packs from 10% of the trial population.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Low risk	Access to study protocol, excluding reporting bias
Other bias	Low risk	No evidence of other bias

Villar 2020

Study characteristics	
Methods	Study design: randomised Study grouping: 2 parallel groups Number of sites: 17 centres The study was conducted from 28 March 2013 to 31 December 2018.
Participants	Adults (age 18 years or more, n = 277) with ARDS, of whom 67 had sepsis and 118 CAP and septic shock Setting: intensive care unit Study location: Spain
Interventions	• Dexamethasone (20 mg intravenous bolus once daily from day 1 to day 5, which was reduced to 10 mg once daily from day 6 to day 10) • Usual care
Outcomes	**Primary outcome** • Number of ventilator-free days at 28 days after randomisation **Secondary outcomes** • 60-day all cause mortality

Villar 2020 *(Continued)*

	• ICU mortality
	• Hospital mortality
	• Duration of mechanical ventilation in ICU survivors
	• Adverse events
	  ◦ Hyperglycaemia in ICU
	  ◦ New infections in ICU
	  ◦ Barotrauma

Identification	Sponsorship: Fondation Mutua Madrinela & Instituto de Carlos III Contact: Jesus Villar, jesus.villar54@gmail.com; Hospital Universitario DE G.C. DR NEGRIN; Barranco de la Ballena, 35019 Las Palmas de GC, Spain
Notes	Separate data for patients with sepsis or septic shock were provided by the trial primary author.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated
Allocation concealment (selection bias)	Low risk	Concealment through central randomisation and masked treatments
Blinding of participants and personnel (performance bias) All outcomes	High risk	This was an open-label trial.
Blinding of outcome assessment (detection bias) All outcomes	High risk	This was an open-label trial.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No patient lost to follow-up
Selective reporting (reporting bias)	Low risk	Reported outcomes matched protocol
Other bias	Unclear risk	Separate data were obtained from the primary author, but sepsis was not an a priori defined subgroup.

Wani 2020

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: parallel group Number of sites: 1 The study was conducted from April 2018 to June 2019.

Wani 2020 *(Continued)*

Participants	Adults (age 18 years or more, n = 100) with a diagnosis of sepsis and septic shock with a serum lactate level of > 2 mmol/L, based on the Third International Consensus Definitions for Sepsis and Septic Shock Setting: Internal Medicine Unit, ICU and Emergency Medicine Unit Study location: India
Interventions	• Hydrocortisone (50 mg intravenous bolus every 6 hours 7 days) + vitamin C (1500 mg every 6 hours 4 days) + thiamine (200 mg every 12 hours up to 4 days) • Usual care
Outcomes	**Primary outcome** • Not mentioned **Secondary outcomes** • In-hospital mortality • 30-day mortality • Duration of vasopressor use • Lactate clearance • Length of stay in the hospital • Change in serum lactate over the first 4 days • Change in the SOFA score over the first 4 days
Identification	Sponsorship source: The study was supported by a research grant by SKIMS, Soura. This work was supported by the Sheri Kashmir Institute of Medical Sciences. Comments: no Author's name: Parvaiz A. Koul Institution: Sheri Kashmir Institute of Medical Sciences Email: parvaizk@gmail.com Address: Department of Internal and Pulmonary Medicine, Sheri Kashmir Institute of Medical Sciences, Srinagar 190011, J&K, India
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement comment: Computer-generated randomisation
Allocation concealment (selection bias)	Low risk	Judgement comment: Although some details are unclear, likely low given the computer-generated sequence and no details to suggest concealment may have been compromised
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: This was an open-label study.
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: This was an open-label study.
Incomplete outcome data (attrition bias)	Unclear risk	Judgement comment: There are no data available to ascertain whether there was significant attrition/dropouts/incomplete outcome data.

Wani 2020 *(Continued)*

All outcomes		
Selective reporting (reporting bias)	Low risk	Judgement comment: No study protocol identified, but all outcomes stated in methods are presented appropriately in sufficient detail.
Other bias	Unclear risk	Judgement comment: No CONSORT diagram provided; unclear if any dropouts/attrition. Sampling was done over a > 1-year span in an 850-bed hospital, so it is unlikely that all consecutive patients with sepsis were actually enrolled. Thus, one may suspect that the authors did not report the exact number of patients who were not included. It is unclear if a per protocol or ITT approach was performed.

Wittermans 2021

Study characteristics	
Methods	Study design: stratified, randomised, double-blind, placebo-controlled trial Number of sites: 4 centres The study was conducted from 23 December 2012 to 28 November 2018.
Participants	Adults (aged 18 years or more, n = 412) with suspicion of community-acquired pneumonia defined as presence of new opacities on chest radiography, two of the following signs and symptoms: cough, production of sputum, temperature > 38.0 °C or < 36.0 °C, abnormalities at auscultation consistent with pneumonia, C-reactive protein (CRP) > 15 mg/L, white blood cell count > 10×10^9 or < 4×10^9 cells/L, or > 10% of bands in leukocyte differentiation Setting: general wards Study location: the Netherlands
Interventions	• Dexamethasone (6 mg orally once a day for 4 days) • Placebo (tablet orally (Tiofarma)
Outcomes	**Primary outcome** • Length of stay at the hospital, measured in 0.5 days, calculated from time of emergency department presentation to the day of discharge, day of death or day of ICU admission (study medication was stopped after ICU admission because patients are regularly treated with corticosteroids in the ICU) **Secondary outcomes** • Admission to the ICU after initial admission to the general ward • 30-day all-cause mortality
Identification	Contact information: Esther Wittermans e.wittermans@antoniusziekenhuis.nl This work was supported by St Antonius Hospital, St Antonius Research Fund via an earmarked donation from Verwelius Construction Corporation.
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**

Wittermans 2021 *(Continued)*

Random sequence generation (selection bias)	Low risk	Computer-generated list
Allocation concealment (selection bias)	Low risk	Computer-generated, and use of an indiscernable active drug and placebo
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Use of indiscernable active drug and placebo
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Use of indiscernable active drug and placebo
Incomplete outcome data (attrition bias) All outcomes	Low risk	6/209 and 5/203 patients were withdrawn in the experimental and comparator arms, respectively, mainly due to consent withdrawal.
Selective reporting (reporting bias)	Low risk	Reported outcomes matched the trial protocol list
Other bias	Low risk	No evidence of other sources of bias

Yildiz 2002

Study characteristics	
Methods	Study design: randomised controlled trial Number of sites: 1 centre The study was conducted from May 1997 to April 1999.
Participants	Adults (age 18 years or more, n = 40) with sepsis (n = 14), severe sepsis (n = 17) and septic shock (n = 9) Subgroups based on adrenal status as assessed by the 250 µg ACTH test Setting: intensive care unit and infectious disease department Study location: Turkey
Interventions	• Prednisolone (2 intravenous boluses: 5 mg at 06:00 and 2.5 mg at 18:00 for 10 days) • Placebo

Outcomes	**Primary**
	
	• 28-day all-cause mortality
	
	Secondary
	
	• Hospital mortality
	• Safety
	  ◦ Complications of drug therapy
	  ◦ Progression of the initial infection
	  ◦ Superinfections

Yildiz 2002 *(Continued)*

Identification	
Notes	We contacted the study authors and obtained details of randomisation and blinding procedures, along with additional information about mortality, hospital length of stay and adverse events. Funding source: not declared

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation scheme
Allocation concealment (selection bias)	Low risk	Randomisation list kept confidential by the pharmacist
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Pharmacist masked study drugs and all other stakeholders remained blinded.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Pharmacist masked study drugs and all other stakeholders remained blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Unclear risk	No access to protocol
Other bias	Unclear risk	No data to exclude selection bias

Yildiz 2011

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 centre The study was conducted from April 2005 to May 2008.
Participants	Adults (n = 55) with sepsis or septic shock Subgroups based on adrenal status as assessed by the 250 µg ACTH test Setting: intensive care unit and infectious disease department Study location: Turkey
Interventions	• Prednisolone (intravenous 3 times a day at 06:00 (10 mg), 14:00 (5 mg) and 22:00 (5 mg) for 10 days) • Placebo (normal saline)

Yildiz 2011 *(Continued)*

Outcomes	**Primary** • 28-day all-cause mortality **Secondary** • Reversal of organ failure • Length of stay • Safety Outcomes were also assessed in relation to adrenal insufficiency.
Identification	
Notes	We contacted the study authors, who did not reply. Funding source: not declared

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random numbers used
Allocation concealment (selection bias)	Low risk	Randomisation list kept by the pharmacist
Blinding of participants and personnel (performance bias) All outcomes	Low risk	The pharmacists masked the study drugs and all other stakeholders remained blinded.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	The pharmacists masked the study drugs and all other stakeholders remained blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Lost to follow-up: none
Selective reporting (reporting bias)	Unclear risk	No information
Other bias	Unclear risk	No information

Zhou 2015

Study characteristics

Methods	Study design: randomised controlled trial Study grouping: 2 parallel groups Number of sites: 1 centre The study was conducted from January 2008 to January 2013.

Zhou 2015 *(Continued)*

Participants	Adults (age 18 years or more, n = 46) with community-acquired pneumonia and ARDS Setting: intensive care unit Study location: China
Interventions	• Methylprednisolone (120 mg intravenously per day for 7 days) • Usual care
Outcomes	**Primary outcome** • Not mentioned **Secondary outcomes** • Duration of mechanical ventilation • Length of stay in the hospital
Identification	
Notes	We contacted the study authors, who did not reply. Funding source: not declared

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No information
Allocation concealment (selection bias)	Unclear risk	No information
Blinding of participants and personnel (performance bias) All outcomes	High risk	This was an open-label trial.
Blinding of outcome assessment (detection bias) All outcomes	High risk	This was an open-label trial.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No information
Selective reporting (reporting bias)	Unclear risk	No information
Other bias	Unclear risk	No information

ACTH: adrenocorticotropic hormone; APACHE III: acute physiology, age, chronic health evaluation III; AKI: acute kidney injury; ARDS: acute respiratory distress syndrome; ATS: American Thoracic Society; CAP: community-acquired pneumonia; CIRCI: critical illness-related corticosteroid insufficiency; CRP: C-reactive protein; d: days; ECMO: extracorporeal membrane oxygenation; h: hours; HCAP: health care-associated pneumonia; ICU: intensive care unit; IDSA: Infectious Diseases Society of America; IMV: invasive mechanical ventilation; ITT: intention-to-treat; IV: intravenous; K-M: Kaplan-Meier; LTC: long-term care; MAP: mean arterial pressure; MODS: multiple organ dysfunction score; MV: mechanical ventilation; N/A: not applicable; PICU: paediatric intensive care unit; RT-PCR: reverse transcription polymerase chain reaction; SOFA: Sequential Organ Failure Assessment and DSOFA: delta SOFA; VA: Veterans Administration; WHO: World Health Organization

Characteristics of excluded studies *[ordered by study ID]*

Study	Reason for exclusion
ACTRN12621000247875 2021	This trial will compare the combination of hydrocortisone plus vitamin C plus vitamin B1 to hydrocortisone alone in children with sepsis.
Asehnoune 2014	This study has compared hydrocortisone plus fludrocortisone vs placebo in patients with traumatic brain injury and not in sepsis.
Bernard 1987	Mixed population of patients with refractory hypoxia; separate data on patients with sepsis not available.
Cicarelli 2006	Mixed population of critically ill patients; separate data on patients with sepsis not available.
CTRI/2021/11/038268 2021	This trial will compare, in surgical patients with septic shock, the combination of hydrocortisone plus vitamin C plus thiamine to hydrocortisone alone.
Fan 2019	Inappropriate design - after recruiting 18 patients to receive the combination of hydrocortisone plus vitamin C and thiamine or usual care, owing to apparent survival benefits, subsequent patients were all given active treatment.
Fujii 2020	Inappropriate comparator - the trial compared the combination of hydrocortisone plus vitamin C and thiamine versus hydrocortisone alone.
Hahn 1951	Patients with acute streptococcal infection This trial investigated the effects of hydrocortisone on fever, anti-streptolysin titres and onset of rheumatic fever. No data are reported for the analysis of the various outcomes considered in this systematic review.
Huang 2014a	This study has compared in adults with community-acquired pneumonia related to *Mycoplasma pneumoniae* the effects of early (within 12 hours) versus late (within 72 hours) administration of methylprednisolone on the clinical course.
Huang 2015	This study included severely burned patients without sepsis.
Hughes 1984	Only acute effects (within 1 hour) of methylprednisolone and/or naloxone on haemodynamic data were available; no data for any of the outcomes considered in this systematic review were reported.
Hussein 2021	This study has compared the combination of hydrocortisone plus vitamin C plus thiamine versus hydrocortisone alone in septic shock.
Kaufman 2008	In this study, participants were randomly assigned to receive hydrocortisone or its placebo for 24 hours only. Then, treatment with open-labelled hydrocortisone was given at physicians' discretion. This study was aimed at exploring the effects of hydrocortisone on immune cell function.
Khan 2021	This trial included adults with persistent ARDS and there is no information about patients meeting sepsis criteria.
Klastersky 1971	This study was not a randomised trial. Investigators did not describe how participants were allocated to experimental treatment.

Study	Reason for exclusion
Lan 2015	This study focused on methylprednisolone effects on cytokine levels in the bronchoalveolar lavage fluid in children with *Mycoplasma pneumoniae* pneumonia without providing information about outcomes relevant to this review.
Lloyd 2019	This trial has evaluated the effectiveness of a bundle of evidence-supported treatments under conditions of routine care in a representative population hospitalised for community-acquired pneumonia. The bundle included prednisolone acetate, 50 mg/d, for 7 days (in the absence of any contraindication) and de-escalated from parenteral to oral antibiotics according to standardised criteria. Algorithm-guided early mobilisation and malnutrition screening and treatment.
Lucas 1984	This study was not a randomised trial. Participants were allocated to experimental treatment according to their hospital number.
Luo 2014	This study compared prednisolone as an adjunct therapy to antibiotics vs antibiotics alone in children with refractory pneumonia due to *Mycoplasma pneumoniae*. Researchers focused on short-term improvement in clinical symptoms and on variations in serum ferritin and LDH levels. None of the outcomes relevant to this review were assessed in this study.
Malaska 2022	This trial has compared two regimens of dexamethasone therapy, 20 mg versus 6 mg per day, in adults with COVID ARDS.
Marik 1993	This study compared the effects of a short course (single bolus) of high-dose (10 mg/kg) hydrocortisone given before antibiotics to adults with community-acquired pneumonia. The trial focused on treatment effects on circulating levels of tumour necrosis factor and on a short-term clinical course without providing information about outcomes relevant to this review.
McKee 1983	Mixed population of critically ill patients; separate data on septic shock not available.
Meduri 1998b	This trial included participants with late acute respiratory distress syndrome phase - not those with sepsis.
Mikami 2007	This study included participants with community-acquired pneumonia and explicitly excluded patients with sepsis, those needing admission to the intensive care unit and those requiring mechanical ventilation.
Munch 2021	This trial was terminated early after recruiting 30 out of 1000 adults with critical COVID, following external evidence of survival benefit from corticosteroids in severe COVID and WHO recommendation.
NCT02602210 2015	This trial was terminated early due to a low recruitment rate and no data are available.
NCT03335124 2017	This trial was terminated due to a low recruitment rate and no data are available.
NCT03828929 2019	This trial was terminated for futility due to external information, and no data are available.
NCT04134403 2019	This trial has a cross-over design.
Newberry 2017	This study assessed the effects of adjuvant prednisolone treatment in HIV-exposed infants aged 2 to 6 months. The study suggested that corticosteroids significantly reduced mortality at hospital discharge and at 6 months.
Peeters 2018	This trial compared ACTH responses to 100 µg IV CRH and placebo in 3 cohorts of 40 matched patients in the acute (ICU day 3 to 6), subacute (ICU day 7 to 16) or prolonged phase (ICU day 17 to 28) of critical illness, with 20 demographically matched healthy subjects. CRH or placebo was injected in random order on 2 consecutive days.

Study	Reason for exclusion
Raghu 2021	This trial used a quasi-randomised design.
Rogers 1970	Study published only as an abstract; no contact with study authors was possible; incomplete information on primary and secondary outcomes.
Roquilly 2011	This study has assessed the effects of hydrocortisone on the onset of ventilator-associated pneumonia in adults with multiple trauma and not in sepsis.
Schwingshackl 2016	This trial included children within 72 hours of onset of acute respiratory distress syndrome of various aetiologies. Separate data on children with sepsis are not available. The trial focused on methylprednisolone effects on circulating levels of several cytokines without providing information on outcomes that are relevant to this review.
Steinberg 2006	This trial included participants with late acute respiratory distress syndrome phase - not those with sepsis.
Tam 2012	This study assessed the effects of oral prednisolone on viraemia in patients with dengue.
Thompson 1976	Study published only as an abstract; no contact with study authors was possible; incomplete information on primary and secondary outcomes.
van Woensel 2003	This study evaluated a short course (48 hours) of dexamethasone in children with viral pneumonia requiring mechanical ventilation. Viral pneumonia is a condition that differs in many ways from bacterial sepsis.
Venet 2015	This study included severely burned patients and not patients with sepsis.
Wagner 1955	This study was not a randomised trial. Participants were allocated to experimental treatment according to hospital numbers.
Weigelt 1985	Mixed population of critically ill patients; separate data on septic shock not available.
Xi 2021	This trial has investigated the effect of methylprednisolone pretreatment on the incidence of urosepsis post percutaneous nephrolithotomy in patients with high-risk factors.

ACTH: adrenocorticotropic hormone; ARDS: acute respiratory distress syndrome; CRH: corticotropin-releasing hormone; d: day; ICU: intensive care unit; IV: intravenous; LDH: lactate dehydrogenase; WHO: World Health Organization

Characteristics of studies awaiting classification *[ordered by study ID]*

Corral Gudino 2023

Methods	Randomised, open-label, controlled, 2-arm, parallel-group trial 5 centres Dates the study was conducted: from April to July 2020
Participants	Adults (n = 64) hospitalised with a laboratory confirmed diagnosis of SARS-CoV-2 infection, and mild to moderate disease not requiring mechanical ventilation Setting: non-ICU wards Study location: Spain
Interventions	• Intravenous methylprednisolone 40 mg twice a day for 3 days and then 20 mg twice a day for 3 more days

Corral Gudino 2023 *(Continued)*

	• Standard of care based on the Spanish Ministry of Health and WHO recommendations

Outcomes	Primary outcome
	
	• A composite endpoint that included in-hospital all-cause mortality, escalation to ICU admission, or progression of respiratory insufficiency that required non-invasive ventilation
	
	Secondary outcomes
	
	• Effects on the individual components of the composite endpoint
	• Variation of the laboratory biomarkers between baseline and 6 days after inclusion (time window 4 to 8 days)
	• Prespecified adverse events were
	  ◦ Hyperglycaemia (> 180 mg/dL based on the fasting blood glucose test) in the first 6 days after being included in the study
	  ◦ Nosocomial infection (infection confirmed with positive cultures that was not present at the time trial inclusion but occurring within 3 to 28 days after the first dose of methylprednisolone)

Notes	

Franco-Moreno 2024

Methods	Randomised, controlled, open-label, parallel-group trial 3 centres Dates the study was conducted: June 2021 and January 2022
Participants	Adults (n = 126) hospitalised with confirmed COVID-19 based on a positive RT-PCR test or rapid antigen test for SARS-CoV-2 RNA in a respiratory specimen (nasopharyngeal or nasal swab), a chest imaging study compatible with pneumonia, a SpO2 ≥ 94% and < 22 bpm breathing on room air, and the presence of at least 2 of the following inflammatory biomarkers: LDH > 245 U/L, CRP > 100 mg/L, or lymphocyte count < 0.80 × 109/L Setting: non-ICU wards Study location: Spain
Interventions	• 6 mg of dexamethasone once daily for 7 days. Participants unable to take oral drugs received intravenous dexamethasone • Standard of care
Outcomes	Primary outcome • A composite of moderate or more severe ARDS development and all-cause 30-day mortality after enrolment Secondary outcomes • Requirement for non-invasive or invasive mechanical ventilation • Hospital length of stay • 90-day all-cause mortality
Notes	

Laikitmongkhon 2024

Methods	Randomised, controlled, open-label trial 2 centres Dates the study was conducted: 18 October 2021 and 31 October 2022
Participants	Adults (n = 50) hospitalised with moderate COVID-19 pneumonia within 48 h with a resting oxygen saturation of 90% to 94% and not requiring mechanical ventilation Setting: non-ICU wards Study location: Thailand
Interventions	• Intravenous 250 mg of methylprednisolone daily for 3 days, followed by oral or intravenous 20 mg of dexamethasone daily on days 4 and 5, followed by 10 mg of dexamethasone daily on days 6 to 10 • Intravenous 20 mg of dexamethasone daily for three days, followed by oral or intravenous 20 mg of dexamethasone daily on days 4 and 5, followed by 10 mg of dexamethasone daily on days 6 to 10
Outcomes	Primary outcome • WHO clinical progression scales on day 5 Secondary outcomes • Difference in the WHO clinical progression scales on day 10 • Oxygenation improvement • Transfer to the ICU • Respiratory failure • Length of hospitalisation • Mortality rate • Inflammatory markers • SOFA score
Notes	

Mohanty 2024

Methods	Randomised, placebo-controlled study performed on 2 parallel groups 1 centre Dates study was conducted: not provided
Participants	Adults (n = 46) hospitalised with sepsis and adrenal insufficiency Setting: ICU Study location: India
Interventions	• Intravenous 50 mg of hydrocortisone every 6 hours for 7 days • Placebo
Outcomes	Primary outcome • Unclear

Mohanty 2024 *(Continued)*

	Secondary outcomes • Development of septic shock • Effect on total WBC count • Survival at 28 days/death
Notes	

Sahraei 2023

Methods	Randomised, controlled, parallel-group, open-label trial 1 centre Dates the study was conducted: from April 2021 to
Participants	Adults (n = 95) hospitalised for mild to moderate COVID-19 ARDS not requiring invasive mechanical ventilation and without shock Setting: referral centre for COVID-19 patients Study location: Iran
Interventions	• Intravenous methylprednisolone at 1 mg/kg/12 hours daily for 10 days • A 3-day pulse regimen of methylprednisolone at 1000 mg followed by methylprednisolone at 1 mg/kg/12 hours daily for 10 days
Outcomes	Primary outcome • Unclear Secondary outcomes • Length of hospital stay • Rate of invasive ventilation requirement • Need for haemoperfusion • Need for immunological intervention • Mortality
Notes	

Salhotra 2024

Methods	Randomised, controlled, double-blind trial 1 centre Dates the study was conducted: January 2021 and July 2022
Participants	Adults (n = 40) hospitalised with septic shock as per the Sepsis 3 definition Setting: ICU Study location: India

Salhotra 2024 *(Continued)*

Interventions	• Continuous intravenous infusion of hydrocortisone (200 mg/day diluted in 50 mL normal saline) plus boluses of normal saline every 6 hours for 5 days • Boluses of hydrocortisone (200 mg/day administered every 6 hours as 50 mg IV diluted to 5 mL) plus continuous intravenous infusion of normal saline for 5 days
Outcomes	Primary outcome • Blood glucose control (average blood glucose over the study period Secondary outcomes • Number of hyperglycaemic and hypoglycaemic episodes • Average daily insulin requirement over the period of steroid therapy and number of insulin adjustments/day • Shock reversal (stable mean arterial pressure of > 65 mmHg for at least 24 hours without norepinephrine support) • Nursing workload (total number of dosage changes in insulin infusion during the study period) • SOFA score, APACHE II score, SAPS II score • Sodium levels • Lactate levels • P/F ratio at the time of ICU admission and on all the subsequent days until 5 days of steroid therapy • Length of ICU stay • In-hospital mortality
Notes	

Salton 2023

Methods	Randomised, controlled, open-label trial 26 centres Study dates: 14 April 2021 to 4 May 2022
Participants	Adults (n = 677) hospitalised with real-time PCR-positive for SARS-CoV-2 on at least one upper respiratory swab or bronchoalveolar lavage and respiratory failure ($PaO2 \leq 60$ mmHg or $SpO2 \leq 90\%$, or on high-flow nasal cannula, CPAP or non-invasive ventilation Setting: internal medicine units, infectious diseases units, emergency medicine departments and respiratory high dependency units Study location: Italy
Interventions	• Methylprednisolone on day 1, a loading dose of 80 mg was administered intravenously for 30 minutes, followed by a continuous daily infusion of 80 mg in 240 mL normal saline at 10 mL per hour for 8 days, then tapered over 14 days • Dexamethasone 6 mg intravenously in 30 minutes or orally, from day 1 to day 10 or until hospital discharge (if sooner)
Outcomes	Primary outcome • Mortality proportion on day 28 Secondary outcomes • Number of days free from mechanical ventilation (either NIV or IMV) by study day 28 • Proportion of patients requiring admission to ICU

Salton 2023 *(Continued)*

	• Number of days of hospitalisation amongst survivors • Proportion of patients requiring tracheostomy • C-reactive protein level (CRP; mg/L) on study days 3, 7 and 14 • PaO2/FIO2 ratio (mmHg) on study days 3, 7 and 14 • World Health Organization (WHO) Clinical Progression Scale on study days 3, 7 and 14
Notes	

Schlapbach 2024

Methods	Investigator-initiated pragmatic, multicentre, randomised, open-label, parallel-group pilot trial 6 centres
Participants	Children of age between 28 days and 18 years (n = 66) requiring vasoactive drugs for septic shock Setting: paediatric ICU Study location: Australia, New Zealand
Interventions	• Intravenous 1 mg/kg (maximum 50 mg per dose) hydrocortisone every 6 hours, 30 mg/kg ascorbic acid (maximum 1500 mg per dose as sodium ascorbate) every 6 hours, and 4 mg/kg (maximum 200 mg per dose) thiamin every 12 hours • Usual care

Outcomes	Primary feasibility outcomes
	
	• Recruitment rates
	• Time to initiate the intervention
	• Proportion of patients receiving hydrocortisone outside the study protocol
	• Adherence to the study protocol
	
	Primary efficacy outcome
	
	• Survival free of organ dysfunction censored on day 28 after randomisation
	• Secondary efficacy outcomes
	• Survival free of inotrope support for 7 days
	• Survival free of multiple organ dysfunction in 7 days
	• Mortality
	• PICU-free survival
	• Survival free of acute kidney injury
	• PICU and hospital length of stay
	• Paediatric Overall Performance Category (POPC) and Functional Status Score (FSS) at 28 days using phone or face-to-face interviews
	• Proxy measures of intervention efficacy
	  ○ Shock reversal
	  ○ Normalisation of lactate (< 2 mmol/L)
	  ○ Time to reversal of tachycardia (defined by age-specific thresholds for Systemic Inflammatory Response Syndrome

Notes	

Sharma 2024

Methods	Randomised controlled trial performed on 2 parallel groups 1 centre Dates the study was conducted: April 2020 till May 2021
Participants	Adults (n = 86) hospitalised with septic shock according to the sepsis-3 definition requiring vasopressors and duration of septic shock not more than 24 hours Setting: ICU Study location: Nepal
Interventions	• Intravenous vitamin C (1.5 g every 6 hours for a maximum of 4 days or until ICU discharge or death) plus intravenous hydrocortisone (50 mg every 6 hours for a maximum of 4 days or until ICU discharge or death) and intravenous thiamine (100 mg every 6 hours for a maximum of 4 days or until ICU discharge or death) • Intravenous hydrocortisone (50 mg IV every 6 hours until vasopressors were stopped, the patient was shifted out of the ICU or died while still on a vasopressor
Outcomes	Primary outcome • ICU mortality Secondary outcomes • Duration free of vasopressor administration on day 7 • Total vasopressor requirement on day 7 • Length of ICU stay • Change in SOFA score after 3 days (72 hours) • Requirements for RRT during ICU stay and duration of mechanical ventilation in the intervention and control groups
Notes	

Taher 2023

Methods	Randomised, double-blind, 3-arm trial 1 centre Dates the study was conducted: from August 2020 to February 2022
Participants	Adults aged 18 to 75 years (n = 101) hospitalised with diagnosis of COVID-19 infection confirmed by positive nasopharyngeal PCR testing (RT-PCR) for SARS-CoV-2 infection on nasopharyngeal swab specimens and COVID-19 severe pneumonia and mild to moderate ARDS Setting: non-ICU wards Study location: Iran
Interventions	• Intravenous dexamethasone at a dose of 6 mg once a day for 10 days • Intravenous methylprednisolone at a dose of 16 mg twice a day for 10 days • Intravenous hydrocortisone at a dose of 50 mg thrice a day for 10 days
Outcomes	Primary outcomes • All-cause 28-day mortality

Taher 2023 *(Continued)*

	• Clinical status on day 28 assessed using the WHO 8-category ordinal clinical scale
	• Recovered patients, defined as patients who achieved a clinical status ≤ 3 on the WHO 8-category ordinal clinical scale on day 28
	• Change in PaO2:FiO2 ratios during the first 5 days
	
	Secondary outcomes
	
	• Number of patients needing ICU care
	• Number of patients needing invasive mechanical ventilation
	• Number of ventilator-free days within 28 days
	• Duration of the ICU stay
	• Duration of hospitalisation
	• Incidence of serious adverse events
	  ◦ Secondary infections
	  ◦ Hyperglycaemia
	  ◦ Clinically important gastrointestinal bleeding
	  ◦ Hypertension

Notes	

Walsham 2024

Methods	Investigator-initiated, multi-centre, open-label, randomised, phase II clinical trial 9 centres Dates the study was conducted: from April 2021 and April 2023
Participants	Adults (n = 153) hospitalised with documented or strongly suspected infection; at least 2 of the 4 Systemic Inflammatory Response Syndrome criteria and treated with both ventilatory and vasopressor support and treatment with adjunctive hydrocortisone at a dose of 200 mg/day for the management of septic shock Setting: ICU Study location: Australia and New Zealand
Interventions	• Intravenous hydrocortisone 200 mg per day continuously • Intravenous hydrocortisone 200 mg per day continuously plus enteral fludrocortisone 50 µg once daily, either orally or via a gastric tube • Intravenous hydrocortisone 200 mg per day continuously plus enteral fludrocortisone 50 µg twice daily (100 µg/day) either orally or via a gastric tube • Intravenous hydrocortison 200 mg per day continuously plus enteral fludrocortisone 50 µg 4 times daily (200 µg/day) either orally or via a gastric tube Treatment was continued for a maximum of 7 days or until intensive care unit (ICU) discharge, cessation of hydrocortisone by the clinician or death, whichever occurred first.
Outcomes	Primary outcome • Time to resolution of shock, defined as the time from randomisation to the attainment of a clinician-prescribed mean arterial pressure target for more than 24-h without the use of vasopressors or inotropes Secondary outcomes

Walsham 2024 *(Continued)*

	• Recurrence of shock (defined as a new requirement for vasopressors or inotropes following resolution of shock) • Ventilator-free days • Maximum sequential organ failure assessment (SOFA) score (SOFA max) • Change in SOFA score (SOFA delta) • Length of intensive care and hospital stay • Death from any cause in the ICU or hospital
Notes	

Wang 2023

Methods	Randomised, double-blind, controlled trial performed on 2 parallel groups 1 centre Dates the study was performed: from 1 February 2019 to 1 January 2020
Participants	Adults aged from 18 to 80 years (n = 27) hospitalised with septic shock with norepinephrine dose of 15 µg/minutes or more Setting: ICU Study location: China
Interventions	• Hydrocortisone (200 mg) was continuously pumped intravenously for 24 hours plus vitamin C (1.5 g) was dissolved in 100 mL of normal saline and infused intravenously. The infusion time was 30 to 60 min, and the interval was 6 hours; 4 doses were used. Vitamin B1 (200 mg) was dissolved in 50 mL of normal saline and infused intravenously. The infusion time was 30 to 60 min, and the interval was 12 hours; 2 doses were used. • Hydrocortisone (200 mg) was continuously pumped intravenously for 24 hours.
Outcomes	Primary outcome • The perfused small vessel density (sPVD) at 24 hours Secondary outcomes • Proportion of perfused small vessels (sPPV), small vessel density (sTVD) and microvascular flow index (MFI) • Renal perfusion (renal contrast-enhanced ultrasound) at 24 hours • Peak intensity (PI), regional blood flow (RBF), time to peak (TTP), mean transit time (MTT) • Lactate level at 24 hours • Norepinephrine dose • ICU stay • Total hospital stay • 28-day mortality
Notes	

APACHE: acute physiology, age, chronic health evaluation; ARDS: acute respiratory distress syndrome; CPAP: continuous positive airway pressure; CRP: C-reactive protein; IMV: invasive mechanical ventilation; ICU: intensive care unit; IV: intravenous; LDH: lactate dehydrogenase; NIV: non-invasive ventilation; PCR: polymerase chain reaction; P/F: ratio of the arterial oxygen tension to the fraction of inspired oxygen; RNA: ribonucleic acid; RRT: renal replacement therapy; RT-PCR: reverse transcription polymerase chain reaction; SAPS 2: simplified acute physiology score 2; SOFA: Sequential Organ Failure Assessment; WBC: white blood cell; WHO: World Health Organization

Characteristics of ongoing studies *[ordered by study ID]*

ChiCTR2000029386 2020

Study name	Effectiveness of glucocorticoid therapy in patients with severe novel coronavirus pneumonia: a randomized controlled trial
Methods	Single-centre, randomised, open-label trial in two parallel groups
Participants	n = 48 **Inclusion criteria** 1. Male or female over 18 years of age 2. Novel coronavirus infection is confirmed by pathogenic detection 3. The diagnosis of severe coronavirus pneumonia will have to meet at least one of the following criteria: (1) Respiratory distress, RR > 30 times/minute; (2) In the state of no oxygen at rest, the patient's SpO2 ≤ 93%; (3) Oxygenation Index (PaO2/FiO2) ≤ 300 mmHg (1 mmHg = 0.133 kPa); (4) Respiratory failure requiring mechanical ventilation; (5) Sepsis; (6) Other organ failure requiring ICU care 4. Be willing to give informed consent **Exclusion criteria** 1. Patients are allergic or intolerant to therapeutic drugs 2. Pregnancy or lactating women 3. Researchers believe that patients may have other factors affecting the efficacy or safety evaluation of this study
Interventions	**Experimental arm** Methylprednisolone, intravenous injection, 1 to 2 mg/kg/d for 3 days **Comparator** Usual care
Outcomes	**Primary outcome measure** SOFA score, no mention of timeframe **Secondary outcomes measures** Mortality Length of hospital stay Need for mechanical ventilation
Starting date	29 January 2020
Contact information	Yaokai Chen yaokaichen@hotmail.com Chongqing Public Health Medical Center
Notes	

ChiCTR2000035902 2020

Study name	Clinical study on glucocorticoid therapy for septic shock in children
Methods	—
Participants	**Inclusion criteria** 1. Aged 29 days to 18 years 2. Meet the diagnostic criteria for septic shock: 2005 International Diagnostic Standard for children with septic shock 3. Patients with PICU hospitalisation time less than 7 days; 4. Patients diagnosed with septic shock within 24 hours
Interventions	**Experimental arm 1** Hydrocortisone 5 mg/kg/d, divided into 2 static points for 3 to 5 days in total **Experimental arm 2** Methylprednisolone 1 mg/kg/d, divided into 2 static points for 3 to 5 days in total **Comparator** Placebo (glucose)
Outcomes	**Primary outcome measure** All-cause mortality **Secondary outcomes measures** All-cause mortality Shock recovery time Days alive and out of the PICU Days alive and out of the hospital CRRT-free days to day 28 Ventilator-free days to day 28 Vasopressor-free days to day 28 Organ failure-free days to day 28
Starting date	
Contact information	
Notes	

CTRI/2021/05/033523 2021

Study name	A comparative study of high dose corticosteroids pulse versus standard of care in severe COVID19 pneumonia
Methods	Multicenter, randomised, open-label trial in 2 parallel groups

CTRI/2021/05/033523 2021 *(Continued)*

Participants	n = 200 **Inclusion criteria** COVID-19 confirmed case (with nasal or oral swab RT-PCR or rapid antigen positive) Chest radiograph suggestive of pneumonia Severity scale according to WHO ordinal scale score of either 4 or 5 or 6 (oxygen requirement or HFNC requirement or NIV requirement or invasive mechanical ventilation) Able to and willing to provide informed consent **Exclusion criteria** Patients belonging to category 7 (requiring ECMO or vasopressor support) Pregnant subjects Patients participating in other clinical trials Known case of any psychiatric illness Treating physician assessed high likelihood of mortality in 72 hours of randomisation Subjects who have received steroid dose of over 0.4 mg/kg/day of dexamethasone or 2 mg/kg/day of methylprednisolone for over 2 days in the past 14 days Co-existing condition mandating the use of high-dose steroids Life-threatening comorbidities like CKD requiring dialysis Life-threatening new onset arrhythmia Hypotension Evidence or suspicion of active bacterial infection Evidence or suspicion of acute onset renal or liver failure Child C category of chronic liver disease Co-existing autoimmune disease requiring immunosuppressive drugs Concurrent use of tocilizumab/baricitinib/itolizumab
Interventions	**Experimental arm** Methylprednisolone, no mention of dose and duration **Comparator** Usual care
Outcomes	Primary outcome measure Time to recovery, where recovery is defined as a fall of 2 points on the WHO 8-point ordinal scale or liberation from oxygen therapy Timepoints: 0, 3rd, 7th, 14th, 28th day
Starting date	10 May 2021
Contact information	Dhruva Chaudhry dhruvachaudhry@yahoo.co.in

CTRI/2021/05/033523 2021 *(Continued)*

Notes	

CTRI/2021/09/036799 2021

Study name	To see the effect of corticosteroids along with standard treatment in pneumonia
Methods	Randomised controlled trial in 2 parallel groups
Participants	**Inclusion criteria** In children aged 2 months to 15 years with severe community-acquired pneumonia All cases of severe pneumonia will be diagnosed by the following criteria: 1. 2 months to 59 months: WHO criteria for severe pneumonia will be used 2. 5 to 15 years: child needs to have tachypnoea RR > 20/min, fever > 38 °C, SPo2 < 94% and radiological changes in CXR to be included as severe pneumonia in the study
Interventions	**Experimental arm** IV dexamethasone at a dose of 0.4 mg/kg IV 12 hourly, starting within 2 hours of antibiotic therapy, and continuing for a total duration of 72 hours **Comparator** Placebo
Outcomes	**Primary outcome measure** Clinical failure in the first 72 hours **Secondary outcomes measures** 1. Mortality 2. Duration of the hospital stay 3. Need for mechanical ventilation 4. Readmission rate within 1 month 5. Complications of pneumonia between 2 groups 6. Severity of inflammation by assessing (ESR, hsCRP, IL-6,IL 8, IL 10, procalcitonin) at time of admission and day 4 (after stopping the steroid therapy) or at the time of discharge
Starting date	Unknown
Contact information	
Notes	

CTRI/2022/02/040039 2022

Study name	Effectiveness study of steroid in severe community acquired pneumonia
Methods	Randomised trial, open-label, in 2 parallel groups
Participants	**Inclusion criteria**

CTRI/2022/02/040039 2022 *(Continued)*

	All consecutive children aged 1 month to 60 months admitted to PICU Severe pneumonia requiring oxygen therapy Informed consent from parents
Interventions	**Experimental arm** Intravenous dexamethasone at a dose of: day 1: 0.6 mg/kg, day 2: 0.6 mg/kg; day 3: 0.3 mg/kg; day 4: 0.15 mg/kg **Comparator** Usual care
Outcomes	**Primary outcome measure** Duration of oxygen therapy (in hours) during the first 5 days of admission **Secondary outcome measures** 1. Rise in secondary infections 2. Hyperglycaemia 3. Gastrointestinal bleeding 4. Hypertension
Starting date	Unknown
Contact information	Unknown
Notes	

CTRI/2022/12/048583 2022

Study name	Corticosteroid effectiveness in treating severe community acquired pneumonia
Methods	Randomised, placebo-controlled, double-blind trial in 2 parallel groups
Participants	**Inclusion criteria** 1. Age more than 18 years 2. Clinical symptoms suggesting community-acquired pneumonia (cough, fever, pleuritic chest pain and/or shortness of breath) 3. Severe community-acquired pneumonia as per the criteria ATS IDSA 2007 4. Informed consent
Interventions	**Experimental arm** Intravenous methylprednisolone at a dose of 0.5 mg/kg every 12 hours for 5 days Comparator Placebo: 0.9%/L sodium solution
Outcomes	**Primary outcome measure** Reduction in the rates of treatment failure (early, late and/or both) at day 0, 3, 6 Secondary outcomes measures

CTRI/2022/12/048583 2022 *(Continued)*

	Attainment of clinical stability at day 0, 3, 6 Length of stay in hospital Mortality at hospital and at 28 days post discharge
Starting date	Unknown
Contact information	Unknown
Notes	

IRCT20220219054058N1 2022

Study name	The effect of hydrocortisone, vitamin C and thiamine in treatment of patients with severe sepsis/septic shock
Methods	Randomised, open-label trial in two parallel groups
Participants	**Inclusion criteria** Age 18 or higher **Exclusion criteria** COVID-19 Pregnancy G6PD deficiency Initial diagnosis of: stroke, acute coronary syndrome, active gastrointestinal bleeding, burns and trauma Use of vasopressors more than 24 hours before entering the study
Interventions	**Experimental arm** Hydrocortisone 50 mg every 6 hours intravenously (Jaber Bin Hayan Pharmaceutical Company, Iran) Thiamine 200 mg in 50 mL of normal saline every 12 hours intravenously (Pouyan daroo, Iran) Vitamin C 1500 mg every 6 hours in 100 mL of normal saline intravenously (Osweh Pharmaceutical Company, Iran) For 3 days **Comparator** Usual care
Outcomes	**Primary outcome measure** Mortality rate **Secondary outcomes measures** Duration of intubation Duration of stay in the intensive care unit up to 7 days

IRCT20220219054058N1 2022 *(Continued)*

	Duration of hospital stay Duration of vasopressor up to 7 days Mortality at the end of 7 days Intubation period up to 7 days Mean daily disease intensity up to 3 days Mean CRP level on the first, third and fifth day Mean level lactate on the first, third and fifth day
Starting date	Unknown
Contact information	Unknown
Notes	

ISRCTN15076735 2023

Study name	GuARDS Trial
Methods	Randomised, parallel-group, allocation-concealed, open-label, pragmatic group-sequential-design clinical and cost-effectiveness study with internal pilot
Participants	n = 1708 **Inclusion criteria** 1. Provision of informed consent 2. Aged 16 years or older 3. Admitted to intensive care unit or high dependency unit (ICU) 4. Receiving respiratory support via invasive mechanical ventilation or non-invasive ventilatory support (non-invasive ventilatory support includes mask or helmet) or high flow nasal cannula (HFNC) > 30 L/min 5. Within 72 hours of diagnosis of ARDS with moderate to severe hypoxaemia defined as 6. Known acute clinical insult or new or worsening respiratory dysfunction (Note: this includes new deterioration at any time-point during the ICU stay), and 6.1. Opacities on chest imaging not fully explained by effusions, lobar/lung collapse/atelectasis, or nodules, and 6.2. Respiratory failure not fully explained by cardiac failure or fluid overload, and 6.3. Assessment of hypoxaemia done with either PaO2/FiO2 ratio < 26.7 kPa from arterial blood gases, or SpO2/FiO2 < 235 with SaO2 < 97% Lower age limit: 16 years Sex: both **Exclusion criteria** 1. ARDS due to microbiologically confirmed SARS-Co-V2 infection (COVID-19 ARDS)

ISRCTN15076735 2023 *(Continued)*

	2. Major upper gastrointestinal bleeding during current hospital admission, defined as requiring endoscopy and transfusion for two or more units of packed red blood cells. This exclusion criterion will exclude patients with contraindications to glucocorticoids on safety grounds. 3. High-dose glucocorticoids are required for a separate proven clinical indication at the time of randomisation as withholding treatments that have been deemed clinically effective would be unethical. Note: Low-dose glucocorticoid treatment for clinical indications (defined as maximum daily dose of 200 mg hydrocortisone or equivalent other steroids) is not an exclusion criterion. 4. Known hypersensitivity to dexamethasone 5. Infections that are not being effectively treated as determined by the treating medical team. Note: Once infections are considered as effectively treated by the treating medical team, they are eligible for the trial. 6. Planned intensive care treatment withdrawal within next 24 hours as determined by the treating medical team 7. Patients who are known to be pregnant 8. Previous enrolment in the GuARDS trial
Interventions	**Experimental arm** On days 1 to 5, 20 mg/day of intravenous (IV) dexamethasone will be administered to patients. On days 6 to 10, the IV dexamethasone will drop to 10 mg/day. The intervention is a 10-day treatment regime whilst patients are in ICU. **Comparator** Usual care
Outcomes	**Primary outcome measure** All-cause mortality measured using medical records at 60 days from randomisation **Secondary outcome measures** Our proposed secondary outcomes (endpoints) are from the CoVENT core outcome set (COS) for ventilation trials and are measured using medical records as and when they occur: 1. First successful extubation, defined as the time from randomisation until the first successful extubation or the patient's death occurs, measured using a record of the date/time of all periods of ventilation up to day 60. Successful extubation is being free from all tubes, endotracheal tube, and tracheostomy with success being defined as remaining free from tubes at 48 hours. If discharged from the hospital before the 48-hour success period, successful extubation is assumed. 2. Duration of mechanical ventilation - Unassisted breathing, defined as no inspiratory support (includes time receiving invasive mechanical ventilation and non-invasive ventilation) or extracorporeal lung support. Success is defined as remaining to breathe unassisted at 48 hours. Defined as the time from randomisation until the first successful unassisted breathing or the patient's death occurs. Death prior to the end of mechanical ventilation or within the 48-hour period after the end of mechanical ventilation is considered censored. 3. Reintubation - All reintubation events with date/time to report the total number of re-intubations after planned extubation in each group and the average number of reintubation events/participants in each group. The COS recommends reintubation rates at 60 days. To capture the risk of delayed re-intubation we will collect this outcome for up to 180 days from randomisation, censored at hospital discharge.

ISRCTN15076735 2023 *(Continued)*

	4. Duration of ICU and hospital stay at the time from randomisation until participant first leaves the relevant facility or death. 5. Health-related quality of life (HRQoL) measured using the EQ-5D (www.euroqol.org) and is participant reported at 60 and 180 days post-randomisation. 6. Mortality - will record the event date/time of the event, as well as the date/time of randomisation to enable a survival analysis at 90 and 180 days post-randomisation. 7. Health service use since hospital discharge - will be a telephone questionnaire completed with a research nurse at 90 and 180 days. It will include questions on rehospitalisations and health service usage. Data will also be collected via data linkage.
Starting date	13 June 2023
Contact information	Dr Kay Russell Dr Manu Shankar-Hari Level 2, IRR South, Edinburgh BioQuarter, 4-5 Little France Drive Edinburgh, EH16 4UU, United Kingdom +44 (0)131 651 8167 guards@ed.ac.uk
Notes	

NCT03592693 2018

Study name	Vitamin C, hydrocortisone, and thiamine for septic shock (CORVICTES)
Methods	Randomised, multicentre, parallel-group, placebo-controlled
Participants	Adults (n = 400) with septic shock
Interventions	• 1500 mg vitamin C every 6 hours for 4 days after randomisation and stress-dose hydrocortisone for 4 days (250 mg on day 1; and 200 mg on days 2, 3 and 4) after randomisation • Placebo
Outcomes	**Primary** • Hospital mortality **Secondary** • 60-dday mortality (timeframe: 60 days): death before day 60 post randomisation • 28-day mortality (timeframe: 28 days): death before day 28 post randomisation • Procalcitonin (PCT) clearance (timeframe: 4 days) will be defined as baseline PCT minus PCT at 96 hours post--randomisation, divided by initial PCT and multiplied by 100 • Delta SOFA score (timeframe: 4 days) will be defined as initial SOFA score minus day 4 post-randomisation SOFA score. The SOFA score is the sum of 6 subscores that range from 0 to 4 and provides an assessment of the function of the following organs or systems: respiratory, nervous, cardiovascular, liver, coagulation and renal. Increasing SOFA subscore (from 0 to 1, 2 and 4) indicates worsening function, culminating into failure of the corresponding organ or system. Maximum possible total SOFA score equals 24. SOFA score ≥ 15 has been previously associated with a mortality rate > 90%. • Neurological failure-free days (defined as daily follow-up Glasgow Coma Score > 9) within first 28 days of follow-up (timeframe: 28 days) will be defined as the number of days with a (daily) follow-up Glasgow Coma Score > 9 within the first 28 days of follow-up.

NCT03592693 2018 *(Continued)*

	• ICU mortality (timeframe: 90 days): death before ICU discharge. • ICU-free days to day 28 (timeframe: 28 days) will be defined as the number of days alive and out of the ICU until follow-up day 28. • ICU length of stay (timeframe: 90 days): duration of the need for intensive care after randomisation. • Hospital length of stay (timeframe: 90 days): duration of hospitalisation after randomisation.
Starting date	September 2018
Contact information	SD Mentzelopoulos; sdmentzelopoulos@yahoo.com
Notes	

NCT04280497 2023

Study name	RECORDS
Methods	The study is divided into two distinctive stages, a run-in prospective observational cohort followed by a biomarker-guided adaptive randomised controlled trial. Randomisation is first stratified by SARS-CoV-2 status. Patients tested positive for SARS-CoV-2 are then randomised with a 1:1 allocation ratio to receive either dexamethasone and fludrocortisone (FC) or dexamethasone and FC placebo. Subsequently, patients tested negative for SARS-CoV-2 are tested for influenza. Patients tested positive for influenza are then randomised for treatment (hydrocortisone (HC) and FC or their respective placebos). Patients negative for SARS-CoV-2 and influenza are first randomly assigned to a biomarker/signature stratum among all the biomarkers/signatures available at the time of randomisation for the patient; then, those patients are randomised with a 1:1 allocation ratio to either HC and FC or their respective placebos, using the randomisation list from this stratified biomarker/signature result. Patients positive to other respiratory viruses are not mandatorily assigned to the arms corresponding to their infections (other respiratory viruses (+)) as they are randomly assigned to any biomarker/signature arms of the two-step randomisation.
Participants	n = 1800 **Inclusion criteria** 1. Intensive care unit (ICU) patients aged ≥ 18 years 2. Proven or suspected infection as the main diagnosis (Sepsis-3 definition) 3. CAP (as defined by the IDSA/ATS CAP severity criteria or vasopressor-dependent sepsis (require vasopressor to maintain mean blood pressure of 65 mmHg and lactate level of < 2 mmol/L), or septic shock or infection-triggered ARDS (Berlin definition) 4. Tested for one or more trial-specific biomarkers 5. Affiliated to the French social security or benefiting from universal health coverage. Patients under guardianship or curatorship may be included. **Exclusion criteria** 1. Expected death or withdrawal of life-sustaining treatments within 48 hours 2. Previously enrolled in the study 3. Formal indication for corticosteroids according to most recent international guidelines 4. Vaccination with live virus within the past 6 months 5. Hypersensitivity to HC or FC, or microsined betamethasone dipropionate, or any of their excipients (Summary of Product Characteristics (SPC)) 6. Pregnancy, women of childbearing potential not using contraception 7. Nursing women
Interventions	**Experimental arm**

NCT04280497 2023 *(Continued)*

	Experimental treatments HC hemisuccinate (SERB, Paris) is administered as a 50 mg intravenous bolus every 6 hours for 7 days FC or its placebo (HAC Pharma, Caen, France) is administered as a 50 µg tablet via a nasogastric tube once a day in the morning, for 7 days Patients with COVID-19 are to be treated with open-label dexamethasone (6 mg) once a day over 10 days and 50 µg FC or its placebo via a nasogastric tube or orally for patients not requiring a nasogastric tube, once a day in the morning, for 7 days **Comparator** Placebo IV Placebo tablet
Outcomes	**Primary outcome measure** The primary outcome for the sequential analyses will be the number of vasopressor-free days at day 28 as the measure of efficacy, and the occurrence of severe adverse events within the first 28 days as the measure of toxicity. The primary outcome of the terminal analysis, assessed at 90 days, is a composite of death or persistent organ dysfunction—defined as SOFA score > 6 or a continued dependency on either mechanical ventilation, new renal replacement therapy or vasopressors. **Secondary outcomes measures** 1. Mortality at 7, 14 and 28 days and 6 months 2. Vasopressor-free days: defined as the number of days with permanent haemodynamic stability in the absence of any vasopressor agent including norepinephrine, phenylephrine, epinephrine, dopamine, vasopressin or its analogues. When a patient dies while receiving vasopressors, the number of vasopressor-free days is arbitrarily set at 0. 3. Mechanical ventilation-free days: defined as the number of days with permanent appropriate oxygenation while the patient is extubated and breathing spontaneously, that is, no need for non-invasive ventilation, high-flow oxygen or continuous positive airway pressure. Other uses of non-invasive ventilation (eg, chronic night-time use for chronic obstructive pulmonary disease) are not compatibilised. When a patient dies while receiving mechanical ventilation or is discharged home while receiving mechanical ventilation, the number of mechanical ventilation-free days is arbitrarily set at 0. 4. Organ dysfunction-free days: organ function (including renal function) will be assessed using the SOFA score. Organ dysfunction will be defined by a SOFA score > 6. Organ dysfunction-free days is defined by the number of days with a total SOFA score of 6 or less. When a patient dies before the SOFA decreases to 6 or less, the number of organ dysfunction-free days is arbitrarily set at 0. 5. 6-month health-related quality of life (HRQoL) in survivors assessed using the EQ-5D-5L. 6. Fatigue, ability to partake in social activities, physical function, emotional distress, depression, anxiety and cognitive function are assessed using the PROMIS short-form questionnaire. 7. Proportion of patients with a decision to withhold and/or withdraw active treatments; (8) ICU and hospital length of stay; (9) rate of ICU readmission up to 180 days after randomisation. Safety endpoints include: proportion of patients affected by any serious adverse event associated with corticosteroids; (1) hospital-acquired infection (Center for Diseases Control [CDC], Healthcare-Associated Infections progress report, 2020) within 90 days of randomisation; (2) hyperglycaemia (blood glucose level > 150 mg/dL) and hypernatraemia (serum sodium > 145 mmol/L) during the ICU stay (or up to day 90, whichever occurs first); (3) gastroduodenal bleeding requiring transfusion or haemostatic treatment during the ICU stay (or up to day 90, whichever occurs first); (4) neurological disorder (coma, stroke or muscle weakness, as defined below) up to 180 days from randomisation. Coma is defined by a Glasgow Coma Score < 8 in the absence of sedation. Neuromuscular sequelae are assessed using the Muscular Disability Rating Scale, with a score of 1 indi-

NCT04280497 2023 *(Continued)*

	cating no deficit, 2 minor deficit with no functional disability, 3 distal motor deficit, 4 mild-to-moderate proximal motor deficit and 5 severe proximal motor deficit.
Starting date	9 April 2020
Contact information	Dr Djillali Annane djillali.annane@aphp.fr
Notes	

NCT05001854 2021

Study name	FLUDROSEPSIS
Methods	Primary purpose: treatment Allocation: randomised Interventional model: parallel assignment Masking: quadruple (participant, care provider, investigator, outcomes assessor)
Participants	**Inclusion criteria:** • Age > 18 years Patient with septic shock for less than 48 hours, defined by the combination of: • Sepsis defined as organ dysfunction caused by an inappropriate host response to infection (increase in SOFA score of at least 2 points secondary to infection), • Persistent hypotension requiring vasopressor drugs to maintain mean arterial pressure ≥ 65 mmHg • Haemodynamic stability for > 30 min with mean arterial pressure ≥ 65 mmHg and noradrenaline dose ≤ 0.5 µg/kg/min • Consent signed by the patient, family member or legal representative or inclusion under emergency procedure • Negative to a diagnostic test for SARS-CoV-2 less than 72 hours old; the test used must be on the list published on the Ministry of Solidarity and Health website **Exclusion criteria:** • Current treatment with steroids or other treatment that may affect the hypothalamic-pituitary-adrenal axis • Anaesthetic induction with etomidate within 6 hours prior to randomisation given its selective inhibitory effect on 11 β-hydroxylase • Known hypersensitivity to fludrocortisone (or any of its excipients), or to tetracosactide (Synacthen®) • Disturbed gastric emptying (gastric residue > 500 mL) in the absence of an alternative route of administration available (naso-jejunal tube or jejunostomy) • Pregnant or nursing woman • Patient participating in another trial, possibly interfering with the study procedures • Person not affiliated to a social security system • Known situation of deprivation of freedom (safeguard of justice), guardianship or curatorship • Patient with a life expectancy of less than 24 hours • Patient with known or suspected SARS-CoV2 pneumonia • Patients under court protection will be excluded as soon as the investigator is aware of their status

NCT05001854 2021 *(Continued)*

	• Haemodynamic worsening with noradrenaline dose > 1.5 µg/kg/min before evaluation of the primary endpoint • Catecholamine withdrawal before evaluation of the primary endpoint
Interventions	Experimental: fludrocortisone 100 µg every 6 hours of fludrocortisone per os Placebo comparator: control Tablet every 6 hours of placebo oral
Outcomes	**Primary outcome measures** Mean arterial blood pressure (mmHg) Mean arterial blood pressure (mmHg) after a dose escalation of norepinephrine in increments of 0.2 µg/kg/min to a maximum dose of 1.5 µg/kg/min 1.5 hours after the second administration of fludrocortisone or placebo (7.5 hours) **Secondary outcome measures** Heart rate, mean arterial pressure, systolic arterial pressure, diastolic arterial pressure Cardiac output, Cardiac index, Indexed Systemic Vascular Resistances (ISVR), Indexed Global Tele-diastolic Volume (IGVT), Ejection Volume Variability (EVV), Extravascular Pulmonary Water (EVW) Baseline, before the first administration of the drug (fludrocortisone/placebo) 1.5 hours after the second administration of fludrocortisone or placebo (7.5 hours) After the 3rd administration of fludrocortisone or placebo (12 hours) After the 4th administration of fludrocortisone or placebo (18 hours) Arterial pressure Day 28 mortality Day 90 mortality Length of stay in intensive care unit Length of stay in care unit Time to wean from catecholamines Duration of mechanical ventilation (MV) Inflammation markers IFNγ, TNF α, IL1β, IL6, IL10 dosage Baseline, before the first administration of the experimental drug (fludrocortisone/placebo) 1.5 hours after the second administration of fludrocortisone or placebo (7.5 hours) Blood and urinary sodium and potassium Baseline, before the first administration of the experimental drug (fludrocortisone/placebo) 1.5 hours after the second administration of fludrocortisone or placebo (7.5 hours) Pharmacokinetics of fludrocortisone: area under the plasma concentration-time curve (plasma concentrations of fludrocortisone at 5 times) in the experimental group only

NCT05001854 2021 *(Continued)*

	After 4th dose of fludrocortisone (18 hours) **Other outcome measures** The dose-response relationship (mean arterial blood pressure to doses of norepinephrine) will be modelled according to a sigmoid Emax model with determination of the maximum effect (Emax) obtained on blood pressure, estimation of the dose of norepinephrine inducing half of the Emax (ED50), and the Hill coefficient γ reflecting the sigmoidality of the curve. 1.5 hours after the second administration of fludrocortisone or placebo (7.5 hours)
Starting date	31 March 2022
Contact information	Study Chair: Bruno Laviolle, MD, PhD, Rennes University Hospital Principal Investigator: Harmonie Perrichet, MD, Rennes University Hospital
Notes	

NCT05136560 2021

Study name	DEXA-SEPSIS
Methods	Multicentre, double-blinded, randomised pilot study
Participants	n = 102 **Inclusion criteria** Age of 19 to 89 years Systolic blood pressure of < 90 mmHg or plasma lactate level of > 2 mmol/L Delta SOFA score of ≥ 2 points consequent to the infection **Exclusion criteria** Decision to limit or withdraw active treatment (i.e. do not resuscitate) Steroid use within 4 weeks Chemotherapy for underlying malignancy within 4 weeks Immunosuppressive therapy within 4 weeks Underlying disease with a life expectancy of < 90 days Patient was hospitalised at another hospital for sepsis and referred (not including those transferred from another ED) Sepsis diagnosed 24 hours after ED admission Etomidate use during hospitalisation Pregnancy or lactation Informed written consent cannot be obtained from a patient or legal representative Any other condition considered harmful in conjunction with the use of steroids by the physician
Interventions	**Experimental arm 1**

NCT05136560 2021 *(Continued)*

	0.1 mg/kg of dexamethasone bolus **Experimental arm 2** 0.2 mg/kg of dexamethasone bolus **Comparator** Placebo (0.9% saline)
Outcomes	**Primary outcome** 28-day mortality **Secondary outcomes** 90-day all-cause mortality Time to septic shock Shock reversal Number of vasopressor-free days Use of continuous renal-replacement therapy Number of ventilator-free days Length of stay in the intensive care unit and/or hospital Delta SOFA score on days 3 and 7 Superinfection (defined as a secondary infection within 28 days) Gastrointestinal bleeding within 14 days Hyperglycaemia (defined by a serum glucose level of > 150 mg/dL within 7 days) Hypernatraemia (defined by a serum sodium level of > 150 mmol/L within 7 days)
Starting date	15 January 2021
Contact information	Kyuseok Kim +82-10-4780-8321 dreinstein70@gmail.com
Notes	

NCT05354778 2022

Study name	HYDRO-SHIP
Methods	Primary purpose: treatment Allocation: randomised Interventional model: parallel assignment Interventional model description: this will be a multicentre, randomised, open-label clinical trial with 2 parallel groups: hydrocortisone and placebo. Data will be analysed as "intention-to-treat". Masking: none (open-label)

NCT05354778 2022 *(Continued)*

Participants	n = 180 **Inclusion criteria:** • 18 years of age or older • Suspected or confirmed case of bacterial nosocomial pneumonia (including ventilator-associated pneumonia) • Intensive care unit stay • Signed consent form (by the patient or a legal guardian) **Exclusion criteria:** • Women who are pregnant, have recently given birth or are breastfeeding • Patients who are moribund or do not have a treatment perspective • Patients with community acquired pneumonia • Patients with other types of pneumonia (viral - including COVID-19, fungal etc.) • Those with pulmonary infiltrates in chest image which are not compatible with bacterial pneumonia • Patients with adrenal insufficiency • Patients who have a condition that demands the use of corticosteroids (acute or chronic) • Patients allergic to hydrocortisone • Patients with refractory septic shock (those receiving more than 0.5 µg/kg/min of norepinephrine)
Interventions	Active comparator: hydrocortisone Hydrocortisone 100 mg + normal saline 100 mL every 8 hours for 5 days or until patient dies or is discharged from the intensive care unit Drug: hydrocortisone Placebo comparator: placebo Normal saline 100 mL every 8 hours for 5 days or until patient dies or is discharged from the intensive care unit Drug: placebo
Outcomes	**Primary outcome measures** Early Clinical Failure defined as a composite outcome: Death OR Respiratory worsening OR Cardiovascular worsening between days 3 and 7 **Secondary outcome measures** Survival in the intensive care unit and in the hospital Survival at days 7 and 28 Need for mechanical ventilation at 7 days Need for vasoactive drugs at 7 days Time of mechanical ventilation Number of days patients require invasive ventilation support up to 7 days Positive end expiratory pressure (PEEP) at 7 days Inspired oxygen fraction (FiO2) at 7 days Partial pressure of oxygen (PaO2) at 7 days

NCT05354778 2022 *(Continued)*

	Horowitz index for lung function (P/F ratio) - obtained from PaO2/FiO2 at 7 days Time of vasoactive drugs use Dosage and types of vasoactive drugs at 7 days Length of stay In the Intensive Care Unit and in the hospital up to 28 days Need for dialysis at 7 days Progression of image in the chest X-ray image at 7 days Adverse effects at 7 days: psychosis, insomnia, hyperglycaemia, hypernatraemia, rhabdomyolysis, gastrointestinal bleeding and critical myopathy illness
Starting date	16 October 2022
Contact information	Dante Raglione 5511984471792 danraglione@gmail.com
Notes	

PACTR202111481740832 2021

Study name	SONIA Trial
Methods	Parallel: different groups receive different interventions at same time during study. Simple randomisation using a randomisation table created by a computer software program; sealed, opaque envelopes Phase 4
Participants	n = 2180 **Inclusion criteria** 1. Adults aged 18 years or over 2. Admitted to hospital with a diagnosis of community-acquired pneumonia. Pneumonia will be based on a clinical definition as follows: the presence of at least 2 of the following signs and symptoms for less than 14 days: cough, fever, dyspnoea, haemoptysis, chest pain or crackles on chest examination. 3. Admitted to hospital within the previous 48 hours 4. Provides written informed consent **Exclusion criteria** 1. Hospital-acquired pneumonia - defined as pneumonia in a patient who has been in hospital for > 48 hours who did not have the symptoms at admission 2. Patients who, in the opinion of the attending clinician, require to be treated with steroids 3. Known or suspected condition which, in the opinion of attending clinician, requires treatment with steroids, including but not limited to chronic obstructive pulmonary disease, asthma, adrenal insufficiency, pneumocystis Jirovecii pneumonia (PCP) 4. If the clinician strongly suspects COVID-19 and wants to provide steroids to the patient because of this suspicion, then the patient will be excluded from the trial 5. Pregnancy or breastfeeding 6. Any contraindication to steroid administration

PACTR202111481740832 2021 *(Continued)*

	7. Diagnosis of COVID-19 confirmed via PCR of NP/OP swabs (this only applies if test result known at the point of enrollment) Age minimum: 19 year(s) Age maximum: 44 year(s) Gender: both
Interventions	**Experimental arm** Steroid plus standard pneumonia treatment - no mention of molecule, dose and duration **Comparator** Standard treatment for pneumonia
Outcomes	**Primary outcome** The primary outcome for the trial will be mortality at 30 days after randomisation. **Secondary outcomes** 1. Mortality at 7, 14 and 21 days following randomisation 2. In-hospital mortality 3. Mortality after discharge from hospital (up to 30 days post-randomisation) 4. Duration of hospital stay 5. Admission to ICU/referral to higher level hospital
Starting date	15 November 2021
Contact information	Ruth Lucinde Nairobi, Kenya Telephone: 0702929295 rlucinde@kemri-wellcome.org Affiliation: Investigator Stella Mwakio Kilifi, Kenya Telephone: 254724832853 smwakio@kemri-wellcome.org Affiliation: Project coordinator
Notes	

TCTR20190219003 2019

Study name	Adjunctive corticosteroid therapy for pediatric catecholamine-dependent septic shock: a pilot study
Methods	Randomised, placebo-controlled trial in 2 parallel groups
Participants	All children 6 months to 18 years of age with septic shock who require vasoactive-inotropic support in the paediatric ICU
Interventions	**Experimental arm** Hydrocortisone 100 mg/m^2 of BSA/dose (max 200 mg/dose) IV bolus will be administered then reduced to 25 mg/m^2 of BSA /dose (max 50 mg/dose) IV every 6 h

TCTR20190219003 2019 *(Continued)*

	If patients do not require fluid bolus and/or increase in vasoactive-inotropic support, the dosing frequency will be reduced to every 8 h The drug will be discontinued if the patients do not require vasoactive-inotropic support for at least 12 hours. The total course of therapy will be a maximum of 7 days. **Comparator** Placebo
Outcomes	**Primary outcome measure** Time to shock reversal until discharge from ICU Time to maintain SBP > 5th percentile + without vasoactive-inotropic support for at least 24 h **Secondary outcomes measures** 28-day mortality Adverse effects at 7 days after enrolment Blood product requirement until discharge from ICU Continuous renal replacement therapy requirement until discharge from ICU Length of hospital stay Length of PICU stay Organ dysfunction-free days
Starting date	Unknown
Contact information	Unknown
Notes	

TCTR20200715008 2020

Study name	Continuous infusion vs. intermittent bolus hydrocortisone administration in children with catecholamine-dependent septic shock: a randomized controlled trial
Methods	Randomised, open-label trial in 2 parallel groups
Participants	**Inclusion criteria** Children 1 month to 18 years of age admitted to PICU with catecholamine-dependent septic shock who are at risk for adrenal insufficiency as follows and received loading dose of hydrocortisone; risk for adrenal insufficiency: physical signs e.g. purpura fulminans or diagnosed of Waterhouse-Friderichsen syndrome; history of hypothalamic-pituitary-adrenal axis abnormalities e.g. congenital adrenal hyperplasia, pituitary gland disease; history of corticosteroid use either long term (longer than 2 weeks within last 6 months) or short term (not longer than 2 weeks within last 4 month); history of etomidate use; history of antifungals use within last 7 days
Interventions	**Experimental arm** Continuous infusion of hydrocortisone at the dose of 100 mg/m^2/day continuous drip in 24 h until shock reversal with a maximum of 7 days

TCTR20200715008 2020 *(Continued)*

	Comparator Intermittent bolus of hydrocortisone at the dose of 25 mg/m^2/day every 6 h until shock reversal with a maximum of 7 days
Outcomes	**Primary outcome measure** Shock reversal Time to shock reversal from diagnosis of septic shock **Secondary outcomes measures** Vasoactive-inotropic drug use at day 28 after diagnosis of septic shock Vasoactive-inotropic drug-free days to day 28
Starting date	Unknown
Contact information	Unknown
Notes	

ARDS: acute respiratory distress syndrome; ATS: American Thoracic Society; BSA: body surface area; CKD: chronic kidney disease; CRRT: continuous renal replacement therapy; CXR: chest X-ray; ECMO: extracorporeal membrane oxygenation; ED: emergency department; ESR: erythrocyte sedimentation rate; h: hours; HFNC: high-flow oxygen nasal canula; hsCRP: high-sensitivity C-reactive protein; ICU: intensive care unit; IDSA: Infectious Diseases Society of America; IV: intravenous; NIV: non-invasive ventilation; PCT: procalcitonin; PICU: paediatric intensive care unit; RR: respiratory rate; RT-PCR: reverse transcription polymerase chain reaction; SBP: systolic blood pressure; SOFA: Sequential Organ Failure Assessment; WHO: World Health Organization

DATA AND ANALYSES

Comparison 1. Corticosteroids versus placebo or usual care

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.1 28-day all-cause mortality	72	22915	Risk Ratio (M-H, Random, 95% CI)	0.89 [0.84, 0.95]
1.2 28-day all-cause mortality - sensitivity analysis based on methodological quality	60		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
1.2.1 Adequate generation of allocation sequence	54	21160	Risk Ratio (M-H, Random, 95% CI)	0.91 [0.85, 0.97]
1.2.2 Adequate allocation concealment	55	21467	Risk Ratio (M-H, Random, 95% CI)	0.90 [0.84, 0.96]
1.2.3 Blinded trials	46	13577	Risk Ratio (M-H, Random, 95% CI)	0.91 [0.84, 0.98]
1.2.4 Studies judged at low risk of bias	25	11125	Risk Ratio (M-H, Random, 95% CI)	0.89 [0.81, 0.99]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.3 28-day all-cause mortality by subgroups based on study drug	69	22699	Risk Ratio (M-H, Random, 95% CI)	0.90 [0.84, 0.96]
1.3.1 Hydrocortisone	29	7961	Risk Ratio (M-H, Random, 95% CI)	0.88 [0.79, 0.99]
1.3.2 Hydrocortisone plus fludrocortisone	4	1714	Risk Ratio (M-H, Random, 95% CI)	0.88 [0.79, 0.98]
1.3.3 Prednisone/prednisolone	6	1572	Risk Ratio (M-H, Random, 95% CI)	0.94 [0.70, 1.27]
1.3.4 Methylprednisolone	14	2037	Risk Ratio (M-H, Random, 95% CI)	0.82 [0.63, 1.06]
1.3.5 Dexamethasone	8	7775	Risk Ratio (M-H, Random, 95% CI)	0.85 [0.73, 0.97]
1.3.6 Hydrocortisone plus vitamin C plus thiamine	8	1640	Risk Ratio (M-H, Random, 95% CI)	1.01 [0.85, 1.19]
1.4 28-day all-cause mortality by subgroups based on treatment dose/duration	66	22494	Risk Ratio (M-H, Random, 95% CI)	0.90 [0.84, 0.96]
1.4.1 Long course of low-dose corticosteroids	61	21584	Risk Ratio (M-H, Random, 95% CI)	0.89 [0.84, 0.95]
1.4.2 Short course of high-dose corticosteroids	5	910	Risk Ratio (M-H, Random, 95% CI)	0.90 [0.59, 1.36]
1.5 28-day all-cause mortality based on mode of drug administration	66	21259	Risk Ratio (M-H, Random, 95% CI)	0.89 [0.84, 0.96]
1.5.1 Intravenous bolus	43	13964	Risk Ratio (M-H, Random, 95% CI)	0.91 [0.85, 0.98]
1.5.2 Continuous infusion	23	7295	Risk Ratio (M-H, Random, 95% CI)	0.82 [0.69, 0.98]
1.6 28-day all-cause mortality based on mode of drug termination	68	22005	Risk Ratio (M-H, Random, 95% CI)	0.89 [0.83, 0.96]
1.6.1 Without taper off	43	18237	Risk Ratio (M-H, Random, 95% CI)	0.90 [0.84, 0.97]
1.6.2 With taper off	25	3768	Risk Ratio (M-H, Random, 95% CI)	0.85 [0.73, 0.99]
1.7 28-day all-cause mortality by subgroups based on age	67	22129	Risk Ratio (M-H, Random, 95% CI)	0.89 [0.83, 0.95]
1.7.1 Adults	62	21851	Risk Ratio (M-H, Random, 95% CI)	0.89 [0.83, 0.95]

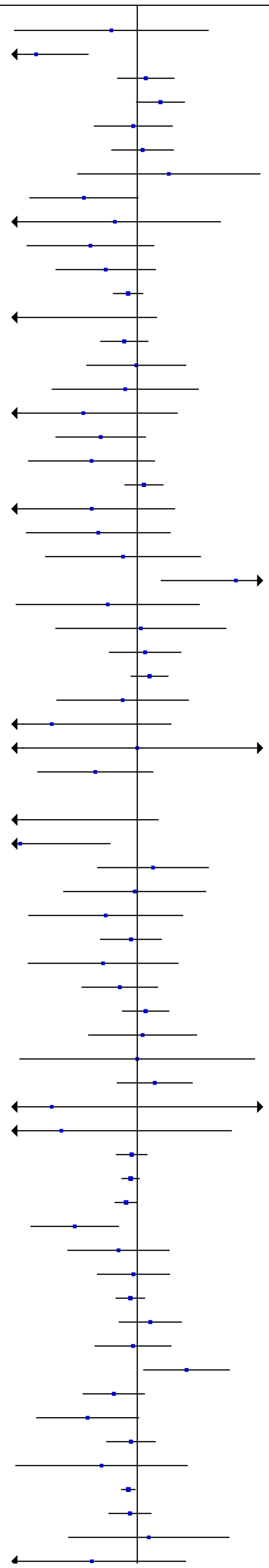
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.7.2 Children	5	278	Risk Ratio (M-H, Random, 95% CI)	1.08 [0.56, 2.09]
1.8 28-day all-cause mortality by subgroups based on targeted population	71	22774	Risk Ratio (M-H, Random, 95% CI)	0.90 [0.84, 0.96]
1.8.1 Sepsis	14	1825	Risk Ratio (M-H, Random, 95% CI)	1.08 [0.90, 1.28]
1.8.2 Septic shock only	29	8871	Risk Ratio (M-H, Random, 95% CI)	0.93 [0.86, 1.00]
1.8.3 Sepsis and ARDS	5	496	Risk Ratio (M-H, Random, 95% CI)	0.59 [0.42, 0.83]
1.8.4 Sepsis and community-acquired pneumonia	16	3818	Risk Ratio (M-H, Random, 95% CI)	0.68 [0.54, 0.86]
1.8.5 COVID-19	7	7764	Risk Ratio (M-H, Random, 95% CI)	0.88 [0.81, 0.95]
1.9 28-day mortality in participants with versus without critical illness-related corticosteroid insufficiency	13	1971	Risk Ratio (M-H, Random, 95% CI)	0.88 [0.80, 0.97]
1.9.1 Patients with critical illness associated with corticosteroid insufficiency	12	1101	Risk Ratio (M-H, Random, 95% CI)	0.90 [0.80, 1.00]
1.9.2 Patients without critical illness associated with corticosteroid insufficiency	9	870	Risk Ratio (M-H, Random, 95% CI)	0.85 [0.71, 1.02]
1.10 Long-term mortality	12	8468	Risk Ratio (M-H, Random, 95% CI)	0.97 [0.91, 1.03]
1.11 Hospital mortality	40	17459	Risk Ratio (M-H, Random, 95% CI)	0.90 [0.84, 0.97]
1.12 90-day all-cause mortality	13	8360	Risk Ratio (M-H, Random, 95% CI)	0.89 [0.82, 0.97]
1.13 Intensive care unit mortality	24	8866	Risk Ratio (M-H, Random, 95% CI)	0.90 [0.83, 0.98]
1.14 Length of intensive care unit stay for all participants	25	8069	Mean Difference (IV, Random, 95% CI)	-0.86 [-1.67, -0.05]
1.15 Length of intensive care unit stay for survivors	10	778	Mean Difference (IV, Random, 95% CI)	-2.19 [-3.93, -0.46]
1.16 Length of hospital stay for all participants	31	16954	Mean Difference (IV, Random, 95% CI)	-1.09 [-1.85, -0.34]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.17 Length of hospital stay for survivors	9	710	Mean Difference (IV, Random, 95% CI)	-4.11 [-8.50, 0.28]
1.18 Number of participants with shock reversal at day 7	18	6938	Risk Ratio (M-H, Random, 95% CI)	1.22 [1.13, 1.33]
1.19 Number of participants with shock reversal at 28 days	16	7415	Risk Ratio (M-H, Random, 95% CI)	1.05 [1.03, 1.07]
1.20 SOFA score at day 7	17	3220	Mean Difference (IV, Random, 95% CI)	-1.27 [-1.63, -0.92]
1.21 Number of participants with adverse events	46		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
1.21.1 Gastroduodenal bleeding	32	6978	Risk Ratio (M-H, Random, 95% CI)	1.00 [0.82, 1.23]
1.21.2 Superinfection	36	7961	Risk Ratio (M-H, Random, 95% CI)	0.96 [0.86, 1.07]
1.21.3 Hyperglycaemia	29	10933	Risk Ratio (M-H, Random, 95% CI)	1.26 [1.13, 1.42]
1.21.4 Hypernatraemia	12	6587	Risk Ratio (M-H, Random, 95% CI)	1.44 [1.08, 1.93]
1.21.5 Muscle weakness	7	6729	Risk Ratio (M-H, Random, 95% CI)	1.09 [0.78, 1.53]
1.21.6 Neuropsychiatric event	10	7926	Risk Ratio (M-H, Random, 95% CI)	1.02 [0.63, 1.65]
1.21.7 Stroke	8	4742	Risk Ratio (M-H, Random, 95% CI)	0.87 [0.51, 1.51]
1.21.8 Cardiac event	14	12488	Risk Ratio (M-H, Random, 95% CI)	0.98 [0.77, 1.25]

Analysis 1.1. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 1: 28-day all-cause mortality

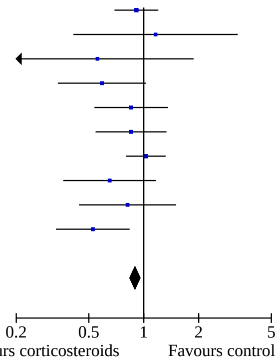
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F	G
McHardy 1972	3	40	9	86	0.3%	0.72 [0.20 , 2.51]	?	+	-	-	+	?	?
Schumer 1976	9	86	33	86	0.8%	0.27 [0.14 , 0.53]	-	-	?	?	+	?	?
Sprung 1984	33	43	11	16	2.1%	1.12 [0.77 , 1.61]	+	-	-	-	+	?	?
Bone 1987	65	191	48	190	2.7%	1.35 [0.98 , 1.84]	+	+	+	+	+	?	+
VASSCSG 1987	23	112	24	111	1.3%	0.95 [0.57 , 1.58]	+	+	+	+	+	?	?
Luce 1988	22	38	20	37	1.9%	1.07 [0.72 , 1.60]	+	+	+	+	-	?	?
Slusher 1996	6	36	4	36	0.3%	1.50 [0.46 , 4.87]	?	?	?	?	+	?	?
Bollaert 1998	7	22	12	19	0.8%	0.50 [0.25 , 1.02]	+	+	+	+	+	+	+
Briegel 1999	3	20	4	20	0.2%	0.75 [0.19 , 2.93]	+	+	+	+	+	+	+
Chawla 1999	6	23	10	21	0.6%	0.55 [0.24 , 1.25]	+	+	+	+	+	+	+
Yildiz 2002	8	20	12	20	0.9%	0.67 [0.35 , 1.27]	+	+	+	+	+	?	?
Annane 2002	82	151	91	149	4.4%	0.89 [0.73 , 1.08]	+	+	+	+	+	+	+
Confalonieri 2005	0	23	6	23	0.1%	0.08 [0.00 , 1.29]	+	+	+	+	+	+	-
Tandan 2005	11	14	13	14	2.7%	0.85 [0.62 , 1.15]	+	+	+	+	?	?	?
Oppert 2005	10	23	11	25	0.9%	0.99 [0.52 , 1.88]	+	+	+	+	-	+	+
Rinaldi 2006	6	26	7	26	0.4%	0.86 [0.33 , 2.21]	+	+	-	-	+	+	+
El Ghamrawy 2006	3	17	6	17	0.3%	0.50 [0.15 , 1.68]	?	?	?	?	+	?	?
Cicarelli 2007	7	14	12	15	1.0%	0.63 [0.35 , 1.12]	+	+	+	+	?	?	?
Meduri 2007	10	63	8	28	0.6%	0.56 [0.25 , 1.26]	+	+	+	+	+	+	-
Sprung 2008	86	251	78	248	3.5%	1.09 [0.85 , 1.40]	+	+	+	+	?	+	-
Aboab 2008	3	10	7	13	0.3%	0.56 [0.19 , 1.63]	+	+	+	+	+	+	+
Kurungundla 2008	4	11	6	10	0.4%	0.61 [0.24 , 1.54]	?	?	?	?	?	?	?
Valoor 2009	5	19	6	19	0.4%	0.83 [0.31 , 2.27]	+	+	?	?	+	?	?
Meduri 2009	22	48	4	31	0.4%	3.55 [1.35 , 9.32]	+	+	+	+	+	+	?
Hu 2009	4	38	6	39	0.3%	0.68 [0.21 , 2.23]	?	?	?	?	+	?	?
Snijders 2010	6	104	6	109	0.3%	1.05 [0.35 , 3.15]	+	+	+	+	+	+	?
Yildiz 2011	16	27	15	28	1.5%	1.11 [0.69 , 1.76]	+	+	+	+	+	?	?
Arabi 2011	33	39	26	36	3.6%	1.17 [0.92 , 1.49]	+	+	+	+	?	+	-
Meijvis 2011	9	151	11	153	0.5%	0.83 [0.35 , 1.94]	+	+	+	+	+	+	+
Sabry 2011	2	40	6	40	0.2%	0.33 [0.07 , 1.55]	?	?	+	+	+	?	?
Fernández-Serrano 2011	1	28	1	28	0.1%	1.00 [0.07 , 15.21]	+	+	+	+	-	+	?
Liu 2012	5	12	10	14	0.7%	0.58 [0.28 , 1.23]	+	?	?	?	?	?	?
Nagy 2013	0	29	0	30		Not estimable	+	?	?	?	+	+	+
Rezk 2013	0	18	3	9	0.1%	0.08 [0.00 , 1.32]	?	?	-	-	+	?	?
Nafae 2013	4	60	6	20	0.3%	0.22 [0.07 , 0.71]	?	?	+	+	?	?	?
Blum 2015	16	402	13	400	0.7%	1.22 [0.60 , 2.51]	+	+	+	+	?	+	+
Gordon 2014	7	31	7	30	0.5%	0.97 [0.39 , 2.43]	+	+	+	+	+	+	+
Huang 2014	4	20	12	40	0.4%	0.67 [0.25 , 1.81]	+	?	?	?	+	?	?
Mirea 2014	44	117	22	54	1.9%	0.92 [0.62 , 1.37]	+	?	-	-	+	+	+
Torres 2015	6	61	9	59	0.4%	0.64 [0.24 , 1.70]	+	+	+	+	+	+	+
Tongyoo 2016	22	104	27	102	1.4%	0.80 [0.49 , 1.31]	+	+	+	+	+	+	+
Gordon 2016	62	208	57	213	2.8%	1.11 [0.82 , 1.51]	+	+	+	+	+	+	+
Keh 2016	15	190	14	190	0.8%	1.07 [0.53 , 2.16]	+	+	+	?	+	+	+
Li 2016	3	29	3	29	0.2%	1.00 [0.22 , 4.55]	?	?	?	?	?	?	?
Lv 2017	23	58	19	60	1.4%	1.25 [0.77 , 2.04]	+	+	+	+	+	+	+
Tagaro 2017	0	30	1	30	0.0%	0.33 [0.01 , 7.87]	+	+	+	+	+	+	+
Menon 2017	1	23	3	26	0.1%	0.38 [0.04 , 3.38]	+	+	+	+	+	+	+
Doluee 2018	54	80	58	80	4.2%	0.93 [0.76 , 1.14]	?	+	+	+	+	+	?
Venkatesh 2018	410	1853	448	1860	5.9%	0.92 [0.82 , 1.03]	+	+	+	+	+	+	+
Annane 2018	207	614	244	627	5.3%	0.87 [0.75 , 1.00]	+	+	+	+	+	+	+
Villar 2020	14	96	29	89	1.1%	0.45 [0.25 , 0.79]	+	+	-	-	+	+	?
Chang 2020	11	40	14	40	0.8%	0.79 [0.41 , 1.52]	+	+	-	-	+	+	+
Wani 2020	20	50	21	50	1.5%	0.95 [0.59 , 1.52]	+	+	-	-	?	+	?
Tomazini 2020	85	151	91	148	4.5%	0.92 [0.76 , 1.11]	+	+	-	-	+	+	+
Moskowitz 2020	35	101	29	99	1.9%	1.18 [0.79 , 1.78]	+	+	+	+	+	+	+
Mohamed 2020	18	45	19	45	1.4%	0.95 [0.58 , 1.55]	+	+	-	-	+	+	+
Iglesias 2020	26	68	14	69	1.1%	1.88 [1.08 , 3.29]	+	+	+	+	+	+	+
Angus 2020	55	278	27	101	1.9%	0.74 [0.50 , 1.10]	+	+	-	-	+	+	+
Dequin 2020	11	76	20	73	0.8%	0.53 [0.27 , 1.02]	+	+	+	+	+	+	+
Sevransky 2021	56	252	60	249	2.6%	0.92 [0.67 , 1.27]	+	+	+	+	+	+	+
Ghanei 2021	4	116	12	220	0.3%	0.63 [0.21 , 1.92]	?	+	-	-	?	+	+
Horby 2021	482	2104	1110	4321	6.4%	0.89 [0.81 , 0.98]	+	+	-	-	+	+	?
Sai 2021	41	75	45	75	3.1%	0.91 [0.69 , 1.20]	?	?	+	+	?	?	?
Corral-Gudino 2021	7	35	5	29	0.4%	1.16 [0.41 , 3.27]	+	+	-	-	-	+	?
Wittermans 2021	4	203	7	198	0.3%	0.56 [0.17 , 1.87]	+	+	+	+	+	+	+

Risk Ratio
M-H, Random, 95% CI



Analysis 1.1. (Continued)

							A	B	C	D	E	F	G
Sai 2021	41	75	45	75	3.1%	0.91 [0.69 , 1.20]	?	?	+	+	?	?	?
Corral-Gudino 2021	7	35	5	29	0.4%	1.16 [0.41 , 3.27]	+	+	−	−	−	+	?
Wittermans 2021	4	203	7	198	0.3%	0.56 [0.17 , 1.87]	+	+	+	+	+	+	+
Birudaraju 2022	9	21	16	22	1.1%	0.59 [0.34 , 1.03]	+	+	+	+	+	?	?
Meduri 2022	30	297	34	287	1.5%	0.85 [0.54 , 1.36]	+	+	+	+	+	+	?
Agarwal 2022	22	61	25	59	1.6%	0.85 [0.54 , 1.33]	+	+	−	−	+	+	+
Lyu 2022	79	213	77	213	3.5%	1.03 [0.80 , 1.32]	+	+	+	+	+	+	+
Mohamed 2023	13	53	20	53	1.0%	0.65 [0.36 , 1.17]	+	+	−	−	+	+	?
Pierre de Oliveira 2023	14	58	16	54	1.0%	0.81 [0.44 , 1.51]	?	+	+	+	+	?	?
Dequin 2023	25	400	47	395	1.5%	0.53 [0.33 , 0.84]	+	+	+	+	+	+	+
Total (Wald[a])		**10460**		**12455**	**100.0%**	**0.89 [0.84 , 0.95]**							
Total events:	2449		3207										



Test for overall effect: Z = 3.41 (P = 0.0007)
Test for subgroup differences: Not applicable
Heterogeneity: Tau² (DL[b]) = 0.01; Chi² = 98.15, df = 70 (P = 0.01); I² = 29%

Footnotes

[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.

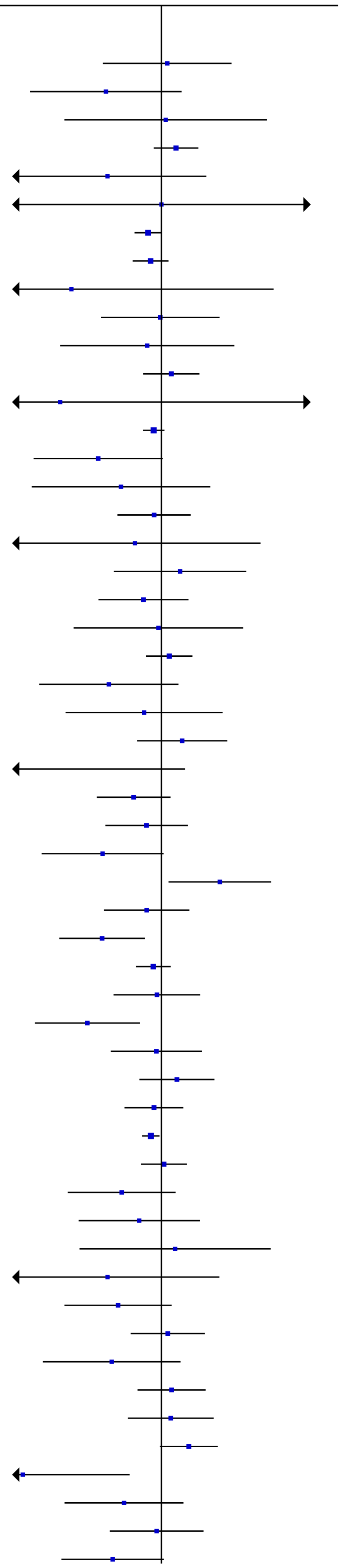
Risk of bias legend

(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 1.2. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 2: 28-day all-cause mortality - sensitivity analysis based on methodological quality

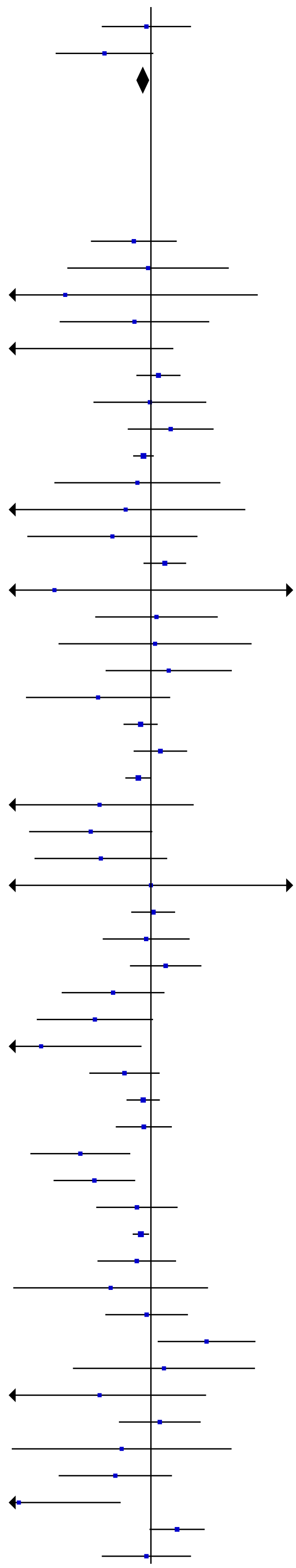
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
1.2.1 Adequate generation of allocation sequence						
Keh 2016	15	177	14	176	0.8%	1.07 [0.53 , 2.14]
Chawla 1999	6	23	10	21	0.6%	0.55 [0.24 , 1.25]
Snijders 2010	6	104	6	109	0.3%	1.05 [0.35 , 3.15]
Arabi 2011	33	39	26	36	4.3%	1.17 [0.92 , 1.49]
Aboab 2008	3	10	7	13	0.4%	0.56 [0.19 , 1.63]
Fernández-Serrano 2011	1	28	1	28	0.1%	1.00 [0.07 , 15.21]
Annane 2018	207	614	244	627	7.0%	0.87 [0.75 , 1.00]
Annane 2002	82	151	91	149	5.5%	0.89 [0.73 , 1.08]
Menon 2017	1	23	3	26	0.1%	0.38 [0.04 , 3.38]
Oppert 2005	10	23	11	25	0.9%	0.99 [0.52 , 1.88]
Rinaldi 2006	6	26	7	26	0.5%	0.86 [0.33 , 2.21]
Gordon 2016	62	208	57	213	3.2%	1.11 [0.82 , 1.51]
Tagaro 2017	0	30	1	30	0.0%	0.33 [0.01 , 7.87]
Venkatesh 2018	410	1853	448	1860	8.1%	0.92 [0.82 , 1.03]
Bollaert 1998	7	22	12	19	0.8%	0.50 [0.25 , 1.02]
Torres 2015	6	61	9	59	0.4%	0.64 [0.24 , 1.70]
Mirea 2014	44	117	22	54	2.1%	0.92 [0.62 , 1.37]
Briegel 1999	3	20	4	20	0.2%	0.75 [0.19 , 2.93]
Blum 2015	16	402	13	400	0.8%	1.22 [0.60 , 2.51]
Tongyoo 2016	22	98	27	99	1.5%	0.82 [0.50 , 1.34]
Gordon 2014	7	31	7	30	0.5%	0.97 [0.39 , 2.43]
Sprung 2008	86	251	78	248	4.1%	1.09 [0.85 , 1.40]
Meduri 2007	10	42	8	19	0.7%	0.57 [0.27 , 1.20]
Meijvis 2011	9	151	11	153	0.5%	0.83 [0.35 , 1.94]
Lv 2017	23	58	19	60	1.5%	1.25 [0.77 , 2.04]
Confalonieri 2005	0	23	6	23	0.1%	0.08 [0.00 , 1.29]
Angus 2020	55	278	27	101	2.1%	0.74 [0.50 , 1.10]
Agarwal 2022	22	61	25	59	1.8%	0.85 [0.54 , 1.33]
Dequin 2020	11	76	20	73	0.9%	0.53 [0.27 , 1.02]
Iglesias 2020	26	68	14	69	1.2%	1.88 [1.08 , 3.29]
Meduri 2022	30	297	34	287	1.7%	0.85 [0.54 , 1.36]
Dequin 2023	25	400	47	395	1.7%	0.53 [0.33 , 0.84]
Tomazini 2020	85	151	91	148	5.7%	0.92 [0.76 , 1.11]
Wani 2020	20	50	21	50	1.6%	0.95 [0.59 , 1.52]
Villar 2020	14	96	29	89	1.2%	0.45 [0.25 , 0.79]
Mohamed 2020	18	45	19	45	1.5%	0.95 [0.58 , 1.55]
Moskowitz 2020	35	101	29	99	2.1%	1.18 [0.79 , 1.78]
Sevransky 2021	56	252	60	249	3.0%	0.92 [0.67 , 1.27]
Horby 2021	482	2104	1110	4321	9.0%	0.89 [0.81 , 0.98]
Lyu 2022	79	213	77	213	4.2%	1.03 [0.80 , 1.32]
Mohamed 2023	13	53	20	53	1.1%	0.65 [0.36 , 1.17]
Chang 2020	11	40	14	40	0.9%	0.79 [0.41 , 1.52]
Corral-Gudino 2021	7	35	5	29	0.4%	1.16 [0.41 , 3.27]
Wittermans 2021	4	203	7	198	0.3%	0.56 [0.17 , 1.87]
Cicarelli 2007	7	14	12	15	1.1%	0.63 [0.35 , 1.12]
Luce 1988	22	38	20	37	2.1%	1.07 [0.72 , 1.60]
Liu 2012	5	12	10	14	0.7%	0.58 [0.28 , 1.23]
Sprung 1984	33	43	11	16	2.4%	1.12 [0.77 , 1.61]
Yildiz 2011	16	27	15	28	1.6%	1.11 [0.69 , 1.76]
Bone 1987	65	191	48	190	3.1%	1.35 [0.98 , 1.84]
Nafae 2013	4	60	6	20	0.3%	0.22 [0.07 , 0.71]
Yildiz 2002	8	20	12	20	0.9%	0.67 [0.35 , 1.27]
VASSCSG 1987	23	112	24	111	1.4%	0.95 [0.57 , 1.58]
Birudaraju 2022	9	21	16	22	1.2%	0.59 [0.34 , 1.03]

Risk Ratio M-H, Random, 95% CI



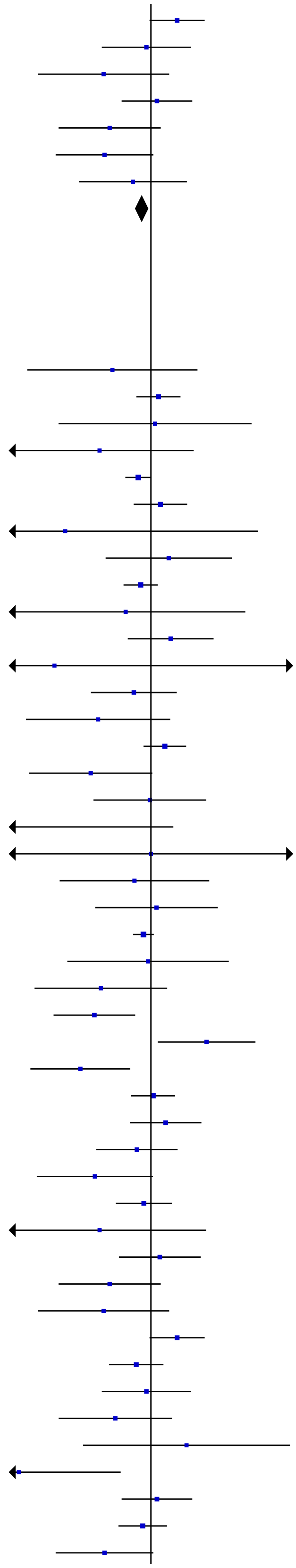
Analysis 1.2. (Continued)

VASSCSG 1987	23	112	24	111	1.4%	0.95 [0.57 , 1.58]
Birudaraju 2022	9	21	16	22	1.2%	0.59 [0.34 , 1.03]
Subtotal (Wald$_a$)		**9646**		**11514**	**100.0%**	**0.91 [0.85 , 0.97]**
Total events:	2260		2965			
Test for overall effect: Z = 2.81 (P = 0.005)						
Heterogeneity: Tau² (DL$_b$) = 0.01; Chi² = 69.56, df = 53 (P = 0.06); I² = 24%						
1.2.2 Adequate allocation concealment						
Tongyoo 2016	22	98	27	99	1.6%	0.82 [0.50 , 1.34]
Gordon 2014	7	31	7	30	0.5%	0.97 [0.39 , 2.43]
Menon 2017	1	23	3	26	0.1%	0.38 [0.04 , 3.38]
Meijvis 2011	9	151	11	153	0.6%	0.83 [0.35 , 1.94]
Confalonieri 2005	0	23	6	23	0.1%	0.08 [0.00 , 1.29]
Sprung 2008	86	251	78	248	4.3%	1.09 [0.85 , 1.40]
Oppert 2005	10	23	11	25	1.0%	0.99 [0.52 , 1.88]
Lv 2017	23	58	19	60	1.6%	1.25 [0.77 , 2.04]
Venkatesh 2018	410	1853	448	1860	7.9%	0.92 [0.82 , 1.03]
Rinaldi 2006	6	26	7	26	0.5%	0.86 [0.33 , 2.21]
Briegel 1999	3	20	4	20	0.2%	0.75 [0.19 , 2.93]
Torres 2015	6	61	9	59	0.5%	0.64 [0.24 , 1.70]
Arabi 2011	33	39	26	36	4.5%	1.17 [0.92 , 1.49]
Tagaro 2017	0	30	1	30	0.0%	0.33 [0.01 , 7.87]
Keh 2016	15	177	14	176	0.9%	1.07 [0.53 , 2.14]
Snijders 2010	6	104	6	109	0.4%	1.05 [0.35 , 3.15]
Blum 2015	16	402	13	400	0.8%	1.22 [0.60 , 2.51]
Chawla 1999	6	23	10	21	0.6%	0.55 [0.24 , 1.25]
Annane 2002	82	151	91	149	5.6%	0.89 [0.73 , 1.08]
Gordon 2016	62	208	57	213	3.4%	1.11 [0.82 , 1.51]
Annane 2018	207	614	244	627	6.9%	0.87 [0.75 , 1.00]
Aboab 2008	3	10	7	13	0.4%	0.56 [0.19 , 1.63]
Bollaert 1998	7	22	12	19	0.9%	0.50 [0.25 , 1.02]
Meduri 2007	10	42	8	19	0.7%	0.57 [0.27 , 1.20]
Fernández-Serrano 2011	1	28	1	28	0.1%	1.00 [0.07 , 15.21]
Lyu 2022	79	213	77	213	4.3%	1.03 [0.80 , 1.32]
Mohamed 2020	18	45	19	45	1.6%	0.95 [0.58 , 1.55]
Moskowitz 2020	35	101	29	99	2.2%	1.18 [0.79 , 1.78]
Mohamed 2023	13	53	20	53	1.2%	0.65 [0.36 , 1.17]
Dequin 2020	11	76	20	73	1.0%	0.53 [0.27 , 1.02]
Chang 2020	3	22	10	21	0.3%	0.29 [0.09 , 0.90]
Angus 2020	55	278	27	101	2.3%	0.74 [0.50 , 1.10]
Tomazini 2020	85	151	91	148	5.7%	0.92 [0.76 , 1.11]
Sevransky 2021	56	252	60	249	3.1%	0.92 [0.67 , 1.27]
Villar 2020	14	96	29	89	1.3%	0.45 [0.25 , 0.79]
Dequin 2023	25	400	47	395	1.8%	0.53 [0.33 , 0.84]
Meduri 2022	30	297	34	287	1.8%	0.85 [0.54 , 1.36]
Horby 2021	482	2104	1110	4321	8.7%	0.89 [0.81 , 0.98]
Agarwal 2022	22	61	25	59	1.9%	0.85 [0.54 , 1.33]
Ghanei 2021	4	116	12	220	0.4%	0.63 [0.21 , 1.92]
Wani 2020	20	50	21	50	1.7%	0.95 [0.59 , 1.52]
Iglesias 2020	26	68	14	69	1.3%	1.88 [1.08 , 3.29]
Corral-Gudino 2021	7	35	5	29	0.4%	1.16 [0.41 , 3.27]
Wittermans 2021	4	203	7	198	0.3%	0.56 [0.17 , 1.87]
Yildiz 2011	16	27	15	28	1.8%	1.11 [0.69 , 1.76]
McHardy 1972	3	40	9	86	0.3%	0.72 [0.20 , 2.51]
Yildiz 2002	8	20	12	20	1.0%	0.67 [0.35 , 1.27]
Nafae 2013	4	60	6	20	0.3%	0.22 [0.07 , 0.71]
Bone 1987	65	191	48	190	3.2%	1.35 [0.98 , 1.84]
VASSCSG 1987	23	112	24	111	1.5%	0.95 [0.57 , 1.58]



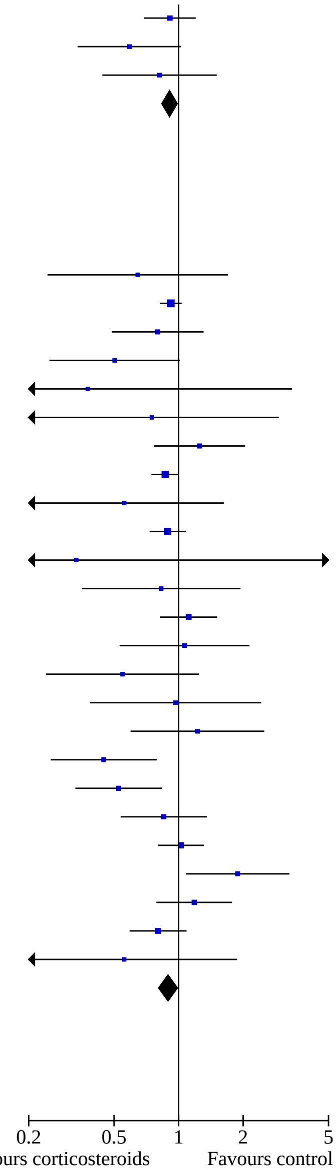
Analysis 1.2. (Continued)

Bone 1987	65	191	48	190	3.2%	1.35 [0.98 , 1.84]
VASSCSG 1987	23	112	24	111	1.5%	0.95 [0.57 , 1.58]
Liu 2012	5	12	10	14	0.8%	0.58 [0.28 , 1.23]
Luce 1988	22	38	20	37	2.2%	1.07 [0.72 , 1.60]
Cicarelli 2007	7	14	12	15	1.2%	0.63 [0.35 , 1.12]
Birudaraju 2022	9	21	16	22	1.3%	0.59 [0.34 , 1.03]
Pierre de Oliveira 2023	14	58	16	54	1.1%	0.81 [0.44 , 1.51]
Subtotal (Wald[a])		**9682**		**11785**	**100.0%**	**0.90 [0.84 , 0.96]**
Total events:	2196		2965			
Test for overall effect: Z = 3.05 (P = 0.002)						
Heterogeneity: Tau^2 (DL[b]) = 0.01; Chi^2 = 72.82, df = 54 (P = 0.04); I^2 = 26%						
1.2.3 Blinded trials						
Torres 2015	6	61	9	59	0.7%	0.64 [0.24 , 1.70]
Sprung 2008	86	251	78	248	5.0%	1.09 [0.85 , 1.40]
Snijders 2010	6	104	6	109	0.5%	1.05 [0.35 , 3.15]
Aboab 2008	3	10	7	13	0.5%	0.56 [0.19 , 1.63]
Annane 2018	207	614	244	627	7.2%	0.87 [0.75 , 1.00]
Gordon 2016	62	208	57	213	4.1%	1.11 [0.82 , 1.51]
Menon 2017	1	23	3	26	0.1%	0.38 [0.04 , 3.38]
Blum 2015	16	402	13	400	1.1%	1.22 [0.60 , 2.51]
Annane 2002	82	151	91	149	6.1%	0.89 [0.73 , 1.08]
Briegel 1999	3	20	4	20	0.3%	0.75 [0.19 , 2.93]
Lv 2017	23	58	19	60	2.2%	1.25 [0.77 , 2.04]
Tagaro 2017	0	30	1	30	0.1%	0.33 [0.01 , 7.87]
Tongyoo 2016	22	98	27	99	2.2%	0.82 [0.50 , 1.34]
Chawla 1999	6	23	10	21	0.9%	0.55 [0.24 , 1.25]
Arabi 2011	33	39	26	36	5.1%	1.17 [0.92 , 1.49]
Bollaert 1998	7	22	12	19	1.2%	0.50 [0.25 , 1.02]
Oppert 2005	10	23	11	25	1.4%	0.99 [0.52 , 1.88]
Confalonieri 2005	0	23	6	23	0.1%	0.08 [0.00 , 1.29]
Fernández-Serrano 2011	1	28	1	28	0.1%	1.00 [0.07 , 15.21]
Meijvis 2011	9	151	11	153	0.8%	0.83 [0.35 , 1.94]
Keh 2016	15	177	14	176	1.2%	1.07 [0.53 , 2.14]
Venkatesh 2018	410	1853	448	1860	7.9%	0.92 [0.82 , 1.03]
Gordon 2014	7	31	7	30	0.7%	0.97 [0.39 , 2.43]
Meduri 2007	10	42	8	19	1.0%	0.57 [0.27 , 1.20]
Dequin 2023	25	400	47	395	2.3%	0.53 [0.33 , 0.84]
Iglesias 2020	26	68	14	69	1.8%	1.88 [1.08 , 3.29]
Villar 2020	14	96	29	89	1.7%	0.45 [0.25 , 0.79]
Lyu 2022	79	213	77	213	5.0%	1.03 [0.80 , 1.32]
Moskowitz 2020	35	101	29	99	2.8%	1.18 [0.79 , 1.78]
Meduri 2022	30	297	34	287	2.3%	0.85 [0.54 , 1.36]
Dequin 2020	11	76	20	73	1.3%	0.53 [0.27 , 1.02]
Sevransky 2021	56	252	60	249	3.9%	0.92 [0.67 , 1.27]
Wittermans 2021	4	203	7	198	0.4%	0.56 [0.17 , 1.87]
Yildiz 2011	16	27	15	28	2.3%	1.11 [0.69 , 1.76]
Cicarelli 2007	7	14	12	15	1.6%	0.63 [0.35 , 1.12]
Liu 2012	5	12	10	14	1.1%	0.58 [0.28 , 1.23]
Bone 1987	65	191	48	190	3.9%	1.35 [0.98 , 1.84]
Tandan 2005	11	14	13	14	4.0%	0.85 [0.62 , 1.15]
VASSCSG 1987	23	112	24	111	2.0%	0.95 [0.57 , 1.58]
Yildiz 2002	8	20	12	20	1.4%	0.67 [0.35 , 1.27]
Slusher 1996	6	36	4	36	0.5%	1.50 [0.46 , 4.87]
Nafae 2013	4	60	6	20	0.5%	0.22 [0.07 , 0.71]
Luce 1988	22	38	20	37	2.9%	1.07 [0.72 , 1.60]
Sai 2021	41	75	45	75	4.5%	0.91 [0.69 , 1.20]
Birudaraju 2022	9	21	16	22	1.8%	0.59 [0.34 , 1.03]



Analysis 1.2.   (Continued)

Sai 2021	41	75	45	75	4.5%	0.91 [0.69 , 1.20]
Birudaraju 2022	9	21	16	22	1.8%	0.59 [0.34 , 1.03]
Pierre de Oliveira 2023	14	58	16	54	1.5%	0.81 [0.44 , 1.51]
Subtotal (Wald[a])		**6826**		**6751**	**100.0%**	**0.91 [0.84 , 0.98]**
Total events:	1536		1671			
Test for overall effect: Z = 2.35 (P = 0.02)						
Heterogeneity: Tau² (DL[b]) = 0.02; Chi² = 66.05, df = 45 (P = 0.02); I² = 32%						
1.2.4 Studies judged at low risk of bias						
Torres 2015	6	61	9	59	1.0%	0.64 [0.24 , 1.70]
Venkatesh 2018	410	1853	448	1860	15.6%	0.92 [0.82 , 1.03]
Tongyoo 2016	22	104	27	102	3.4%	0.80 [0.49 , 1.31]
Bollaert 1998	7	22	12	19	1.8%	0.50 [0.25 , 1.02]
Menon 2017	1	23	3	26	0.2%	0.38 [0.04 , 3.38]
Briegel 1999	3	20	4	20	0.5%	0.75 [0.19 , 2.93]
Lv 2017	23	58	19	60	3.4%	1.25 [0.77 , 2.04]
Annane 2018	207	614	244	627	13.8%	0.87 [0.75 , 1.00]
Aboab 2008	3	10	7	13	0.8%	0.56 [0.19 , 1.63]
Annane 2002	82	151	91	149	11.3%	0.89 [0.73 , 1.08]
Tagaro 2017	0	30	1	30	0.1%	0.33 [0.01 , 7.87]
Meijvis 2011	9	151	11	153	1.3%	0.83 [0.35 , 1.94]
Gordon 2016	62	208	57	213	6.9%	1.11 [0.82 , 1.51]
Keh 2016	15	177	14	176	1.8%	1.07 [0.53 , 2.14]
Chawla 1999	6	23	10	21	1.4%	0.55 [0.24 , 1.25]
Gordon 2014	7	31	7	30	1.1%	0.97 [0.39 , 2.43]
Blum 2015	16	402	13	400	1.7%	1.22 [0.60 , 2.51]
Villar 2020	14	96	29	89	2.6%	0.45 [0.25 , 0.79]
Dequin 2023	25	400	47	395	3.7%	0.53 [0.33 , 0.84]
Meduri 2022	30	297	34	287	3.7%	0.85 [0.54 , 1.36]
Lyu 2022	79	213	77	213	8.8%	1.03 [0.80 , 1.32]
Iglesias 2020	26	68	14	69	2.7%	1.88 [1.08 , 3.29]
Moskowitz 2020	35	101	29	99	4.6%	1.18 [0.79 , 1.78]
Sevransky 2021	56	252	69	249	6.9%	0.80 [0.59 , 1.09]
Wittermans 2021	4	203	7	198	0.6%	0.56 [0.17 , 1.87]
Subtotal (Wald[a])		**5568**		**5557**	**100.0%**	**0.89 [0.81 , 0.99]**
Total events:	1148		1283			
Test for overall effect: Z = 2.25 (P = 0.02)						
Heterogeneity: Tau² (DL[b]) = 0.01; Chi² = 33.27, df = 24 (P = 0.10); I² = 28%						



Footnotes

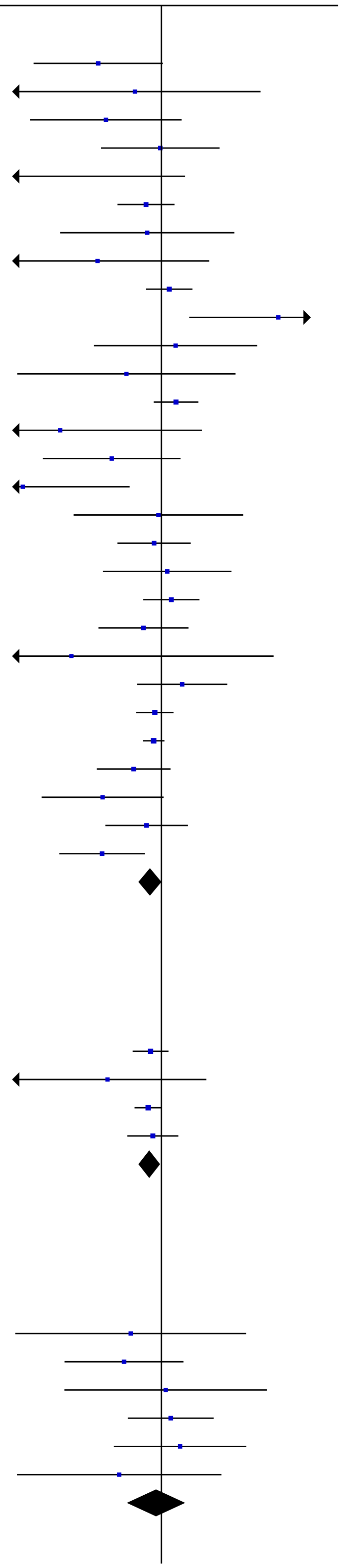
[a]CI calculated by Wald-type method.

[b]Tau² calculated by DerSimonian and Laird method.

Analysis 1.3. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 3: 28-day all-cause mortality by subgroups based on study drug

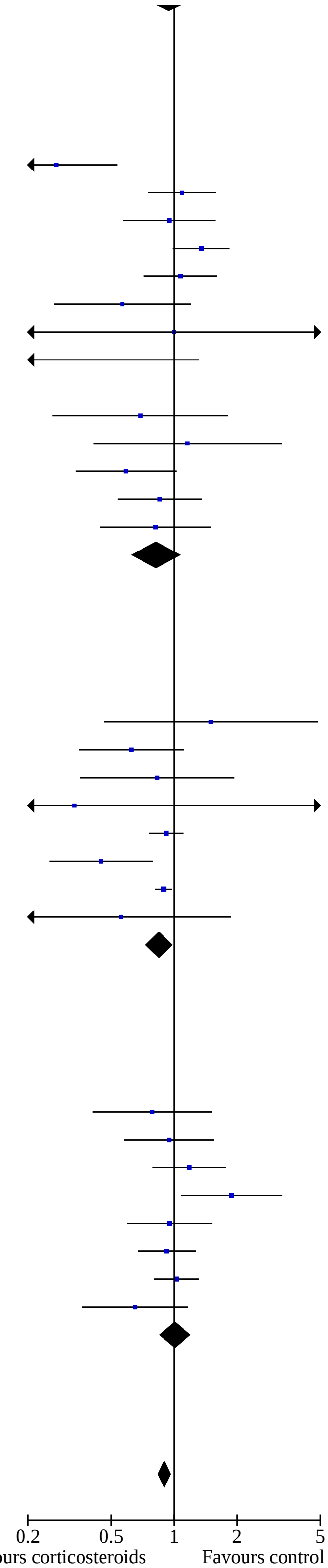
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
1.3.1 Hydrocortisone						
Bollaert 1998	7	22	12	19	0.8%	0.50 [0.25 , 1.02]
Briegel 1999	3	20	4	20	0.2%	0.75 [0.19 , 2.93]
Chawla 1999	6	23	10	21	0.6%	0.55 [0.24 , 1.25]
Oppert 2005	10	23	11	25	0.9%	0.99 [0.52 , 1.88]
Confalonieri 2005	0	23	6	23	0.1%	0.08 [0.00 , 1.29]
Tandan 2005	11	14	13	14	2.7%	0.85 [0.62 , 1.15]
Rinaldi 2006	6	26	7	26	0.4%	0.86 [0.33 , 2.21]
El Ghamrawy 2006	3	17	6	17	0.3%	0.50 [0.15 , 1.68]
Sprung 2008	86	251	78	248	3.5%	1.09 [0.85 , 1.40]
Meduri 2009	22	48	4	31	0.4%	3.55 [1.35 , 9.32]
Valoor 2009	7	19	6	19	0.5%	1.17 [0.48 , 2.83]
Hu 2009	4	38	6	39	0.3%	0.68 [0.21 , 2.23]
Arabi 2011	33	39	26	36	3.6%	1.17 [0.92 , 1.49]
Sabry 2011	2	40	6	40	0.2%	0.33 [0.07 , 1.55]
Liu 2012	5	12	10	14	0.7%	0.58 [0.28 , 1.23]
Nafae 2013	4	60	6	20	0.3%	0.22 [0.07 , 0.71]
Gordon 2014	7	31	7	30	0.5%	0.97 [0.39 , 2.43]
Mirea 2014	44	117	22	54	2.0%	0.92 [0.62 , 1.37]
Keh 2016	15	171	14	170	0.8%	1.07 [0.53 , 2.14]
Gordon 2016	62	208	57	213	2.8%	1.11 [0.82 , 1.51]
Tongyoo 2016	22	98	27	99	1.4%	0.82 [0.50 , 1.34]
Menon 2017	1	23	3	26	0.1%	0.38 [0.04 , 3.38]
Lv 2017	23	58	19	60	1.4%	1.25 [0.77 , 2.04]
Doluee 2018	54	80	58	80	4.2%	0.93 [0.76 , 1.14]
Venkatesh 2018	410	1853	448	1860	5.8%	0.92 [0.82 , 1.03]
Angus 2020	55	278	27	101	1.9%	0.74 [0.50 , 1.10]
Dequin 2020	11	76	20	73	0.9%	0.53 [0.27 , 1.02]
Agarwal 2022	22	61	25	59	1.6%	0.85 [0.54 , 1.33]
Dequin 2023	25	400	47	395	1.5%	0.53 [0.33 , 0.84]
Subtotal (Wald[a])		**4129**		**3832**	**40.5%**	**0.88 [0.79 , 0.99]**
Total events:	960		985			
Test for overall effect: Z = 2.13 (P = 0.03)						
Heterogeneity: Tau^2 (DL[b]) = 0.03; Chi^2 = 45.14, df = 28 (P = 0.02); I^2 = 38%						
1.3.2 Hydrocortisone plus fludrocortisone						
Annane 2002	82	151	91	149	4.4%	0.89 [0.73 , 1.08]
Aboab 2008	3	10	7	13	0.4%	0.56 [0.19 , 1.63]
Annane 2018	207	614	244	627	5.2%	0.87 [0.75 , 1.00]
Sai 2021	41	75	45	75	3.1%	0.91 [0.69 , 1.20]
Subtotal (Wald[a])		**850**		**864**	**13.1%**	**0.88 [0.79 , 0.98]**
Total events:	333		387			
Test for overall effect: Z = 2.40 (P = 0.02)						
Heterogeneity: Tau^2 (DL[b]) = 0.00; Chi^2 = 0.81, df = 3 (P = 0.85); I^2 = 0%						
1.3.3 Prednisone/prednisolone						
McHardy 1972	3	40	9	86	0.3%	0.72 [0.20 , 2.51]
Yildiz 2002	8	20	12	20	0.9%	0.67 [0.35 , 1.27]
Snijders 2010	6	104	6	109	0.3%	1.05 [0.35 , 3.15]
Yildiz 2011	16	27	15	28	1.5%	1.11 [0.69 , 1.76]
Blum 2015	16	402	13	400	0.7%	1.22 [0.60 , 2.51]
Ghanei 2021	4	116	12	220	0.3%	0.63 [0.21 , 1.92]
Subtotal (Wald[a])		**709**		**863**	**4.1%**	**0.94 [0.70 , 1.27]**
Total events:	53		67			
Test for overall effect: Z = 0.39 (P = 0.70)						

Risk Ratio M-H, Random, 95% CI



Analysis 1.3. (Continued)

Total events:	53		67			
Test for overall effect: Z = 0.39 (P = 0.70)						
Heterogeneity: Tau^2 (DL[b]) = 0.00; Chi^2 = 2.79, df = 5 (P = 0.73); I^2 = 0%						
1.3.4 Methylprednisolone						
Schumer 1976	9	86	33	86	0.8%	0.27 [0.14 , 0.53]
Sprung 1984	33	44	11	16	2.2%	1.09 [0.75 , 1.58]
VASSCSG 1987	23	112	24	111	1.3%	0.95 [0.57 , 1.58]
Bone 1987	65	191	48	190	2.7%	1.35 [0.98 , 1.84]
Luce 1988	22	38	20	37	1.9%	1.07 [0.72 , 1.60]
Meduri 2007	10	42	8	19	0.7%	0.57 [0.27 , 1.20]
Fernández-Serrano 2011	1	28	1	28	0.1%	1.00 [0.07 , 15.21]
Rezk 2013	0	18	3	9	0.1%	0.08 [0.00 , 1.32]
Nagy 2013	0	29	0	30		Not estimable
Torres 2015	6	59	9	61	0.4%	0.69 [0.26 , 1.82]
Corral-Gudino 2021	7	35	5	29	0.4%	1.16 [0.41 , 3.27]
Birudaraju 2022	9	21	16	22	1.2%	0.59 [0.34 , 1.03]
Meduri 2022	30	297	34	287	1.6%	0.85 [0.54 , 1.36]
Pierre de Oliveira 2023	14	58	16	54	1.0%	0.81 [0.44 , 1.51]
Subtotal (Wald[a])		**1058**		**979**	**14.2%**	**0.82 [0.63 , 1.06]**
Total events:	229		228			
Test for overall effect: Z = 1.52 (P = 0.13)						
Heterogeneity: Tau^2 (DL[b]) = 0.11; Chi^2 = 27.91, df = 12 (P = 0.006); I^2 = 57%						
1.3.5 Dexamethasone						
Slusher 1996	6	36	4	36	0.3%	1.50 [0.46 , 4.87]
Cicarelli 2007	7	14	12	15	1.1%	0.63 [0.35 , 1.12]
Meijvis 2011	9	151	11	153	0.5%	0.83 [0.35 , 1.94]
Tagaro 2017	0	30	1	30	0.0%	0.33 [0.01 , 7.87]
Tomazini 2020	85	151	91	148	4.5%	0.92 [0.76 , 1.11]
Villar 2020	14	96	29	89	1.1%	0.45 [0.25 , 0.79]
Horby 2021	482	2104	1110	4321	6.2%	0.89 [0.81 , 0.98]
Wittermans 2021	4	203	7	198	0.3%	0.56 [0.17 , 1.87]
Subtotal (Wald[a])		**2785**		**4990**	**14.0%**	**0.85 [0.73 , 0.97]**
Total events:	607		1265			
Test for overall effect: Z = 2.33 (P = 0.02)						
Heterogeneity: Tau^2 (DL[b]) = 0.01; Chi^2 = 8.71, df = 7 (P = 0.27); I^2 = 20%						
1.3.6 Hydrocortisone plus vitamin C plus thiamine						
Chang 2020	11	40	14	40	0.9%	0.79 [0.41 , 1.52]
Mohamed 2020	18	45	19	45	1.4%	0.95 [0.58 , 1.55]
Moskowitz 2020	35	101	29	99	1.9%	1.18 [0.79 , 1.78]
Iglesias 2020	26	68	14	69	1.2%	1.88 [1.08 , 3.29]
Wani 2020	20	50	21	50	1.5%	0.95 [0.59 , 1.52]
Sevransky 2021	56	252	60	249	2.6%	0.92 [0.67 , 1.27]
Lyu 2022	79	213	77	213	3.5%	1.03 [0.80 , 1.32]
Mohamed 2023	13	53	20	53	1.1%	0.65 [0.36 , 1.17]
Subtotal (Wald[a])		**822**		**818**	**14.0%**	**1.01 [0.85 , 1.19]**
Total events:	258		254			
Test for overall effect: Z = 0.10 (P = 0.92)						
Heterogeneity: Tau^2 (DL[b]) = 0.01; Chi^2 = 8.59, df = 7 (P = 0.28); I^2 = 19%						
Total (Wald[a])		**10353**		**12346**	**100.0%**	**0.90 [0.84 , 0.96]**
Total events:	2440		3186			
Test for overall effect: Z = 3.25 (P = 0.001)						
Test for subgroup differences: Chi^2 = 3.33, df = 5 (P = 0.65), I^2 = 0%						
Heterogeneity: Tau^2 (DL[b]) = 0.02; Chi^2 = 96.86, df = 67 (P = 0.010); I^2 = 31%						



Analysis 1.3. (Continued)

Heterogeneity: Tau^2 (DL_b) = 0.02; Chi^2 = 96.86, df = 67 (P = 0.010); I^2 = 31%

Footnotes

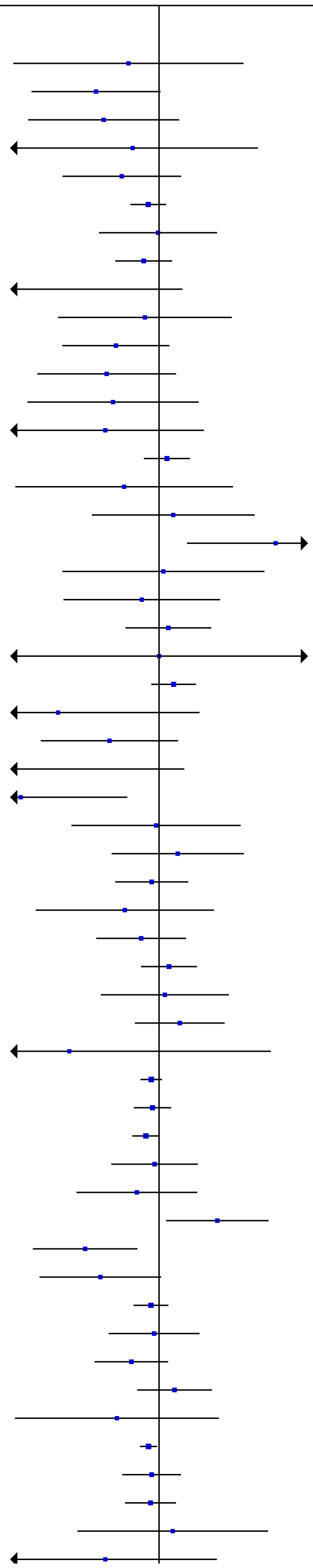
[a]CI calculated by Wald-type method.

[b]Tau^2 calculated by DerSimonian and Laird method.

Analysis 1.4. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 4: 28-day all-cause mortality by subgroups based on treatment dose/duration

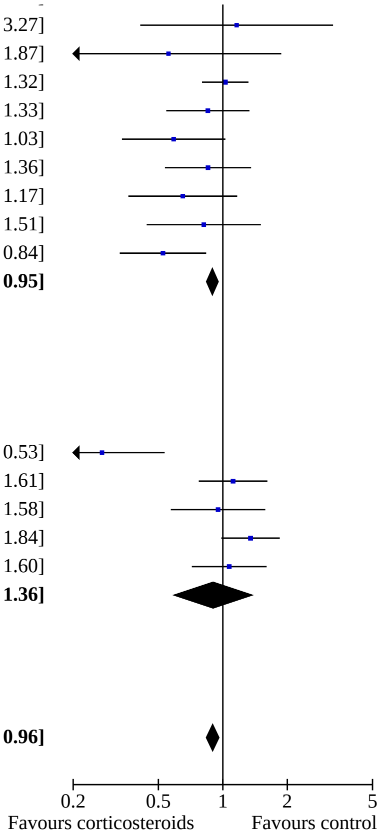
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
1.4.1 Long course of low-dose corticosteroids						
McHardy 1972	3	40	9	86	0.3%	0.72 [0.20 , 2.51]
Bollaert 1998	7	22	12	19	0.8%	0.50 [0.25 , 1.02]
Chawla 1999	6	23	10	21	0.6%	0.55 [0.24 , 1.25]
Briegel 1999	3	20	4	20	0.2%	0.75 [0.19 , 2.93]
Yildiz 2002	8	20	12	20	0.9%	0.67 [0.35 , 1.27]
Annane 2002	82	151	91	149	4.3%	0.89 [0.73 , 1.08]
Oppert 2005	10	23	11	25	0.9%	0.99 [0.52 , 1.88]
Tandan 2005	11	14	13	14	2.7%	0.85 [0.62 , 1.15]
Confalonieri 2005	0	23	6	23	0.1%	0.08 [0.00 , 1.29]
Rinaldi 2006	6	26	7	26	0.5%	0.86 [0.33 , 2.21]
Cicarelli 2007	7	14	12	15	1.1%	0.63 [0.35 , 1.12]
Meduri 2007	10	42	8	19	0.7%	0.57 [0.27 , 1.20]
Kurungundla 2008	4	11	6	10	0.5%	0.61 [0.24 , 1.54]
Aboab 2008	3	10	7	13	0.4%	0.56 [0.19 , 1.63]
Sprung 2008	86	251	78	248	3.5%	1.09 [0.85 , 1.40]
Hu 2009	4	38	6	39	0.3%	0.68 [0.21 , 2.23]
Valoor 2009	7	19	6	19	0.5%	1.17 [0.48 , 2.83]
Meduri 2009	22	48	4	31	0.4%	3.55 [1.35 , 9.32]
Snijders 2010	6	104	6	109	0.3%	1.05 [0.35 , 3.15]
Meijvis 2011	9	151	11	153	0.6%	0.83 [0.35 , 1.94]
Yildiz 2011	16	27	15	28	1.6%	1.11 [0.69 , 1.76]
Fernández-Serrano 2011	1	28	1	28	0.1%	1.00 [0.07 , 15.21]
Arabi 2011	33	39	26	36	3.6%	1.17 [0.92 , 1.49]
Sabry 2011	2	40	6	40	0.2%	0.33 [0.07 , 1.55]
Liu 2012	5	12	10	14	0.7%	0.58 [0.28 , 1.23]
Rezk 2013	0	18	3	9	0.1%	0.08 [0.00 , 1.32]
Nafae 2013	4	60	6	20	0.3%	0.22 [0.07 , 0.71]
Gordon 2014	7	31	7	30	0.5%	0.97 [0.39 , 2.43]
Blum 2015	16	402	13	400	0.8%	1.22 [0.60 , 2.51]
Mirea 2014	44	117	22	54	2.0%	0.92 [0.62 , 1.37]
Torres 2015	6	59	9	61	0.4%	0.69 [0.26 , 1.82]
Tongyoo 2016	22	98	27	99	1.4%	0.82 [0.50 , 1.34]
Gordon 2016	62	208	57	213	2.8%	1.11 [0.82 , 1.51]
Keh 2016	15	171	14	170	0.8%	1.07 [0.53 , 2.14]
Lv 2017	23	58	19	60	1.4%	1.25 [0.77 , 2.04]
Menon 2017	1	23	3	26	0.1%	0.38 [0.04 , 3.38]
Venkatesh 2018	410	1853	448	1860	5.7%	0.92 [0.82 , 1.03]
Doluee 2018	54	80	58	80	4.2%	0.93 [0.76 , 1.14]
Annane 2018	207	614	244	627	5.2%	0.87 [0.75 , 1.00]
Wani 2020	20	50	21	50	1.5%	0.95 [0.59 , 1.52]
Chang 2020	11	40	14	40	0.9%	0.79 [0.41 , 1.52]
Iglesias 2020	26	68	14	69	1.2%	1.88 [1.08 , 3.29]
Villar 2020	14	96	29	89	1.1%	0.45 [0.25 , 0.79]
Dequin 2020	11	76	20	73	0.9%	0.53 [0.27 , 1.02]
Tomazini 2020	85	151	91	148	4.4%	0.92 [0.76 , 1.11]
Mohamed 2020	18	45	19	45	1.4%	0.95 [0.58 , 1.55]
Angus 2020	55	278	27	101	2.0%	0.74 [0.50 , 1.10]
Moskowitz 2020	35	101	29	99	1.9%	1.18 [0.79 , 1.78]
Ghanei 2021	4	116	12	220	0.3%	0.63 [0.21 , 1.92]
Horby 2021	482	2104	1110	4321	6.1%	0.89 [0.81 , 0.98]
Sevransky 2021	56	252	60	249	2.6%	0.92 [0.67 , 1.27]
Sai 2021	41	75	45	75	3.1%	0.91 [0.69 , 1.20]
Corral-Gudino 2021	7	35	5	29	0.4%	1.16 [0.41 , 3.27]
Wittermans 2021	4	203	7	198	0.3%	0.56 [0.17 , 1.87]

Risk Ratio
M-H, Random, 95% CI



Analysis 1.4. (Continued)

Corral-Gudino 2021	7	35	5	29	0.4%	1.16 [0.41 , 3.27]
Wittermans 2021	4	203	7	198	0.3%	0.56 [0.17 , 1.87]
Lyu 2022	79	213	77	213	3.5%	1.03 [0.80 , 1.32]
Agarwal 2022	22	61	25	59	1.7%	0.85 [0.54 , 1.33]
Birudaraju 2022	9	21	16	22	1.2%	0.59 [0.34 , 1.03]
Meduri 2022	30	297	34	287	1.6%	0.85 [0.54 , 1.36]
Mohamed 2023	13	53	20	53	1.1%	0.65 [0.36 , 1.17]
Pierre de Oliveira 2023	14	58	16	54	1.0%	0.81 [0.44 , 1.51]
Dequin 2023	25	400	47	395	1.6%	0.53 [0.33 , 0.84]
Subtotal (Wald[a])		**9781**		**11803**	**91.0%**	**0.89 [0.84 , 0.95]**
Total events:	2283		3045			
Test for overall effect: Z = 3.61 (P = 0.0003)						
Heterogeneity: Tau² (DL[b]) = 0.01; Chi² = 75.37, df = 60 (P = 0.09); I² = 20%						
1.4.2 Short course of high-dose corticosteroids						
Schumer 1976	9	86	33	86	0.8%	0.27 [0.14 , 0.53]
Sprung 1984	33	43	11	16	2.2%	1.12 [0.77 , 1.61]
VASSCSG 1987	23	112	24	111	1.4%	0.95 [0.57 , 1.58]
Bone 1987	65	191	48	190	2.7%	1.35 [0.98 , 1.84]
Luce 1988	22	38	20	37	1.9%	1.07 [0.72 , 1.60]
Subtotal (Wald[a])		**470**		**440**	**9.0%**	**0.90 [0.59 , 1.36]**
Total events:	152		136			
Test for overall effect: Z = 0.50 (P = 0.62)						
Heterogeneity: Tau² (DL[b]) = 0.17; Chi² = 18.73, df = 4 (P = 0.0009); I² = 79%						
Total (Wald[a])		**10251**		**12243**	**100.0%**	**0.90 [0.84 , 0.96]**
Total events:	2435		3181			
Test for overall effect: Z = 3.27 (P = 0.001)						
Test for subgroup differences: Chi² = 0.00, df = 1 (P = 0.97), I² = 0%						
Heterogeneity: Tau² (DL[b]) = 0.02; Chi² = 95.83, df = 65 (P = 0.008); I² = 32%						



Footnotes

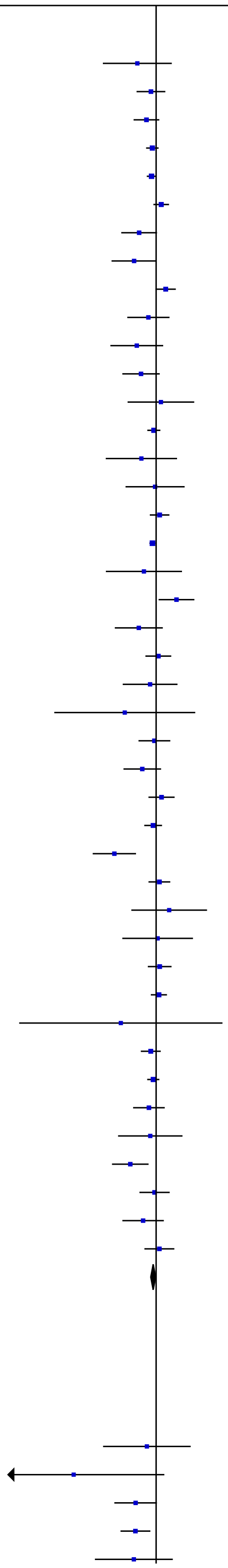
[a]CI calculated by Wald-type method.

[b]Tau² calculated by DerSimonian and Laird method.

Analysis 1.5. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 5: 28-day all-cause mortality based on mode of drug administration

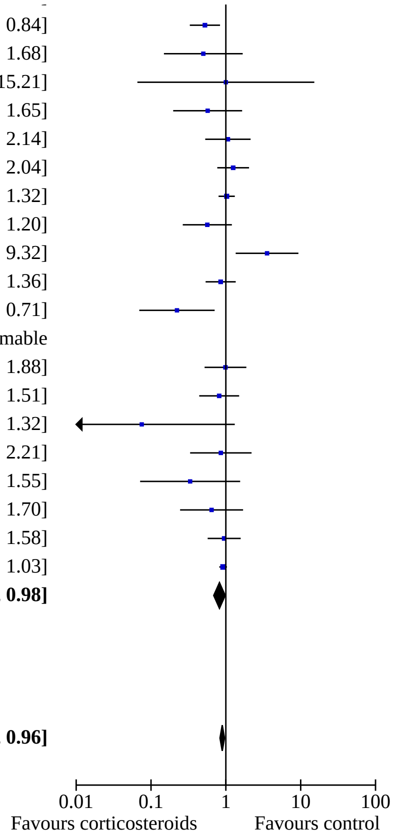
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
1.5.1 Intravenous bolus						
Aboab 2008	3	10	7	13	0.4%	0.56 [0.19 , 1.63]
Agarwal 2022	22	61	25	59	1.7%	0.85 [0.54 , 1.33]
Angus 2020	55	278	27	101	2.0%	0.74 [0.50 , 1.10]
Annane 2002	82	151	91	149	4.4%	0.89 [0.73 , 1.08]
Annane 2018	207	614	244	627	5.1%	0.87 [0.75 , 1.00]
Arabi 2011	33	39	26	36	3.7%	1.17 [0.92 , 1.49]
Birudaraju 2022	9	21	16	22	1.2%	0.59 [0.34 , 1.03]
Bollaert 1998	7	22	12	19	0.8%	0.50 [0.25 , 1.02]
Bone 1987	65	191	48	190	2.8%	1.35 [0.98 , 1.84]
Chang 2020	11	40	14	40	0.9%	0.79 [0.41 , 1.52]
Chawla 1999	6	23	10	21	0.6%	0.55 [0.24 , 1.25]
Cicarelli 2007	7	14	12	15	1.2%	0.63 [0.35 , 1.12]
Corral-Gudino 2021	7	35	5	29	0.4%	1.16 [0.41 , 3.27]
Doluee 2018	54	80	58	80	4.2%	0.93 [0.76 , 1.14]
Ghanei 2021	4	116	12	220	0.4%	0.63 [0.21 , 1.92]
Gordon 2014	7	31	7	30	0.5%	0.97 [0.39 , 2.43]
Gordon 2016	62	208	57	213	2.9%	1.11 [0.82 , 1.51]
Horby 2021	482	2104	1110	4321	6.0%	0.89 [0.81 , 0.98]
Hu 2009	4	38	6	39	0.3%	0.68 [0.21 , 2.23]
Iglesias 2020	26	68	14	69	1.2%	1.88 [1.08 , 3.29]
Liu 2012	5	12	10	14	0.7%	0.58 [0.28 , 1.23]
Luce 1988	22	38	20	37	2.0%	1.07 [0.72 , 1.60]
Meijvis 2011	9	151	11	153	0.6%	0.83 [0.35 , 1.94]
Menon 2017	1	23	3	26	0.1%	0.38 [0.04 , 3.38]
Mohamed 2020	18	45	19	45	1.5%	0.95 [0.58 , 1.55]
Mohamed 2023	13	53	20	53	1.1%	0.65 [0.36 , 1.17]
Moskowitz 2020	35	101	29	99	2.0%	1.18 [0.79 , 1.78]
Sai 2021	41	75	45	75	3.2%	0.91 [0.69 , 1.20]
Schumer 1976	9	86	33	86	0.9%	0.27 [0.14 , 0.53]
Sevransky 2021	56	252	50	249	2.6%	1.11 [0.79 , 1.55]
Slusher 1996	6	36	4	36	0.3%	1.50 [0.46 , 4.87]
Snijders 2010	6	104	6	109	0.4%	1.05 [0.35 , 3.15]
Sprung 1984	33	43	11	16	2.3%	1.12 [0.77 , 1.61]
Sprung 2008	86	251	78	248	3.5%	1.09 [0.85 , 1.40]
Tagaro 2017	0	30	1	30	0.0%	0.33 [0.01 , 7.87]
Tandan 2005	11	14	13	14	2.8%	0.85 [0.62 , 1.15]
Tomazini 2020	85	151	91	148	4.5%	0.92 [0.76 , 1.11]
Tongyoo 2016	22	104	27	102	1.5%	0.80 [0.49 , 1.31]
Valoor 2009	5	19	6	19	0.4%	0.83 [0.31 , 2.27]
Villar 2020	14	96	29	89	1.2%	0.45 [0.25 , 0.79]
Wani 2020	20	50	21	50	1.6%	0.95 [0.59 , 1.52]
Yildiz 2002	8	20	12	20	1.0%	0.67 [0.35 , 1.27]
Yildiz 2011	16	27	15	28	1.6%	1.11 [0.69 , 1.76]
Subtotal (Wald[a])		**5925**		**8039**	**76.9%**	**0.91 [0.85 , 0.98]**
Total events:	1674		2355			
Test for overall effect: Z = 2.42 (P = 0.02)						
Heterogeneity: Tau² (DL[b]) = 0.01; Chi² = 61.88, df = 42 (P = 0.02); I² = 32%						
1.5.2 Continuous infusion						
Briegel 1999	3	20	4	20	0.2%	0.75 [0.19 , 2.93]
Confalonieri 2005	0	23	6	23	0.1%	0.08 [0.00 , 1.29]
Dequin 2020	11	76	20	73	0.9%	0.53 [0.27 , 1.02]
Dequin 2023	25	400	47	395	1.6%	0.53 [0.33 , 0.84]
El Ghamrawy 2006	3	17	6	17	0.3%	0.50 [0.15 , 1.68]

Risk Ratio M-H, Random, 95% CI



Analysis 1.5. (Continued)

Dequin 2023	25	400	47	395	1.6%	0.53 [0.33 , 0.84]
El Ghamrawy 2006	3	17	6	17	0.3%	0.50 [0.15 , 1.68]
Fernández-Serrano 2011	1	28	1	28	0.1%	1.00 [0.07 , 15.21]
Huang 2014	4	20	7	20	0.4%	0.57 [0.20 , 1.65]
Keh 2016	15	177	14	176	0.8%	1.07 [0.53 , 2.14]
Lv 2017	23	58	19	60	1.5%	1.25 [0.77 , 2.04]
Lyu 2022	79	213	77	213	3.6%	1.03 [0.80 , 1.32]
Meduri 2007	10	42	8	19	0.7%	0.57 [0.27 , 1.20]
Meduri 2009	22	48	4	31	0.5%	3.55 [1.35 , 9.32]
Meduri 2022	30	297	34	287	1.7%	0.85 [0.54 , 1.36]
Nafae 2013	4	60	6	20	0.3%	0.22 [0.07 , 0.71]
Nagy 2013	0	29	0	30		Not estimable
Oppert 2005	10	23	11	25	1.0%	0.99 [0.52 , 1.88]
Pierre de Oliveira 2023	14	58	16	54	1.1%	0.81 [0.44 , 1.51]
Rezk 2013	0	18	3	9	0.1%	0.08 [0.00 , 1.32]
Rinaldi 2006	6	26	7	26	0.5%	0.86 [0.33 , 2.21]
Sabry 2011	2	40	6	40	0.2%	0.33 [0.07 , 1.55]
Torres 2015	6	61	9	59	0.5%	0.64 [0.24 , 1.70]
VASSCSG 1987	23	112	24	111	1.4%	0.95 [0.57 , 1.58]
Venkatesh 2018	410	1853	448	1860	5.7%	0.92 [0.82 , 1.03]
Subtotal (Wald[a])		**3699**		**3596**	**23.1%**	**0.82 [0.69 , 0.98]**
Total events:	701		777			
Test for overall effect: Z = 2.20 (P = 0.03)						
Heterogeneity: Tau² (DL[b]) = 0.05; Chi² = 35.50, df = 21 (P = 0.02); I² = 41%						
Total (Wald[a])		**9624**		**11635**	**100.0%**	**0.89 [0.84 , 0.96]**
Total events:	2375		3132			
Test for overall effect: Z = 3.18 (P = 0.001)						
Test for subgroup differences: Chi² = 1.17, df = 1 (P = 0.28), I² = 14.6%						
Heterogeneity: Tau² (DL[b]) = 0.02; Chi² = 97.71, df = 64 (P = 0.004); I² = 35%						



Footnotes

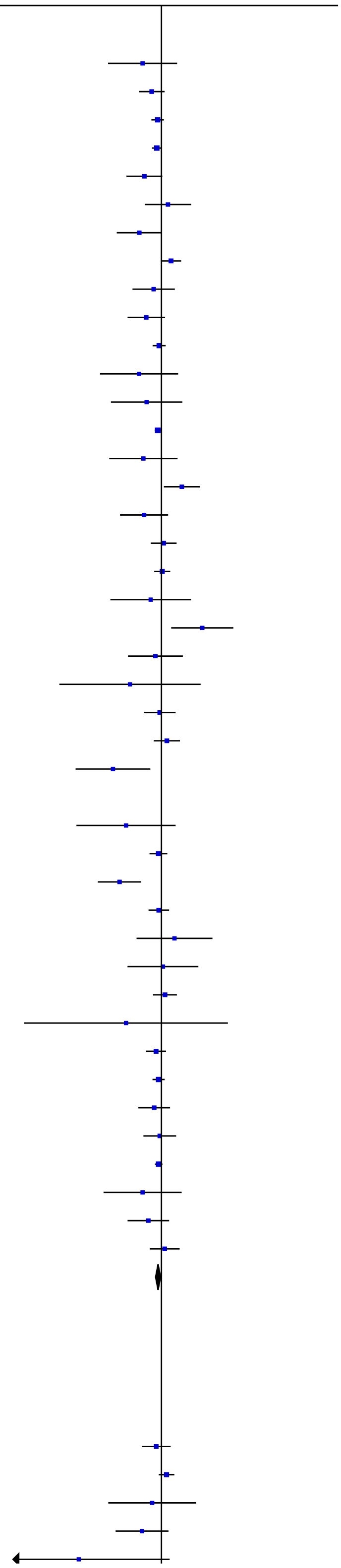
[a]CI calculated by Wald-type method.

[b]Tau² calculated by DerSimonian and Laird method.

Analysis 1.6. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 6: 28-day all-cause mortality based on mode of drug termination

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
1.6.1 Without taper off						
Aboab 2008	3	10	7	13	0.4%	0.56 [0.19 , 1.63]
Angus 2020	55	278	27	101	2.0%	0.74 [0.50 , 1.10]
Annane 2002	82	151	91	149	4.4%	0.89 [0.73 , 1.08]
Annane 2018	207	614	244	627	5.3%	0.87 [0.75 , 1.00]
Birudaraju 2022	9	21	16	22	1.2%	0.59 [0.34 , 1.03]
Blum 2015	16	402	13	400	0.8%	1.22 [0.60 , 2.51]
Bollaert 1998	7	22	12	19	0.8%	0.50 [0.25 , 1.02]
Bone 1987	65	191	48	190	2.8%	1.35 [0.98 , 1.84]
Chang 2020	11	40	14	40	0.9%	0.79 [0.41 , 1.52]
Cicarelli 2007	7	14	12	15	1.1%	0.63 [0.35 , 1.12]
Doluee 2018	54	80	58	80	4.3%	0.93 [0.76 , 1.14]
El Ghamrawy 2006	3	17	6	17	0.3%	0.50 [0.15 , 1.68]
Ghanei 2021	4	116	12	220	0.4%	0.63 [0.21 , 1.92]
Horby 2021	482	2104	1110	4321	6.2%	0.89 [0.81 , 0.98]
Huang 2014	4	20	7	20	0.4%	0.57 [0.20 , 1.65]
Iglesias 2020	26	68	14	69	1.2%	1.88 [1.08 , 3.29]
Liu 2012	5	12	10	14	0.7%	0.58 [0.28 , 1.23]
Luce 1988	22	38	20	37	2.0%	1.07 [0.72 , 1.60]
Lyu 2022	79	213	77	213	3.6%	1.03 [0.80 , 1.32]
McHardy 1972	3	40	9	86	0.3%	0.72 [0.20 , 2.51]
Meduri 2009	22	48	4	31	0.5%	3.55 [1.35 , 9.32]
Meijvis 2011	9	151	11	153	0.6%	0.83 [0.35 , 1.94]
Menon 2017	1	23	3	26	0.1%	0.38 [0.04 , 3.38]
Mohamed 2020	18	45	19	45	1.5%	0.95 [0.58 , 1.55]
Moskowitz 2020	35	101	29	99	2.0%	1.18 [0.79 , 1.78]
Nafae 2013	4	60	6	20	0.3%	0.22 [0.07 , 0.71]
Nagy 2013	0	29	0	30		Not estimable
Sabry 2011	2	40	6	40	0.2%	0.33 [0.07 , 1.55]
Sai 2021	41	75	45	75	3.2%	0.91 [0.69 , 1.20]
Schumer 1976	9	86	33	86	0.9%	0.27 [0.14 , 0.53]
Sevransky 2021	56	252	60	249	2.7%	0.92 [0.67 , 1.27]
Slusher 1996	6	36	4	36	0.3%	1.50 [0.46 , 4.87]
Snijders 2010	6	104	6	109	0.4%	1.05 [0.35 , 3.15]
Sprung 1984	33	43	11	16	2.3%	1.12 [0.77 , 1.61]
Tagaro 2017	0	30	1	30	0.0%	0.33 [0.01 , 7.87]
Tandan 2005	11	14	13	14	2.8%	0.85 [0.62 , 1.15]
Tomazini 2020	85	151	91	148	4.5%	0.92 [0.76 , 1.11]
Tongyoo 2016	22	104	27	102	1.5%	0.80 [0.49 , 1.31]
VASSCSG 1987	23	112	24	111	1.4%	0.95 [0.57 , 1.58]
Venkatesh 2018	410	1853	448	1860	5.8%	0.92 [0.82 , 1.03]
Wittermans 2021	4	203	7	198	0.3%	0.56 [0.17 , 1.87]
Yildiz 2002	8	20	12	20	1.0%	0.67 [0.35 , 1.27]
Yildiz 2011	16	27	15	28	1.6%	1.11 [0.69 , 1.76]
Subtotal (Wald[a])		**8058**		**10179**	**73.0%**	**0.90 [0.84 , 0.97]**
Total events:	1965		2682			
Test for overall effect: Z = 2.68 (P = 0.007)						
Heterogeneity: Tau² (DL[b]) = 0.01; Chi² = 62.28, df = 41 (P = 0.02); I² = 34%						
1.6.2 With taper off						
Agarwal 2022	22	61	25	59	1.7%	0.85 [0.54 , 1.33]
Arabi 2011	33	39	26	36	3.7%	1.17 [0.92 , 1.49]
Briegel 1999	3	20	4	20	0.2%	0.75 [0.19 , 2.93]
Chawla 1999	6	23	10	21	0.6%	0.55 [0.24 , 1.25]
Confalonieri 2005	0	23	6	23	0.1%	0.08 [0.00 , 1.29]

Risk Ratio M-H, Random, 95% CI



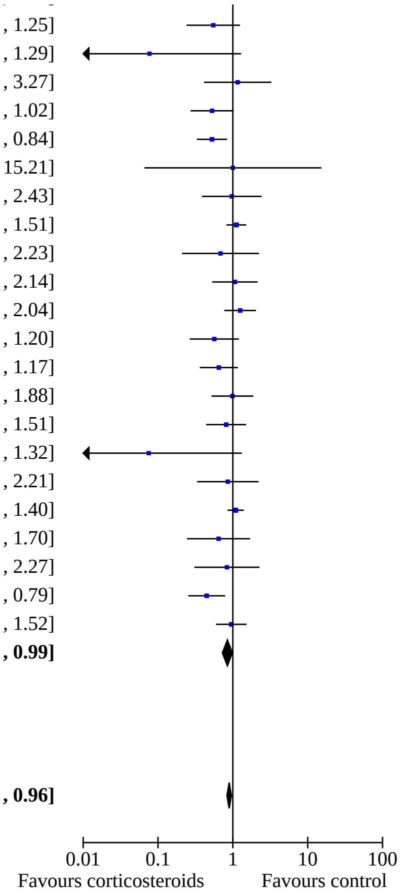
Analysis 1.6. (Continued)

Chawla 1999	6	23	10	21	0.6%	0.55 [0.24 , 1.25]
Confalonieri 2005	0	23	6	23	0.1%	0.08 [0.00 , 1.29]
Corral-Gudino 2021	7	35	5	29	0.4%	1.16 [0.41 , 3.27]
Dequin 2020	11	76	20	73	0.9%	0.53 [0.27 , 1.02]
Dequin 2023	25	400	47	396	1.6%	0.53 [0.33 , 0.84]
Fernández-Serrano 2011	1	28	1	28	0.1%	1.00 [0.07 , 15.21]
Gordon 2014	7	31	7	30	0.5%	0.97 [0.39 , 2.43]
Gordon 2016	62	208	57	213	2.9%	1.11 [0.82 , 1.51]
Hu 2009	4	38	6	39	0.3%	0.68 [0.21 , 2.23]
Keh 2016	15	177	14	176	0.8%	1.07 [0.53 , 2.14]
Lv 2017	23	58	19	60	1.5%	1.25 [0.77 , 2.04]
Meduri 2007	10	42	8	19	0.7%	0.57 [0.27 , 1.20]
Mohamed 2023	13	53	20	53	1.1%	0.65 [0.36 , 1.17]
Oppert 2005	10	23	11	25	1.0%	0.99 [0.52 , 1.88]
Pierre de Oliveira 2023	14	58	16	54	1.0%	0.81 [0.44 , 1.51]
Rezk 2013	0	18	3	9	0.1%	0.08 [0.00 , 1.32]
Rinaldi 2006	6	26	7	26	0.5%	0.86 [0.33 , 2.21]
Sprung 2008	86	251	78	248	3.6%	1.09 [0.85 , 1.40]
Torres 2015	6	61	9	59	0.5%	0.64 [0.24 , 1.70]
Valoor 2009	5	19	6	19	0.4%	0.83 [0.31 , 2.27]
Villar 2020	14	96	29	89	1.2%	0.45 [0.25 , 0.79]
Wani 2020	20	50	21	50	1.6%	0.95 [0.59 , 1.52]
Subtotal (Wald[a])		**1914**		**1854**	**27.0%**	**0.85 [0.73 , 0.99]**
Total events:	403		455			

Test for overall effect: Z = 2.15 (P = 0.03)
Heterogeneity: Tau² (DL[b]) = 0.04; Chi² = 36.85, df = 24 (P = 0.05); I² = 35%

Total (Wald[a])		**9972**		**12033**	**100.0%**	**0.89 [0.83 , 0.96]**
Total events:	2368		3137			

Test for overall effect: Z = 3.28 (P = 0.001)
Test for subgroup differences: Chi² = 0.54, df = 1 (P = 0.46), I² = 0%
Heterogeneity: Tau² (DL[b]) = 0.02; Chi² = 97.75, df = 66 (P = 0.007); I² = 32%

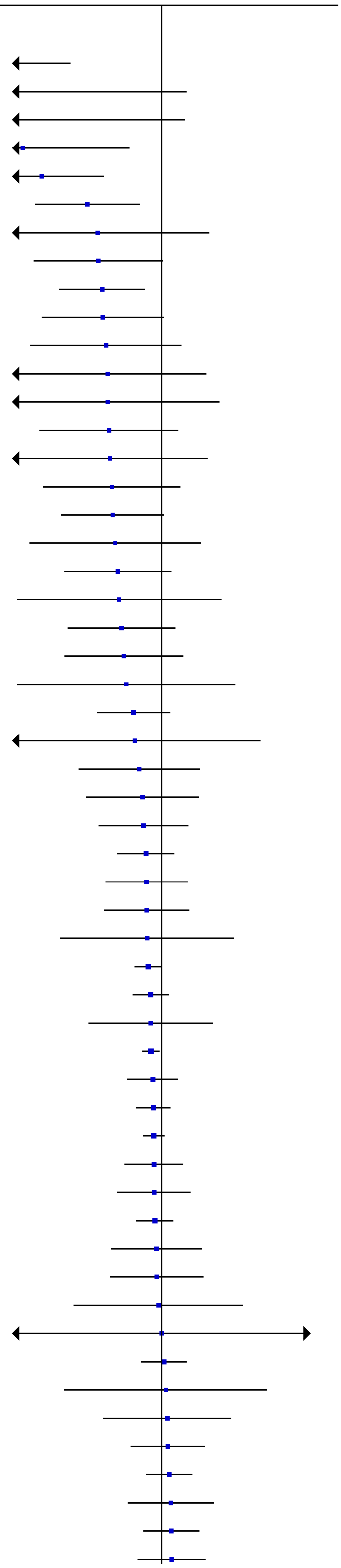


Footnotes
[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.

Analysis 1.7. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 7: 28-day all-cause mortality by subgroups based on age

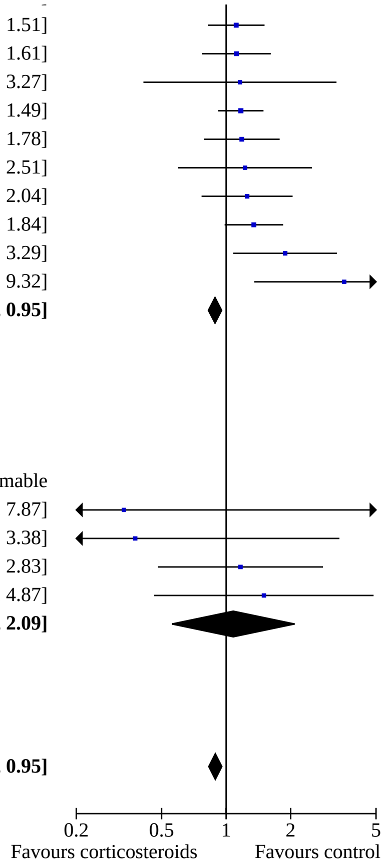
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
1.7.1 Adults						
Wani 2020	0	50	21	50	0.1%	0.02 [0.00 , 0.37]
Rezk 2013	0	18	3	9	0.1%	0.08 [0.00 , 1.32]
Confalonieri 2005	0	23	6	23	0.1%	0.08 [0.00 , 1.29]
Nafae 2013	4	60	6	20	0.3%	0.22 [0.07 , 0.71]
Schumer 1976	9	86	33	86	0.9%	0.27 [0.14 , 0.53]
Villar 2020	14	96	29	89	1.2%	0.45 [0.25 , 0.79]
El Ghamrawy 2006	3	17	6	17	0.3%	0.50 [0.15 , 1.68]
Bollaert 1998	7	22	12	19	0.9%	0.50 [0.25 , 1.02]
Dequin 2023	25	400	47	395	1.7%	0.53 [0.33 , 0.84]
Dequin 2020	11	76	20	73	0.9%	0.53 [0.27 , 1.02]
Chawla 1999	6	23	10	21	0.7%	0.55 [0.24 , 1.25]
Aboab 2008	3	10	7	13	0.4%	0.56 [0.19 , 1.63]
Wittermans 2021	4	203	7	198	0.3%	0.56 [0.17 , 1.87]
Meduri 2007	10	42	8	19	0.8%	0.57 [0.27 , 1.20]
Huang 2014	4	20	7	20	0.4%	0.57 [0.20 , 1.65]
Liu 2012	5	12	10	14	0.8%	0.58 [0.28 , 1.23]
Birudaraju 2022	9	21	16	22	1.3%	0.59 [0.34 , 1.03]
Kurungundla 2008	4	11	6	10	0.5%	0.61 [0.24 , 1.54]
Cicarelli 2007	7	14	12	15	1.2%	0.63 [0.35 , 1.12]
Ghanei 2021	4	116	12	220	0.4%	0.63 [0.21 , 1.92]
Mohamed 2023	13	53	20	53	1.2%	0.65 [0.36 , 1.17]
Yildiz 2002	8	20	12	20	1.0%	0.67 [0.35 , 1.27]
Hu 2009	4	38	6	39	0.3%	0.68 [0.21 , 2.23]
Angus 2020	55	278	27	101	2.1%	0.74 [0.50 , 1.10]
Briegel 1999	3	20	4	20	0.3%	0.75 [0.19 , 2.93]
Chang 2020	11	40	14	40	1.0%	0.79 [0.41 , 1.52]
Pierre de Oliveira 2023	14	58	16	54	1.1%	0.81 [0.44 , 1.51]
Tongyoo 2016	22	98	27	99	1.5%	0.82 [0.50 , 1.34]
Tandan 2005	11	14	13	14	2.8%	0.85 [0.62 , 1.15]
Agarwal 2022	22	61	25	59	1.8%	0.85 [0.54 , 1.33]
Meduri 2022	30	297	34	287	1.7%	0.85 [0.54 , 1.36]
Rinaldi 2006	6	26	7	26	0.5%	0.86 [0.33 , 2.21]
Annane 2018	207	614	244	627	4.9%	0.87 [0.75 , 1.00]
Annane 2002	82	151	91	149	4.2%	0.89 [0.73 , 1.08]
Oppert 2005	9	23	11	25	0.9%	0.89 [0.45 , 1.75]
Horby 2021	482	2104	1110	4321	5.6%	0.89 [0.81 , 0.98]
Sai 2021	41	75	45	75	3.2%	0.91 [0.69 , 1.20]
Tomazini 2020	85	151	91	148	4.3%	0.92 [0.76 , 1.11]
Venkatesh 2018	410	1853	448	1860	5.3%	0.92 [0.82 , 1.03]
Sevransky 2021	56	252	60	249	2.7%	0.92 [0.67 , 1.27]
Mirea 2014	44	117	22	54	2.1%	0.92 [0.62 , 1.37]
Doluee 2018	54	80	58	80	4.1%	0.93 [0.76 , 1.14]
Mohamed 2020	18	45	19	45	1.5%	0.95 [0.58 , 1.55]
VASSCSG 1987	23	112	24	111	1.5%	0.95 [0.57 , 1.58]
Gordon 2014	7	31	7	30	0.5%	0.97 [0.39 , 2.43]
Fernández-Serrano 2011	1	28	1	28	0.1%	1.00 [0.07 , 15.21]
Lyu 2022	79	213	77	213	3.5%	1.03 [0.80 , 1.32]
Snijders 2010	6	104	6	109	0.4%	1.05 [0.35 , 3.15]
Keh 2016	15	171	14	170	0.9%	1.07 [0.53 , 2.14]
Luce 1988	22	38	20	37	2.0%	1.07 [0.72 , 1.60]
Sprung 2008	86	251	78	248	3.5%	1.09 [0.85 , 1.40]
Yildiz 2011	16	27	15	28	1.7%	1.11 [0.69 , 1.76]
Gordon 2016	62	208	57	213	2.9%	1.11 [0.82 , 1.51]
Sprung 1984	33	43	11	16	2.3%	1.12 [0.77 , 1.61]

Risk Ratio M-H, Random, 95% CI



Analysis 1.7. (Continued)

Gordon 2016	62	208	57	213	2.9%	1.11 [0.82 , 1.51]
Sprung 1984	33	43	11	16	2.3%	1.12 [0.77 , 1.61]
Corral-Gudino 2021	7	35	5	29	0.4%	1.16 [0.41 , 3.27]
Arabi 2011	33	39	26	36	3.6%	1.17 [0.92 , 1.49]
Moskowitz 2020	35	101	29	99	2.0%	1.18 [0.79 , 1.78]
Blum 2015	16	402	13	400	0.8%	1.22 [0.60 , 2.51]
Lv 2017	23	58	19	60	1.5%	1.25 [0.77 , 2.04]
Bone 1987	65	191	48	190	2.8%	1.35 [0.98 , 1.84]
Iglesias 2020	26	68	14	69	1.3%	1.88 [1.08 , 3.29]
Meduri 2009	22	48	4	31	0.5%	3.55 [1.35 , 9.32]
Subtotal (Wald[a])		**9956**		**11895**	**99.0%**	**0.89 [0.83 , 0.95]**
Total events:	2393		3150			
Test for overall effect: Z = 3.29 (P = 0.001)						
Heterogeneity: Tau² (DL[b]) = 0.02; Chi² = 101.42, df = 61 (P = 0.0009); I² = 40%						
1.7.2 Children						
Nagy 2013	0	29	0	30		Not estimable
Tagaro 2017	0	30	1	30	0.0%	0.33 [0.01 , 7.87]
Menon 2017	1	23	3	26	0.1%	0.38 [0.04 , 3.38]
Valoor 2009	7	19	6	19	0.6%	1.17 [0.48 , 2.83]
Slusher 1996	6	36	4	36	0.3%	1.50 [0.46 , 4.87]
Subtotal (Wald[a])		**137**		**141**	**1.0%**	**1.08 [0.56 , 2.09]**
Total events:	14		14			
Test for overall effect: Z = 0.23 (P = 0.82)						
Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 1.78, df = 3 (P = 0.62); I² = 0%						
Total (Wald[a])		**10093**		**12036**	**100.0%**	**0.89 [0.83 , 0.95]**
Total events:	2407		3164			
Test for overall effect: Z = 3.26 (P = 0.001)						
Test for subgroup differences: Chi² = 0.34, df = 1 (P = 0.56), I² = 0%						
Heterogeneity: Tau² (DL[b]) = 0.02; Chi² = 103.44, df = 65 (P = 0.002); I² = 37%						



Footnotes

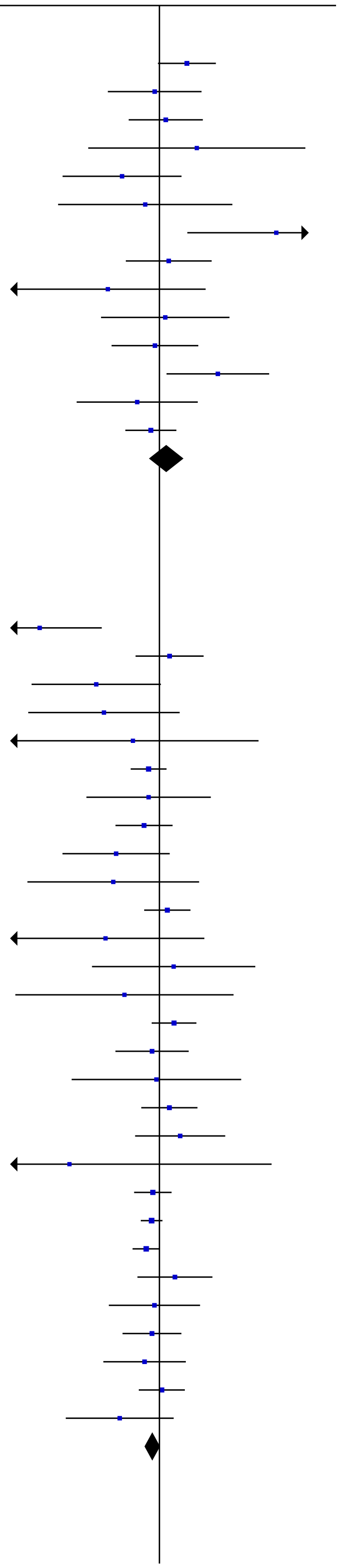
[a]CI calculated by Wald-type method.

[b]Tau² calculated by DerSimonian and Laird method.

Analysis 1.8. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 8: 28-day all-cause mortality by subgroups based on targeted population

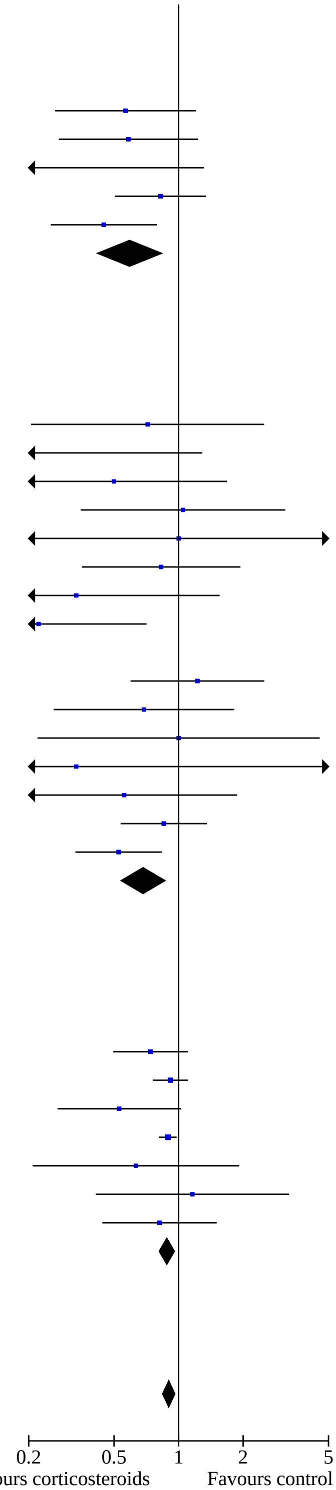
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
1.8.1 Sepsis						
Bone 1987	65	191	48	190	2.7%	1.35 [0.98 , 1.84]
VASSCSG 1987	23	112	24	111	1.3%	0.95 [0.57 , 1.58]
Luce 1988	22	38	20	37	1.9%	1.07 [0.72 , 1.60]
Slusher 1996	6	36	4	36	0.3%	1.50 [0.46 , 4.87]
Yildiz 2002	8	20	12	20	0.9%	0.67 [0.35 , 1.27]
Rinaldi 2006	6	26	7	26	0.4%	0.86 [0.33 , 2.21]
Meduri 2009	22	48	4	31	0.4%	3.55 [1.35 , 9.32]
Yildiz 2011	16	27	15	28	1.5%	1.11 [0.69 , 1.76]
Huang 2014	4	20	7	20	0.4%	0.57 [0.20 , 1.65]
Keh 2016	15	171	14	170	0.8%	1.07 [0.53 , 2.14]
Wani 2020	20	50	21	50	1.5%	0.95 [0.59 , 1.52]
Iglesias 2020	26	68	14	69	1.1%	1.88 [1.08 , 3.29]
Chang 2020	11	40	14	40	0.9%	0.79 [0.41 , 1.52]
Sai 2021	41	75	45	75	3.1%	0.91 [0.69 , 1.20]
Subtotal (Wald[a])		**922**		**903**	**17.2%**	**1.08 [0.90 , 1.28]**
Total events:	285		249			
Test for overall effect: Z = 0.82 (P = 0.41)						
Heterogeneity: Tau^2 (DL[b]) = 0.03; Chi^2 = 18.94, df = 13 (P = 0.12); I^2 = 31%						
1.8.2 Septic shock only						
Schumer 1976	9	86	33	86	0.8%	0.27 [0.14 , 0.53]
Sprung 1984	33	43	11	16	2.2%	1.12 [0.77 , 1.61]
Bollaert 1998	7	22	12	19	0.8%	0.50 [0.25 , 1.02]
Chawla 1999	6	23	10	21	0.6%	0.55 [0.24 , 1.25]
Briegel 1999	3	20	4	20	0.2%	0.75 [0.19 , 2.93]
Annane 2002	82	151	91	149	4.5%	0.89 [0.73 , 1.08]
Oppert 2005	9	23	11	25	0.8%	0.89 [0.45 , 1.75]
Tandan 2005	11	14	13	14	2.7%	0.85 [0.62 , 1.15]
Cicarelli 2007	7	14	12	15	1.1%	0.63 [0.35 , 1.12]
Kurungundla 2008	4	11	6	10	0.4%	0.61 [0.24 , 1.54]
Sprung 2008	86	251	78	248	3.5%	1.09 [0.85 , 1.40]
Aboab 2008	3	10	7	13	0.3%	0.56 [0.19 , 1.63]
Valoor 2009	7	19	6	19	0.5%	1.17 [0.48 , 2.83]
Hu 2009	4	38	6	39	0.3%	0.68 [0.21 , 2.23]
Arabi 2011	33	39	26	36	3.6%	1.17 [0.92 , 1.49]
Mirea 2014	44	117	22	54	1.9%	0.92 [0.62 , 1.37]
Gordon 2014	7	31	7	30	0.5%	0.97 [0.39 , 2.43]
Gordon 2016	62	208	57	213	2.8%	1.11 [0.82 , 1.51]
Lv 2017	23	58	19	60	1.4%	1.25 [0.77 , 2.04]
Menon 2017	1	23	3	26	0.1%	0.38 [0.04 , 3.38]
Doluee 2018	54	80	58	80	4.3%	0.93 [0.76 , 1.14]
Venkatesh 2018	410	1853	448	1860	6.0%	0.92 [0.82 , 1.03]
Annane 2018	207	614	244	627	5.4%	0.87 [0.75 , 1.00]
Moskowitz 2020	35	101	29	99	1.9%	1.18 [0.79 , 1.78]
Mohamed 2020	18	45	19	45	1.4%	0.95 [0.58 , 1.55]
Sevransky 2021	56	252	60	249	2.6%	0.92 [0.67 , 1.27]
Agarwal 2022	22	61	25	59	1.6%	0.85 [0.54 , 1.33]
Lyu 2022	79	213	77	213	3.5%	1.03 [0.80 , 1.32]
Mohamed 2023	13	53	20	53	1.0%	0.65 [0.36 , 1.17]
Subtotal (Wald[a])		**4473**		**4398**	**56.8%**	**0.93 [0.86 , 1.00]**
Total events:	1335		1414			
Test for overall effect: Z = 2.01 (P = 0.04)						
Heterogeneity: Tau^2 (DL[b]) = 0.01; Chi^2 = 35.95, df = 28 (P = 0.14); I^2 = 22%						

Risk Ratio
M-H, Random, 95% CI



Analysis 1.8. (Continued)

Heterogeneity: Tau^2 (DL[b]) = 0.01; Chi^2 = 35.95, df = 28 (P = 0.14); I^2 = 22%						
1.8.3 Sepsis and ARDS						
Meduri 2007	10	42	8	19	0.7%	0.57 [0.27 , 1.20]
Liu 2012	5	12	10	14	0.7%	0.58 [0.28 , 1.23]
Rezk 2013	0	18	3	9	0.1%	0.08 [0.00 , 1.32]
Tongyoo 2016	22	98	27	99	1.4%	0.82 [0.50 , 1.34]
Villar 2020	14	96	29	89	1.1%	0.45 [0.25 , 0.79]
Subtotal (Wald[a])		**266**		**230**	**3.9%**	**0.59 [0.42 , 0.83]**
Total events:	51		77			
Test for overall effect: Z = 3.03 (P = 0.002)						
Heterogeneity: Tau^2 (DL[b]) = 0.02; Chi^2 = 4.72, df = 4 (P = 0.32); I^2 = 15%						
1.8.4 Sepsis and community-acquired pneumonia						
McHardy 1972	3	40	9	86	0.3%	0.72 [0.20 , 2.51]
Confalonieri 2005	0	23	6	23	0.1%	0.08 [0.00 , 1.29]
El Ghamrawy 2006	3	17	6	17	0.3%	0.50 [0.15 , 1.68]
Snijders 2010	6	104	6	109	0.3%	1.05 [0.35 , 3.15]
Fernández-Serrano 2011	1	28	1	28	0.1%	1.00 [0.07 , 15.21]
Meijvis 2011	9	151	11	153	0.5%	0.83 [0.35 , 1.94]
Sabry 2011	2	40	6	40	0.2%	0.33 [0.07 , 1.55]
Nafae 2013	4	60	6	20	0.3%	0.22 [0.07 , 0.71]
Nagy 2013	0	29	0	30		Not estimable
Blum 2015	16	402	13	400	0.7%	1.22 [0.60 , 2.51]
Torres 2015	6	59	9	61	0.4%	0.69 [0.26 , 1.82]
Li 2016	3	29	3	29	0.2%	1.00 [0.22 , 4.55]
Tagaro 2017	0	30	1	30	0.0%	0.33 [0.01 , 7.87]
Wittermans 2021	4	203	7	198	0.3%	0.56 [0.17 , 1.87]
Meduri 2022	30	297	34	287	1.5%	0.85 [0.54 , 1.36]
Dequin 2023	25	400	47	395	1.5%	0.53 [0.33 , 0.84]
Subtotal (Wald[a])		**1912**		**1906**	**6.6%**	**0.68 [0.54 , 0.86]**
Total events:	112		165			
Test for overall effect: Z = 3.21 (P = 0.001)						
Heterogeneity: Tau^2 (DL[b]) = 0.00; Chi^2 = 13.07, df = 14 (P = 0.52); I^2 = 0%						
1.8.5 COVID-19						
Angus 2020	55	278	27	101	1.9%	0.74 [0.50 , 1.10]
Tomazini 2020	85	151	91	148	4.6%	0.92 [0.76 , 1.11]
Dequin 2020	11	76	20	73	0.8%	0.53 [0.27 , 1.02]
Horby 2021	482	2104	1110	4321	6.5%	0.89 [0.81 , 0.98]
Ghanei 2021	4	116	12	220	0.3%	0.63 [0.21 , 1.92]
Corral-Gudino 2021	7	35	5	29	0.4%	1.16 [0.41 , 3.27]
Pierre de Oliveira 2023	14	58	16	54	1.0%	0.81 [0.44 , 1.51]
Subtotal (Wald[a])		**2818**		**4946**	**15.5%**	**0.88 [0.81 , 0.95]**
Total events:	658		1281			
Test for overall effect: Z = 3.10 (P = 0.002)						
Heterogeneity: Tau^2 (DL[b]) = 0.00; Chi^2 = 3.93, df = 6 (P = 0.69); I^2 = 0%						
Total (Wald[a])		**10391**		**12383**	**100.0%**	**0.90 [0.84 , 0.96]**
Total events:	2441		3186			
Test for overall effect: Z = 3.23 (P = 0.001)						
Test for subgroup differences: Chi^2 = 16.05, df = 4 (P = 0.003), I^2 = 75.1%						
Heterogeneity: Tau^2 (DL[b]) = 0.01; Chi^2 = 96.25, df = 69 (P = 0.02); I^2 = 28%						



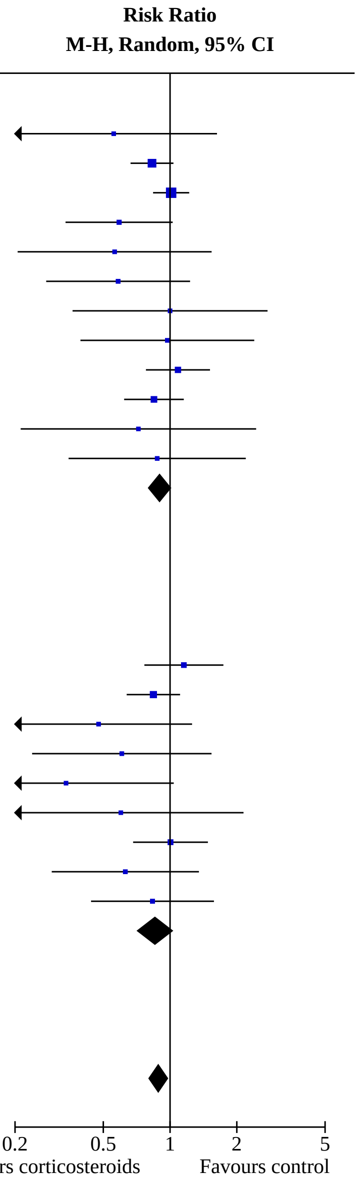
Footnotes

[a]CI calculated by Wald-type method.

[b]Tau^2 calculated by DerSimonian and Laird method.

Analysis 1.9. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 9: 28-day mortality in participants with versus without critical illness-related corticosteroid insufficiency

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
1.9.1 Patients with critical illness associated with corticosteroid insufficiency						
Aboab 2008	3	10	7	13	0.8%	0.56 [0.19 , 1.63]
Annane 2002	60	114	73	115	18.0%	0.83 [0.66 , 1.04]
Annane 2018	101	198	115	228	25.3%	1.01 [0.84 , 1.22]
Birudaraju 2022	9	21	16	22	2.9%	0.59 [0.34 , 1.03]
Bollaert 1998	3	8	6	9	0.9%	0.56 [0.21 , 1.54]
Liu 2012	5	12	10	14	1.6%	0.58 [0.28 , 1.23]
Meduri 2007	5	10	3	6	0.9%	1.00 [0.36 , 2.75]
Oppert 2005	5	12	6	14	1.1%	0.97 [0.39 , 2.40]
Sprung 2008	49	125	39	108	8.0%	1.09 [0.78 , 1.51]
Tandan 2005	11	14	13	14	9.3%	0.85 [0.62 , 1.15]
Yildiz 2002	2	5	5	9	0.6%	0.72 [0.21 , 2.44]
Yildiz 2011	3	6	8	14	1.1%	0.88 [0.35 , 2.20]
Subtotal (Wald[a])		**535**		**566**	**70.3%**	**0.90 [0.80 , 1.00]**
Total events:	256		301			
Test for overall effect: Z = 1.90 (P = 0.06)						
Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 8.74, df = 11 (P = 0.65); I² = 0%						
1.9.2 Patients without critical illness associated with corticosteroid insufficiency						
Annane 2002	22	36	18	34	5.3%	1.15 [0.77 , 1.74]
Annane 2018	61	184	67	170	11.6%	0.84 [0.64 , 1.11]
Bollaert 1998	4	14	6	10	0.9%	0.48 [0.18 , 1.26]
Kurungundla 2008	4	11	6	10	1.0%	0.61 [0.24 , 1.54]
Meduri 2007	5	53	5	18	0.7%	0.34 [0.11 , 1.04]
Oppert 2005	2	6	5	9	0.5%	0.60 [0.17 , 2.14]
Sprung 2008	34	118	39	136	5.9%	1.00 [0.68 , 1.48]
Yildiz 2002	6	15	7	11	1.5%	0.63 [0.29 , 1.35]
Yildiz 2011	10	21	8	14	2.2%	0.83 [0.44 , 1.58]
Subtotal (Wald[a])		**458**		**412**	**29.7%**	**0.85 [0.71 , 1.02]**
Total events:	148		161			
Test for overall effect: Z = 1.73 (P = 0.08)						
Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 8.21, df = 8 (P = 0.41); I² = 3%						
Total (Wald[a])		**993**		**978**	**100.0%**	**0.88 [0.80 , 0.97]**
Total events:	404		462			
Test for overall effect: Z = 2.55 (P = 0.01)						
Test for subgroup differences: Chi² = 0.20, df = 1 (P = 0.66), I² = 0%						
Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 17.09, df = 20 (P = 0.65); I² = 0%						



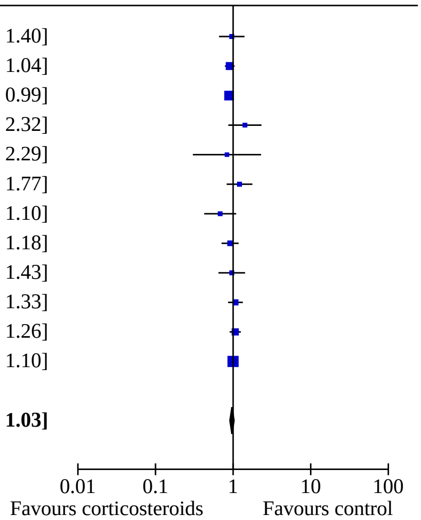
Footnotes

[a]CI calculated by Wald-type method.

[b]Tau² calculated by DerSimonian and Laird method.

Analysis 1.10. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 10: Long-term mortality

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Angus 2020	73	284	27	101	2.4%	0.96 [0.66 , 1.40]
Annane 2002	102	150	112	149	15.0%	0.90 [0.78 , 1.04]
Annane 2018	285	614	328	627	22.2%	0.89 [0.79 , 0.99]
Blum 2015	35	361	25	366	1.4%	1.42 [0.87 , 2.32]
Briegel 1999	5	20	6	20	0.3%	0.83 [0.30 , 2.29]
Keh 2016	45	177	37	176	2.4%	1.21 [0.83 , 1.77]
Meduri 2007	23	63	15	28	1.6%	0.68 [0.42 , 1.10]
Meduri 2022	79	260	84	253	5.3%	0.92 [0.71 , 1.18]
Mohamed 2023	25	53	26	53	2.2%	0.96 [0.65 , 1.43]
Sevransky 2021	102	252	94	249	7.0%	1.07 [0.86 , 1.33]
Sprung 2008	137	251	127	248	11.6%	1.07 [0.90 , 1.26]
Venkatesh 2018	571	1853	574	1860	28.5%	1.00 [0.91 , 1.10]
Total (Wald[a])		**4338**		**4130**	**100.0%**	**0.97 [0.91 , 1.03]**
Total events:	1482		1455			



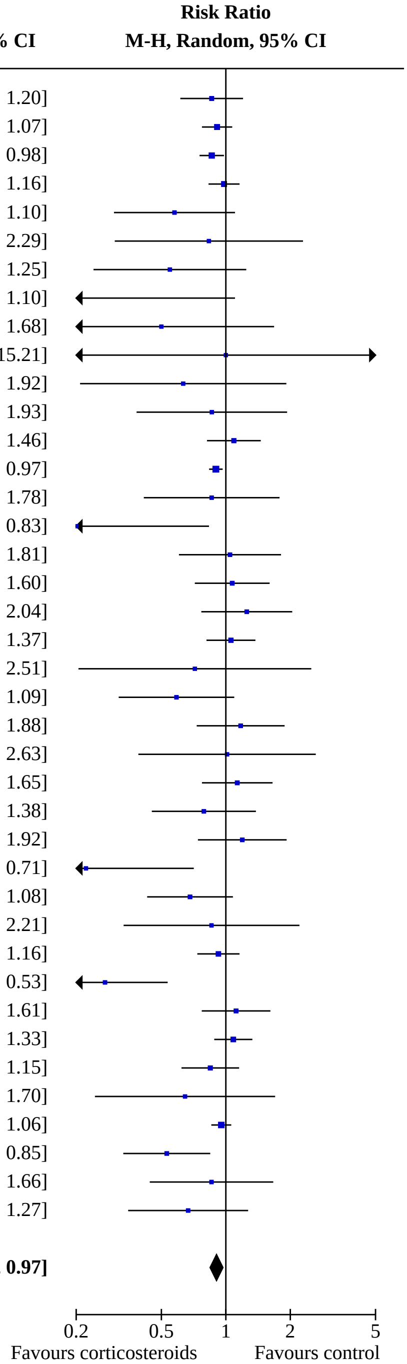
Test for overall effect: Z = 1.09 (P = 0.27)
Test for subgroup differences: Not applicable
Heterogeneity: Tau^2 (DL[b]) = 0.00; Chi^2 = 11.87, df = 11 (P = 0.37); I^2 = 7%

Footnotes

[a]CI calculated by Wald-type method.
[b]Tau^2 calculated by DerSimonian and Laird method.

Analysis 1.11. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 11: Hospital mortality

	Treatment		Control			Risk Ratio
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Random, 95% CI
Angus 2020	78	278	33	101	3.1%	0.86 [0.61 , 1.20]
Annane 2002	95	151	103	149	7.4%	0.91 [0.77 , 1.07]
Annane 2018	239	614	284	627	8.6%	0.86 [0.75 , 0.98]
Arabi 2011	34	39	32	36	7.2%	0.98 [0.83 , 1.16]
Bollaert 1998	8	22	12	19	1.0%	0.58 [0.30 , 1.10]
Briegel 1999	5	20	6	20	0.4%	0.83 [0.30 , 2.29]
Chawla 1999	6	23	10	21	0.7%	0.55 [0.24 , 1.25]
Confalonieri 2005	0	23	7	23	0.1%	0.07 [0.00 , 1.10]
El Ghamrawy 2006	3	17	6	17	0.3%	0.50 [0.15 , 1.68]
Fernández-Serrano 2011	1	28	1	28	0.1%	1.00 [0.07 , 15.21]
Ghanei 2021	4	116	12	220	0.4%	0.63 [0.21 , 1.92]
Gordon 2014	8	31	9	30	0.7%	0.86 [0.38 , 1.93]
Gordon 2016	66	208	62	213	3.9%	1.09 [0.82 , 1.46]
Horby 2021	688	2104	1573	4321	11.1%	0.90 [0.84 , 0.97]
Iglesias 2020	11	68	13	69	0.8%	0.86 [0.41 , 1.78]
Jamshidi 2021	2	29	10	29	0.2%	0.20 [0.05 , 0.83]
Keh 2016	23	190	22	190	1.4%	1.05 [0.60 , 1.81]
Luce 1988	22	38	20	37	2.3%	1.07 [0.72 , 1.60]
Lv 2017	23	58	19	60	1.7%	1.25 [0.77 , 2.04]
Lyu 2022	75	213	71	213	4.4%	1.06 [0.81 , 1.37]
McHardy 1972	3	40	9	86	0.3%	0.72 [0.20 , 2.51]
Meduri 2007	13	42	10	19	1.1%	0.59 [0.32 , 1.09]
Meduri 2022	34	291	28	281	1.8%	1.17 [0.73 , 1.88]
Meijvis 2011	8	151	8	153	0.5%	1.01 [0.39 , 2.63]
Mohamed 2020	26	45	23	45	2.6%	1.13 [0.77 , 1.65]
Mohamed 2023	15	53	19	53	1.3%	0.79 [0.45 , 1.38]
Moskowitz 2020	28	101	23	99	1.8%	1.19 [0.74 , 1.92]
Nafae 2013	4	60	6	20	0.3%	0.22 [0.07 , 0.71]
Pierre de Oliveira 2023	19	58	26	54	1.9%	0.68 [0.43 , 1.08]
Rinaldi 2006	6	26	7	26	0.5%	0.86 [0.33 , 2.21]
Sai 2021	48	75	52	75	5.3%	0.92 [0.74 , 1.16]
Schumer 1976	9	86	33	86	0.9%	0.27 [0.14 , 0.53]
Sprung 1984	33	43	11	16	2.7%	1.12 [0.77 , 1.61]
Sprung 2008	111	251	100	245	5.9%	1.08 [0.88 , 1.33]
Tandan 2005	11	14	13	14	3.5%	0.85 [0.62 , 1.15]
Torres 2015	6	61	9	59	0.5%	0.64 [0.24 , 1.70]
Venkatesh 2018	478	1853	504	1860	9.7%	0.95 [0.86 , 1.06]
Villar 2020	20	96	35	89	1.8%	0.53 [0.33 , 0.85]
Wani 2020	12	50	14	50	1.0%	0.86 [0.44 , 1.66]
Yildiz 2002	8	20	12	20	1.0%	0.67 [0.35 , 1.27]
Total (Wald[a])		**7686**		**9773**	**100.0%**	**0.90 [0.84 , 0.97]**
Total events:	2283		3247			



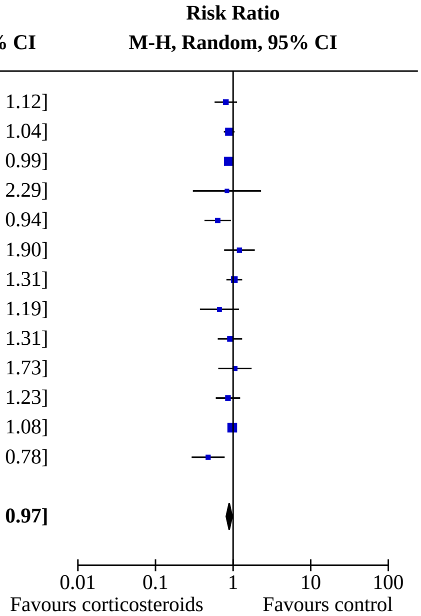
Test for overall effect: Z = 2.89 (P = 0.004)
Test for subgroup differences: Not applicable
Heterogeneity: Tau² (DL[b]) = 0.01; Chi² = 56.57, df = 39 (P = 0.03); I² = 31%

Footnotes

[a] CI calculated by Wald-type method.
[b] Tau² calculated by DerSimonian and Laird method.

Analysis 1.12. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 12: 90-day all-cause mortality

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Angus 2020	77	284	34	101	5.7%	0.81 [0.58 , 1.12]
Annane 2002	95	150	106	149	15.7%	0.89 [0.76 , 1.04]
Annane 2018	264	614	308	627	20.1%	0.88 [0.78 , 0.99]
Briegel 1999	5	20	6	20	0.7%	0.83 [0.30 , 2.29]
Dequin 2023	36	388	57	389	4.3%	0.63 [0.43 , 0.94]
Keh 2016	34	177	28	176	3.3%	1.21 [0.77 , 1.90]
Lyu 2022	86	213	83	213	9.8%	1.04 [0.82 , 1.31]
Meduri 2007	18	63	12	28	2.1%	0.67 [0.37 , 1.19]
Meduri 2022	47	286	50	277	5.0%	0.91 [0.63 , 1.31]
Mohamed 2020	19	45	18	45	2.9%	1.06 [0.64 , 1.73]
Tongyoo 2016	34	98	40	99	5.0%	0.86 [0.60 , 1.23]
Venkatesh 2018	511	1853	526	1860	22.4%	0.98 [0.88 , 1.08]
Villar 2020	18	96	35	89	2.9%	0.48 [0.29 , 0.78]
Total (Wald[a])		**4287**		**4073**	**100.0%**	**0.89 [0.82 , 0.97]**
Total events:	1244		1303			



Test for overall effect: Z = 2.58 (P = 0.010)

Test for subgroup differences: Not applicable

Heterogeneity: Tau^2 (DL[b]) = 0.01; Chi^2 = 16.91, df = 12 (P = 0.15); I^2 = 29%

Footnotes

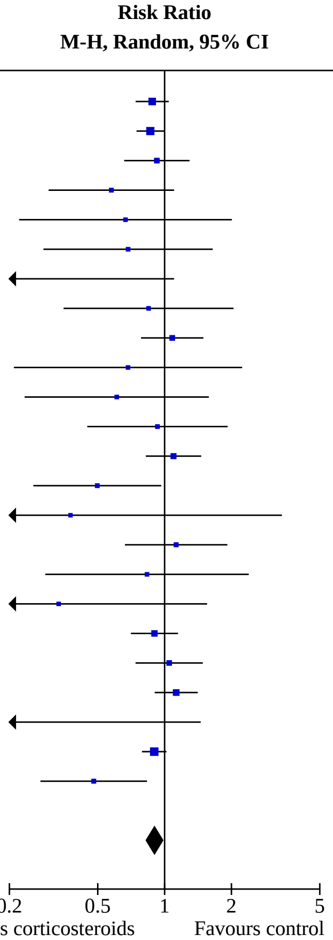
[a]CI calculated by Wald-type method.

[b]Tau^2 calculated by DerSimonian and Laird method.

Analysis 1.13. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 13: Intensive care unit mortality

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Annane 2002	90	151	101	149	13.1%	0.88 [0.74 , 1.04]
Annane 2018	217	614	257	627	15.8%	0.86 [0.75 , 0.99]
Arabi 2011	24	39	24	36	5.1%	0.92 [0.66 , 1.30]
Bollaert 1998	8	22	12	19	1.6%	0.58 [0.30 , 1.10]
Briegel 1999	4	20	6	20	0.6%	0.67 [0.22 , 2.01]
Chawla 1999	6	23	8	21	0.9%	0.68 [0.28 , 1.65]
Confalonieri 2005	0	23	7	23	0.1%	0.07 [0.00 , 1.10]
Gordon 2014	7	31	8	30	0.9%	0.85 [0.35 , 2.04]
Gordon 2016	56	208	53	213	5.5%	1.08 [0.78 , 1.50]
Hu 2009	4	38	6	39	0.5%	0.68 [0.21 , 2.23]
Iglesias 2020	6	68	10	69	0.8%	0.61 [0.23 , 1.58]
Keh 2016	13	190	14	190	1.3%	0.93 [0.45 , 1.92]
Lyu 2022	68	213	62	213	6.6%	1.10 [0.82 , 1.46]
Meduri 2007	11	42	10	19	1.5%	0.50 [0.26 , 0.97]
Menon 2017	1	23	3	26	0.1%	0.38 [0.04 , 3.38]
Moskowitz 2020	23	101	20	99	2.3%	1.13 [0.66 , 1.92]
Rinaldi 2006	5	26	6	26	0.6%	0.83 [0.29 , 2.39]
Sabry 2011	2	40	6	40	0.3%	0.33 [0.07 , 1.55]
Sai 2021	45	75	50	75	8.4%	0.90 [0.70 , 1.15]
Sevransky 2021	52	252	49	249	4.9%	1.05 [0.74 , 1.49]
Sprung 2008	102	251	89	247	9.5%	1.13 [0.90 , 1.41]
Torres 2015	1	42	6	46	0.2%	0.18 [0.02 , 1.45]
Venkatesh 2018	357	1853	399	1860	17.3%	0.90 [0.79 , 1.02]
Villar 2020	15	96	29	89	2.2%	0.48 [0.28 , 0.83]
Total (Wald[a])		**4441**		**4425**	**100.0%**	**0.90 [0.83 , 0.98]**
Total events:	1117		1235			

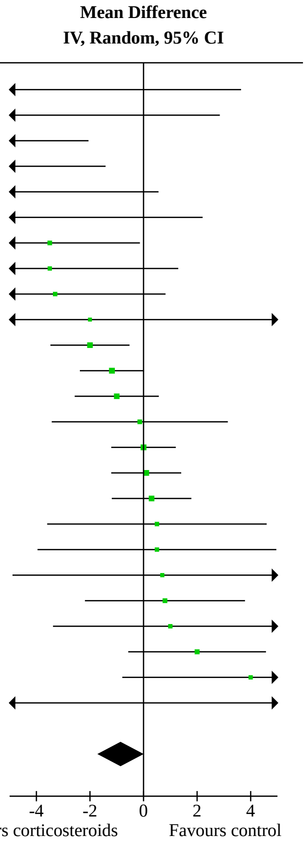
Test for overall effect: Z = 2.44 (P = 0.01)
Test for subgroup differences: Not applicable
Heterogeneity: Tau² (DL[b]) = 0.01; Chi² = 28.07, df = 23 (P = 0.21); I² = 18%



Footnotes
[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.

Analysis 1.14. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 14: Length of intensive care unit stay for all participants

Study or Subgroup	Treatment Mean	Treatment SD	Treatment Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Random, 95% CI
Briegel 1999	29	16	20	38	24	20	0.4%	-9.00 [-21.64 , 3.64]
Meduri 2007	7	14	42	14.5	21	19	0.6%	-7.50 [-17.85 , 2.85]
Confalonieri 2005	7.2	7.1	23	14.2	9.8	23	2.2%	-7.00 [-11.95 , -2.05]
Pierre de Oliveira 2023	12.5	10.45	58	18.76	15.1	54	2.3%	-6.26 [-11.10 , -1.42]
Meduri 2009	6	6	48	12	18	31	1.4%	-6.00 [-12.56 , 0.56]
Gordon 2014	14.3	14.6	31	19.5	14.9	30	1.1%	-5.20 [-12.61 , 2.21]
Fernández-Serrano 2011	9	5.5	28	12.5	7.23	28	4.0%	-3.50 [-6.86 , -0.14]
Annane 2002	22	24	151	25.5	18	149	2.3%	-3.50 [-8.30 , 1.30]
Chawla 1999	7.4	6.2	23	10.7	7.6	21	3.0%	-3.30 [-7.42 , 0.82]
Rinaldi 2006	19	21	26	21	19	26	0.5%	-2.00 [-12.89 , 8.89]
Venkatesh 2018	10	18.5	1853	12	26.7	1860	8.8%	-2.00 [-3.48 , -0.52]
Hu 2009	4.18	2.86	38	5.36	2.48	39	9.8%	-1.18 [-2.38 , 0.02]
Keh 2016	8	7.4	190	9	8.2	190	8.5%	-1.00 [-2.57 , 0.57]
Mirea 2014	11.53	5.67	117	11.67	11.71	54	4.1%	-0.14 [-3.43 , 3.15]
Gordon 2016	6	5.9	208	6	6.7	213	9.8%	0.00 [-1.21 , 1.21]
Iglesias 2020	4.76	4.3	68	4.66	3.45	69	9.4%	0.10 [-1.21 , 1.41]
Agarwal 2022	10.4	4.1	61	10.1	4.2	59	8.8%	0.30 [-1.19 , 1.79]
Torres 2015	8.2	10.7	42	7.7	8.7	46	3.0%	0.50 [-3.60 , 4.60]
Menon 2017	8.3	8.4	23	7.8	7.4	26	2.6%	0.50 [-3.96 , 4.96]
Lv 2017	10.9	17.5	58	10.2	13.1	60	1.8%	0.70 [-4.89 , 6.29]
Arabi 2011	10.5	6.7	39	9.7	6.5	36	4.6%	0.80 [-2.19 , 3.79]
Sprung 2008	19	31	251	18	17	248	2.7%	1.00 [-3.38 , 5.38]
Annane 2018	19	25	614	17	21	627	5.5%	2.00 [-0.57 , 4.57]
Mohamed 2020	12.44	14.12	45	8.44	8.16	43	2.3%	4.00 [-0.79 , 8.79]
Bollaert 1998	26	24	22	19	18	19	0.4%	7.00 [-5.89 , 19.89]
Total (Wald[a])			**4079**			**3990**	**100.0%**	**-0.86 [-1.67 , -0.05]**

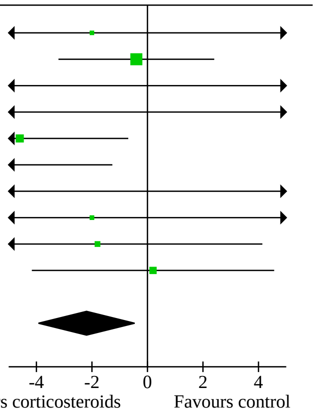


Test for overall effect: Z = 2.07 (P = 0.04)
Test for subgroup differences: Not applicable
Heterogeneity: Tau² (DL[b]) = 1.39; Chi² = 44.09, df = 24 (P = 0.007); I² = 46%

Footnotes
[a] CI calculated by Wald-type method.
[b] Tau² calculated by DerSimonian and Laird method.

Analysis 1.15. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 15: Length of intensive care unit stay for survivors

Study or Subgroup	Treatment Mean	Treatment SD	Treatment Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Random, 95% CI
Annane 2002	31	30	60	33	25	48	2.8%	-2.00 [-12.38 , 8.38]
Arabi 2011	9.2	6.4	39	9.6	6	36	38.1%	-0.40 [-3.21 , 2.41]
Bollaert 1998	29	24	14	36	33	8	0.4%	-7.00 [-33.10 , 19.10]
Briegel 1999	29	16	16	38	24	14	1.4%	-9.00 [-23.82 , 5.82]
Chawla 1999	6.8	4.7	17	11.4	5.9	13	19.6%	-4.60 [-8.51 , -0.69]
Confalonieri 2005	7.2	7.1	23	14.5	10.8	16	8.2%	-7.30 [-13.34 , -1.26]
Meduri 2007	12	13	31	20	19	9	1.7%	-8.00 [-21.23 , 5.23]
Rinaldi 2006	21	16	21	23	16	25	3.5%	-2.00 [-11.28 , 7.28]
Sprung 2008	18.7	33.2	149	20.5	16.8	158	8.5%	-1.80 [-7.74 , 4.14]
Torres 2015	8.4	10.8	41	8.2	9.2	40	15.7%	0.20 [-4.17 , 4.57]
Total (Wald[a])			**411**			**367**	**100.0%**	**-2.19 [-3.93 , -0.46]**

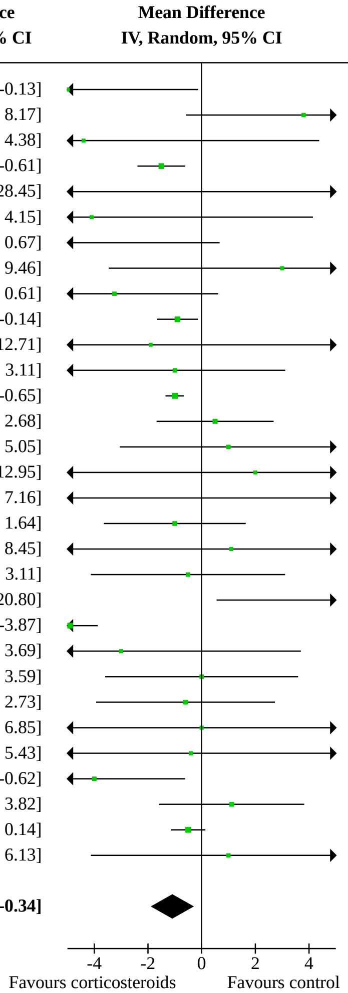


Test for overall effect: Z = 2.48 (P = 0.01)
Test for subgroup differences: Not applicable
Heterogeneity: Tau^2 (DL[b]) = 0.00; Chi^2 = 8.63, df = 9 (P = 0.47); I^2 = 0%

Footnotes
[a]CI calculated by Wald-type method.
[b]Tau^2 calculated by DerSimonian and Laird method.

Analysis 1.16. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 16: Length of hospital stay for all participants

Study or Subgroup	Treatment Mean	Treatment SD	Treatment Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Random, 95% CI
Annane 2002	20	21	151	25	22	149	2.0%	-5.00 [-9.87 , -0.13]
Annane 2018	32.6	38.2	614	28.8	40.4	627	2.3%	3.80 [-0.57 , 8.17]
Arabi 2011	22	13.4	39	26.4	23.6	36	0.7%	-4.40 [-13.18 , 4.38]
Blum 2015	8.5	5.8	392	10	6.9	393	9.3%	-1.50 [-2.39 , -0.61]
Bollaert 1998	35	31	22	24	26	19	0.2%	11.00 [-6.45 , 28.45]
Chawla 1999	16.9	13.3	23	21	14.5	21	0.8%	-4.10 [-12.35 , 4.15]
Confalonieri 2005	17.7	9.9	23	25	16.8	23	0.8%	-7.30 [-15.27 , 0.67]
Corral-Gudino 2021	23	12	35	20	14	29	1.2%	3.00 [-3.46 , 9.46]
Fernández-Serrano 2011	11.75	7.22	28	15	7.53	28	2.8%	-3.25 [-7.11 , 0.61]
Ghanei 2021	5.5	3.1	116	6.4	3.8	220	9.6%	-0.90 [-1.66 , -0.14]
Gordon 2014	34	32.8	31	35.9	25	30	0.3%	-1.90 [-16.51 , 12.71]
Gordon 2016	15	20	208	16	23	213	2.6%	-1.00 [-5.11 , 3.11]
Horby 2021	12	7	2104	13	6	4321	10.4%	-1.00 [-1.35 , -0.65]
Iglesias 2020	11.5	6.8	68	11	6.2	69	5.7%	0.50 [-1.68 , 2.68]
Keh 2016	26	22.2	190	25	17.8	190	2.6%	1.00 [-3.05 , 5.05]
Lv 2017	23.7	36.8	58	21.7	21.7	60	0.5%	2.00 [-8.95 , 12.95]
Meduri 2007	13	19	42	20.5	30	19	0.3%	-7.50 [-22.16 , 7.16]
Meijvis 2011	6.5	9.3	151	7.5	13.8	153	4.6%	-1.00 [-3.64 , 1.64]
Menon 2017	10.7	15.2	23	9.6	10.2	26	1.0%	1.10 [-6.25 , 8.45]
Mirea 2014	17.42	10.08	117	17.93	11.71	54	3.1%	-0.51 [-4.13 , 3.11]
Mohamed 2020	31.58	31.06	45	20.9	15.01	43	0.5%	10.68 [0.56 , 20.80]
Nagy 2013	11.5	1.7	29	16.4	2.3	30	8.9%	-4.90 [-5.93 , -3.87]
Pierre de Oliveira 2023	23	17	58	26	19	54	1.1%	-3.00 [-9.69 , 3.69]
Sevransky 2021	14	10	250	14	27	246	3.1%	0.00 [-3.59 , 3.59]
Snijders 2010	10	12	104	10.6	12.8	109	3.5%	-0.60 [-3.93 , 2.73]
Sprung 2008	34	41	251	34	37	248	1.1%	0.00 [-6.85 , 6.85]
Torres 2015	14.5	14.5	60	14.9	17.6	58	1.5%	-0.40 [-6.23 , 5.43]
Venkatesh 2018	39	52.6	1853	43	52.6	1860	3.4%	-4.00 [-7.38 , -0.62]
Wani 2020	11.82	7.36	50	10.7	6.39	50	4.5%	1.12 [-1.58 , 3.82]
Wittermans 2021	4.5	3.6	203	5	2.9	198	9.9%	-0.50 [-1.14 , 0.14]
Yildiz 2002	14	9.6	20	13	6.7	20	1.8%	1.00 [-4.13 , 6.13]
Total (Wald[a])			**7358**			**9596**	**100.0%**	**-1.09 [-1.85 , -0.34]**



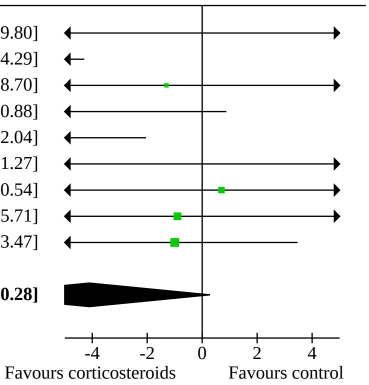
Test for overall effect: Z = 2.83 (P = 0.005)
Test for subgroup differences: Not applicable
Heterogeneity: Tau² (DL[b]) = 1.40; Chi² = 89.01, df = 30 (P < 0.00001); I² = 66%

Footnotes

[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.

Analysis 1.17. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 17: Length of hospital stay for survivors

Study or Subgroup	Treatment Mean	Treatment SD	Treatment Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Random, 95% CI
Annane 2002	60	60	55	48	28	46	5.1%	12.00 [-5.80 , 29.80]
Arabi 2011	27.2	12.8	39	43.3	34	36	9.5%	-16.10 [-27.91 , -4.29]
Bollaert 1998	42.7	24.2	14	44	36.7	7	2.0%	-1.30 [-31.30 , 28.70]
Chawla 1999	11.9	8.2	17	18.3	10.4	11	16.5%	-6.40 [-13.68 , 0.88]
Confalonieri 2005	17.7	9.9	23	29.4	17.9	16	12.2%	-11.70 [-21.36 , -2.04]
Meduri 2007	22	31	29	41	43	9	2.0%	-19.00 [-49.27 , 11.27]
Sprung 2008	42.8	45.2	140	42.1	39.2	145	12.0%	0.70 [-9.14 , 10.54]
Torres 2015	15.4	15	54	16.3	18.8	49	17.9%	-0.90 [-7.51 , 5.71]
Yildiz 2002	7	5	12	8	5	8	22.9%	-1.00 [-5.47 , 3.47]
Total (Wald[a])			**383**			**327**	**100.0%**	**-4.11 [-8.50 , 0.28]**

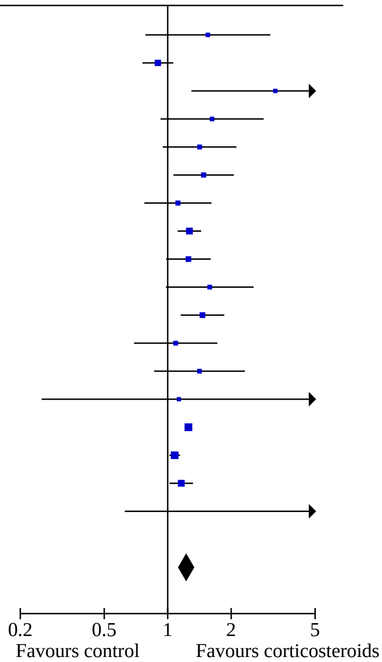


Test for overall effect: Z = 1.83 (P = 0.07)
Test for subgroup differences: Not applicable
Heterogeneity: Tau² (DL[b]) = 16.69; Chi² = 14.16, df = 8 (P = 0.08); I² = 43%

Footnotes
[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.

Analysis 1.18. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 18: Number of participants with shock reversal at day 7

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Sprung 1984	25	43	6	16	1.3%	1.55 [0.78 , 3.06]
Bone 1987	85	130	83	114	9.6%	0.90 [0.76 , 1.06]
Bollaert 1998	15	22	4	19	0.7%	3.24 [1.30 , 8.10]
Chawla 1999	16	23	9	21	1.8%	1.62 [0.92 , 2.85]
Briegel 1999	17	20	12	20	3.2%	1.42 [0.95 , 2.12]
Annane 2002	60	151	40	149	4.4%	1.48 [1.06 , 2.06]
Oppert 2005	14	18	16	23	3.7%	1.12 [0.78 , 1.61]
Sprung 2008	186	251	145	248	11.6%	1.27 [1.12 , 1.44]
Hu 2009	33	38	27	39	6.6%	1.25 [0.98 , 1.60]
Arabi 2011	24	39	14	36	2.4%	1.58 [0.98 , 2.55]
Sabry 2011	38	40	26	40	6.7%	1.46 [1.15 , 1.85]
Huang 2014	12	20	22	40	2.6%	1.09 [0.69 , 1.72]
Gordon 2014	19	31	13	30	2.3%	1.41 [0.86 , 2.32]
Menon 2017	3	23	3	26	0.3%	1.13 [0.25 , 5.06]
Venkatesh 2018	1594	1898	1274	1902	15.9%	1.25 [1.21 , 1.30]
Annane 2018	501	614	474	627	15.2%	1.08 [1.02 , 1.14]
Iglesias 2020	64	68	56	69	11.6%	1.16 [1.02 , 1.32]
Mohamed 2020	5	45	0	45	0.1%	11.00 [0.63 , 193.25]
Total (Wald[a])		**3474**		**3464**	**100.0%**	**1.22 [1.13 , 1.33]**
Total events:	2711		2224			

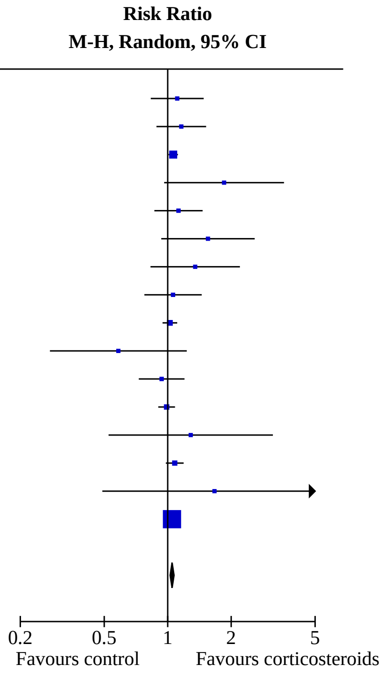


Test for overall effect: Z = 4.91 (P < 0.00001)
Test for subgroup differences: Not applicable
Heterogeneity: Tau² (DL[b]) = 0.01; Chi² = 46.77, df = 17 (P = 0.0001); I² = 64%

Footnotes
[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.

Analysis 1.19. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 19: Number of participants with shock reversal at 28 days

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Agarwal 2022	39	61	34	59	0.5%	1.11 [0.83 , 1.48]
Annane 2002	67	151	57	149	0.6%	1.16 [0.88 , 1.52]
Annane 2018	522	614	502	627	17.0%	1.06 [1.01 , 1.12]
Bollaert 1998	15	22	7	19	0.1%	1.85 [0.96 , 3.56]
Briegel 1999	18	20	16	20	0.6%	1.13 [0.86 , 1.46]
Chawla 1999	17	23	10	21	0.2%	1.55 [0.93 , 2.58]
Doluee 2018	27	80	20	80	0.2%	1.35 [0.83 , 2.20]
Gordon 2014	23	31	21	30	0.5%	1.06 [0.78 , 1.45]
Gordon 2016	179	208	179	213	7.0%	1.02 [0.95 , 1.11]
Liu 2012	5	12	10	14	0.1%	0.58 [0.28 , 1.23]
Lv 2017	38	58	42	60	0.7%	0.94 [0.73 , 1.20]
Lyu 2022	172	213	174	213	5.4%	0.99 [0.90 , 1.08]
Mohamed 2020	9	45	7	45	0.1%	1.29 [0.52 , 3.15]
Sprung 2008	200	251	183	248	4.7%	1.08 [0.98 , 1.19]
Tandan 2005	5	14	3	14	0.0%	1.67 [0.49 , 5.67]
Venkatesh 2018	1651	1898	1579	1902	62.4%	1.05 [1.02 , 1.08]
Total (Wald[a])		**3701**		**3714**	**100.0%**	**1.05 [1.03 , 1.07]**
Total events:	2987		2844			



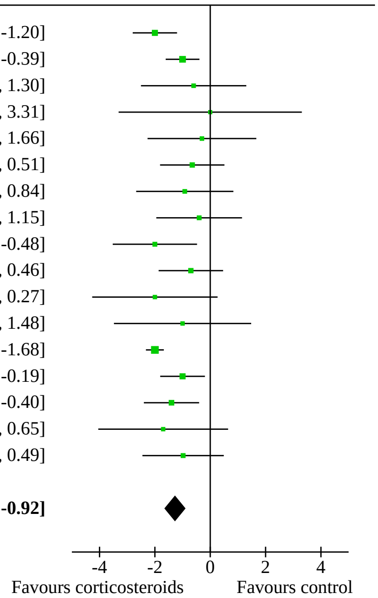
Test for overall effect: Z = 4.44 (P < 0.00001)
Test for subgroup differences: Not applicable
Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 13.87, df = 15 (P = 0.54); I² = 0%

Footnotes
[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.

Analysis 1.20. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 20: SOFA score at day 7

	Treatment			Control				Mean Difference
Study or Subgroup	**Mean**	**SD**	**Total**	**Mean**	**SD**	**Total**	**Weight**	**IV, Random, 95% CI**
Annane 2002	7.5	3	151	9.5	4	149	10.0%	-2.00 [-2.80 , -1.20]
Annane 2018	6	4	440	7	5	414	12.8%	-1.00 [-1.61 , -0.39]
Arabi 2011	11.7	4.2	39	12.3	4.2	36	2.9%	-0.60 [-2.50 , 1.30]
Cicarelli 2007	9	4	15	9	5	14	1.1%	0.00 [-3.31 , 3.31]
Gordon 2014	6.2	4.3	31	6.5	3.5	30	2.8%	-0.30 [-2.26 , 1.66]
Iglesias 2020	4.93	3.14	68	5.58	3.78	69	6.4%	-0.65 [-1.81 , 0.51]
Mirea 2014	7.69	6.02	117	8.61	5.16	54	3.4%	-0.92 [-2.68 , 0.84]
Mohamed 2020	8.9	3.6	45	9.3	3.8	43	4.1%	-0.40 [-1.95 , 1.15]
Mohamed 2023	8	4	53	10	4	53	4.3%	-2.00 [-3.52 , -0.48]
Moskowitz 2020	4.4	4.1	101	5.1	4.3	99	6.3%	-0.70 [-1.86 , 0.46]
Oppert 2005	6	4	23	8	4	25	2.2%	-2.00 [-4.27 , 0.27]
Rinaldi 2006	1	4	20	2	4	20	1.8%	-1.00 [-3.48 , 1.48]
Sabry 2011	1	0.53	40	3	0.9	40	17.7%	-2.00 [-2.32 , -1.68]
Sprung 2008	6.1	4.4	251	7.1	4.8	248	9.9%	-1.00 [-1.81 , -0.19]
Tomazini 2020	6.1	4	127	7.5	4	120	7.8%	-1.40 [-2.40 , -0.40]
Villar 2020	4.9	6	96	6.6	9.7	89	2.0%	-1.70 [-4.05 , 0.65]
Wani 2020	5.64	3.55	50	6.62	3.94	50	4.5%	-0.98 [-2.45 , 0.49]
Total (Wald[a])			**1667**			**1553**	**100.0%**	**-1.27 [-1.63 , -0.92]**

Mean Difference IV, Random, 95% CI



Test for overall effect: Z = 7.10 (P < 0.00001)
Test for subgroup differences: Not applicable
Heterogeneity: Tau^2 (DL[b]) = 0.16; Chi^2 = 25.21, df = 16 (P = 0.07); I^2 = 37%

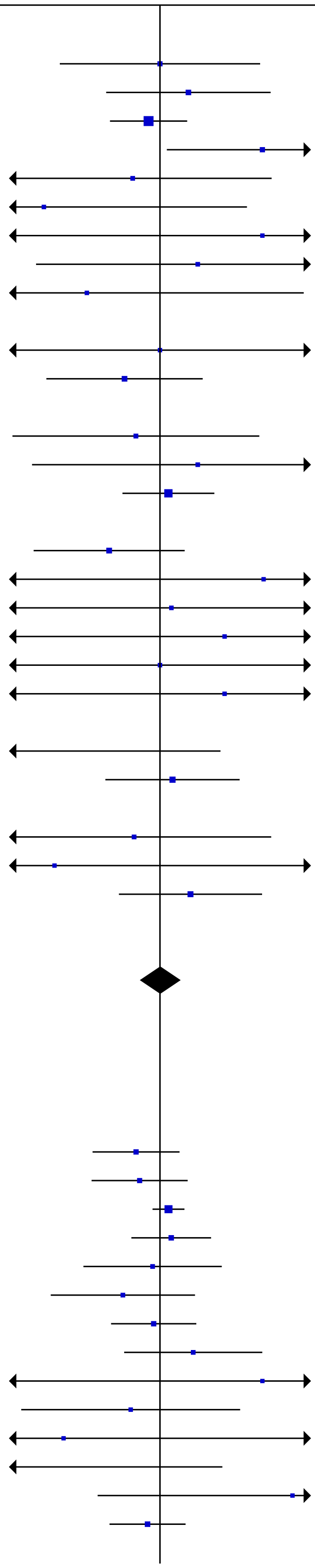
Footnotes

[a]CI calculated by Wald-type method.
[b]Tau^2 calculated by DerSimonian and Laird method.

Analysis 1.21. Comparison 1: Corticosteroids versus placebo or usual care, Outcome 21: Number of participants with adverse events

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
1.21.1 Gastroduodenal bleeding						
Agarwal 2022	6	61	6	61	3.7%	1.00 [0.34 , 2.93]
Annane 2002	11	151	8	149	5.4%	1.36 [0.56 , 3.28]
Annane 2018	39	614	45	627	24.6%	0.89 [0.58 , 1.34]
Arabi 2011	13	39	4	36	4.0%	3.00 [1.08 , 8.36]
Blum 2015	3	402	4	400	1.9%	0.75 [0.17 , 3.31]
Bollaert 1998	1	22	3	19	0.9%	0.29 [0.03 , 2.54]
Briegel 1999	1	20	0	20	0.4%	3.00 [0.13 , 69.52]
Chang 2020	3	40	2	40	1.4%	1.50 [0.26 , 8.50]
Chawla 1999	1	23	2	21	0.8%	0.46 [0.04 , 4.68]
Cicarelli 2007	0	14	0	15		Not estimable
Confalonieri 2005	1	23	1	23	0.6%	1.00 [0.07 , 15.04]
Dequin 2023	9	400	13	395	6.0%	0.68 [0.30 , 1.58]
Gordon 2014	0	31	0	30		Not estimable
Jiang 2019	5	41	3	19	2.4%	0.77 [0.21 , 2.90]
Keh 2016	3	190	2	190	1.3%	1.50 [0.25 , 8.88]
Luce 1988	18	37	16	36	17.4%	1.09 [0.67 , 1.79]
Meduri 2007	0	42	0	19		Not estimable
Meduri 2022	9	297	15	287	6.4%	0.58 [0.26 , 1.30]
Meijvis 2011	1	151	0	153	0.4%	3.04 [0.12 , 74.03]
Menon 2017	2	23	2	26	1.2%	1.13 [0.17 , 7.39]
Mohamed 2023	2	53	1	53	0.8%	2.00 [0.19 , 21.40]
Sabry 2011	2	40	2	40	1.2%	1.00 [0.15 , 6.76]
Schumer 1976	2	86	1	86	0.7%	2.00 [0.18 , 21.65]
Snijders 2010	0	104	0	109		Not estimable
Sprung 1984	1	43	2	16	0.8%	0.19 [0.02 , 1.91]
Sprung 2008	15	234	13	232	8.1%	1.14 [0.56 , 2.35]
Tagaro 2017	0	30	0	30		Not estimable
Tongyoo 2016	3	98	4	99	2.0%	0.76 [0.17 , 3.30]
Torres 2015	0	61	1	59	0.4%	0.32 [0.01 , 7.76]
VASSCSG 1987	14	112	10	111	7.2%	1.39 [0.64 , 2.99]
Yildiz 2002	0	20	0	20		Not estimable
Yildiz 2011	0	27	0	28		Not estimable
Subtotal (Wald[a])		**3529**		**3449**	**100.0%**	**1.00 [0.82 , 1.23]**
Total events:	165		160			
Test for overall effect: Z = 0.03 (P = 0.98)						
Heterogeneity: Tau^2 (DL[b]) = 0.00; Chi^2 = 15.32, df = 24 (P = 0.91); I^2 = 0%						
1.21.2 Superinfection						
Agarwal 2022	20	61	25	59	4.4%	0.77 [0.49 , 1.23]
Annane 2002	22	151	27	149	3.7%	0.80 [0.48 , 1.35]
Annane 2018	191	614	178	627	16.2%	1.10 [0.92 , 1.30]
Arabi 2011	22	39	18	36	5.1%	1.13 [0.74 , 1.73]
Blum 2015	13	402	14	400	1.9%	0.92 [0.44 , 1.94]
Bollaert 1998	7	22	9	19	1.8%	0.67 [0.31 , 1.46]
Bone 1987	29	152	30	147	4.5%	0.93 [0.59 , 1.48]
Briegel 1999	10	20	7	20	1.9%	1.43 [0.68 , 3.00]
Chang 2020	1	40	0	40	0.1%	3.00 [0.13 , 71.51]
Chawla 1999	4	23	5	21	0.8%	0.73 [0.23 , 2.36]
Cicarelli 2007	0	14	1	15	0.1%	0.36 [0.02 , 8.07]
Confalonieri 2005	0	23	4	23	0.1%	0.11 [0.01 , 1.95]
Corral-Gudino 2021	5	35	1	29	0.3%	4.14 [0.51 , 33.49]
Dequin 2023	39	400	44	395	5.5%	0.88 [0.58 , 1.32]
Gordon 2014	0	31	0	30		Not estimable

Risk Ratio
M-H, Random, 95% CI



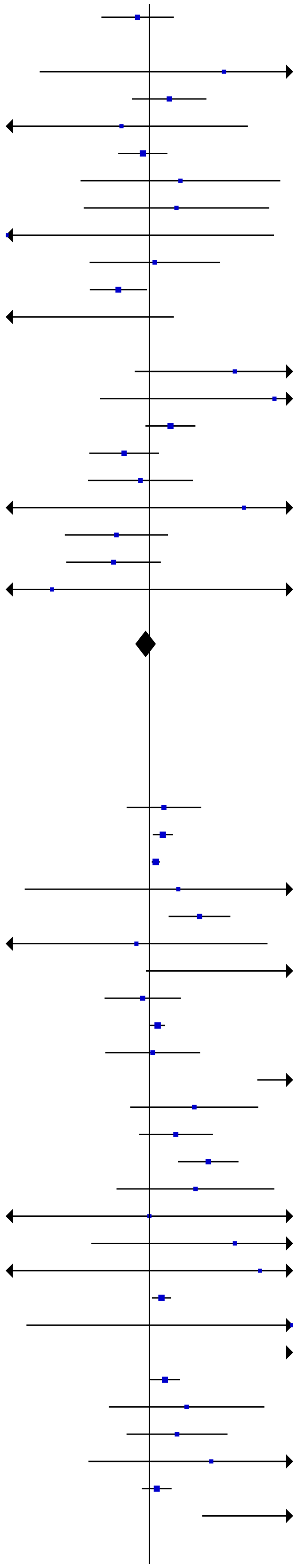
Analysis 1.21. (Continued)

Dequin 2023	39	400	44	395	5.5%	0.88 [0.58 , 1.32]
Gordon 2014	0	31	0	30		Not estimable
Jiang 2019	5	41	1	19	0.3%	2.32 [0.29 , 18.49]
Keh 2016	40	190	32	190	5.2%	1.25 [0.82 , 1.90]
Luce 1988	3	37	4	36	0.5%	0.73 [0.18 , 3.03]
Meduri 2022	73	297	76	287	9.6%	0.93 [0.70 , 1.22]
Meijvis 2011	7	151	5	153	0.9%	1.42 [0.46 , 4.37]
Menon 2017	6	23	5	26	1.0%	1.36 [0.48 , 3.86]
Mohamed 2023	0	53	2	53	0.1%	0.20 [0.01 , 4.07]
Moskowitz 2020	13	101	12	99	2.0%	1.06 [0.51 , 2.21]
Pierre de Oliveira 2023	28	58	37	54	7.8%	0.70 [0.51 , 0.97]
Rezk 2013	0	18	3	9	0.1%	0.08 [0.00 , 1.32]
Schumer 1976	0	86	0	86		Not estimable
Snijders 2010	10	104	4	109	0.9%	2.62 [0.85 , 8.09]
Sprung 1984	11	43	1	16	0.3%	4.09 [0.57 , 29.20]
Sprung 2008	78	234	61	232	9.4%	1.27 [0.96 , 1.68]
Tomazini 2020	33	151	43	148	5.8%	0.75 [0.51 , 1.11]
Tongyoo 2016	17	98	19	99	2.9%	0.90 [0.50 , 1.63]
Torres 2015	1	61	0	59	0.1%	2.90 [0.12 , 69.87]
VASSCSG 1987	16	112	23	111	3.0%	0.69 [0.39 , 1.23]
Villar 2020	18	96	25	89	3.5%	0.67 [0.39 , 1.14]
Yildiz 2002	0	20	1	20	0.1%	0.33 [0.01 , 7.72]
Yildiz 2011	0	27	0	28		Not estimable
Subtotal (Wald[a])		**4028**		**3933**	**100.0%**	**0.96 [0.86 , 1.07]**
Total events:	722		717			

Test for overall effect: Z = 0.79 (P = 0.43)
Heterogeneity: Tau^2 (DL[b]) = 0.01; Chi^2 = 36.62, df = 32 (P = 0.26); I^2 = 13%

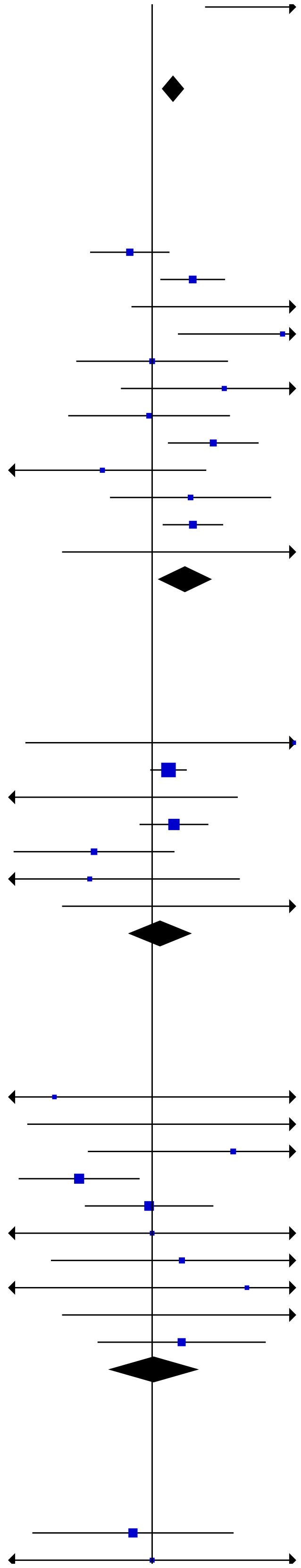
1.21.3 Hyperglycaemia

Agarwal 2022	28	61	23	59	4.5%	1.18 [0.77 , 1.79]
Annane 2002	130	150	111	149	10.2%	1.16 [1.04 , 1.30]
Annane 2018	547	614	520	627	11.1%	1.07 [1.03 , 1.12]
Arabi 2011	3	39	2	36	0.4%	1.38 [0.25 , 7.82]
Blum 2015	76	402	43	400	5.6%	1.76 [1.24 , 2.49]
Bollaert 1998	3	22	3	19	0.6%	0.86 [0.20 , 3.79]
Corral-Gudino 2021	9	35	0	29	0.2%	15.83 [0.96 , 260.96]
Jiang 2019	24	41	12	19	4.4%	0.93 [0.60 , 1.42]
Keh 2016	169	190	154	190	10.6%	1.10 [1.01 , 1.19]
Luce 1988	16	37	15	36	3.3%	1.04 [0.61 , 1.77]
Lyu 2022	27	213	0	213	0.2%	55.00 [3.38 , 895.87]
Meduri 2007	22	42	6	19	2.1%	1.66 [0.81 , 3.41]
Meduri 2022	46	297	33	287	4.6%	1.35 [0.89 , 2.04]
Meijvis 2011	67	151	35	153	5.7%	1.94 [1.38 , 2.73]
Moskowitz 2020	12	101	7	99	1.5%	1.68 [0.69 , 4.09]
Schumer 1976	1	86	1	86	0.2%	1.00 [0.06 , 15.73]
Snijders 2010	5	104	2	109	0.5%	2.62 [0.52 , 13.21]
Sprung 1984	4	43	0	16	0.2%	3.48 [0.20 , 61.18]
Sprung 2008	186	234	161	232	10.3%	1.15 [1.03 , 1.28]
Tagaro 2017	2	30	0	30	0.1%	5.00 [0.25 , 99.95]
Tomazini 2020	47	151	0	148	0.2%	93.12 [5.79 , 1496.86]
Tongyoo 2016	79	98	67	99	9.1%	1.19 [1.01 , 1.41]
Torres 2015	11	61	7	59	1.5%	1.52 [0.63 , 3.66]
VASSCSG 1987	23	111	17	112	3.0%	1.37 [0.77 , 2.41]
Venkatesh 2018	6	1853	3	1860	0.7%	2.01 [0.50 , 8.02]
Villar 2020	75	96	64	89	9.1%	1.09 [0.92 , 1.28]
Wittermans 2021	14	203	1	198	0.3%	13.66 [1.81 , 102.86]
Yildiz 2002	0	20	0	20		Not estimable



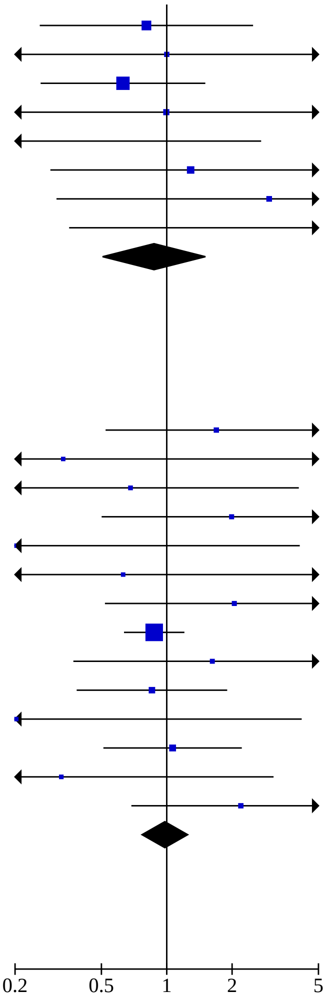
Analysis 1.21. (Continued)

Wittermans 2021	14	203	1	198	0.5%	13.66 [1.81 , 102.86]
Yildiz 2002	0	20	0	20		Not estimable
Yildiz 2011	0	27	0	28		Not estimable
Subtotal (Wald$_a$)		**5512**		**5421**	**100.0%**	**1.26 [1.13 , 1.42]**
Total events:	1632		1287			
Test for overall effect: Z = 3.97 (P < 0.0001)						
Heterogeneity: Tau² (DL$_b$) = 0.03; Chi² = 98.76, df = 26 (P < 0.00001); I² = 74%						
1.21.4 Hypernatraemia						
Agarwal 2022	21	61	27	61	14.6%	0.78 [0.50 , 1.22]
Annane 2002	54	150	34	149	16.6%	1.58 [1.10 , 2.27]
Briegel 1999	6	20	1	20	1.8%	6.00 [0.79 , 45.42]
Chang 2020	13	40	3	40	4.7%	4.33 [1.34 , 14.05]
Keh 2016	10	190	10	190	7.5%	1.00 [0.43 , 2.35]
Lyu 2022	9	213	4	213	4.8%	2.25 [0.70 , 7.19]
Meduri 2022	9	297	9	287	6.9%	0.97 [0.39 , 2.40]
Mirea 2014	56	117	13	54	13.2%	1.99 [1.19 , 3.31]
Mohamed 2023	4	53	7	53	4.8%	0.57 [0.18 , 1.84]
Moskowitz 2020	11	101	7	99	7.0%	1.54 [0.62 , 3.81]
Sprung 2008	67	234	42	232	17.2%	1.58 [1.13 , 2.22]
Venkatesh 2018	3	1853	0	1860	0.9%	7.03 [0.36 , 135.93]
Subtotal (Wald$_a$)		**3329**		**3258**	**100.0%**	**1.44 [1.08 , 1.93]**
Total events:	263		157			
Test for overall effect: Z = 2.51 (P = 0.01)						
Heterogeneity: Tau² (DL$_b$) = 0.10; Chi² = 20.39, df = 11 (P = 0.04); I² = 46%						
1.21.5 Muscle weakness						
Annane 2002	2	150	0	149	1.2%	4.97 [0.24 , 102.59]
Annane 2018	153	614	130	627	48.0%	1.20 [0.98 , 1.48]
Confalonieri 2005	0	23	3	23	1.3%	0.14 [0.01 , 2.62]
Keh 2016	46	190	36	190	33.1%	1.28 [0.87 , 1.88]
Meduri 2022	7	297	13	287	11.3%	0.52 [0.21 , 1.29]
Sprung 2008	2	234	4	232	3.8%	0.50 [0.09 , 2.68]
Venkatesh 2018	3	1853	0	1860	1.3%	7.03 [0.36 , 135.93]
Subtotal (Wald$_a$)		**3361**		**3368**	**100.0%**	**1.09 [0.78 , 1.53]**
Total events:	213		186			
Test for overall effect: Z = 0.51 (P = 0.61)						
Heterogeneity: Tau² (DL$_b$) = 0.05; Chi² = 8.65, df = 6 (P = 0.19); I² = 31%						
1.21.6 Neuropsychiatric event						
Annane 2002	0	150	1	150	2.2%	0.33 [0.01 , 8.12]
Annane 2018	2	614	0	627	2.4%	5.11 [0.25 , 106.13]
Blum 2015	5	402	2	400	7.6%	2.49 [0.49 , 12.75]
Keh 2016	11	190	25	190	27.3%	0.44 [0.22 , 0.87]
Meduri 2022	14	297	14	287	25.5%	0.97 [0.47 , 1.99]
Schumer 1976	1	86	1	86	2.9%	1.00 [0.06 , 15.73]
Snijders 2010	4	104	3	109	9.1%	1.40 [0.32 , 6.09]
Torres 2015	1	61	0	59	2.2%	2.90 [0.12 , 69.87]
Venkatesh 2018	3	1853	0	1860	2.5%	7.03 [0.36 , 135.93]
Wittermans 2021	10	203	7	198	18.1%	1.39 [0.54 , 3.59]
Subtotal (Wald$_a$)		**3960**		**3966**	**100.0%**	**1.02 [0.63 , 1.65]**
Total events:	51		53			
Test for overall effect: Z = 0.06 (P = 0.95)						
Heterogeneity: Tau² (DL$_b$) = 0.10; Chi² = 10.98, df = 9 (P = 0.28); I² = 18%						
1.21.7 Stroke						
Agarwal 2022	5	61	6	59	23.2%	0.81 [0.26 , 2.50]
Annane 2002	1	150	1	150	3.9%	1.00 [0.06 , 15.84]



Analysis 1.21. (Continued)

Agarwal 2022	5	61	6	59	23.2%	0.81 [0.26 , 2.50]
Annane 2002	1	150	1	150	3.9%	1.00 [0.06 , 15.84]
Annane 2018	8	614	13	627	39.0%	0.63 [0.26 , 1.51]
Blum 2015	2	402	2	400	7.8%	1.00 [0.14 , 7.03]
Dequin 2023	0	400	3	395	3.4%	0.14 [0.01 , 2.72]
Meduri 2022	4	297	3	287	13.4%	1.29 [0.29 , 5.71]
Sprung 2008	3	251	1	248	5.8%	2.96 [0.31 , 28.30]
Wittermans 2021	3	203	0	198	3.4%	6.83 [0.35 , 131.35]
Subtotal (Wald[a])		**2378**		**2364**	**100.0%**	**0.87 [0.51 , 1.51]**
Total events:	26		29			
Test for overall effect: Z = 0.49 (P = 0.63)						
Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 5.30, df = 7 (P = 0.62); I² = 0%						
1.21.8 Cardiac event						
Agarwal 2022	7	61	4	59	4.2%	1.69 [0.52 , 5.48]
Annane 2002	0	150	1	150	0.6%	0.33 [0.01 , 8.12]
Annane 2018	2	614	3	627	1.8%	0.68 [0.11 , 4.06]
Blum 2015	6	402	3	400	3.1%	1.99 [0.50 , 7.90]
Dequin 2023	0	400	2	395	0.6%	0.20 [0.01 , 4.10]
Ghanei 2021	0	116	1	220	0.6%	0.63 [0.03 , 15.33]
Gordon 2016	6	208	3	213	3.1%	2.05 [0.52 , 8.08]
Horby 2021	52	2104	122	4321	57.1%	0.88 [0.64 , 1.21]
Jiang 2019	7	41	2	19	2.7%	1.62 [0.37 , 7.08]
Meduri 2022	11	251	12	234	9.2%	0.85 [0.38 , 1.90]
Meijvis 2011	0	151	2	153	0.6%	0.20 [0.01 , 4.19]
Sprung 2008	14	251	13	248	10.9%	1.06 [0.51 , 2.22]
Tomazini 2020	1	151	3	148	1.2%	0.33 [0.03 , 3.11]
Wittermans 2021	9	203	4	198	4.3%	2.19 [0.69 , 7.01]
Subtotal (Wald[a])		**5103**		**7385**	**100.0%**	**0.98 [0.77 , 1.25]**
Total events:	115		175			
Test for overall effect: Z = 0.16 (P = 0.87)						
Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 9.59, df = 13 (P = 0.73); I² = 0%						



Test for subgroup differences: Chi² = 17.70, df = 7 (P = 0.01), I² = 60.4%

Footnotes

[a] CI calculated by Wald-type method.

[b] Tau² calculated by DerSimonian and Laird method.

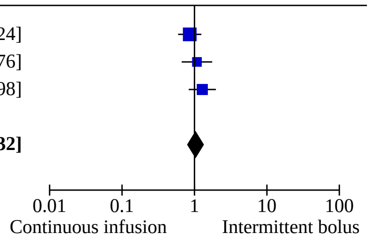
Comparison 2. Continuous infusion versus bolus administration of corticosteroids

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.1 28-day all-cause mortality	3	310	Risk Ratio (M-H, Random, 95% CI)	1.03 [0.81, 1.32]
2.2 Long-term mortality	1	70	Risk Ratio (M-H, Random, 95% CI)	1.36 [1.02, 1.81]
2.3 Hospital mortality	3	352	Risk Ratio (M-H, Random, 95% CI)	0.92 [0.71, 1.19]
2.4 90-day all-cause mortality	1	123	Risk Ratio (M-H, Random, 95% CI)	0.87 [0.61, 1.22]
2.5 Intensive care unit mortality	4	358	Risk Ratio (M-H, Random, 95% CI)	1.00 [0.79, 1.28]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.6 Length of intensive care unit stay for all participants	4	422	Mean Difference (IV, Random, 95% CI)	-0.56 [-3.44, 2.32]
2.7 Length of hospital stay for all participants	4	422	Mean Difference (IV, Random, 95% CI)	-0.21 [-4.72, 4.30]
2.8 Number of participants with shock reversal at day 7	4	358	Risk Ratio (M-H, Random, 95% CI)	0.80 [0.59, 1.10]
2.9 Number of participants with shock reversal at day 28	1	70	Risk Ratio (M-H, Random, 95% CI)	0.78 [0.45, 1.34]
2.10 SOFA score at day 7	3	260	Mean Difference (IV, Random, 95% CI)	1.01 [-0.30, 2.31]
2.11 Number of participants with adverse events	3		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
2.11.1 Gastrointestinal bleeding	2	193	Risk Ratio (M-H, Random, 95% CI)	0.79 [0.10, 6.37]
2.11.2 Superinfection	2	193	Risk Ratio (M-H, Random, 95% CI)	1.12 [0.37, 3.33]
2.11.3 Hyperglycaemia	3	310	Risk Ratio (M-H, Random, 95% CI)	0.89 [0.47, 1.71]
2.11.4 Hypernatraemia	2	187	Risk Ratio (M-H, Random, 95% CI)	0.74 [0.34, 1.61]
2.11.5 Muscle weakness	1	70	Risk Ratio (M-H, Random, 95% CI)	0.89 [0.13, 5.98]
2.11.6 Neuropsychiatric event	1	70	Risk Ratio (M-H, Random, 95% CI)	1.56 [0.50, 4.86]
2.11.7 Stroke	1	70	Risk Ratio (M-H, Random, 95% CI)	Not estimable
2.11.8 Cardiac event	1	70	Risk Ratio (M-H, Random, 95% CI)	Not estimable

Analysis 2.1. Comparison 2: Continuous infusion versus bolus administration of corticosteroids, Outcome 1: 28-day all-cause mortality

	Continuous infusion		Intermittent bolus			Risk Ratio	Risk Ratio
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Random, 95% CI	M-H, Random, 95% CI
Hyvernat 2016	28	63	31	60	43.4%	0.86 [0.59 , 1.24]	
Mirea 2014	22	59	20	58	25.1%	1.08 [0.67 , 1.76]	
Tilouche 2019	23	37	16	33	31.6%	1.28 [0.83 , 1.98]	
Total (Wald[a])		**159**		**151**	**100.0%**	**1.03 [0.81 , 1.32]**	
Total events:	73		67				



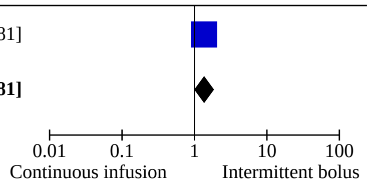
Test for overall effect: Z = 0.26 (P = 0.79)
Test for subgroup differences: Not applicable
Heterogeneity: Tau^2 (DL[b]) = 0.00; Chi^2 = 1.94, df = 2 (P = 0.38); I^2 = 0%

Footnotes
[a]CI calculated by Wald-type method.
[b]Tau^2 calculated by DerSimonian and Laird method.

Analysis 2.2. Comparison 2: Continuous infusion versus bolus administration of corticosteroids, Outcome 2: Long-term mortality

Study or Subgroup	Continuous infusion Events	Continuous infusion Total	Intermittent bolus Events	Intermittent bolus Total	Weight	Risk Ratio M-H, Random, 95% CI
Tilouche 2019	32	37	21	33	100.0%	1.36 [1.02 , 1.81]
Total		**37**		**33**	**100.0%**	**1.36 [1.02 , 1.81]**
Total events:	32		21			

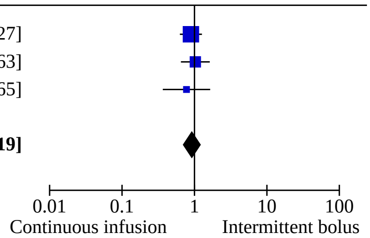
Test for overall effect: Z = 2.09 (P = 0.04)
Test for subgroup differences: Not applicable
Heterogeneity: Not applicable



Analysis 2.3. Comparison 2: Continuous infusion versus bolus administration of corticosteroids, Outcome 3: Hospital mortality

Study or Subgroup	Continuous infusion Events	Continuous infusion Total	Intermittent bolus Events	Intermittent bolus Total	Weight	Risk Ratio M-H, Random, 95% CI
Hyvernat 2016	30	63	32	60	55.5%	0.89 [0.63 , 1.27]
Mirea 2014	23	59	22	58	32.5%	1.03 [0.65 , 1.63]
Ram 2022	10	58	12	54	12.0%	0.78 [0.37 , 1.65]
Total (Wald[a])		**180**		**172**	**100.0%**	**0.92 [0.71 , 1.19]**
Total events:	63		66			

Test for overall effect: Z = 0.63 (P = 0.53)
Test for subgroup differences: Not applicable
Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 0.45, df = 2 (P = 0.80); I² = 0%

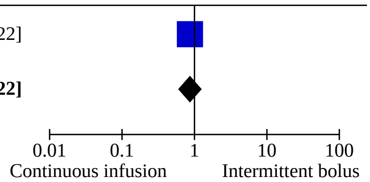


Footnotes
[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.

Analysis 2.4. Comparison 2: Continuous infusion versus bolus administration of corticosteroids, Outcome 4: 90-day all-cause mortality

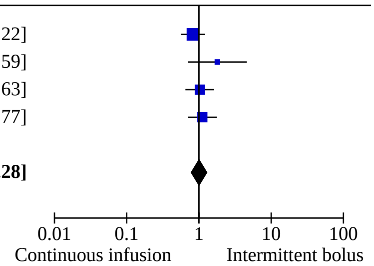
Study or Subgroup	Continuous infusion Events	Continuous infusion Total	Intermittent bolus Events	Intermittent bolus Total	Weight	Risk Ratio M-H, Random, 95% CI
Hyvernat 2016	30	63	33	60	100.0%	0.87 [0.61 , 1.22]
Total		**63**		**60**	**100.0%**	**0.87 [0.61 , 1.22]**
Total events:	30		33			

Test for overall effect: Z = 0.82 (P = 0.41)
Test for subgroup differences: Not applicable
Heterogeneity: Not applicable



Analysis 2.5. Comparison 2: Continuous infusion versus bolus administration of corticosteroids, Outcome 5: Intensive care unit mortality

Study or Subgroup	Continuous infusion Events	Continuous infusion Total	Intermittent bolus Events	Intermittent bolus Total	Weight	Risk Ratio M-H, Random, 95% CI
Hyvernat 2016	26	63	30	60	38.4%	0.83 [0.56 , 1.22]
Loisa 2007	9	24	5	24	6.6%	1.80 [0.71 , 4.59]
Mirea 2014	23	59	22	58	27.6%	1.03 [0.65 , 1.63]
Tilouche 2019	20	37	16	33	27.4%	1.11 [0.70 , 1.77]
Total (Wald[a])		**183**		**175**	**100.0%**	**1.00 [0.79 , 1.28]**
Total events:	78		73			

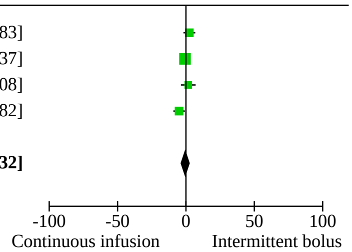


Test for overall effect: Z = 0.02 (P = 0.98)
Test for subgroup differences: Not applicable
Heterogeneity: Tau^2 (DL[b]) = 0.00; Chi^2 = 2.69, df = 3 (P = 0.44); I^2 = 0%

Footnotes
[a]CI calculated by Wald-type method.
[b]Tau^2 calculated by DerSimonian and Laird method.

Analysis 2.6. Comparison 2: Continuous infusion versus bolus administration of corticosteroids, Outcome 6: Length of intensive care unit stay for all participants

Study or Subgroup	Continuous infusion Mean	Continuous infusion SD	Continuous infusion Total	Intermittent bolus Mean	Intermittent bolus SD	Intermittent bolus Total	Weight	Mean Difference IV, Random, 95% CI
Hyvernat 2016	13.3	13.9	63	10.8	10.4	60	22.4%	2.50 [-1.83 , 6.83]
Mirea 2014	11.2	5.7	59	11.9	5.7	58	36.8%	-0.70 [-2.77 , 1.37]
Ram 2022	10.6	16	58	8.9	13	54	17.5%	1.70 [-3.68 , 7.08]
Tilouche 2019	11	7.5	37	16	10	33	23.2%	-5.00 [-9.18 , -0.82]
Total (Wald[a])			**217**			**205**	**100.0%**	**-0.56 [-3.44 , 2.32]**

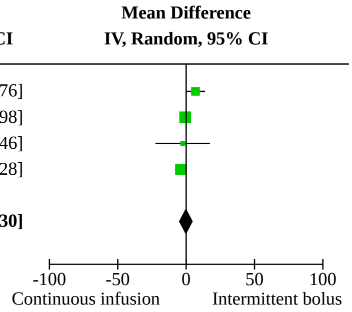


Test for overall effect: Z = 0.38 (P = 0.70)
Test for subgroup differences: Not applicable
Heterogeneity: Tau^2 (DL[b]) = 4.75; Chi^2 = 6.93, df = 3 (P = 0.07); I^2 = 57%

Footnotes
[a]CI calculated by Wald-type method.
[b]Tau^2 calculated by DerSimonian and Laird method.

Analysis 2.7. Comparison 2: Continuous infusion versus bolus administration of corticosteroids, Outcome 7: Length of hospital stay for all participants

	Continuous infusion			Intermittent bolus				Mean Difference
Study or Subgroup	**Mean**	**SD**	**Total**	**Mean**	**SD**	**Total**	**Weight**	**IV, Random, 95% CI**
Hyvernat 2016	24.3	22.8	63	17.5	16.2	60	22.9%	6.80 [-0.16 , 13.76]
Mirea 2014	17	10.4	59	17.7	9.9	58	37.8%	-0.70 [-4.38 , 2.98]
Ram 2022	16.5	58.6	58	19	49	54	4.6%	-2.50 [-22.46 , 17.46]
Tilouche 2019	13	8	37	17	10	33	34.7%	-4.00 [-8.28 , 0.28]
Total (Wald[a])			**217**			**205**	**100.0%**	**-0.21 [-4.72 , 4.30]**

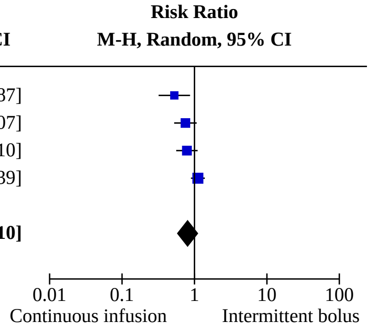


Test for overall effect: Z = 0.09 (P = 0.93)
Test for subgroup differences: Not applicable
Heterogeneity: Tau² (DL[b]) = 10.48; Chi² = 6.75, df = 3 (P = 0.08); I² = 56%

Footnotes
[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.

Analysis 2.8. Comparison 2: Continuous infusion versus bolus administration of corticosteroids, Outcome 8: Number of participants with shock reversal at day 7

	Continuous infusion		Intermittent bolus			Risk Ratio
Study or Subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Random, 95% CI**
Tilouche 2019	13	37	22	33	18.8%	0.53 [0.32 , 0.87]
Loisa 2007	15	24	20	24	24.7%	0.75 [0.52 , 1.07]
Mirea 2014	28	59	35	58	25.5%	0.79 [0.56 , 1.10]
Hyvernat 2016	48	63	41	60	31.0%	1.11 [0.89 , 1.39]
Total (Wald[a])		**183**		**175**	**100.0%**	**0.80 [0.59 , 1.10]**
Total events:	104		118			

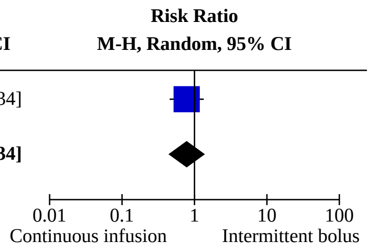


Test for overall effect: Z = 1.38 (P = 0.17)
Test for subgroup differences: Not applicable
Heterogeneity: Tau² (DL[b]) = 0.07; Chi² = 10.09, df = 3 (P = 0.02); I² = 70%

Footnotes
[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.

Analysis 2.9. Comparison 2: Continuous infusion versus bolus administration of corticosteroids, Outcome 9: Number of participants with shock reversal at day 28

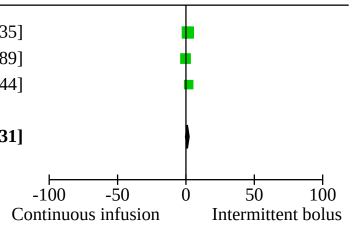
	Continuous infusion		Intermittent bolus			Risk Ratio
Study or Subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Random, 95% CI**
Tilouche 2019	14	37	16	33	100.0%	0.78 [0.45 , 1.34]
Total		**37**		**33**	**100.0%**	**0.78 [0.45 , 1.34]**
Total events:	14		16			



Test for overall effect: Z = 0.90 (P = 0.37)
Test for subgroup differences: Not applicable
Heterogeneity: Not applicable

Analysis 2.10. Comparison 2: Continuous infusion versus bolus administration of corticosteroids, Outcome 10: SOFA score at day 7

Study or Subgroup	Continuous infusion Mean	SD	Total	Intermittent bolus Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Hyvernat 2016	7.4	3.9	40	6	4.5	33	40.4%	1.40 [-0.55 , 3.35]
Mirea 2014	7.5	6.3	59	7.8	5.8	58	32.7%	-0.30 [-2.49 , 1.89]
Tilouche 2019	2	7	37	0	2.7	33	26.9%	2.00 [-0.44 , 4.44]
Total (Wald[a])			**136**			**124**	**100.0%**	**1.01 [-0.30 , 2.31]**



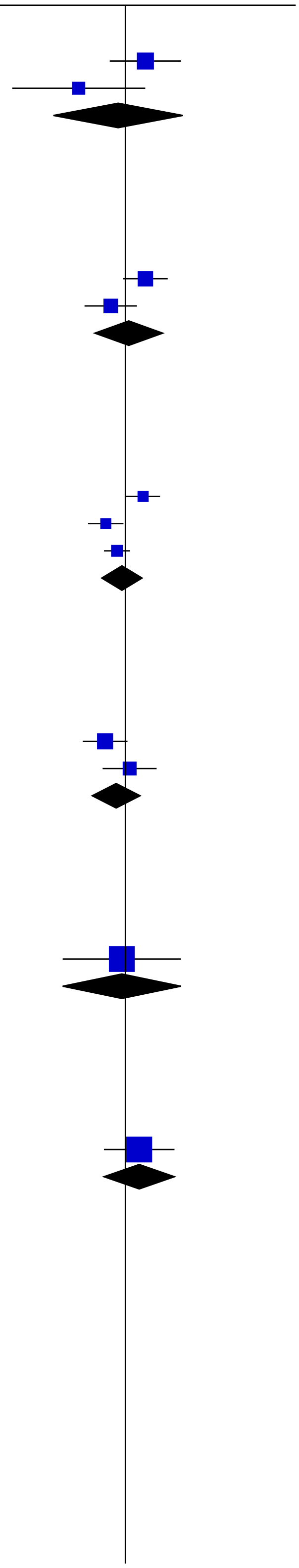
Test for overall effect: Z = 1.51 (P = 0.13)
Test for subgroup differences: Not applicable
Heterogeneity: Tau^2 (DL[b]) = 0.10; Chi^2 = 2.16, df = 2 (P = 0.34); I^2 = 7%

Footnotes
[a]CI calculated by Wald-type method.
[b]Tau^2 calculated by DerSimonian and Laird method.

Analysis 2.11. Comparison 2: Continuous infusion versus bolus administration of corticosteroids, Outcome 11: Number of participants with adverse events

Study or Subgroup	Continuous infusion Events	Continuous infusion Total	Intermittent bolus Events	Intermittent bolus Total	Weight	Risk Ratio M-H, Random, 95% CI
2.11.1 Gastrointestinal bleeding						
Hyvernat 2016	8	63	4	60	59.1%	1.90 [0.60 , 6.00]
Tilouche 2019	1	37	4	33	40.9%	0.22 [0.03 , 1.90]
Subtotal (Wald[a])		**100**		**93**	**100.0%**	**0.79 [0.10 , 6.37]**
Total events:	9		8			
Test for overall effect: Z = 0.22 (P = 0.83)						
Heterogeneity: Tau² (DL[b]) = 1.57; Chi² = 3.05, df = 1 (P = 0.08); I² = 67%						
2.11.2 Superinfection						
Hyvernat 2016	18	63	9	60	52.1%	1.90 [0.93 , 3.90]
Tilouche 2019	7	37	10	33	47.9%	0.62 [0.27 , 1.45]
Subtotal (Wald[a])		**100**		**93**	**100.0%**	**1.12 [0.37 , 3.33]**
Total events:	25		19			
Test for overall effect: Z = 0.20 (P = 0.84)						
Heterogeneity: Tau² (DL[b]) = 0.46; Chi² = 3.90, df = 1 (P = 0.05); I² = 74%						
2.11.3 Hyperglycaemia						
Hyvernat 2016	26	63	14	60	32.4%	1.77 [1.03 , 3.05]
Mirea 2014	13	59	24	58	31.8%	0.53 [0.30 , 0.94]
Tilouche 2019	18	37	21	33	35.8%	0.76 [0.50 , 1.16]
Subtotal (Wald[a])		**159**		**151**	**100.0%**	**0.89 [0.47 , 1.71]**
Total events:	57		59			
Test for overall effect: Z = 0.34 (P = 0.74)						
Heterogeneity: Tau² (DL[b]) = 0.26; Chi² = 9.77, df = 2 (P = 0.008); I² = 80%						
2.11.4 Hypernatraemia						
Mirea 2014	9	59	17	58	54.9%	0.52 [0.25 , 1.07]
Tilouche 2019	9	37	7	33	45.1%	1.15 [0.48 , 2.73]
Subtotal (Wald[a])		**96**		**91**	**100.0%**	**0.74 [0.34 , 1.61]**
Total events:	18		24			
Test for overall effect: Z = 0.75 (P = 0.45)						
Heterogeneity: Tau² (DL[b]) = 0.15; Chi² = 1.88, df = 1 (P = 0.17); I² = 47%						
2.11.5 Muscle weakness						
Tilouche 2019	2	37	2	33	100.0%	0.89 [0.13 , 5.98]
Subtotal		**37**		**33**	**100.0%**	**0.89 [0.13 , 5.98]**
Total events:	2		2			
Test for overall effect: Z = 0.12 (P = 0.91)						
Heterogeneity: Not applicable						
2.11.6 Neuropsychiatric event						
Tilouche 2019	7	37	4	33	100.0%	1.56 [0.50 , 4.86]
Subtotal		**37**		**33**	**100.0%**	**1.56 [0.50 , 4.86]**
Total events:	7		4			
Test for overall effect: Z = 0.77 (P = 0.44)						
Heterogeneity: Not applicable						
2.11.7 Stroke						
Tilouche 2019	0	37	0	33		Not estimable
Subtotal		**37**		**33**		**Not estimable**
Total events:	0		0			
Test for overall effect: Not applicable						
Heterogeneity: Not applicable						
2.11.8 Cardiac event						
Tilouche 2019	0	37	0	33		Not estimable
Subtotal		**37**		**33**		**Not estimable**

Risk Ratio
M-H, Random, 95% CI

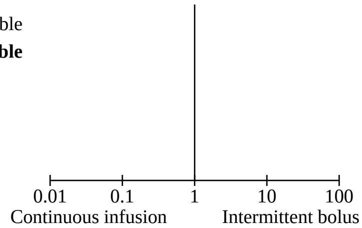


Analysis 2.11. (Continued)

Tilouche 2019	0	37	0	33	Not estimable
Subtotal		**37**		**33**	**Not estimable**
Total events:	0		0		

Test for overall effect: Not applicable
Heterogeneity: Not applicable

Test for subgroup differences: Chi² = 1.27, df = 5 (P = 0.94), I² = 0%



Footnotes
[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.

ADDITIONAL TABLES

Table 1. Main systematic reviews about the use of corticosteroids versus placebo or usual care in sepsis

Author	Number of trials	Primary outcome	No. of patients	Estimate	95% CI
Ni 2018	19	28-day mortality	7035	Sepsis: RR 0.91 Septic shock: RR 0.92	0.85 to 0.98 0.85 to 0.99
Allen 2018	6	Not stated	5689	Only qualitative assessment	Only qualitative assessment
Rochwerg 2018	42	28- to 31-day mortality	10,194	RR 0.93	0.84 to 1.03
Rygard 2018	22	Short-term mortality	7297	RR 0.96	0.91 to 1.02
Fang 2018	37	28-day mortality	9564	RR 0.90	0.82 to 0.98
Pirracchio 2023	17	90-day mortality	7882	Marginal RR 0.93	0.82 to 1.04
Pitre 2023	45	Short-term mortality	9563	RR 0.93	0.88 to 0.99

CI: confidence interval; RR: risk ratio

APPENDICES

Appendix 1. Search strategy for CENTRAL

#1 MeSH descriptor: [Sepsis] explode all trees

#2 MeSH descriptor: [Shock, Septic] explode all trees

#3 MeSH descriptor: [Systemic Inflammatory Response Syndrome] explode all trees

#4 MeSH descriptor: [Central Nervous System Bacterial Infections] explode all trees and with qualifier(s): [blood - BL, complications - CO, drug therapy - DT]

#5 MeSH descriptor: [Pneumonia] explode all trees

#6 MeSH descriptor: [Community-Acquired Infections] explode all trees and with qualifier(s): [complications - CO, drug therapy - DT]

#7 MeSH descriptor: [Respiratory Distress Syndrome, Adult] explode all trees and with qualifier(s): [complications - CO, drug therapy - DT]

#8 MeSH descriptor: [Acute Lung Injury] explode all trees and with qualifier(s): [complications - CO, drug therapy - DT]

#9 sepsis or (septic* NEAR/3 shock*)

#10 (bacterem* or bacteraem* or pyrexia or septicaem* or septicem*)

#11 SIRS or (Inflammatory next Response next Syndrome*)

#12 bacteria* NEAR infect* NEAR (blood* or serum or invas* or severe or systemic)

#13 ((community next acquired) or severe) NEAR pneumonia

#14 (acute or adult) NEAR/2 (respiratory NEAR/2 distress)

#15 (acute or adult) NEAR/2 (lung NEAR/2 injury)

#16 ARDS

#17 #1 or #2 or #3 or #4 or #5 or #6 or #7 or #8 or #9 or #10 or #11 or #12 or #13 or #14 or #15 or #16

#18 MeSH descriptor: [Adrenal Cortex Hormones] explode all trees

#19 MeSH descriptor: [Hydrocortisone] explode all trees

#20 MeSH descriptor: [Cortisone] explode all trees

#21 MeSH descriptor: [Steroids] explode all trees

#22 corticosteroid* or steroid* or cortison* or hydrocortison*

#23 methylprednisolon* or (methyl next prednisolon*) or betamethason* or dexamethason* or glucocorticoid* or fludrocortison* or mineralocorticoid*

#24 #18 or #19 or #20 or #21 or #22 or #23

#25 #17 and #24

#26 #25 in Trials

Appendix 2. Search strategy for MEDLINE (Ovid SP)

1 exp Sepsis/

2 exp Shock, Septic/

3 Systemic Inflammatory Response Syndrome/

4 exp Bacteremia/

5 Bacterial Infections/bl, dt, co

6 Pneumonia/co, dt

7 Community-Acquired Infections/co, dt

8 Respiratory Distress Syndrome, Adult/co, dt

9 Acute Lung Injury/co, dt

10 (sepsis or septic*).mp.

11 (bacter?em* or septic?em* or pyrexia).mp.

12 (SIRS or Inflammatory Response Syndrome*).mp.

13 (bacteria* adj6 infect* adj6 (blood* or serum or invas* or severe or systemic)).mp.

14 ((community-acquired or severe) adj3 pneumonia).mp.

15 ((acute or adult) adj2 (respiratory adj2 distress)).mp.

16 ARDS.mp.

17 ((acute or adult) adj2 (lung adj2 injury)).mp.

18 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17

19 exp Adrenal Cortex Hormones/

20 exp Hydrocortisone/

21 (corticosteroid* or steroid* or cortison* or hydrocortison*).mp.

22 (methylprednisolon* or betamethason* or dexamethason* or glucocorticoid* or fludrocortison* or mineralocorticoid*).mp.

23 19 or 20 or 21 or 22

24 18 and 23

25 ((randomized controlled trial or controlled clinical trial).pt. or randomi?ed.ab. or placebo.ab. or clinical trials as topic.sh. or randomly.ab. or trial.ti.) not (animals not (humans and animals)).sh.

26 24 and 25

Appendix 3. Search strategy for Embase (Ovid SP)

1 exp sepsis/

2 exp septic shock/

3 pneumonia/co, dt [Complication, Drug Therapy]

4 adult respiratory distress syndrome/co, dt [Complication, Drug Therapy]

5 acute lung injury/co, dt [Complication, Drug Therapy]

6 systemic inflammatory response syndrome/co, dt [Complication, Drug Therapy]

7 community acquired infection/co, dt [Complication, Drug Therapy]

8 (sepsis or (septic* adj5 shock) or (bacter?em* or pyrexia or septic?em*) or (SIRS or Inflammatory Response Syndrome*)).mp.

9 (bacteria* adj2 infect* adj2 (blood* or serum or invas* or severe or systemic)).mp.

10 (((community-acquired or severe) adj2 pneumonia) or ((acute or adult) adj1 (respiratory adj1 distress)) or ((acute or adult) adj1 (lung adj1 injury)) or ARDS).mp.

11 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10

12 steroid/

13 corticosteroid/

14 cortisone/

15 hydrocortisone/

16 (corticosteroid* or steroid* or cortison* or hydrocortison* or (methylprednisolon* or methyl prednisolon* or betamethason* or dexamethason* or glucocorticoid* or fludrocortison* or mineralocorticoid*)).mp.

17 12 or 13 or 14 or 15 or 16

18 11 and 17

19 ((placebo or randomized controlled trial).sh. or controlled study.ab. or random*.ti,ab. or trial*.ti,ab.) not (animal not human).sh.

20 18 and 19

Appendix 4. Search strategy for LILACS (via BIREME)

(sepsis OR septic$ OR SEPSIS OR SEPTIC OR SIRS OR "septic shock" OR "SEPTIC SHOCK/" OR SEPTICEMIA OR PNEUMONIA OR bact* OR "adult respiratory distress syndrome" OR "acute lung injury" OR "systemic inflammatory response syndrome" OR "bacterial infection" OR "community acquired infection") (corticosteroid* OR steroid* OR glucocorticoid* OR CORTCOSTEROID* OR GLUCOCORTICOID/ OR STEROID OR MINERALOCORTICOID OR cortison* OR hydrocortison* OR fludrocortison* OR betamethason* OR methylprednisolon* OR prednison* OR dexamethason*)

Appendix 5. Data extraction form template

DATA EXTRACTION FORM

Reviewer ID: I__I__I

Date of data extraction I__I__I/ I__I__I/ I__I__I

Study ID: I__I__I

Corresponding author contacts..

Trial protocol available yes/no I__I

Data sources: I_I

1) Published data only

2) Mixture

3) Unpublished only

4) Unpublished sought but not used

Sources of funding:

...

Centres: I__I

If 3 or 4, number of sites I__I__I__I

1) Monocentre

2) Bicentre

3) Multicentre national

4) Multicentre international

Describe country (counties) where study was conducted: ....

Random sequence generation: I__I

1) Adequate

2) Unclear

3) Inadequate

4) Not valid

Describe the method use: ....

Allocation concealment: I__I

1) Adequate

2) Unclear

3) Inadequate

4) Not valid

Describe the method use: ……

Diagnostic criteria: I__I

1. Explicit
2. Inadequate
3. Not clear

Outcome criteria: I _I

1. Explicit primary outcome criterion
2. Explicit outcome criteria but primary outcome criterion not stated
3. Inadequate
4. Unclear

List of primary and secondary outcomes: ……

Completeness of follow-up: I__I

If 2, % of lost to follow-up I__I__I__I

1. Complete
2. Incomplete
3. Unclear

Blinding:

1. Participants: yes/no I__I
2. Caregivers: yes/no I__I
3. Data collectors: yes/no I__I
4. Outcome assessors: yes/no I__I
5. Data analysts: yes/no I__I

Success of blinding of outcome: I__I

1. No unblinding
2. Partial blinding
3. Not tested

Study design: I__I

If 1, number of arms I__I

1) Parallel groups

2) Crossover design

3) Factorial

4) Cluster-randomisation

5) Other

6) Unclear

Setting: I__I

1) ICU

2) non-ICU ward

Type of participants:

Primary population I__I

1) Septic shock only

2) Sepsis and septic shock

3) ARDS

4) CAP

5) mixed ICU population

6) Covid 19

7) CIRCI/ relative adrenal insufficiency

Age I__I

1) All ages (including children)

2) Only adults

3) Only children

Gender I__I

1) Male

2) Female

3) other

List of inclusion criteria:..List of exclusion criteria: ....

Type of intervention: I__I I__I

1) hydrocortisone

2) methylprednisolone

3) betamethasone

4) dexamethasone

5) others (specify)……

continuous infusion yes/no I__I

bolus yes/no I__I

doses per day: I__I__I__I__I__I__I mg

duration: I__I__I__I__I__I__I hours

Time since shock onset > 48 hours yes/no I__I

Type of control: I__I

1) placebo

2) active treatment

3) usual care

Conclusion: I__I

1) Study included

2) Study not included

describe reason for non-inclusion: ………………………………………………………………………….

DATA EXTRACTION FORM

Number of patients randomised I__II__II__II__I

Number of patients randomised in control arm I__II__II__II__I

Number of patients randomised in active arm 1 I__II__II__II__I

Number of patients randomised in active arm 2 I__II__II__II__I

Number of patients assessed I__II__II__II__I

Number of patients assessed in control arm I__II__II__II__I

Number of patients assessed in active arm 1 I__II__II__II__I

Number of patients assessed in active arm 2 I__II__II__II__I

Base line characteristics Mean # standard deviation or numbers absolute	Control N=	Treatment N=
Age		
Male gender		
SAPS		
Number of OSF		
Time since shock onset, hours		
Number of patients who had positive blood cultures		
Number of patients who had pure Gram negative infections		
Number of patients who had relative adrenal insufficiency		

All patients

	Control	Treatment

(Continued)

Number of patients
28-day mortality
Number of patients assessed
Number of survivors
Number of non-survivors
ICU mortality
Number of patients assessed
Number of survivors
Number of non-survivors
Hospital mortality
Number of patients assessed
Number of survivors
Number of non-survivors
90-day mortality
Number of patients assessed
Number of survivors
Number of non-survivors
Long-term mortality
Number of patients assessed
Number of survivors
Number of non-survivors
SOFA score at day 7
Number of patients assessed
mean
Standard deviation
Length of ICU stay in all patients
Number of patients assessed
mean

(Continued)

Standard deviation
Length of ICU stay in survivors
Number of patients assessed
mean
Standard deviation
Length of hospital stay in all patients
Number of patients assessed
mean
Standard deviation
Length of hospital stay in survivors
Number of patients assessed
mean
Standard deviation
Tolerance
Number of patients assessed
Number of patients with at least one adverse event
Number of patients with GI bleeding
Number of patients with hyperglycaemia
Number of patients with hypernatremia
Number of patients with superinfection
Number of patients with nosocomial bacteraemia
Number of patients with nosocomial pneumonia
Number of patients with nosocomial UTI
Number of patients with nosocomial wounds infection Number of patients with muscle weakness Number of patients with stroke Number of patients with cardiac events

Repeat for each subgroup

Appendix 6. Unpublished data obtained from trial authors

Studies	Type of unpublished data provided by primary authors
Angus 2020	Full access to individual data, details for randomisation and blinding procedures
Annane 2002	Full access to individual data, details for randomisation and blinding procedures
Annane 2010	Full access to individual data, details for randomisation and blinding procedures
Annane 2018	Full access to individual data, details for randomisation and blinding procedures
Arabi 2011	Full access to individual data
Bollaert 1998	Full access to individual data, details for randomisation and blinding procedures. Additional information on adrenal function (data according to the review definition: delta cortisol ≤ 9 µg/dL). Additional information for ICU length of stay and adverse events
Briegel 1999	Full access to individual data, details for randomisation and blinding procedures. Additional information for ICU length of stay and adverse events
Chawla 1999	Details for randomisation and blinding procedures. Additional information for mortality, shock reversal, ICU length of stay and adverse events
Cicarelli 2007	Details for randomisation and blinding procedures
Confalonieri 2005	Full access to individual data, details for randomisation and blinding procedures
Dequin 2020	Full access to individual data, details for randomisation and blinding procedures
Dequin 2023	Full access to individual data, details for randomisation and blinding procedures
Gordon 2014	Full access to individual data, details for randomisation and blinding procedures
Gordon 2016	Full access to individual data, details for randomisation and blinding procedures
Hyvernat 2016	Full access to individual data, details for randomisation and blinding procedures
Keh 2003	Details for randomisation and blinding procedures. Additional information for adverse events
Keh 2016	Full access to individual data, details for randomisation and blinding procedures
Liu 2012	Full access to individual data, details for randomisation and blinding procedures
Meduri 2007	Details for randomisation and blinding procedures. Additional information for subgroups of patients with sepsis or septic shock on mortality, ICU and hospital length of stay, and adverse events
Meduri 2009	Full access to individual data, details for randomisation and blinding procedures
Meduri 2022	Full access to individual data, details for randomisation and blinding procedures
Meijvis 2011	Separate information for patients with sepsis
Mirea 2014	Full access to individual data, details for randomisation and blinding procedures

(Continued)

Oppert 2005	Details for randomisation and blinding procedures. Additional information for mortality, for outcomes of patients randomised and not analysed, shock reversal and adverse events
Rinaldi 2006	Details for randomisation and blinding procedures. Additional information for mortality, for outcomes of patients randomised and not analysed, and adverse events
Sprung 1984	Additional information for 28-day all-cause mortality
Sprung 2008	Full access to individual data, details for randomisation and blinding procedures
Tandan 2005	Details for randomisation and blinding procedures
Tilouche 2019	Additional information for mortality data at 6 months and 1 year, number of patients with shock reversal at day 28, mean and SD value for SOFA score at day 7, number of patients with at least 1 episode of severe glycaemia > 180 mg/dL
Tongyoo 2016	Full access to individual data, details for randomisation and blinding procedures
Torres 2015	Details for randomisation and blinding procedures. Additional information for mortality, shock reversal, SOFA, length of stay and adverse events
Villar 2020	Details for randomisation and blinding procedures. Separate data for the subgroup of patients with sepsis/septic shock
Yildiz 2002	Details for randomisation and blinding procedures. Additional information for mortality, hospital length of stay and adverse events

FEEDBACK

Feedback, 9 May 2013

Summary

Annane et al in their systematic review of corticosteroids for treating severe sepsis and septic shock concluded “a long course of low dose corticosteroids reduced mortality without inducing major complications” (Annane 2004; updated 2010). This was based on a subgroup analysis, as all doses/durations of corticosteroids for septic shock did not show a benefit in reduction of mortality.

Given that the two largest trials have contradicting results (Annane 2002; Sprung 2008), we were interested in looking into these trials. Upon our review, we feel that the risk of bias assessment for these trials has not adequately addressed the issue of incomplete data. In one trial (Sprung 2008), authors used a per-protocol analysis for their adverse event data; this was subsequently used in this review. Neither the trial nor the review addresses the reasons for this approach. Using per-protocol data is not the preferred method of outcome reporting, as it does not allow for preservation of randomization. Section 14.6.1 of the *Cochrane Handbook for Systematic Reviews of Interventions* specifically asks whether any patients were excluded in reporting of adverse events. Participants who experience unfavourable adverse events may drop out of the trial. When the per-protocol analysis is used, adverse effect results may therefore be biased in favour of steroids. The same scenario can be applied for the placebo group.

In another trial (Annane 2002), we are concerned with the way data were collected and reported for adverse events. Events are reported as being “possibly related to steroids” and “possibly related to vasopressors”. We are unsure as to how one would know whether or not an adverse event was related to the intervention. Neither the review nor the trial specifically outlines the adjusting procedure for determining whether or not adverse events were due to steroids. Patients in the intensive care unit have many risk factors for infection - GI bleeding, psychiatric disorders (e.g. delirium) - therefore it seems inappropriate to try to ascertain whether or not the event was secondary to steroids. Section 14.6.1 of the *Cochrane Handbook for Systematic Reviews of Interventions* explains how clinical trials may have a well-designed method for collecting data for the primary outcome, but in fact may take a retrospective, unblinded approach to collecting adverse event data. An extension of the CONSORT statement for harm also echoes this, recommending that clinical trials should explicitly define how data for adverse events were defined, collected, and analysed (Ioannidis 2004).

As Annane 2002 was a randomized controlled trial, adverse events would not require assessment of whether or not they were thought to be due to treatment. Randomization should take care of confounding factors and thus should be able to show differences (if they truly exist) in adverse events. In our opinion, preference should be given to all-cause adverse events for this reason.

Last, we note that in Analysis 1.12 for "superinfection", the percentage of participants with an event was used instead of the actual number of events in Annane 2002. For example, the number of events for the treatment arm should have been "22", but "15" was used instead.

Given that these issues on selective reporting - Annane 2002 - and incomplete data surrounding the two largest trials of this review have not been adequately addressed in the risk of bias assessments, we find it difficult to conclude at this time on the safety profile of corticosteroid use in treating severe sepsis and septic shock. We look forward to hearing your response to our concerns.

Reply

We are grateful to Dr. Harbin and colleagues for their comments on the Cochrane review on corticosteroids for severe sepsis and septic shock.

Dr. Harbin and colleagues questioned the validity of the concluding statement of this review that corticotherapy was overall well tolerated apart from inducing hyperglycaemia and hypernatraemia. Indeed, they pointed out that Sprung and colleagues reported serious adverse events as per-protocol (Sprung 2008). In fact, in this trial, data for adverse events were reported only for 466 of 499 patients. We have now reported this information in the risk of bias table in a revised version of the review.

Annane and colleagues reported in their main paper the number of participants with any serious adverse events in each treatment arm as per intent-to-treat analysis (Annane 2002). All serious adverse events that were observed were reported in each treatment group. Serious adverse events were further classified according to what was known about complications of corticosteroids or catecholamines. For example, all superinfections, gastroduodenal bleeding, metabolic disorders, and psychiatric disorders that occurred at any time from randomization were reported and further classified in a blinded manner as possibly related to corticosteroids. Thus, no manipulation of data occurred. All serious adverse events were carefully scrutinized and reported in the manuscript, and additional unpublished information (i.e. raw data) was available during preparation of the Cochrane review.

Finally, Dr. Harbin and colleagues highlighted an error in numbers used for Analysis 1.12 (number of participants with superinfection) that is now corrected in the revised version of the review. This modification did not significantly alter the direction and the magnitude of the pooled estimate for evaluation of serious adverse events. Thus, we believe that the conclusion statement - that treatment with corticosteroids in patients with sepsis or septic shock is well tolerated apart from metabolic disorders - is still valid.

Contributors

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Reply:

Djillali Annane

Feedback, 9 November 2017

Summary

1. Annane and colleagues, in their systematic review (Annane 2015), concluded that “a long course of low dose corticosteroids reduced mortality without inducing major complications”. This was based on a subgroup analysis, analysis of outcome 1.4, so we pursued it in greater detail. Upon our review, we found that the forest plot was created using a fixed-effect model. However, the largest trial - Sprung 2008 - carried only 21.5% weight, and the second largest trial - Annane 2002 - was allotted a larger weight of 25.1%. Although we acknowledge that given that I^2 was 16%, a fixed-effect model was chosen, we felt that because these two trials had conflicting results, suggesting clinical heterogeneity, a random-effects model would have been a more conservative and more appropriate approach in this scenario. We took the initiative to re-create the forest plot using a random-effects model, and we found that the treatment effect became not statistically significant. This suggests that perhaps the conclusion is not as simple as “a long course of low dose corticosteroids reduced mortality” in severe sepsis and septic shock.
2. Furthermore, when we reviewed # Summary of findings table 1, we noticed that footnote “a” states that the certainty of evidence for the primary outcome was downgraded due to “1 of the 2 largest trials [showing] no survival benefit”. We feel that this sentence implies that the other largest trial did show a survival benefit, and may mislead the readers to believe so. However, both of the largest trials did not show a statistical benefit for corticosteroids compared to placebo for mortality benefits in sepsis and septic shock (Annane 2002; Sprung 2008). One trial simply showed that there was no increase in mortality, but did not show a reduction in mortality.

3. Based on our findings, we feel that there could be an alternative interpretation of the results of these trials. This review includes studies from 1976 to 2015, and when we look at studies published before 2002, the studies as a group show a more convincing trend toward mortality benefit with corticosteroids. In contrast, studies published after 2002 show more conflicting results. This may suggest that medical therapy today is better for treating sepsis and hence the benefits from corticosteroids are not as apparent as was previously thought. In contrast to the conclusion stated by the review authors, we feel that this review presents inconclusive evidence for mortality benefit with corticosteroid use in septic patients. We look forward to your response.

Reply

1. The planned analysis was to use a fixed-effect model unless heterogeneity across trials could be suspected (i.e. squared I statistic > 30%). We weighted studies by the amount of information they contribute (more specifically, by the inverse variances of their effect estimates). It is also important to highlight that CORTICUS - Sprung 2008 - was terminated prematurely (after 500 participants were recruited out of 800 expected) owing to low recruitment rate and loss of equipoise among investigators. Changing from fixed- to random-effects models did not change the magnitude nor the direction of the point estimate (RR 0.87 vs 0.88) and slightly enlarged the 95% CI 0.78 to 0.97 versus 0.77 to 1.00. Thus, we do not believe that the conclusion from our systematic review was not supported by data analysis, and we disagree about over-interpreting the data.
2. In the trial Annane 2002, the primary outcome was time to death in non-responders to ACTH testing (modified intent-to-treat analysis). The primary analysis for the primary outcome in this trial showed a statistically significant increase in survival time (P = 0.02). The CORTICUS trial - Sprung 2008 - found no significant effect of treatment on mortality. Thus, we do not believe that we have misinterpreted (misreported) findings from the Annane 2002 or Sprung 2008 trial.
3. The pooled RR of dying from trials published before 2002 was 0.90 (95% CI 0.75 to 1.07). The pooled RR of dying from trials published from 2002 was 0.89 (95% CI 0.80 to 1.00). Thus, no evidence suggests that the effect of corticosteroids on mortality differed between trials published before 2002 versus those published since 2002. Finally, new trials have been published since the last update of this review, including Keh et al JAMA 2016; Bi et al PLoSOne 2016; Menon et al Ped CCM 2017; El Nawawy The Pediatric Infectious Disease Journal 2016; Qing-quan Lv et al Am J Emerg Med 2017; and Tongyoo et al. Critical Care 2016; and two large trials are about to be published in the very next future (ADRENAl, n = 3800 and APROCCHSS, n = 1241). Thus, we believe that there is a need to update the review in light of these newly published studies.

Contributors

Summary

Candy Lee, Karen Ng, Shalini Singla, Marco Yeung

Pharmacy Residents

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We do not have any affiliation with or involvement in any organisation with a financial interest in the subject matter of my comment.

Reply

Djillali Annane

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WHAT'S NEW

Date	Event	Description
5 June 2025	New search has been performed	We reran the literature search to 31 December 2023 and identified 26 new eligible trials. The search was updated in December 2024 and 12 trial reports were added to 'Studies awaiting classification' to be assessed and incorporated at the next update.
5 June 2025	New citation required but conclusions have not changed	A new search of the literature revealed 26 additional trials. Cumulative evidence from 72 trials including 24,336 patients found that corticosteroids probably reduced 28-day and in-hospital

Date	Event	Description
		mortality. Corticosteroids may shorten ICU and hospital length of stay. There may be little or no difference in the risk of superinfection or the risk of muscle weakness. Corticosteroids probably increase the risk of hyperglycaemia and hypernatraemia. The effects of continuous versus intermittent bolus administration of corticosteroids are uncertain.

HISTORY

Protocol first published: Issue 3, 2000
Review first published: Issue 1, 2004

Date	Event	Description
25 July 2019	New search has been performed	We reran the search from October 2014 to 31 March 2019. The search was updated again on 25 July 2019.
25 July 2019	New citation required but conclusions have not changed	A new search of the literature revealed 25 additional trials. Cumulative evidence from 48 trials including 11,114 patients confirmed the direction and magnitude of the point estimate for 28-day mortality with narrow confidence interval limits. The review found evidence for a significant reduction in ICU and hospital mortality, and in ICU and hospital length of stay. The review now includes evidence for a reduction in mid-term (90-day) to long-term (up to 1 year) mortality by corticosteroids. We made several changes to the Methods section. • We now include treatment effects on mortality at 90 days, and at longest follow-up beyond 3 months and up to 1 year. • We have included the new Sepsis 3 definition for the selection of type of participants. • We have added new adverse events (i.e. neuropsychiatric events, stroke, and cardiac events). • We have added subgroup analyses based on modes of administration of corticosteroids (e.g. continuous vs bolus administration, mode of drug termination, i.e. stopping with or without taper-off). • We have conducted sensitivity analysis for trials judged at low risk of bias. • We introduced a new comparison that is a direct comparison of continuous infusion vs intermittent bolus of corticosteroids.
14 December 2018	Amended	Editorial team changed to Cochrane Emergency and Critical Care
4 May 2018	Feedback has been incorporated	New Feedback and reply posted in review
18 January 2016	Amended	Typo corrected in plain language summary (it was made clear that corticosteroids decreased the number of organs that were not functioning properly (organ failure))
30 November 2015	New search has been performed	We reran the search from October 2009 to October 2014.
30 November 2015	New citation required and conclusions have changed	A new search of the literature revealed 9 additional trials. Cumulated evidence from 33 trials confirmed the direction and mag-

Date	Event	Description
		nitude of the point estimate for 28-day mortality with narrow confidence interval limits. Thus, this update suggests moderate evidence for reduced 28-day mortality with corticosteroids in the primary analysis. Evidence also confirms significant interactions between the relative risk of dying at 28 days and treatment modalities (lower doses and longer durations yielded better chance of survival) and patient case mix (patients with septic shock, sepsis-related acute respiratory distress syndrome (ARDS) or community-acquired pneumonia may be more likely to benefit from corticosteroids). We decided to change the title to "Corticosteroids for treating sepsis" owing to recent changes in the definition of sepsis, suggesting that the term "severe sepsis" should be avoided. We made several changes to the Methods section. • We now exclude quasi-randomised trials (3 trials). • We changed the definition of "long course" from at least 5 days to at least 3 days. Indeed, and in keeping with the Surviving Sepsis Campaign recommendation, corticosteroids are often given at full dose until cessation of vasopressor therapy, which may occur faster than 5 days. • We changed the definition of "low dose" from 300 mg or less per day to 400 mg or less per day. Indeed, no consensus has been reached on the optimal dose, and several randomised controlled trials testing so-called "low-dose" corticosteroids used variable doses up to 400 mg. • According to findings from meta-regression analysis, changes in the definitions of "low dose" and "long course" might have had a negative impact on the observed survival benefit of corticosteroids. Indeed, we found that both longer duration and lower dose were associated with better survival rates. Sensitivity analyses based on methodological quality are now restricted to the primary outcome. We used random-effects models only in cases of heterogeneity with an I^2 statistic > 30%. Otherwise, we used fixed-effect models.
14 August 2013	Feedback has been incorporated	Feedback was submitted and was responded to. An error in numbers used for Analysis 1.12 (number of participants with superinfection) has been corrected in the amended version of this review. 'Risk of bias' tables and 'Summary of findings' tables have also been amended.
1 November 2010	New search has been performed	• We reran the searches from August 2003 to October 2009. • We found 21 new trials. Of those 21 trials, we included 9 randomised controlled trials in this update (Annane 2010; Cicarelli 2007; Confalonieri 2005; Huh 2007; Meduri 2007; Oppert 2005; Rinaldi 2006; Sprung 2008; Tandan 2005); we excluded 3 (Cicarelli 2006; Kaufman 2008; Mikami 2007), and 9 are ongoing (IRSCTN99675218 2006; NCT00127985 2005; NCT00149123 2005; NCT00368381 2008; NCT00471640 2008; NCT00562835 2008; NCT00625209 2008; NCT00670254 2008; NCT00732277 2008) • Two (Oppert 2002; Sprung 2002) of the 3 previous ongoing studies (Oppert 2002; Sprung 2002; Tayer 2002) have now been

Date	Event	Description
		published and are included in this update as Oppert 2005 and Sprung 2008. The third trial has never been completed, and no data are available. • In total, this updated review now describes 25 included studies, 10 excluded studies, and 9 ongoing studies. • The additional included studies did not change the conclusions of this review. • We included 'Risk of bias' and 'Summary of findings' tables in this updated version. • Search strategies changed from Silver Platter to Ovid. • We changed the statistical analysis by using the random-effects model rather than the fixed-effect model, and we included meta-regression analysis to explore the influence of dose and duration of corticosteroids on risk of death.
25 March 2008	Amended	Converted to new review format.

CONTRIBUTIONS OF AUTHORS

Conceiving the review: Djillali Annane (DA), Eric Bellissant (EB), Pierre Edouard Bollaert (PEB), Josef Briegel (JB), Didier Keh (DK), Yizhak Kupfer (YK), Romain Piracchio (RP), Bram Rochwerg (BR).

Co-ordinating the review: DA.

Undertaking manual searches: DA, PEB, JB, DG, DK, RP, BR.

Screening search results: DA, PEB, JB, DG, DK, YK, RP, BR.

Organising retrieval of papers: DA, PEB, JB, DG, DK, YK, RP, BR.

Screening retrieved papers against inclusion criteria: DA, PEB, JB, DG, DK, YK, RP, BR.

Appraising the quality of papers: DA, PEB, JB, DG, DK, RP, BR, YK.

Abstracting data from papers: DA, JB, DG, RP, BR.

Writing to authors of papers to ask for additional information: DA.

Providing additional data about papers: DA, PEB, JB, DK, YK.

Obtaining and screening data on unpublished studies: DA, PEB, JB, DG, DK, RP, BR, YK.

Managing data for the review: DA, EB, DG, RP, BR.

Entering data into Review Manager (RevMan 2024): DA.

Analysing RevMan statistical data: DA, EB, DG, RP, BR.

Performing other statistical analyses not using RevMan: DA.

Performing double entry of data (data entered by person one: DA; data entered by person two: Laurenne Authelet). For this update, this was done using Covidence with double entry of data by DA, JB, DG.

Interpreting data: DA, EB, PEB, JB, DG, DK, YK, RP, BR.

Making statistical inferences: EB.

Writing the review: DA, EB, PEB, JB, DG, DK, YK, RP, BR.

Securing funding for the review: DA.

Performing previous work that served as the foundation of the present study: DA, EB, PEB, JB, DK, YK.

Serving as guarantor for the review (one review author): DA.

Taking responsibility for reading and checking the review before submission: DA.

Pierre Edouard Bollaert died on 30 January 2024. He contributed to the literature search and to data interpretation up to December 2023. After his death, living authors submitted the updated review and made critical revisions requested thereafter.

DECLARATIONS OF INTEREST

Djillali Annane is the author of the following studies included in this review: Aboab 2008; Annane 2002; Annane 2010; Annane 2018; Sprung 2008. He obtained funds from the French Ministry of Health to conduct the following trials: Annane 2002; Annane 2010; Annane 2018. He has been the chair of the international task force for elaborating 2017 guidelines for the diagnosis and treatment of critical illness-related corticosteroid insufficiency. He is a member of the Cochrane Neuromuscular review group editorial board, but had no role in the editorial processing of this review.

Eric Bellissant is the author of the following studies included in this review: Annane 2002; Annane 2018.

Pierre Edouard Bollaert (author deceased) is the author of the following studies included in this review: Annane 2002; Bollaert 1998. He obtained public funds from the University of Nancy to conduct the trial Bollaert 1998. No declaration of interest form available.

Josef Briegel is the author of the following studies included in this review: Briegel 1999; Keh 2016; Sprung 2008. He obtained public funds to conduct the trial Briegel 1999. He contributed to the international task force for elaborating 2017 guidelines for the diagnosis and treatment of critical illness-related corticosteroid insufficiency. He has participated in the European Society of Intensive Care Medicine, the Deutsche Interdisziplinäre Vereinigung Intensivmedizin, and the Deutsche Gesellschaft für Anästhesie und Intensivmedizin, and he has given lectures and talks on hydrocortisone treatment for septic shock.

David Granton reports no conflicts of interest.

Didier Keh is the author of the following studies included in this review: Keh 2003; Keh 2016; Sprung 2008. He obtained public funds from Charité-Universitätsmedizin Berlin and from the German Federal Ministry of Education and Research to conduct the following trials: Keh 2003; Keh 2016.

Yizhak Kupfer is the author of the following study included in this review: Chawla 1999. He is a member of the Pfizer/BMS speakers' bureau for epixaban. This product has no relationship to steroids in sepsis. He obtained funds from his institution to conduct the trial Chawla 1999.

Romain Pirracchio received funding for International Mobility from the Fulbright Foundation and from the Assistance Publique – Hôpitaux de Paris (APHP). He contributed to the following study: Pirracchio 2023.

Bram Rochwerg is supported by McMaster University Department of Medicine early career research awards. He has contributed to the international task force for elaborating 2017 guidelines for the diagnosis and treatment of critical illness-related corticosteroid insufficiency. He is a methodologist for the American Thoracic Society, European Society of Intensive Care Medicine and the American Society of Haematology.

SOURCES OF SUPPORT

Internal sources

- Hopital Raymond Poincaré, Garches, France

  Help with secretarial work
- University of Versailles Saint Quentin en Yvelines, France

  Logistical support for literature search

External sources

- Department for International Development, UK

  Technical assistance with searching literature and using RevMan web
- Agence Nationale de la Recherche, France

  Within the national Programme d'Investissement d'Avenir, DA obtained funding through the RHU programme: ANR-18-RHUS-0004, and within the national programme France 2030, DA obtained funding through the IHU programme: IAHU-0004-PROMETHEUS
- ERA Per Med, Other

DA and JB obtained funding from the European call ERA Per MEd (JTC 2021): ANR-21-PERM-0005

DIFFERENCES BETWEEN PROTOCOL AND REVIEW

Update 2024

We have made the following changes from the protocol.

- We used Covidence for screening, selecting trials and abstracting data.
- We included the ICEMAN tool to evaluate the credibility of subgroup analyses.
- We included evaluation of a combination of hydrocortisone plus vitamin C and thiamine.
- We included COVID-19-related sepsis as a subgroup of interest.

Update 2019

We have made the following changes from the protocol.

- We now include treatment effects on mortality at 90 days and in the long term (i.e. at longest follow-up available beyond three months). Indeed, with the decline in sepsis-related short-term mortality observed in the past decade, more attention is given to mortality in the longer term. The two most recent and largest trials of corticosteroids for septic shock had all-cause mortality at 90 days as the primary outcome and followed up patients to six months.
- We included the new Sepsis 3 definition in the selection of types of participants. For this update, we found no trial that used the Sepsis 3 definition, as all studies were designed before the new definition was available.
- We added new adverse events (i.e. neuropsychiatric events, stroke and cardiac events). Indeed, with more patients surviving in the short term, adverse events from treatment that may impact patients' long-term outcomes have been given greater attention in sepsis trials. Such adverse events may include muscle weakness, neuropsychiatric events, stroke and cardiac events, and we have added them to this review.
- We added a subgroup analysis based on modes of administration of corticosteroids (i.e. continuous vs bolus administration, stopping with or without taper off). Indeed, over the past decades, increasing use of corticosteroids for sepsis has raised new issues of controversy based on how corticosteroids are administered.
- We conducted sensitivity analysis based on methodological quality by conducting an analysis of all studies judged to be at low risk of bias.
- We introduced a new comparison that is a direct comparison of continuous infusion versus intermittent bolus of corticosteroids.
- We modified the search strategy to increase its sensitivity by adding the search terms "pneumonia", "acute respiratory distress syndrome" and "acute lung injury".

NOTES

This review was initially developed within the Cochrane Infectious Diseases Group and was transferred to the Cochrane Anaesthesia Group in May 2005.

Update 2010

The review was updated in 2010 (Annane 2004). At that time, Cochrane updates did not earn a new citation unless they had new review authors or included a change to the conclusions. Review authors found 21 new trials in 2010. Of those 21 trials, they included nine randomised controlled trials in the 2010 update. Additional included studies did not change the conclusions of this review. Therefore, the 2010 update did not earn a new citation.

Update 2015

The new search of the literature identified nine additional trials. Accumulated evidence from 33 trials confirmed the direction and magnitude of the point estimate for 28-day mortality with narrow confidence interval limits. Thus, this update suggested moderate evidence for reduced 28-day mortality with corticosteroids in the primary analysis and confirmed significant interactions between the relative risk of dying at 28 days and treatment modalities used (lower doses and longer duration yielded better chance of survival) and the patient case mix included (patients with septic shock, sepsis-related ARDS or community-acquired pneumonia may be more likely to benefit from corticosteroids).

We decided to change the title to "Corticosteroids for treating sepsis" owing to recent changes in the definition of sepsis, suggesting that the term "severe sepsis" should be avoided.

We made several changes to the Methods section.

- We excluded quasi-randomised trials (three trials).

- We changed the definition of "long course" from at least five days to at least three days. Indeed, and in keeping with the Surviving Sepsis Campaign recommendation, corticosteroids are often given at full dose until cessation of vasopressor therapy, which may occur sooner than five days. We changed the definition of "low dose" from 300 mg or less per day to 400 mg or less per day. Indeed, no consensus has been reached about what should be the optimal dose, and several RCTs testing so-called "low-dose" corticosteroids used variable doses up to 400 mg. According to findings from the meta-regression analysis, changes in the definitions of "low dose" and "long course" might have had a negative impact on observed survival benefits of corticosteroids. Indeed, we found that both longer duration and lower dose were associated with better survival rates.
- We incorporated information on how we used the GRADE system and how we selected outcomes for the 'Summary of findings' table.

Sensitivity analyses based on methodological quality are now restricted to the primary outcome.

Random-effects models are used only in cases of heterogeneity, with I^2 statistic > 30%. Otherwise, fixed-effect models are used.

INDEX TERMS

Medical Subject Headings (MeSH)

*Adrenal Cortex Hormones [administration & dosage] [adverse effects] [therapeutic use]; Bias; Cause of Death; Length of Stay; Placebos [therapeutic use]; Randomized Controlled Trials as Topic; *Sepsis [drug therapy] [mortality]

MeSH check words

Adult; Child; Humans